Discrete Choice Models

Lecture Slides

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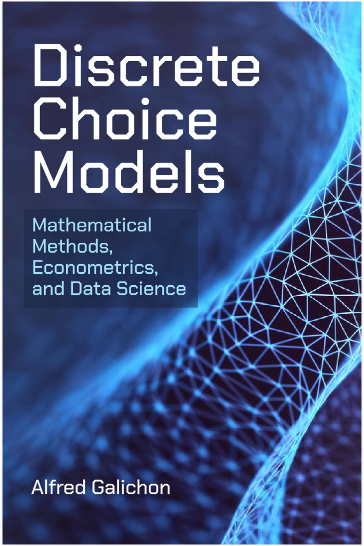

These slides are adapted from Alfred Galichon's book *Discrete Choice Models*, published by Princeton University Press (2026). They include direct excerpts, Python code and figures from that book. All rights to reproduce or transmit this material in any form are reserved. For permissions, contact the book's author. These slides aim to offer a mathematical perspective on discrete choice models and related topics. At the start of each section, we indicate the corresponding pages in the book.

Book page at Princeton University Press

Note on Mathematical Appendices

These slides include, at the end, the mathematical appendices featured in *Discrete Choice Models*, which the reader can refer to for the required mathematical background (college-level mathematics is assumed throughout this course). A basic level of Python is also assumed, as coding is an integral part of the learning experience.

Why should we study discrete choice models?

Discrete choice models (DCM) are used to model situations where the consumer needs to choose one option from among several. They were historically applied to the choice of transportation mode, but the methodology has proved useful in understanding many other decisions. They are based on the traditional idea of individual rationality, where choices result from a simple utility maximization problem that is then aggregated across individuals, which allows us to express the demand (or "market share") of every option.

Why should we study discrete choice models?

However, utilities are most often not observed by the analyst, but can be inferred from the observed choice data (i.e. the "market shares"). DCM are thus non-linear econometric models predicting a categorical outcome—which option is chosen. An attractive feature is the possibility to conduct counterfactual analysis: as demand is estimated, it is possible to predict the change in aggregate demand due to, for instance, the introduction of a new product in a market.

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Overall content

Chapter 1. Theory of random utility models

***Discrete Choice Models*, ch. 1, pp. 15–39**

Content

Random utility models are used to model discrete consumer choices. This is the case when each consumer has to make one choice from a given menu of options.

The econometrician has partial information on consumers, and, therefore, uses a probability model to capture the distribution of their tastes. The purpose of these models is to predict choices consumers will make, analyze the consumer welfare, and therefore, predicting the market shares, using data on observed choices to estimate the models.

Notable applications include: transport mode choice, occupational choices (retirement, type of work...), consumer choice in empirical marketing, IO...

Section 1.1. Setting

Section content

- Random utility model
- Stochastic choice problem
- Welfare function

Definition: random utility model

In a random utility model (RUM), a consumer $i \in [I] = \{1, \ldots, I\}$ has to pick one option $y \in [Y] = \{1, \ldots, Y\}$. The utility of option y for consumer i is

$$U_y + \varepsilon_{iy},$$

where U_y is option-specific, but not consumer-specific and is called *systematic utility*, to which an individual-specific random term ε_{iy} gets added. Consumer i draws $(\varepsilon_{iy})_y$ from a given distribution $\mathcal{P}$.

ε is unknown to the econometrician (and thus called “random”). However, consumer i knows $(\varepsilon_{iy})_y$ with certainty.

Stochastic choice problem

Often we shall add an outside option $y = 0$ to the Y existing ones. In that case the set of options is $[0 : Y] = \{0, ..., Y\}$. We typically normalize $U_0 = 0$.

For a realization of utility shocks ε, the *indirect utility* of the consumer is the utility of the optimal choice,

$$u(\varepsilon) = \max_{y \in [Y] \cup \{0\}} \{U_y + \varepsilon_y\} = \max_{y \in [Y]} \{U_y + \varepsilon_y, \varepsilon_0\}.$$

The welfare function is the sum of the individual welfare across the population. As the total number of individuals is normalized to one, this sum is an expectation over the distribution $\mathcal{P}$. Following McFadden, we view it as a function of the vector of systematic utilities U.

The welfare function

For a given U, the *welfare function* of a consumer is

$$G(U) = \mathbb{E}_{\mathcal{P}}\Big[\max_{y\in[Y]}\{U_y + \varepsilon_y, \varepsilon_0\}\Big]. \tag{1}$$

As an important remark, G is a convex function of U. Indeed, each of the indirect utilities $u(\varepsilon)$ is convex in U. As summation preserves convexity, the integral $G(U)$ is still convex in U.

The welfare function: the binomial case

Generally, G has no closed-form expression. But in the binary case, one can give a compact formula:

Example 1.1: binomial choice model

The binomial choice model usually allows for simpler expressions.

In this case $Y = 1$ and we normalize $\varepsilon_0 = 0$. U and ε are of dimension 1.

Let f and F be the respective p.d.f. and c.d.f. of $\mathcal{P}$.

Then, $G(U_1) = \mathbb{E}[\max\{U_1 + \varepsilon_1, 0\}] = \int_{-U_1}^{\infty} (U_1 + e)\, f(e)\, de$

Integrating by parts and using $\lim_{e \to \infty} e(1 - F(e)) = 0$ yields

$$G(U_1) = \int_{-\infty}^{U_1} (1 - F(-e))\, de.$$

Section 1.2. Demand map and welfare function

Section content

- Demand map
- The Daly-Zachary-Williams theorem

Demand map

We consider the model with $Y+1$ options, one default 0 and Y alternatives $y \in [Y]$.

Take a simple example when $U_y = 0$ for all option $y \in [0:Y]$ and the realization of ε is $(0, \ldots, 0)$. Then the consumer is indifferent between all alternatives, and therefore the demand is ambiguous. Instructed by this example, we want to ensure that indifference between two options happens with zero probability.

Therefore, we need:

Assumption 1.1 (continuity)

The distribution $\mathcal{P}$ of the vector of utility shocks ε has a density with respect to the Lebesgue measure on $\mathbb{R}^{Y+1}$.

As a result, the probability of indifference is zero and the demand map is well-defined.

Market shares

Assumption 1.1 just made allows one to define the *market share map* or *demand map* which maps the systematic utilities to the choice probabilities.

Definition 1.2.1 (market share map)

Under assumption 1.1, the market share of alternative y is the fraction of consumers who choose y, as a function of U. That is:

$$\boldsymbol{\pi}_y(U) := \mathcal{P}(U_y + \varepsilon_y \geq U_z + \varepsilon_z, \ \forall z \in [0{:}Y]).$$

By the continuity assumption, weak inequalities can be replaced by strict ones.

If $\boldsymbol{\pi}(U) = \pi$ for a given $\pi \in \mathbb{R}^{Y+1}$, then $\pi_0 = 1 - \sum_{y \in [Y]} \pi_y$. With $U_0 = 0$, we only need to define $\boldsymbol{\pi}$ as a map from $\mathbb{R}^Y$ to $\mathbb{R}^Y$.

Going back to the binomial example, one can easily express the market share map in that case where there are only two options. We have:

Example 1.1 (continued): binomial choice model

Let F be the *c.d.f.* of ε_1.

$$\pi_1(U_1) = \mathcal{P}(U_1 + \varepsilon_1 \geq 0) = \mathcal{P}(\varepsilon_1 \geq -U_1) = 1 - F(-U_1).$$

The following result is one of the main results of chapter 1. It exhibits a close relationship between the welfare function $G(U)$ and the demand map $\pi(U)$.

Theorem 1.2.1 (Daly-Zachary-Williams)

Under assumption 1.1, the welfare function G is C^1, and its gradient is the market share map $\boldsymbol{\pi}$, that is

$$\boldsymbol{\pi}(U) = \nabla G(U).$$

In words, $\pi_y(U)$, which is the market share of option y, is the derivative of the social welfare $G(U)$ with respect to U_y. We shall provide intuition for this result.

Intuition for the Daly-Zachary-Williams theorem

Suppose that for one given option $y \in [Y]$ we move U_y to $U_y + \delta$ for a small $\delta > 0$, while the other components of U are the same. Then, there are three possibilities for consumers:

(i) Alternative y was not chosen before, and is not chosen after. No welfare impact.

(ii) Alternative y was chosen before, and is chosen after. Impact on overall welfare is the number of consumers impacted times welfare change, i.e.

$$\boldsymbol{\pi}_y(U)\delta.$$

(iii) Alternative y was not chosen before, but is chosen after. Share of the switchers is of order δ, so the change in welfare is of order δ^2.

All in all, one has $G(U + \delta \mathbf{e}_Y^y) = G(U) + \boldsymbol{\pi}_y(U)\delta + O(\delta^2)$, QED.

Example 1.2 : the logit model

Assume (ε_{iy}) are independent across i and y. ε_{iy} follows a centered Gumbel distribution, that is $F(x) = \exp(-\exp(-x))$.
In this case, $\max_y \left\{U_y + \varepsilon_y\right\}$ is distributed as $\log(\sum_y \exp U_y) + \eta$ where $\eta \sim \mathcal{G}$.
With the normalization $U_0 = 0$, we get

$$G(U) = \log\left(1 + \sum_{y\in[Y]} \exp U_y\right).$$

By the Daly-Zachary-Williams theorem,

$$\boldsymbol{\pi}_y(U) = \frac{\exp U_y}{1 + \sum_{z\in[Y]} \exp(U_z)}.$$

Independence of irrelevant alternatives

A particular feature of the Logit Model is the *Independence of Irrelevant Alternatives* (IIA), which can be expressed, among other statements, as:

$$\boldsymbol{\pi}_y(U)/\boldsymbol{\pi}_{y'}(U) \text{ depends only on } U_y \text{ and } U_{y'}.$$

It can be shown that the logit model is the only random utility model that satisfies this property.

However, this is not necessarily a plausible property, as we usually expect other alternatives to affect the previous ratio through substitution. For instance, if there are three goods $Y = 3$, and if good 1 is a substitute for good 2 only, good 2 for both 1 and 3, and good 3 for good 2 only, then it is likely that π_2/π_1 will decrease with respect to U_3, as making option 3 more attractive will compete against option 2, but not against option 1. Such patterns will be common in the characteristics models we shall see in chapter 4.

The link between G and the market share map suggests that G is an important element of RUMs. We state:

Proposition 1.2.1

The welfare function G is convex and submodular.

Proof:

Convexity: by definition, $G(U) = \mathbb{E}_{\mathcal{P}}[\max_{y \in [Y]}\{U_y + \varepsilon_y, \varepsilon_0\}]$. Each function $U \mapsto \max_y\{U_y + \varepsilon_y, \varepsilon_0\}$ is convex in U, and convexity is preserved by taking the expectation over ε; therefore G is convex in U.

Submodularity: the quantity $\pi_y(U)$ is decreasing in U_z, for $z \neq y$. By DZW, this means that $\partial G(U)/\partial U_y$ is decreasing in U_z for $z \neq y$. Therefore G is submodular.

Example 1.2 (continued): The logit model

In the logit model, one can show that $D^2 G(U) = \operatorname{diag}(\pi) - \pi\pi^\intercal$ with $\pi = \nabla G(U)$. This matrix is symmetric positive definite with negative off-diagonal entries (a Stieltjes matrix), which is consistent with the fact that G is convex and submodular.

Section 1.3. Demand inversion and entropy of choice

Discrete Choice Models, sec. 1.3, pp. 20-25

Section content

- Demand inversion problem
- Inverse demand map
- Entropy of choice

What is demand inversion?

Throughout, we assume $Y+1$ options, one default ($y=0$) and Y alternatives.

So far we have focused on the *theorist's problem*: starting from a given vector of systematic utilities (U_y), predict the vector of market shares (π_y).

We now turn to the *econometrician's problem*: observing the vector of market shares (π_y), recover what vector of systematic utilities (U_y) led to it. More formally, we seek to solve the following problem, as defined, page 20:

Market share inversion problem

Under assumption 1.1, letting (π_y) a vector of nonnegative entries such that $\sum_{y\in[Y]}\pi_y \leq 1$, we would like to solve for (U_y) such that $\boldsymbol{\pi}(U)=\pi$, that is, determine

$$\boldsymbol{\pi}^{-1}(\pi)=\left\{U\in\mathbb{R}^Y : \boldsymbol{\pi}(U)=\pi\right\}.$$

Reformulation as an optimization problem

We reformulate the previous problem as an optimization problem. Given $\pi(U) = \nabla G(U)$, we view the equality $\pi = \nabla G(U)$ as the FOC for the maximization of the concave function $U \mapsto \pi^\intercal U - G(U)$, which allows us to state:

Proposition 1.3.1

$U \in \pi^{-1}(\pi)$ if and only if U is a solution to

$$\max_{\tilde{U} \in \mathbb{R}^Y} \{\pi^\top \tilde{U} - G(\tilde{U})\},$$

or in other words, $\pi^{-1}(\pi) = \arg\max_{\tilde{U}} \{\pi^\top \tilde{U} - G(\tilde{U})\}$.

This proposition is useful for computation and also for the analysis of the structure of inverse demand (convexity and lattice structure of $\boldsymbol{\pi}^{-1}(\pi)$).

Existence and uniqueness in inverse demand

Two important questions with inverse demand are:

- **Existence**: is there at least one $U \in \mathbb{R}^{Y}$ such that $\boldsymbol{\pi}(U) = \pi$? in other words, does $\boldsymbol{\pi}^{-1}(\pi)$ have *at least* one element?
- **Uniqueness**: is there at most one $U \in \mathbb{R}^{Y}$ such that $\boldsymbol{\pi}(U) = \pi$? in other words, does $\boldsymbol{\pi}^{-1}(\pi)$ have *at most* one element?

Full support assumption

Intuition: If we go back to the binomial case, then $\boldsymbol{\pi}_1(U_1) = 1 - F(-U_1)$. Then for a given π, inversion means finding U_1 such that $F(U_1) = 1 - \pi_1$, i.e. computing F^{-1}.

However:

• a solution U_1 **may not exist** if there is a jump in the graph of F – that is if $\mathcal{P}$ is not absolutely continuous.

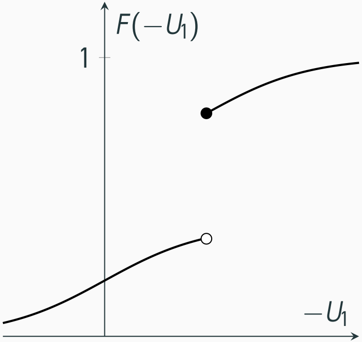
• a solution U_1 **may not be unique** if there is a flat part in the graph of F – that is if $\mathcal{P}$ does not have full support.

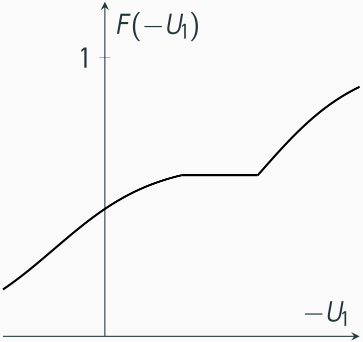
To ensure uniqueness, we will need:

Assumption 1.2 (Full support)

The probability distribution $\mathcal{P}$ of ε has full support.

Existence and uniqueness issues





Left: distribution has a mass point. **Right:** distribution has not full support.

Existence and uniqueness of inverse demand

We state:

Theorem 1.3.1

Let $\pi \in \mathbb{R}^{\mathcal{Y}}$ be such that $\pi_y > 0$ for all $y \in [Y]$ and $\sum_y \pi_y < 1$. Then the set

$$\arg\max_{U \in \mathbb{R}^{\mathcal{Y}}} \{\pi^\top U - G(U)\},$$

- has **at least** one element U under assumption 1.1 (continuity).
- has **at most** one element U under assumption 1.2 (full support).

As a result, under both assumptions 1.1 and 1.2, the map $\boldsymbol{\pi}$ is one-to-one from $\mathbb{R}^{\mathcal{Y}}$ to the set $\{\pi \in \mathbb{R}^{\mathcal{Y}} : \pi_y > 0 \ \forall y \in [Y], \sum_y \pi_y < 1\}$.

We have seen that the inversion of the market share map consisted in looking for the vector U maximizing $\{\pi^\top U - G(U)\}$, which motivates the definition:

Definition 1.3.1 (Generalized entropy of choice)

Let G be the welfare function. The generalized entropy of choice associated with $\mathcal{P}$ is the *Legendre-Fenchel transform of G*, that is:

$$G^*(\pi) = \max_{U \in \mathbb{R}^Y} \left\{ \sum_{y \in [Y]} \pi_y U_y - G(U) \right\}.$$

By the envelope theorem, one has $\nabla G^*(\pi) = U$, where U is the maximizer we are looking for. As a result, $\boldsymbol{\pi}^{-1}(\pi) = \nabla G^*$. More on this later.

Example 1.1 (continued): binomial choice model

In the case of the binomial model, we have $U_1 = -F^{-1}(1-\pi_1)$, and therefore by integration, if F is the c.d.f. of ε_1, one has

$$G^*(\pi_1) = -\int_0^{\pi_1} F^{-1}(1-m)\, dm.$$

In the logit model, the entropy of choice coincides with (minus) the Gibbs entropy. More precisely:

Example 1.1 (continued): logit model

In the logit model, $G^*(\pi)$ is defined for $\pi_y > 0$ such that $\sum_{y \in [Y]} \pi_y < 1$, and we have

$$G^*(\pi) = \sum_{y \in [Y]} \pi_y \log \pi_y + \pi_0 \log \pi_0 = \sum_{y \in [0:Y]} \pi_y \log \pi_y,$$

where π_0 is defined as $\pi_0 = 1 - \sum_{y \in [Y]} \pi_y$. This is (up to a sign flip) the usual definition of the Gibbs entropy associated with the probability distribution (π_y).

The welfare function is recovered from the generalized entropy of choice:

Proposition 1.3.5

We have

$$G(U) = \max_{\pi \in \mathbb{R}^{\mathcal{Y}}} \left\{ \sum_{y \in [Y]} \pi_y U_y - G^*(\pi) \right\},$$

or equivalently by making the domain of G^* explicit,

$$G(U) = \max_{\pi \in \mathbb{R}^{\mathcal{Y}}} \left\{ \sum_{y \in [Y]} \pi_y U_y - G^*(\pi) : \pi_y > 0, \sum_{y \in [Y]} \pi_y < 1 \right\}.$$

Proof. For $U \in \mathbb{R}^{\mathcal{Y}}$ and $\pi \in \mathbb{R}^{\mathcal{Y}}$ in the domain of G^*, $G^*(\pi) \geq \sum_y \pi_y U_y - G(U)$. Thus $G(U) \geq \sum_y \pi_y U_y - G^*(\pi)$, with equality if $\pi = \boldsymbol{\pi}(U)$.

Interpreting the aggregate choice problem

The result in Proposition 1.3.2 recalls a known property in convex analysis: the Legendre-Fenchel transform of the transform recovers the original convex function. This means that the social aggregation of decision-makers' choices with random utility is equivalent to maximizing the sum of systematic utilities $\sum_y \pi_y U_y$, plus a perturbation $G^*(\pi)$ (i.e., minus the entropy of choice). This can be interpreted as the effect of the random utility terms on the aggregate optimization problem.

One can state:

Theorem 1.3.2 (discrete choice inversion)

Let $\pi \in \mathbb{R}^{Y}$ be such that $\pi_y > 0$ and $\sum_{y \in [Y]} \pi_y < 1$. Then under both assumptions 1.1 and 1.2, one has $\pi^{-1}(\pi) = \nabla G^{*}(\pi)$ which is the vector U solution to

$$\max_{U \in \mathbb{R}^{Y}} \{\pi^{\top} U - G(U)\}.$$

Example 1.2 (continued): logit model

In the logit case, we have

$$G^*(\pi) = \begin{cases} \sum_{y\in[Y]} \pi_y \log \pi_y + \pi_0 \log & \text{if } \pi_y > 0 \text{ for all } y \text{ and } \sum_{y\in[Y]} \pi_y < 1 \\ +\infty & \text{otherwise.} \end{cases}$$

where $\pi_0 = 1 - \sum_{y\in[Y]} \pi_y$, and in the case when $G^*(\pi) < +\infty$ with the inequalities above holding strictly, we have

$$(\boldsymbol{\pi}^{-1})_y(\pi) = \frac{\partial G^*(\pi)}{\partial \pi_y} = \log \frac{\pi_y}{\pi_0},$$

where π_0 is defined as $\pi_0 = 1 - \sum_{y\in[Y]} \pi_y$.

An interpretation of G^*

Let U the solution to (4), then:

$$G^*(\pi) = \sum_{y \in [Y]} \pi_y U_y - G(U).$$

Define $y^\star(\varepsilon) \in \arg\max_{y \in [0:Y]} \left\{ U_y + \varepsilon_y \right\}$ as the random variable valued in $[0:Y]$ that is the optimal choice of an agent with a vector of utility shocks ε.

$$G(U) = \mathbb{E}\left[U_{y^\star(\varepsilon)}\right] + \mathbb{E}\left[\varepsilon_{y^\star(\varepsilon)}\right] = \sum_{y \in [Y]} \pi_y U_y + \mathbb{E}\left[\varepsilon_{y^\star(\varepsilon)}\right]$$

Combining these two equations yields

$$G^*(\pi) = -\mathbb{E}\left[\varepsilon_{y^\star(\varepsilon)}\right].$$

Hence, the entropy of choice $G^*(\pi)$ is interpreted as minus the expected amount of heterogeneity needed to rationalize the vector of market shares π.

Galichon and Salanié showed that determining $G^*(\pi)$ is an optimal transport problem:

Theorem 1.3.3 (Inversion theorem)

(i) If $\mathcal{M}(\mathcal{P}, \pi)$ is the set of distributions of r.v.'s $(\varepsilon, \tilde{y})$ s.t. $\varepsilon \sim \mathcal{P}$ and $\tilde{y} \sim \pi$,

$$-G^*(\pi) = \max_{\lambda \in \mathcal{M}(\mathcal{P},\pi)} \mathbb{E}_\lambda \left[\varepsilon_{\tilde{y}}\right],$$

and whose dual formulation is

$$-G^*(\pi) = \min_{u,U} \int u(\varepsilon)\, d\mathcal{P}(\varepsilon) - \sum_{y \in [0:Y]} \pi_y U_y$$

$$s.t.\ u(\varepsilon) - U_y \geq \varepsilon_y \ \forall \varepsilon \in \mathbb{R}^{Y+1}, \forall y \in [0:Y].$$

[*Theorem continued on the next page* ➠]

[➠ *Theorem continued from the previous page*]

Theorem 1.3.3 (continued)

(ii) Under assumption, 1.1 , $U \in \boldsymbol{\pi}^{-1}(\pi)$ iff there exists u such that (u, U) is the unique solution to (1.3.13) s.t. $U_0 = 0$.

Proof. We start from

$$G^*(\pi) = \max_{U \in \mathbb{R}^Y} \left\{ \sum_{y \in [Y]} \pi_y U_y - G(U) \right\}$$

that is

$$G^*(\pi) = \max_{U \in \mathbb{R}^Y} \left\{ \sum_{y \in [Y]} \pi_y U_y - \mathbb{E}_{\mathcal{P}} [\max_{y \in [Y]} \{ U_y + \varepsilon_y \}] \right\}$$

Inversion via optimal transport (3/3)

Proof (continued). However $\max_y\{U_y + \varepsilon_y\}$ is equivalent to $\min\{\tilde{u} \text{ s.t. } \tilde{u} \geq U_y + \varepsilon_y \ \forall y \in [0:Y]\}$. This yields

$$G^*(\pi) = \max_{U,\tilde{u}} \left\{ \mathbb{E}_\pi[U] - \mathbb{E}_{\mathcal{P}}[\tilde{u}(\varepsilon)] \right\}.$$

Finally, letting $V = -U$, we obtain

$$-G^*(\pi) = \inf_{V,\tilde{u}} \left\{ \mathbb{E}_\pi[V] + \mathbb{E}_{\mathcal{P}}[\tilde{u}(\varepsilon)] \right\}$$

which is the solution to the dual of an OT problem with margins π and $\mathcal{P}$.

Section 1.4. Sampling

Section content

- Simulations
- Linear programming formulations

Estimators

Consider a sample of consumers $i \in [I]$ with utility shocks $\varepsilon_i \in \mathbb{R}^{Y+1}$. These are i.i.d draws from the distribution $\mathcal{P}$.

Let $\mathcal{P}_I$ be the empirical distribution associated with this sample, which assigns $1/I$ to each observation. We then define the empirical counterparts to our previous theoretical quantities:

$$\begin{cases} \hat{G}(U) = \frac{1}{I} \sum_{i \in [I]} \max_{y \in [Y]} \left\{ U_y + \varepsilon_{iy}, \varepsilon_{i0} \right\} \\ \hat{\pi}_y(U) = \frac{1}{I} \sum_{i \in [I]} 1 \left\{ U_y + \varepsilon_{iy} \geq U_z + \varepsilon_{iz}, \forall z \in [0{:}Y] \right\}. \end{cases}$$

Simulated demand inversion

We also define $\hat{G}^*(\pi) = \max_U\{\sum_{y\in[Y]} \hat{\pi}_y U_y - \hat{G}(U)\}$ for which Chiong, Galichon and Shum have proven the following result:

Theorem 1.4.1 (Inversion via LP)

Let $\hat{G}^*(\pi)$ be the entropy of choice associated with the empirical distribution $\mathcal{P}_I$.
(i) Then $-\hat{G}^*(\pi)$ is the finite-dimensional optimal transport problem

$$\max_{\lambda_{iy}\geq 0} \quad \sum_{iy\in[I\times 0:Y]} \lambda_{iy}\varepsilon_{iy} \tag{2}$$

$$s.t. \quad \sum_{y\in[0:Y]} \lambda_{iy} = \frac{1}{I} \text{ and } \sum_{i\in[I]} \lambda_{iy} = \pi_y,$$

[*theorem continued on the next page* ➠]

Simulated demand inversion (ctd)

[➠ *theorem continued from the previous page*]

Theorem 1.4.1 (ctd)

whose dual expression is

$$\begin{aligned} \min_{u_i, U_y} \quad & \frac{1}{I} \sum_{i \in [I]} u_i - \sum_{y \in [Y]} \pi_y U_y \\ s.t. \quad & u_i \geq U_y + \varepsilon_{iy}, \quad U_0 = 0. \end{aligned} \tag{3}$$

(ii) The convergence $\hat{G}^* (\pi) \to G^* (\pi)$ holds almost surely as $I \to +\infty$.
(iii) Under assumption 1.1, consider $\pi = \pi (U)$, and let $\left(u^I, U^I\right)$ be any solution to the dual. Then the convergence $U^I \to U$ holds almost surely as $I \to +\infty$.

Section 1.5. Asymptotic theory

Section content

- Large-sample asymptotics
- Central limit theorems

Likelihood

In this section, we investigate the statistical properties of the inversion problem. Assume each consumer i makes a choice y_i after observing a realization of $(\varepsilon_{iy})_y$. As before, the empirical market share of y is simply:

$$\hat{\pi}_y = \frac{1}{I} \sum_{i \in [I]} 1\{y_i = y\}$$

This is a parametric model, with $U \in \mathbb{R}^Y$ as a parameter, for which probability of observing y is $\boldsymbol{\pi}_y(U) = \partial G(U) / \partial U_y$.
The log-likelihood of the sample is:

$$\ell(U) = \sum_{i \in [I]} \log \boldsymbol{\pi}_{y_i}(U) = I \sum_{y \in [0:Y]} \hat{\pi}_y \log \boldsymbol{\pi}_y(U)$$

with $\boldsymbol{\pi}_0(U) = 1 - \sum_{y \in [Y]} \boldsymbol{\pi}_y(U)$.

Maximum likelihood estimation

We obtain the score function by differentiating:

$$\frac{1}{I}\frac{\partial \ell(U)}{\partial U_z} = \sum_{y\in[Y]} \frac{\partial^2 G(U)}{\partial U_y \partial U_z}\left(\frac{\hat{\pi}_y}{\boldsymbol{\pi}_y(U)} - \frac{\hat{\pi}_0}{\boldsymbol{\pi}_0(U)}\right),$$

so $\frac{1}{I}\nabla \ell(U) = D^2 G(U)\,\delta(\hat{\pi}, U)$ where $\delta_y(\hat{\pi}, U) := \frac{\hat{\pi}_y}{\boldsymbol{\pi}_y(U)} - \frac{\hat{\pi}_0}{\boldsymbol{\pi}_0(U)}$, $y \in [Y]$.

The maximum likelihood estimator is such that:$\nabla \ell(U) = 0$, or equivalently $\delta(\hat{\pi}, U) = 0$, that is: $\boldsymbol{\pi}_y(U) = \hat{\pi}_y$. As a result:

Theorem 1.5.1

The maximum likelihood estimator in this model is given by:

$$\hat{U} = \pi^{-1}(\hat{\pi}) = \nabla G^*(\hat{\pi}).$$

Asymptotic properties

To retrieve the asymptotic distribution of $\hat{U}$, we start by finding that of $\hat{\pi}$.

The y_i's are drawn from a multinomial probability distribution with I trials, and $Y+1$ mutually exclusive events associated with probability vector $\pi \in \mathbb{R}^{Y+1}$. By the central limit theorem $\sqrt{I}(\hat{\pi} - \boldsymbol{\pi}) \to \mathcal{N}(0, V_\pi)$, where V_π is obtained by

$$
\begin{aligned}
Cov(\hat{\pi}_y, \hat{\pi}'_y) &= \frac{1}{I^2} \sum_{i,j} Cov(\mathbf{1}\{y_i = y\}, \mathbf{1}\{y_j = y'\}) \\
&= \begin{cases} \frac{1}{I^2} \sum_i V(\mathbf{1}\{y_i = y\}), \text{if } y = y' \\ \frac{1}{I^2} \sum_i Cov(\mathbf{1}\{y_i = y\}, \mathbf{1}\{y_i = y'\}), \text{if not} \end{cases} \\
&= \begin{cases} \frac{1}{I} \boldsymbol{\pi}_y (1 - \boldsymbol{\pi}_y), \text{if y=y'} \\ -\frac{1}{I} \boldsymbol{\pi}_y \boldsymbol{\pi}'_y, \text{if not} \end{cases}
\end{aligned}
$$

That is $V_\pi = diag(\pi) - \pi\pi^\top$.

Asymptotic distribution of inverse demand

We have:

Theorem 1.5.2

Under assumptions 1.1 and 1.2, one has the convergence of $\hat{U}$ towards U at rate $I^{1/2}$ as $I \to +\infty$, and the following convergence in distribution holds

$$I^{1/2}\left(\hat{U} - U\right) \Rightarrow \mathcal{N}\left(0, D^2 G^*\left(\pi\right) V_\pi D^2 G^*\left(\pi\right)\right).$$

This result highlights the role played by the Hessian of the entropy of choice $D^2 G^*$. Given that the map ∇G^* is the inverse map of ∇G, it follows that

$$D^2 G^*\left(\pi\right) = \left(D^2 G\left(U\right)\right)^{-1} \text{ where } \pi = \nabla G\left(U\right),$$

and both these matrices are symmetric positive definite.

Example 1.2 (continued): logit model

In the logit model, for $\pi \in \mathbb{R}^Y$ such that $\pi_y > 0$ for all $y \in [Y]$ and $\sum_{y \in [Y]} \pi_y < 1$, we have

$$\begin{cases} D^2 G(U) = \mathit{diag}(\pi) - \pi\pi^\top \text{ where } \pi = \nabla G(U) \\ D^2 G^*(\pi) = \mathit{diag}(\pi)^{-1} + \mathbf{1}_Y \mathbf{1}_Y^\top / \pi_0, \end{cases}$$

where $\mathbf{1}_Y \in \mathbb{R}^Y$ denotes the column vector of ones of size Y, and $\pi_0 = 1 - \sum_{y \in [Y]} \pi_y$. This shows that in the logit model, one has $D^2 G(U) = V_\pi$. Therefore, the convergence becomes $I^{1/2}(\hat{U} - U) \to \mathcal{N}(0, V_\pi^{-1})$, where $V_\pi^{-1} = \mathit{diag}(\pi)^{-1} + \mathbf{1}_Y \mathbf{1}_Y^\top / \pi_0$.

Section 1.6. Comparative statics

Discrete Choice Models, sec. 1.6, pp. 30-32

Section content

- Weak and strong substitutability
- Isotonicity of inverse demand

Strong substitutes

Substitutability is the idea that if we make an option more attractive, consumers will substitute other options for it, and therefore the demand for other options will decrease. Whether this decrease is strict or weak depends on the assumptions imposed on the model. If the decrease is strict, one speaks of *strong substitutes*. The following result is proposition 1.6.2.

Proposition 1.6.2

Under assumptions 1.1 and 1.2, the *strong substitutes property* holds, which means that for each $y \in [0{:}Y]$, $\pi_y(U)$ is decreasing in U_z for all $z \in [Y]$, $z \neq y$.

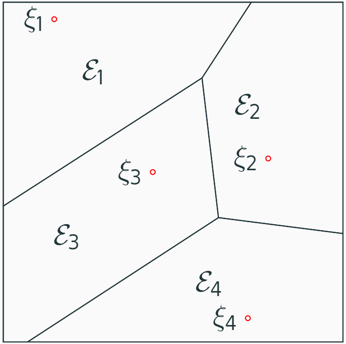
However, assumption 1.2 is sometimes too strong, and does not hold in some important models we shall see, like the pure characteristics model for instance, as seen graphically in the next slide.

As its name indicates, *weak substitutes* is a weaker version of substitute, where "strictly decreasing" is replaced by "weakly decreasing" or "nonincreasing". Weak substitutes is prevalent in every discrete choice models: because consumers choose one item, they necessarily need to substitute one for another one. The following result asserts:

Proposition 1.6.1

Under assumptions 1.1, for each $y \in [Y]$, the market share of y, $\pi_y(U)$ is increasing in U_y and for each $y \in [0:Y]$, $\pi_y(U)$ is nonincreasing in U_z for all $z \in [Y]$, $z \neq y$. That property is called *weak substitutes*.

Strong vs. connected substitutes



Pure characteristics model to be seen in chapter 4: $\varepsilon_y = \epsilon^\top \xi_y$, where $\epsilon = (\epsilon_1, \epsilon_2) \sim \mathcal{U}([0,1])^2$ is uniformly distributed on the square, and ξ_y are vectors in $[0,1]^2$ for each $y \in [4]$. Options $y = 1$ and $y = 2$ are strong substitutes as their demand areas are adjacent. Options $y = 1$ and $y = 4$ are not strong substitutes (no common frontier); however the are connected substitutes since they have joint neighbours, $y = 2$ and $y = 3$.

Inverse isotonicity of demand

We have seen that under the continuity assumption 1.1 and the full support assumption 1.2, the map π satisfied the strong substitutes property. One can show that under the same set of assumptions, it is inverse isotone, which means that for any pair of vectors U and U' in $\mathbb{R}^{Y}$,

$$\pi(U) \leq \pi(U') \implies U \leq U'.$$

The inverse isotonicity of π is an important property because it implies its injectivity (that is, the fact that the inverse demand π^{-1} is defined as a point-valued function). Indeed, $\pi(U) = \pi(U')$ implies inequalities in both directions, and thus $U = U'$. We state:

Theorem 1.6.1

Under assumptions 1.1 and 1.2, the map π is inverse isotone.

We introduce the following assumptions that can be found in the Appendices:

Assumption A.4 (Weak law of aggregate supply)

The aggregate supply $\sum_{x\in[X]} \boldsymbol{q_x}(p)$ is non decreasing in p_x for each $x \in [X]$. Defining $\boldsymbol{q_0}(p) = 1 - \sum_{x\in[X]} \boldsymbol{q_x}(p)$, we can restate the assumption by saying that $\boldsymbol{q_0}(p)$ is non increasing in p_y for each $x \in [X]$.

Assumption A.6 (Connected strong substitutes)

For any $B \subseteq [X]$, and for any p and $p' \in \mathbb{R}^X$ such that $p_x = p'_x$ for all $x \in B$ and $p_x > p'_x$ for $x \notin B$, there exists $x \in B \cup \{0\}$ such that $\boldsymbol{q_x}(p) < \boldsymbol{q_x}(p')$.

Connected substitutes and inverse demand

The connected strong substitutes assumption is a middle ground between weak and strong substitutes. Berry, Gandhi and Haile have proven the following theorem (part (ii) of theorem A.3.9:

Theorem A.3.9 (Berry-Gandhi-Haile)

If π satisfies weak substitutes, the weak law of aggregate supply, and connected strong substitutes, then it is inverse isotone.

The pure characteristics model exemplified in the pure characteristics figure above does not satisfy strong substitutes, but satisfies the assumption of Berry, Gandhi and Haile's theorem. Therefore the demand map is injective in that setting.

Exercises and Quick Recap Chapter 1

Short Exercises

These exercises are selected from *Discrete Choice Models*. The interested reader may find additional exercises and problems at the end of each of the book's chapters.

Exercise 1.1: Consider a choice set with two alternatives ($|\mathcal{Y}| = 2$) and no default option. Assume that the random utility terms coincide, such that $\varepsilon_1 = \varepsilon_2$, and that these terms follow a standard normal distribution, i.e., $\varepsilon_i \sim \mathcal{N}(0, 1)$. Does the distribution P of this random vector satisfy **Assumption 1.1** ? Does it satisfy **Assumption 1.2** ?

Exercise 1.2: Show that the value of the problem $\max_{\pi \in \mathbb{R}_+^Y} \left\{ \sum_{y \in [Y]} \pi_y a_y - \sum_{y \in [Y]} \pi_y \log \left(\frac{\pi_y}{\sum_{\tilde{y} \in [Y]} \pi_{\tilde{y}}} \right) \right\}$ equals 0 if $\sum_{y \in Y} \exp(a_y) \leq 1$, and $+\infty$ otherwise. *(Hint: One can use the expression of G^* in the logit model without a default option.)*

In this first chapter, we introduced Random Utility Models (RUMs), where a consumer's choice is modeled as a function of systematic utility (common to all consumers) and idiosyncratic utility (varying across individuals). Section 1.2 highlighted the Daly-Zachary-Williams theorem, which connects the welfare function's gradient to the market share map. In Section 1.3, we shifted our focus to the demand inversion problem, a key concept in econometrics aiming to recover systematic utilities from observed market shares. This discussion was extended in Section 1.4 with simulation methods for estimating market shares and the empirical welfare function. Finally, Section 1.5 discussed the asymptotic properties of estimators within the RUM framework.

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Chapter 2. The logit model and its generalizations

***Discrete Choice Models*, ch. 2, pp. 39- 63**

Content

The logit model is based on i.i.d. random utility shocks with a Gumbel distribution. This distribution belongs to a more general class of *multivariate extreme value* (MEV) distributions, often called "generalized extreme value" (GEV) in economics since McFadden. This class leads in particular to a closed-form expression for the welfare function $G(U)$.

A prime example of MEV model is the nested logit model, which we will explore in detail. We will cover the two other families of max-stable distributions after the Gumbel, the Fréchet and the Weibull distributions, which we will relate to each other. We shall state the Fisher–Tippett–Gnedenko theorem, which motivates the use of extreme value distributions. Lastly, we will explore the continuous logit model.

Section 2.1. The logit model and the Gumbel distribution

Section content

- The Gumbel distribution
- Max-stability
- The plain logit model

The Gumbel distribution

The standard Gumbel distribution, denoted $\mathcal{G}$, is the distribution with c.d.f.

$$F_{\mathcal{G}}(u) = \exp\left(-\exp\left(-u\right)\right), u \in \mathbb{R}.$$

The mean of a standard Gumbel variable is $\gamma = -\int_0^{+\infty} \exp(-t)\log(t)dt \approx 0.5772...$, the Euler–Mascheroni constant, and its variance is $\pi^2/6$. By extension, the Gumbel family of distributions describes the set of distributions obtained by a location-scale transformation of $\mathcal{G}$.

Max-stability (1/2)

A distribution $\mathcal{P}$ is called *max-stable* if for $\varepsilon_1, ..., \varepsilon_Y$, Y i.i.d. copies of $\mathcal{P}$, there are scalars $a_Y > 0$ and $b_Y \in \mathbb{R}$ such that:

$$\max_{y \in [Y]} \left\{ \varepsilon_y \right\} =_D a_Y \epsilon + b_Y, \text{ where } \epsilon \sim \mathcal{P}.$$

In the case where the errors follow a standard Gumbel distribution, the CDF of $\eta = \max_y \varepsilon_y$ is:

$$F_\eta(u) = \Pr(\varepsilon_y \leq u, \forall y \in [Y]) = F_{\mathcal{G}}(u - \log Y)$$

One can notice that max-stability holds with scaling factor $a_Y = 1$ and shift $b_Y = \log Y$, demonstrating that the Gumbel distribution is max-stable. As a result of the Fisher-Tippett-Gnedenko theorem, the Gumbel is the unique unbounded max-stable distribution.

We state:

Proposition 2.1.1

The standard Gumbel distribution $\mathcal{G}$ is (up to a location-scale transform) the only max-stable distribution whose support is unbounded above and below.

Welfare function in the logit model (1/2)

We thus state:

Proposition 2.1.2

If $\varepsilon_1, \ldots, \varepsilon_Y$ are Y i.i.d. standard Gumbel random variables, then

$$\max_{y \in [Y]} \left\{ U_y + \varepsilon_y \right\} =_D \log \left(\sum_{y \in [Y]} \exp U_y \right) + \epsilon \quad \text{where } \epsilon \sim \mathcal{G}.$$

Proof. If $\eta = \max_y \{U_y + \varepsilon_y\}$, then $F_\eta(x) = \Pr(\forall y : \ \varepsilon_y \leq x - U_y)$, and thus

$$F_\eta(x) = \prod_y \exp(-\exp(U_y - x)) = \exp(-\exp(-x + \mu))$$

where $\mu = \log(\sum_y \exp U_y)$.

We easily deduce the welfare function and the market share map in the case of the i.i.d standard Gumbel distribution.

Theorem 2.1.1

Assume ε_y are Y i.i.d. standard Gumbel random variables. Then the welfare function $G(U) = \mathbb{E}[\max_y\{U_y + \varepsilon_y\}]$ and the market share map $\pi(U) = \nabla G(U)$ are respectively given by

$$G(U) = \log\left(\sum_{y\in[Y]} \exp U_y\right) + \gamma \quad \text{and} \quad \pi_y(U) = \frac{\exp U_y}{\sum_{y\in[Y]} \exp U_y}.$$

Preference for variety

Because each entry ε_y is independent, the logit model implies a “preference for variety.” If there are Y options with the same systematic utility $U_y = U$, the indirect utility is $u(\varepsilon) = U + \log Y + \epsilon$, which increases with Y as the logarithm of Y.

Preference for variety – and the underlying idea that more choice makes consumers more happy overall – is a natural idea, and often empirically grounded.

However, pushing this idea too far can lead to paradoxes, such as the blue bus, red bus paradox in the next slide...

The blue bus, red bus paradox (1/2)

Consider an individual who must choose a transportation mode. They can choose either a train ($y = 0$ with $U_0 = 0$) or a bus ($y = 1$ with U_1). The market share of the train is then $1/(1 + \exp(U_1))$.

Now assume that we segment one of the options according to a criterion that is irrelevant to the choice: we now have a blue bus (option $y = 1b$) and a red bus (option $y = 1r$). Since color does not matter, we give both options the same systematic utility U_1. In this setting, the market share of the train becomes $1/(1 + 2\exp(U_1))$. This is not a desirable feature of the model: we expect the combined market share of both buses to coincide with the market share of the single bus in the original model.

The problem is that we have partitioned one option (bus) with respect to an irrelevant characteristic, and it ended up with twice its actual weight. Effectively, we have increased the systematic utility of taking the bus from U_1 to $U_1 + \log 2$.

This paradox is in reality no paradox; the inherent feature of the logit is a preference for variety. Providing more bus options increases the probability that an agent will draw a sufficiently large utility shock for one type of bus, increasing its total market share. We find a resolution in alternative random utility models, such as the nested logit and characteristics models, which we shall examine later.

Section 2.2. Multivariate Extreme Value generalizations

Section content

- MEV as a factor model
- Definition and properties
- Pickands representation

MEV as a factor model: introduction

The logit model imposes a structure to the random utility shock, in particular, it assumes that the utility shock across two alternatives ϵ_y and $\epsilon_{y'}$ are independent. There are many cases where this is hard to believe, we move by describing a natural class of distribution that embeds logit while removing some of its drawbacks: the MEV, allows for correlation between the random utility terms.

Previous remarks suggest a general way of coupling two or more Gumbel random variables using the entries of η as factors.

MEV as a factor model

Let $(\eta_x)_{x\in[X]}$ an i.i.d draw of a standard Gumbel, and assume $\sum_x \exp U_x = 1$. One has

$$\varepsilon = \max_{x\in[X]} \{U_x + \eta_x\} \sim \mathcal{G}.$$

We may construct another random variable ε' using the same factors η_i, but with a different vector U' such that $\sum_x \exp U'_x = 1$ by

$$\varepsilon' = \max_{x\in[X]} \left\{U'_x + \eta_x\right\} \sim \mathcal{G},$$

We can impose more or less correlation between ε and ε'. Indeed:

- If $U = U'$, then correlation=1.
- If $U_1 = 0$, $U_{-1} = -\infty$, $U'_2 = 0$ and $U'_{-2} = -\infty$: $\varepsilon = \eta_1$ and $\varepsilon' = \eta_2$, correlation=0.
- If U and U' partially overlap, then the correlation is between zero and one.

Example 2.1

In the coding example in appendix B.2.1, we set for all $x \in [X]$

$$U_x = \ln \frac{x^{-\lambda}}{\sum_{x \in [X]} x^{-\lambda}} \text{ and } U'_x = \ln \frac{(1 + X - x)^{-\lambda}}{\sum_{x \in [X]} x^{-\lambda}},$$

and we compute by simulation the correlation of ε and ε'. Exercise 2.5 in DCM p. 55, it is shown that this random utility vector converges in distribution to a single-nest binomial nested logit model, which, as we shall see later, implies that this correlation tends to $1 - \lambda^2$ as $X \to +\infty$.

MEV as a factor model

```
import numpy as np
np.random.seed(7)
O = 1000
X = 10000000
thelambda = .8
epsilon_o = np.zeros(O)
epsilonprime_o = np.zeros(O)

denom = (np.arange(1,X+1) ** (-thelambda) ).sum()
U_x = np.log( (np.arange(1,X+1) ** (-thelambda) )  / denom )
Uprime_x = np.log(  (np.arange(X,0,-1) ** (-thelambda) ) / denom )

rho = 1-thelambda*thelambda
for o in range(O):
    eta_x = - np.log( np.random.uniform(size=X))
    epsilon_o[o] = (U_x + eta_x).max()
    epsilonprime_o[o] = (Uprime_x + eta_x).max()
    if (1+o) % 100 == 0:
        print(np.corrcoef(epsilon_o[:o], epsilonprime_o[:o])[0,1] , rho)
```


Max factors, general case

Let $(\eta_x)_{x\in[X]}$ be i.i.d. Gumbel variables, and consider (U_{xy}) s.t. $\sum_x exp(U_{xy}) = 1$. Defining

$$\varepsilon_y = \max_x \{U_{xy} + \eta_x\},$$

we have $\varepsilon_y \sim \mathcal{G}$ for all $y \in [Y]$.

The dependence and we have

$F_\varepsilon(u) = \Pr(\cap_y \varepsilon_y \leq u_y) = \Pr(\cap_{x,y} U_{xy} + \eta_x \leq u_y) = \Pr(\cap_x \eta_x \leq \min_y \{u_y - U_{xy}\}$, so
$F_\varepsilon(u) = \prod_x \Pr(\eta_x \leq \min_y \{u_y - U_{xy}\}) = \exp(-\sum_x \exp(-\min_y \{u_y - U_{xy}\}))$, so

$$F_\varepsilon(u) = \exp\left(-h\left(\exp(-u)\right)\right) \text{ where } h(b) := \sum_{x\in[X]} \max_{y\in[Y]} \left\{ b_y e^{U_{xy}} \right\}$$

where the function h is positive homogeneous of degree one.

Some remarks

- Note the similarity with the Gaussian case where the add-stability plays the same role as the max-stability in our example.

- We recover the logit case (where the ε_y are i.i.d. Gumbel) when $h(b) = \sum_y b_y$.

Definition 2.2.1 (Multivariate Extreme Value Class)

The distribution of a random vector ε in $\mathbb{R}^{\mathcal{Y}}$ with c.d.f. F_ε belongs to the multivariate extreme value class if $h(b) = -\log F_\varepsilon(-\ln b_1, ..., \ln b_Y)$ is positive homogeneous of degree one. In that case, the function h is called *dependence function* associated with the distribution of ε.

As an example, the function

$$h\,(b) = \sum_{x\in[X]} (\sum_{y\in\mathcal{Y}_x} b_y^{1/\lambda_x})^{\lambda_x}$$

satisfies the previous requirements. This is the dependence function associated with the *nested logit model* (more on this soon).

Computing the c.d.f. of $\max_y \{U_y + \varepsilon_y\}$ and using the positive homogeneity of h, one is led to assert:

Proposition 2.2.1

If the distribution F_ε of the random vector ε in $\mathbb{R}^Y$ belongs to the multivariate extreme value class, with associated dependence function $h(b) = -\log F_\varepsilon(-\log b_1, \ldots, -\log b_Y)$, then $\max_y \{U_y + \varepsilon_y\} = \log h(\exp U) + \epsilon$, where ϵ is a standard Gumbel variable, and therefore

$$\mathbb{E}\left[\max_{y \in [Y]} \left\{U_y + \varepsilon_y\right\}\right] = \log h(\exp(U)) + \gamma.$$

Pickands Representation

By the previous result, $\max_y \{b_y \frac{\exp \varepsilon_y}{\sum_{y'} \exp \varepsilon_{y'}}\} =_D h(b) \frac{\exp \epsilon}{\sum_{y'} \exp \varepsilon_{y'}}$. Defining Z appropriately, one can state:

Proposition 2.2.2

If $F(x) = \exp(-h(\exp x))$ is the distribution of a multivariate extreme value distribution, there is a random vector Z defined on $\mathbb{R}_+^Y$ such that

$$h(b) = \mathbb{E}[\max_{y \in [Y]} \{b_y Z_y\}],$$

which is called the *Pickands representation* of the m.e.v. distribution.

The following result summarizes the applications of the multivariate extreme value distribution class for the random utility models.

Theorem 2.2.1

Assume (ε_y) is a random vector on $\mathbb{R}^Y$ whose distribution $\mathcal{P}$ is multivariate extreme value with a dependence function h. Then the welfare function and the market share map are respectively given by

$$G(U) = \log h(e^U) + \gamma \quad \text{and} \quad \pi_y(U) = \frac{e^{U_y}}{h(e^U)} \frac{\partial h}{\partial b_y}(e^U) .$$

Remark

In the case when the random variables $\varepsilon_y, 1 \leq y \leq Y$ are i.i.d. standard Gumbel variables, we saw that $h(b) = \sum_y b_y$.

The formula $F(x) = \exp(-h(\exp(x)))$ implies that F_ε is the product of functions of x_y, which means that the random variables ε_y are independent. Hence the additivity on $h(b)$ is directly related to the independence between the random utility terms.

If $h(b) = h_1(b) + h_2(b)$ where $h_1(b)$ does not depend on b_y for $y \in A$ and $h_2(b)$ does not depend on b_y for $y \in [Y] \backslash A$, then the random vectors $(\varepsilon_y)_{y \in A}$ and $(\varepsilon_y)_{y \notin A}$ are independent.

Section 2.3. Application: the nested logit model

***Discrete Choice Models*, sec. 2.3, pp. 46-49**

Section content

- Positive stable distributions
- A decomposition lemma
- The nested logit model
- Correlation structure in the nested logit model

Intuition behind the logit model

• Ben-Akiva (1970s): a decision-maker needs to choose between Y options (say travel options) that can be grouped into nests (say transportation mode: train, air, bus, car...)

• The nested logit model assumes that agents pick the transportation mode first, and then pick their final option within the transportation mode.

• In terms of random utility, there is a nest effect, which means that there is positive correlation among the random utilities associated with options in the same nest: random utility associated with airline A is positively correlated with the one associated with airline B, etc.

A decomposition lemma (1/3)

We wish to decompose a standard Gumbel variable $\varepsilon \sim \mathcal{G}$ as:

$$\varepsilon = \alpha\eta + X, \text{ where } \eta \perp\!\!\!\perp X \text{ and } \eta \sim \mathcal{G}$$

where X is a common factor across several Gumbel variables. Say there are three alternatives and (η_1, η_2, η_3) are i.i.d. Gumbel, we can e.g. define $\varepsilon_1 = \alpha\eta_1 + X$, $\varepsilon_2 = \alpha\eta_2 + X$ and $\varepsilon_3 = \eta_3$ so that, denoting $\tilde{U}_{12} = \alpha(\log(\exp(U_1/\alpha) + \exp(U_2/\alpha))$, one has

$$\max\{U_1 + \varepsilon_1, U_2 + \varepsilon_2, U_3 + \varepsilon_3\} = \tilde{U}_{12} + X + \alpha\tilde{\eta} = \max\{\tilde{U}_{12} + \tilde{\varepsilon}_{12}, U_3 + \varepsilon_3\},$$

and $\Pr(1) = \Pr(12|123) \times \Pr(1|12)$, that is

$$\pi_1 = \frac{e^{\tilde{U}_{12}}}{e^{\tilde{U}_{12}} + e^{U_3}} \times \frac{e^{\frac{U_1}{\alpha}}}{e^{\frac{U_1}{\alpha}} + e^{\frac{U_2}{\alpha}}} = \frac{(e^{\frac{U_1}{\alpha}} + e^{\frac{U_2}{\alpha}})^{\alpha}}{(e^{\frac{U_1}{\alpha}} + e^{\frac{U_2}{\alpha}})^{\alpha} + e^{U_3}} \times \frac{e^{\frac{U_1}{\alpha}}}{e^{\frac{U_1}{\alpha}} + e^{\frac{U_2}{\alpha}}}.$$

If indeed we can decompose $\varepsilon \sim \mathcal{G}$ as $\varepsilon = \alpha\eta + X$, where $\eta \perp\!\!\!\perp X$ and $\eta \sim \mathcal{G}$, then the c.d.f. of ε has expression

$$F_{\varepsilon}(u) = \Pr(\alpha\eta + X \leq u) = \mathbb{E}\left[\mathbb{E}[1\{\eta \leq (u - X)/\alpha\} \,|X]\right] = \mathbb{E}[\exp(-\exp(X/\alpha - u/\alpha))],$$

and so letting $Z = \exp(X/\alpha)$ and $t = \exp(-u/\alpha)$, we have $F_{\varepsilon}(u) = \mathbb{E}\exp(-tZ)$, implying that if Z exists, its Laplace transform at t should coincide with the c.d.f. of ε at $u = -\alpha \log t$. Thus we need to have

$$\mathbb{E}\left[\exp(-tZ)\right] = \exp\left(-t^{\alpha}\right),$$

which is precisely the case for the positive stable distribution of parameter α.

The positive stable distribution of parameter α (see appendix A.2.2 in DCM, page 259), denoted $\mathcal{PS}_\alpha$, is such that if $(Z_1, \ldots, Z_I)$ are i.i.d. copies of that distribution, then $Z_1 + \ldots + Z_I =_D I^{1/\alpha} Z$, where $Z \sim \mathcal{PS}_\alpha$. Its Laplace transform is $t \mapsto \exp(-t^\alpha)$.

As a result, we state:

Lemma 2.3.1 (Decomposition lemma)

If $\eta \sim \mathcal{G}$, $Z \sim \mathcal{PS}_\alpha$ for $\alpha \in (0,1]$, and $\eta \perp\!\!\!\perp Z$, then

$$\alpha(\eta + \log Z) \sim \mathcal{G}.$$

Nested logit model: setup

We let Y the total number of options, and X the number of nests. For each option $y \in [Y]$, we denote $x_y \in [X]$ the nest to which y belongs.
For a given $x \in [X]$, we denote $\mathcal{Y}_x$ the set of branches in x. That is the set of $y \in [Y]$ such that $x_y = x$.
For $y \in [Y]$, write

$$\varepsilon_y = \lambda_{x_y} \log Z_{x_y} + \lambda_{x_y} \eta_y,$$

where η_y are i.i.d. Gumbel random variables, $\lambda_x \in (0,1)$, Z_x are i.i.d. random variables distributed as $\mathcal{PS}_{\lambda_x}$, and independent from η.

Correlation of the random utilities in the nested logit model

If y and y' are both in nest x,

$$cov(\varepsilon_y, \varepsilon_{y'}) = \lambda_x^2 var(\log Z_x).$$

Yet

$$var(\varepsilon_y) = \lambda_x^2 (var(\log Z_x) + var(\varepsilon_y)),$$

and therefore $\lambda_x^2 var(\log Z_x) = (1 - \lambda_x^2) var(\varepsilon_y)$. Hence:

Proposition 2.3.1

In the nested logit model, the random utility vector (ε_y) is such that for $y \neq y'$, the correlation between ε_y and $\varepsilon_{y'}$ is $1 - \lambda_x^2$ if y and y' are in the same nest x, and zero if they are in different nests.

The decision maker's indirect utility is $u(\varepsilon) = \max_y \{U_y + \varepsilon_y\}$, which is

$$u(\varepsilon) = \max_{y \in [Y]} \{U_y + \lambda_{x_y} \log Z_{x_y} + \lambda_{x_y} \eta_y\} = \max_{x \in [X]} \left\{ \lambda_x \log Z_x + \max_{y \in \mathcal{Y}_x} \{U_y + \lambda_x \eta_y\} \right\}.$$

By independence between (Z_x) and (η_y), the within-nest x indirect utility equals $\lambda_x \log \sum_{y \in \mathcal{Y}_x} \exp(U_y/\lambda_x) + \lambda_x \log Z_x + \lambda_x \hat{\eta}_x$, where $\hat{\eta}_x$ are i.i.d. standard Gumbel independent from (Z_x). By the decomposition lemma, $\hat{\varepsilon}_x := \lambda_x \log Z_x + \lambda_x \hat{\eta}_x$ are i.i.d. standard Gumbel and

$$u(\varepsilon) = \max_{x \in [X]} \left\{ \lambda_x \log \sum_{y \in \mathcal{Y}_x} \exp(U_y/\lambda_x) + \hat{\varepsilon}_x \right\}$$

which shows that the nests are picked as in a logit model, where the systematic utility associated with nest x is $\lambda_x \log \sum_{y \in \mathcal{Y}_x} \exp(U_y/\lambda_x)$.

Welfare function and entropy of choice, nested logit model (1)

Proposition 2.3.2

(ε_y) has a multivariate extreme value distribution, and

$$G(U) = \log\Big(\sum_{x\in[X]}\Big(\sum_{y\in\mathcal{Y}_x}\exp(U_y/\lambda_x)\Big)^{\lambda_x}\Big)+\gamma \quad \text{and} \quad h(b) = \sum_{x\in[X]}\Big(\sum_{y\in\mathcal{Y}_x} b_y^{1/\lambda_x}\Big)^{\lambda_x},$$

and the corresponding market share map of an alternative y in nest x is

$$\pi_y(U) = \frac{(\sum_{y\in\mathcal{Y}_x}\exp(U_y/\lambda_x))^{\lambda_x}}{\sum_{x'\in[X]}(\sum_{y\in\mathcal{Y}_{x'}}\exp(U_y/\lambda_{x'}))^{\lambda_{x'}}}\frac{\exp(U_y/\lambda_x)}{\sum_{y'\in\mathcal{Y}_x}\exp(U_{y'}/\lambda_x)}.$$

[*proposition continued on the next page* ➠]

Welfare function and entropy of choice, nested logit model (2)

[➠ *proposition continued from the previous page*]

Proposition 2.3.2 (ctd)

The entropy of choice G^* in the nested logit model is given by

$$G^* (\pi) = \sum_{x \in [X]} \lambda_x \sum_{y \in \mathcal{Y}_x} \pi_y \log \pi_y + \sum_{x \in [X]} (1 - \lambda_x) \Big(\sum_{z \in \mathcal{Y}_x} \pi_z \Big) \log \Big(\sum_{z \in \mathcal{Y}_x} \pi_z \Big)$$

if $\pi_y \geq 0$ and $\sum_{x \in [X]} \sum_{y \in \mathcal{Y}_x} \pi_y = 1$, $G^* (\pi) = +\infty$ otherwise. Taking the normalization $G^*(\pi) = 0$, the inverse market share map for an alternative y in nest x is given by

$$\pi_y^{-1} (\pi) = \lambda_x \log \pi_y + (1 - \lambda_x) \log \Big(\sum_{z \in \mathcal{Y}_x} \pi_z \Big).$$

Python code for inverting the nested logit model

Inverting the nested logit model

```
import numpy as np
Y = 10
X = 3
pi_y = np.array([0.05,0.12,0.08,0.21,0.17,0.02,0.14,0.10,0.07, 0.04])
nest_y = np.array([0,2,1,1,0,2,1,2,2,0])
lambda_x = np.array([0.3, 0.5, 0.8])
# computing the fixed effect matrix
in_x_y = 1* (np.arange(X)[:,None] == nest_y[None,:])
# computing the market share of nests
pix_y = (in_x_y @ pi_y) @ in_x_y
lambdax_y = lambda_x @ in_x_y
# inverse market share map:
U_y = lambdax_y * np.log(pi_y) + (1 - lambdax_y) * np.log( pix_y  )
# entropy of choice:
Gstar = lambdax_y * pi_y * np.log(pi_y) + (1 - lambdax_y) * pix_y * np.log( pix_y  )
# displaying results
print('G*=',Gstar)
print('U_y=',U_y)
```

Section 2.4. The three families of max-stable distributions

Discrete Choice Models, sec. 2.4, pp. 49-52

Section content

- The three max-stable families
- The Fisher-Tippett-Gnedenko theorem
- Links between the three families

Other max-stable distributions

We have previously seen with proposition 2.1.1 that the Gumbel distribution is the only max-stable distribution whose support is unbounded both above and below. There are other max-stable distributions whose support is bounded above or bounded below, and it is easy to construct these distributions from Gumbel random variables.
For instance, if the random variable ε has a Gumbel distribution, the distribution of $c + \exp(k\varepsilon)$ for $k > 0$ is max-stable with support bounded below; it is called a Fréchet distribution.

The following result is the basic theorem of extreme value theory.

Theorem 2.4.1 (Fisher–Tippett–Gnedenko)

If $X_1, X_2, \ldots$ are i.i.d. real random variables, and $M_n = \max(X_1, X_2, \ldots, X_n)$ is their running maximum, and if there exist constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that $(M_n - b_n)/a_n$ converges in distribution as $n \to \infty$ to a non-degenerate limiting distribution, then the c.d.f. F of that limiting distribution is (up to a location-scale transform) one of the three max-stable distributions.

The three max-stable families

The three type of max-stable families are:

- Type I (Gumbel): $F(x) = \exp\left(-\exp\left(-x\right)\right), \quad x \in \mathbb{R}$
- Type II (Fréchet): $F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ \exp\left(-x^{-\alpha}\right) & \text{for } x > 0 \end{cases}, \quad \alpha > 0$
- Type III (Weibull): $F(x) = \begin{cases} \exp\left(-(-x)^{\alpha}\right) & \text{for } x \leq 0 \\ 1 & \text{for } x > 0 \end{cases}, \quad \alpha > 0.$

Note that the Gumbel distribution has unbounded support (above and below); the Fréchet distribution has support bounded below but not above; and the Weibull distribution has support bounded below but not above. One can easily transform one into the other.

Functional relationships among random variables with distributions of interests

We provide relationships between $U \sim \mathcal{U}([0,1])$ the uniform, $E \sim \mathcal{E}$ the exponential of parameter 1, $G \sim \mathcal{G}$ the standard Gumbel, F the Fréchet of parameter $\alpha > 0$, and W the Weibull of parameter $\alpha > 0$:

	w.r.t. U	**w.r.t.** E	**w.r.t.** G	**w.r.t.** F	**w.r.t.** W
U	$= U$	$= e^{-E}$	$= e^{-e^{-G}}$	$= e^{-F^{-\alpha}}$	$= \exp\left(-(-W)^{\alpha}\right)$
E	$= -\ln U$	$= E$	$= e^{-G}$	$= F^{-\alpha}$	$= (-W)^{\alpha}$
G	$= -\ln(-\ln U)$	$= -\ln E$	$= G$	$= \alpha \ln F$	$= -\alpha \ln(-W)$
F	$= (-\ln U)^{-1/\alpha}$	$= E^{-1/\alpha}$	$= e^{G/\alpha}$	$= F$	$= -\frac{1}{W}$
W	$= -(-\ln U)^{1/\alpha}$	$= -E^{1/\alpha}$	$= -e^{-G/\alpha}$	$= -\frac{1}{F}$	$= W$

Example 2.2

- The exponential distribution of parameter $\lambda > 0$, which has c.d.f. $f(x) = \lambda e^{-\lambda x}$, $x \in \mathbb{R}_+$, lies in the domain of attraction of the Gumbel distribution.
- The Pareto distribution of parameter $\alpha > 0$, which has c.d.f. $f(x) = \alpha x^{-\alpha-1},\ x \in [1, +\infty)$, lies in the domain of attraction of the Fréchet distribution.
- The uniform distribution on the unit interval, which has c.d.f. $f(x) = 1$ for $x \in [0, 1]$, lies in the domain of attraction of the Weibull class.

Note on Fréchet-based models

Fréchet variables are widely used in international trade to model random productivities. By exponentiating the standard Gumbel max-stability condition, we obtain the **Fréchet magic formula** (which corresponds to a Constant Elasticity of Substitution (CES) aggregation of the parameters V_y)::

$$\max_{y \in [Y]} \left(V_y \eta_y \right) =_D \left(\sum_{y \in [Y]} V_y^{\alpha} \right)^{1/\alpha} \tilde{\eta},$$

where $\eta_y \sim$ Fréchet(α) are i.i.d., V_y are deterministic shifters, and $\tilde{\eta} \sim$ Fréchet(α).

Section 2.5. Continuous logit model

Section content

- Continuous logit model
- Dagsvik-Cosslett's approach

The continuous logit model

The continuous logit model seeks to capture situations where the choice set $\mathcal{Y}$ is no longer the finite set $[Y]$, but is a continuous set, typically $\mathbb{R}^d$. E.g. amount of time worked, for example; or of the age, income or other characteristics of one's marital partner; etc.

The usual logit formulas have obvious extensions to that case, i.e.

$$G(U) = \log\left(\int_{\mathcal{Y}} \exp(U(y))dy\right) + \gamma \quad \text{and} \quad \pi(y) = \frac{\exp(U(y))}{\int_{\mathcal{Y}} \exp(U(y))dy},$$

where dy is the Lebesgue measure $\pi(y)$ is now the density with respect to dy.

However, writing a random utility model to micro-found these formulas is far from obvious. What would a continuum infinite of independent realizations of Gumbel variables mean? further, the indirect utility would be infinite with probability one.

Dagsvik-Cosslett's approach

Assume that the decision-maker draws a random subset of $\mathcal{Y}$, an "opportunity set" of options available to her, along with corresponding random utility shocks she associates with each of these opportunities.

We denote $\{(y_k, \varepsilon_k), k \in \mathbb{N}\}$ a Poisson point process where y_k is some alternative, and ε_k is the corresponding utility. One can verifie that if this Poisson process has intensity $dy \otimes \exp(-\epsilon)d\epsilon$, then we can define $\varepsilon(y)$ by $\varepsilon(y) = \varepsilon_k$ if $y = y_k$, and $\varepsilon(y) = -\infty$ else, so that the random choice problem

$$\max_{y \in \mathcal{Y}} \{U(y) + \varepsilon(y)\} = \max_{k \in \mathbb{N}} \{U(y_k) + \varepsilon_k\},$$

leads to the continuous logit formulas.

Exercises and Quick Recap Chapter 2

Exercise 2.1 (Nested Logit inversion):

Assume that there are three options: one default $y = 0$ and two alternatives $y = 1$ and $y = 2$. We normalize the systematic utility of the default alternative, so $U_0 = 0$. Assume the vector of market shares is $\pi = (\pi_1, \pi_2) = (0.3, 0.5)$.

(i) Compute the systematic utilities (U_y) in the standard logit model.

(ii) Same question if we assume that $y = 1$ and $y = 2$ are in the same nest, and $\lambda = 0.3$.

(iii) Assuming that $y = 1$ and $y = 2$ are in the same nest, but without fixing the value of λ, what is the set of utilities that U can take?

Short exercises

Exercise 2.2: Verify the claims in Example 2.2, showing that the normalizing constants a_n and b_n in the statement of Theorem 2.4.1 are given by:

(i) For exponential random variables with parameter λ:

$$a_n = \frac{1}{\lambda} \quad \text{and} \quad b_n = \frac{\log n}{\lambda}.$$

(ii) For the Pareto distribution with $\alpha > 1$ on $[1, +\infty)$:

$$a_n = n^{1/\alpha} \quad \text{and} \quad b_n = 0.$$

(iii) For the uniform distribution:

$$a_n = \frac{1}{n} \quad \text{and} \quad b_n = 1.$$

Quick recap of Chapter 2

In this chapter, we dealt with the logit model and its probabilistic foundation (Section 2.1) and the theory of extreme values (Section 2.2), which studies limits arising from the maximum of i.i.d. distributed terms. We introduced the nested logit model (Section 2.3) to address the Independence of Irrelevant Alternatives (IIA) property shown by the blue bus, red bus paradox.

We provided (Section 2.4) the fundamental theorem of extreme value theory, Theorem 2.4.1, which asserts that the maximum of i.i.d. random values, after scale transformation, can converge to three possible distributions. We finally explained the challenges of extending the logit to a continuum of alternatives; we dealt with this by assuming that agents draw options with their corresponding utilities according to a Poisson point process.

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Chapter 2: The Logit Model a,d Its Generalization

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Chapter 3. Fundamentals of logistic regression

***Discrete Choice Models*, ch. 3, pp. 63- 87**

Content

In this chapter, we analyze the determinants of individual choices by considering a parametric family of systematic utilities. In the case of the logit model, this will lead us to a fundamental topic: the logistic regression. We will explore existence, computation, asymptotic properties, model selection, as well as various interpretations and links with statistical frameworks.

Besides its interest per se, the logistic regression will be an important building block on which we shall construct equilibrium models introduced later in this book. Many of the ideas we introduce in this chapter shall extend then.

Section 3.1. Parametric random utility models

Section content

- Parametric random utility model
- Choice likelihood and maximum likelihood estimation in choice problems
- Method of moments estimation and entropy methods

Parametric random utility model

Until now, we have assumed that all individuals share the same systematic utilities $(U_y)_y$ and differ only in utility shocks $(\varepsilon_{iy})_{iy}$. We go beyond this by assuming that each consumer i has indirect utility

$$u_i = \max_{y \in [Y]} \{U_{iy} + \varepsilon_{iy}\}$$

there are Y choices (no default option here), and where

- the systematic utility $(U_{iy})_{iy}$ depends both on the consumer and the product, and
- the idiosyncratic utility ε is drawn from a continuous distribution $\mathcal{P}$.

We assume that U_{iy} is linearly parameterized by a K-dimensional vector $\lambda \in \mathbb{R}^K$:

$$U_{iy}(\lambda) = \sum_{k \in [K]} \phi_{iyk} \lambda_k = (\Phi\lambda)_{iy}$$

where Φ is the $(IY) \times K$ matrix, rows indexed by $iy \in [I \times Y]$ and columns by $k \in [K]$.

Example 3.1 (Greene and Hensher)

Greene and Hensher (1997) model travel mode choice from Sydney to Canberra. The options are $Y = \{\text{air, train, bus, car}\}$. The observables may include the income of i, travel duration of y, impact of y on the environment as seen by i. Notice that observables can characterize i or y but also their interactions. For instance, we could include three regressors ($K = 3$) where:

- the first regressor ϕ_{iy1} is the in-vehicle travel time of mode y for individual i,
- the second one ϕ_{iy2} is the in-vehicle travel time times income of individual i, and
- the third one ϕ_{iy3} is the generalized cost of the transportation mode y for agent i.

Likelihood of choice

We aim at estimating the parameter vector λ. Let G be the welfare function associated to $\mathcal{P}$. The probability that i chooses y is

$$\pi_{iy}^{\lambda} := \boldsymbol{\pi}_y\left((\Phi\lambda)_{i.}\right) = \frac{\partial G\left((\Phi\lambda)_{i.}\right)}{\partial U_y},$$

We let y_i denote the choice of i, and define $\hat{\pi}_{iy} = 1\{y = y_i\}$ as the dummy for i chooses y. The log-likelihood that i chooses y_i is $\log \pi_{iy_i}^{\lambda} = \sum_y \hat{\pi}_{iy} \log \pi_{iy}^{\lambda}$, and the sample log-likelihood is

$$\ell(\lambda) := \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \log \pi_{iy}^{\lambda}.$$

Definition 3.1.1 (Maximum Likelihood Estimator)

The maximum likelihood estimator $\hat{\lambda}_{ML}$ is the solution to

$$\max_{\lambda \in \mathbb{R}^K} \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \log \pi_{iy}^{\lambda}.$$

Using the chain rule, the corresponding first order conditions are

$$\sum_{iyz \in [I \times Y \times Y]} \frac{\hat{\pi}_{iy}}{\pi_{iy}^{\lambda}} \frac{\partial^2 G\left((\Phi\lambda)_{i.}\right)}{\partial U_y \partial U_z} \phi_{izk} = 0, \quad \forall k \in [K].$$

Since computing MLE it requires solving a maximization problem which may not be necessarily concave. We investigate an alternative method, the *method of moments*.

Another method of estimating parameters θ consists of equating an empirical moment $\frac{1}{n}\sum_i z_i^s$ with its theoretical counterpart $\mathbb{E}_\theta[Z^s]$ for $s \in \{1, ..., S\}$. In our context, we let:

Definition 3.1.2 (Method of Moments Estimator)

The method of moments estimator $\hat{\lambda}_{MM}$ is the parameter vector $\lambda \in \mathbb{R}^K$ such that

$$\sum_{iy \in [I \times Y]} \pi_{iy}^{\lambda} \phi_{iyk} = \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iyk}, \quad k \in [K].$$

This system of equations can be seen as first order conditions to a convex optimization problem.

The method of moments is computationally attractive because the previous equation can be interpreted as the optimality condition of a convex optimization problem, for which $\hat{\lambda}_{\mathrm{MM}}$ is the solution. We recall that $\pi_{iy}^{\lambda} = \frac{\partial G((\Phi\lambda)_{i\cdot})}{\partial U_y}$, and therefore $\sum_y \pi_{iy}^{\lambda} \phi_{iyk} = \sum_y \frac{\partial G((\Phi\lambda)_{i\cdot})}{\partial U_y} \phi_{iyk}$ is the derivative of $G((\Phi\lambda)_{i\cdot})$ w.r.t λ_k.

Then, the term $((\pi^{\lambda} - \hat{\pi})^{\intercal}\Phi)_k$ arises as the derivative of the convex function $\mathcal{L}(\lambda) = \sum_i G((\Phi\lambda)_{i\cdot}) - \hat{\pi}^{\intercal}\Phi\lambda$ with respect to λ_k.This allows us to formulate the following theorem.

Theorem 3.1.1

The following claims hold under assumption (1.1) (continuity) on $\mathcal{P}$:

(i) the method of moments estimator $\hat{\lambda}_{MM}$ is the solution to

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\pi}^\top \Phi \lambda - \sum_{i \in [I]} G\left((\Phi \lambda)_{i.} \right) \right\},$$

(ii) the value of problem (3.1.7) is equal to the value of its dual

$$\min_{\pi_{iy} \geq 0} \left\{ \sum_{i \in [I]} G^* \left(\pi_{i.} \right) \text{ } s.t. \text{ } \Phi^\top (\pi - \hat{\pi}) = 0 \right\},$$

and $\hat{\lambda}_{MM}$ is the vector of Lagrange multipliers for the latter constraints.

Proof of MME as an optimization problem

Sketch of proof

Since $\pi_{iy}^{\lambda} = \partial G\left((\Phi\lambda)_{i.}\right) / \partial U_y$. Then $\sum_y \pi_{iy}^{\lambda} \phi_{iyk} = \sum_y (\partial G\left((\Phi\lambda)_{i.}\right) / \partial U_y) \phi_{iyk}$ is the derivative of $G\left((\Phi\lambda)_{i.}\right)$ with respect to λ_k. As a result, $((\hat{\pi} - \pi^{\lambda})^{\top}\Phi)_k$ is the derivative of $\hat{\pi}^{\top}\Phi\lambda - \sum_i G\left((\Phi\lambda)_{i.}\right)$ with respect to λ_k.

We know from proposition 1.3.2 that $G(U) = \max_{\pi} \sum_y \pi_y U_y - G^{*}(\pi)$. The Lagrangian for the problem is $\max_{\lambda_k} \min_{\pi_{iy} \geq 0} \left\{ (\hat{\pi} - \pi)^{\top}\Phi\lambda + \sum_{i \in [I]} G^{*}\left(\pi_{i.}\right) \right\}$. Under strong duality, the latter is equal to $\min_{\pi_{iy} \geq 0} \max_{\lambda_k} \left\{ \sum_{i \in [I]} G^{*}\left(\pi_{i.}\right) + (\hat{\pi} - \pi)^{\top}\Phi\lambda \right\}$. The value of inner problem is infinite unless $\Phi^{\top}(\pi - \hat{\pi}) = 0$.

Some remarks

The optimization problem in theorem 3.1.1 defines a *multivariate generalized linear model*. More on generalized linear models soon...

The dual form indicates that the estimation procedure consists of minimizing the sum of the generalized entropies of choice subject to moment conditions. As we define entropy with a minus sign, this procedure is a generalization of the Maximum Entropy Principle (MaxEnt), developed by physicist E.T. Jaynes.

Section 3.2. Multinomial logistic regression

Section content

- The conditional logit model
- The multinomial logistic regression
- Asymptotic properties

The conditional logit model

When the components of ε are independent across options and individuals and follow a centered Gumbel distributed, we get McFadden's conditional logit model. The resulting choice probabilities are

$$\pi_{iy}^{\lambda} = \frac{\exp(\Phi\lambda)_{iy}}{\sum_{z\in[Y]}\exp(\Phi\lambda)_{iz}} \tag{4}$$

and the sample log-likelihood is

$$\ell(\lambda) = \hat{\pi}^{\top}\Phi\lambda - \sum_{i\in[I]}\log\left(\sum_{z\in[Y]}\exp\left(\Phi\lambda\right)_{iz}\right) = \hat{\pi}^{\top}\Phi\lambda - \sum_{i\in[I]} G\left((\Phi\lambda)_{i.}\right)$$

which-*in the logit case only*-is **both** the maximum likelihood estimator and the method of moments one.

The multinomial logistic regression

The multinomial logistic regression is the maximum likelihood estimation procedure (or equivalently here, the moment matching procedure) in the conditional logit model, thus

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\pi}^\top \left(\Phi\lambda - \sum_{i \in [I]} \log \left(\sum_{z \in [Y]} \exp \left(\Phi\lambda \right)_{iy} \right) \right) \right\}.$$

The first order conditions are the ones of the method of moments, i.e.

$$\sum_{iy \in [I \times Y]} \pi_{iy}^{\lambda} \phi_{iyk} = \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iyk}, \forall k \in [K],$$

where we recall the expression of π^λ from equation (4) in the previous slide

$$\pi_{iy}^{\lambda} = \frac{\exp(\Phi\lambda)_{iy}}{\sum_{z \in [Y]} \exp(\Phi\lambda)_{iz}}.$$

Asymptotic properties

Define a $K \times K$ matrix $\hat{B}$ by

$$\hat{B}_{kl} = \frac{1}{I} \sum_{iy \in [I \times Y]} \pi^{\lambda}_{iy} \phi_{iyk} \phi_{iyl} - \frac{1}{I} \sum_{iyz \in [I \times Y \times Y]} \pi^{\lambda}_{iy} \pi^{\lambda}_{iz} \phi_{iyk} \phi_{izl},$$

so that $\hat{B} = I^{-1} \sum_i \mathbb{V}(\tilde{\phi}_i | i)$, where $\tilde{\phi}_i$ is the K-dimensional random vector of regressors $(\phi_{iyk})_k$ for a fixed observation i and when y is distributed according to the true choice distribution π^{λ}_{iy}. As $I \to +\infty$, $\hat{B} \to B = \mathbb{E}[\mathbb{V}(\tilde{\phi})]$ is the Fisher information matrix. As I gets large, the MLE $\hat{\lambda}$ gets close to the true parameter vector, and with expanding the equation from 3.1.2, yields to:

$$\sum_{l\in[K]} B_{kl}(\hat{\lambda}_l - \lambda_l) \approx \frac{1}{I} \sum_{iy\in[I\times Y]} \phi_{iyk}(\hat{\pi}_{iy} - \pi_{iy}), \quad \forall k \in [K]$$

or, in vectorized form,

$$B(\hat{\lambda} - \lambda) \approx \Phi^{\intercal} \frac{\hat{\pi} - \pi}{I}.$$

But as $\mathrm{Cov}(\hat{\pi}_{iy}, \hat{\pi}_{iz}) = \pi^{\lambda}_{iy} \left(\nVdash_{\{y=z\}} - \pi^{\lambda}_{iz}\right)$, and as the average of the vector $\hat{\pi}$ over the observations is asymptotically Gaussian, the asymptotic distribution is $\sqrt{I}(\hat{\lambda} - \lambda) \xrightarrow{d} \mathcal{N}\left(0, B^{-1}\right)$. We then state:

Theorem 3.2.1

As $I \to +\infty$, the convergence $\hat{\lambda} \to \lambda$ holds almost surely; and the convergence

$$\sqrt{I}(\hat{\lambda} - \lambda) \Rightarrow \mathcal{N}\left(0, B^{-1}\right)$$

Section 3.3. Computation with gradient descent

Section content

- Derivatives of the log-likelihood
- Gradient descent

Computation of logistic regression by gradient descent

Recall that

$$\ell(\lambda) = \hat{\pi}^\top \left(\Phi\lambda - \sum_{i\in[I]} \log \left(\sum_{z\in[Y]} \exp (\Phi\lambda)_{iz} \right) \right)$$

and letting $\boldsymbol{\Delta}_\pi := \textit{diag}\left(\pi^\lambda\right)$, one has

$$\nabla\ell(\lambda) = \Phi^\top \left(\hat{\pi} - \pi^\lambda\right).$$

Gradient descent

For an initial choice of λ^0, and a learning rate $\alpha > 0$, iterate over $t \geq 0$

$$\lambda^{t+1} = \lambda^t + \alpha\nabla\ell(\lambda^t) = \lambda^t + \alpha\Phi^\top(\hat{\pi} - \pi^{\lambda^t}),$$

until $\|\nabla\ell(\lambda^t)\|$ falls below some required tolerance level.

Convergence of gradient descent for logistic regression

The learning rate α needs to be appropriately chosen.

- If α is too large, then gradient descent may fail to converge and oscillate forever
- If α is too small, then convergence will be too slow.

The following result provides guidance as to the choice of α:

Proposition 3.3.1

Assume the objective function $\lambda \mapsto \ell(\lambda)$ has a maximizer, and let $\alpha = 1/\left\|\Phi\right\|^2$. Then the gradient descent converges to a maximizer $\hat{\lambda}$ of ℓ.

Section 3.4 Logistic regression as a GLM

Section content

- Poisson regression and generalized linear models
- Fixed effects
- The "Poisson trick"
- Computation using `scikit-learn`

Facts on generalized linear models (1/2)

GLMs are presented in appendix A.2.8

We have Ω observations with $\hat{\mu} = (\tilde{\mu}_\omega)$ = dependent variables and $R = (\tilde{\rho}_{\omega,p})$ = the P explanatory variables. We assign weight w_ω to observation ω and set $W = diag(w)$.

Letting F be a real function, F^* be its convex conjugate, and f be its derivative, the GLM estimator $\hat{\theta}$ is obtained as the solution to

$$\max_{\theta \in \mathbb{R}^P} \{ \hat{\mu}^\top W R \theta - \sum_{\omega \in [\Omega]} w_\omega F^* \left((R\theta)_\omega \right) \},$$

whose dual expression is

$$\begin{aligned} \min_{\mu \geq 0} \quad & \sum_{\omega \in [\Omega]} w_\omega F \left(\frac{\mu_\omega}{w_\omega} \right) \\ s.t. \quad & \mu^\top R = \hat{\mu}^\top R. \end{aligned}$$

Facts on generalized linear models (2/2)

By first order conditions, the estimator $\hat{\theta}$ solution to the primal problem satisfies

$$\sum_{\omega \in [\Omega]} w_\omega \left(f^{-1}((R\hat{\theta})_\omega) - \hat{\mu}_\omega \right) R_{\omega p} = 0 \text{ for all } p \in [P]$$

and the solution $\bar{\mu}_\omega$ to the dual problem satisfies

$$\bar{\mu}_\omega = f^{-1}((R\hat{\theta})_\omega).$$

f is called the *link function*: its inverse provides a predictor $\bar{\mu}_\omega$ based on a score $(R\hat{\theta})_\omega$ that is formed by applying f^{-1} to a linear combination of the regressors.

Setting $f(x) = x$ recovers the (weighted) least square regression.

The Poisson regression as a generalized linear model

The Poisson regression consists of assuming that conditional on the explanatory variables, the dependent variable μ_ω obeys a Poisson distribution of parameter $\lambda_\omega = \exp((R\theta)_\omega)$. (A discrete variable $\tilde{\mu}$ follows a Poisson distribution of parameter λ if for all integer m, $\Pr(\tilde{\mu} = m) = e^{-\lambda}\lambda^m/m!$, in which case $\mathbb{E}[\tilde{\mu}] = \lambda$.)

We recall from appendix A.2.9 that the weighted MLE in the Poisson regression is

$$\max_{\theta \in \mathbb{R}^P} \{\mu^\top W R\theta - \sum_{\omega \in \Omega} w_\omega \exp\left((R\theta)_\omega\right)\}$$

which is a generalized linear model with *log-link function*, i.e. $f(x) = \log x$.

The "Poisson trick" (1/2)

We can interpret the logistic regression problem as a Poisson regression with fixed effects (a special case of GLM). For this, we consider:

Each pair (i, y) as an individual observation: $\Omega = YI$.

The dependent variable is $\tilde{\mu_{iy}} = \pi_{iy}$ (dummy for i chooses y).

The explanatory variable are $\sum_p \tilde{\rho}_{iyp}\theta_p = \sum_k \phi_{iyk}\lambda_k - u_i$, where u_i is an i- fixed effect.

The parameter of the Poisson regression is thus $\theta = (\lambda^T, u^T)^T \in \mathbb{R}^P$ with $P = K + I$.

Recall that for $u \in \mathbb{R}^I$, $((\mathbf{I}_I \otimes \mathbf{1}_Y)u)_{iy} = u_i$, so the design matrix corresponding with the i-fixed effects is $\mathbf{I}_I \otimes \mathbf{1}_Y$. As a result, the matrix of regressors in the Poisson regression is $R = (\Phi, -\mathbf{I}_I \otimes \mathbf{1}_Y)$, where $\otimes$ is the Kronecker product, and $\sum_p \tilde{\rho}_{iyp}\theta_p = (R\theta)_{iy}$.

The "Poisson trick" (2/2)

With this specification, the Poisson regression problem writes:

$$\max_{\lambda\in\mathbb{R}^K, u\in\mathbb{R}^I} \left\{ \hat{\pi}^\top \Phi\lambda - \sum_{i\in[I]} u_i - \sum_{iy\in[I\times Y]} \exp\left((\Phi\lambda)_{iy} - u_i \right) \right\}.$$

Fix λ and maximize over u. By f.o.c., $u_i = \log(\sum_y \exp((\Phi\lambda)_{iy}))$, which is the indirect utility of consumer i. The Poisson regression problem becomes

$$\max_{\lambda\in\mathbb{R}^K} \left\{ \hat{\pi}^\top \Phi\lambda - \sum_{i\in[I]} \log\left(\sum_{y\in[Y]} \exp(\Phi\lambda,)_{iy} \right) \right\}$$

and we have therefore:

Theorem 3.4.1

The multinomial logistic regression can be estimated as a Poisson regression with an individual fixed effect.

Logistic regression, dual formulation

Since we know how to write the dual problem of a Poisson regression, we can retrieve the dual of a logistic regression:

$$
\begin{aligned}
\min_{\pi_{iy} \geq 0} \quad & \sum_{iy \in [I \times Y]} \pi_{iy} \log \pi_{iy} \\
s.t. \quad & \sum_{y \in [Y]} \pi_{iy} = 1 \quad [u_i], \\
& \sum_{iy \in [I \times Y]} \pi_{iy} \phi_{iyk} = \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iyk} \quad [\lambda_k].
\end{aligned}
$$

Note that u_i (indirect utility of i) arises as the Lagrange multiplier of the scarcity constraint $\sum_y \pi_{iy} = 1$, which imposes one choice per consumer.

Section 3.5. Identification

Section content

- Identification

Identification

Logistic regression is "identified" if the regression problem has at most one solution. Essentially, one would like to ensure that $(\Phi\lambda)_{iy} - u_i = (\Phi\lambda')_{iy} - u'_i$ for all $iy \in [I \times Y]$ implies $\lambda = \lambda'$ and $u = u'$.

We state:

Theorem 3.5.1

The logistic regressor $\hat{\lambda}$ is identified if and only if $rank\,(\Phi) = K$, and $Im\Phi$ does not contains any vector of the form $a \otimes 1_Y$ for $a \in \mathbb{R}^I \backslash \{0\}$.

Section 3.6. Existence and the zero-cell problem

Discrete Choice Models, sec. 3.6, pp. 72-74

Section content

- Coercivity
- Zero-cell problem

Coercivity issues

In this section we are concerned with the problem of the existence of a logistic regressor, that is a solution λ to the convex optimization problem

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\pi}^\top \Phi \lambda - \sum_{i \in [I]} \log \left(\sum_{y \in [Y]} \exp(\Phi\lambda)_{iy} \right) \right\}.$$

Until now, we have taken for granted that an optimal λ existed. But all concave functions (even those bounded from above) don't necessarily have a maximizer...

An example of nonexistence

Consider as an example two observations ($I = 2$), two options ($Y = 2$), and one regressor $\Phi_{iy} = 1\{y = 2\}$. In this case, $\hat{\pi}_2 = \sum_{i \in [2]} \hat{\pi}_{i2}$, and the logistic regression becomes

$$\max_{\lambda \in \mathbb{R}} \left\{ \hat{\pi}_2 \lambda - I \log \left(1 + \exp \lambda\right) \right\},$$

and we see that if $\hat{\pi}_2 = 0$, meaning that the second option is not observed, there is no solution to the problem and the logistic regressor does not exist. This comes from the fact that, although concave, the objective function is decreasing for $\lambda \to +\infty$, but not increasing for $\lambda \to -\infty$; the objective is not *coercive*, and as a result, the optimal λ would be $-\infty$, predicting no occurrence of $y = 2$.

Theorem 3.6.1

The following conditions are equivalent:

(i) the logistic estimator (3.2.3) exists,

(ii) there exists $\bar{\pi} \in \mathbb{R}^{I \times Y}$ such that $\bar{\pi}_{iy} > 0$ for all $iy \in [I \times Y]$ and

$$\sum_{i \in [I]} \bar{\pi}_{iy} = \sum_{i \in [I]} \hat{\pi}_{iy} \ \forall y \in [Y] \quad \text{and} \quad \sum_{iy \in [I \times Y]} \bar{\pi}_{iy} \phi_{iyk} = \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iyk} \ \forall k \in [K],$$

[*theorem continued on the next page* ➠]

Theorem 3.6.1

The two previous conditions are also equivalent with:

(iii) the value of the following linear program is finite and strictly positive:

$$
\begin{aligned}
V \quad = \quad & \max_{t \geq 0, \bar{\pi}_{iy} \geq 0} \; t \\
s.t. \quad & \bar{\pi}_{iy} \geq t \\
& \sum_{y \in [Y]} \bar{\pi}_{iy} = \sum_{y \in [Y]} \hat{\pi}_{iy} \; \forall i \in [I] \\
& \sum_{iy \in [I \times Y]} \bar{\pi}_{iy} \phi_{iyk} = \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iyk} \; \forall k \in [K]
\end{aligned}
\tag{5}
$$

Note on linear programming formulation

The next slide presents the linear programming formulation of problem (5). We see from the theorem that stacking new regressors onto the existing ones amounts to adding constraints to program (5). Consequently, the value V of the program decreases. This implies that the less parsimonious the model is, the more likely it is that the logistic estimator does not exist.

Python code for coercivity detector

```
import numpy as np, scipy.sparse as sp, gurobipy as grb
np.random.seed(7)
I,Y,K = 100,5,3
muhat_i_y = np.random.multinomial(1, [1/Y]*Y, size=I)
phi_i_y_k = np.random.normal( size = (I,Y,K))
# Computing:
m=grb.Model()
t = m.addMVar(1)
mubar_i_y = m.addMVar( (I,Y))
m.setObjective(t, sense=grb.GRB.MAXIMIZE)
m.addConstr( mubar_i_y >= t )
m.addConstr( mubar_i_y.sum(axis=1) == muhat_i_y.sum(axis=1) )
m.addConstr( (mubar_i_y[:,:,None]*phi_i_y_k).sum(axis=(0,1) ) == (muhat_i_y[:,:,None]*phi_i_y_k
     ).sum(axis=(0,1) ) )

m.optimize()
# Displaying results:
print('V=', t.X)
```

Section 3.7. Shrinkage and regularization

Section content

- Shrinkage
- LASSO
- Proximal gradient methods

Overfitting

In regression settings, finding the optimal parameters depends on the data used (training data). However, in some cases, including too many features can lead to an estimate that fits the training data well but performs poorly on new datasets, especially when the training data lack variation. That is, we account too much for the specific properties of one dataset, failing to obtain general estimates that reflect the underlying distributions rather than the particular data. This undesirable phenomenon is called **overfitting**.

This typically takes place when there are too many parameters for too little data or not enough variation in training data.

Shrinkage

A common method to prevent over-fitting is to force the estimates to remain contained in certain sets that are not too large. That is by imposing further restrictions on the estimation problem. *Shrinkage estimators* search among **bounded** parameter vectors, $\lambda \in \mathbb{R}^k$, that are "localized", not "too large" in the sense that they maximize the same objective function as the multinomial logistic regression, but on a restricted set of parameters:

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\pi}^\top \Phi \lambda - \sum_{i \in [I]} \log \left(\sum_{y \in [Y]} \exp(\Phi\lambda)_{iy} \right) \right\}$$

$$s.t. \; \varphi(\lambda) \leq t$$

where φ is a convex function, often based on a norm, called *regularizing function*. The resulting estimator is called shrinkage estimator.

Choosing the regularization function

Common choices are:

- the squared L2 norm (a.k.a. Euclidian norm) $|.|_2$, which is such that $\varphi(\lambda) = |\lambda|_2^2 = \sum_k \lambda_k^2$, this is the called the *ridge regularization*, or *Tikhonov regularization*.
- the L1 norm $|.|_1$ such that $|\lambda|_1 = \sum_k |\lambda_k|$, which is called the *least absolute shrinkage and selection operator* (LASSO) *regularization*.
- a combination between the previous two: the *elastic net regularization* is such that $\varphi(\lambda) = \kappa \, |\lambda|_2^2 + (1 - \kappa) \, |\lambda|_1$.

Regularized logistic regression

We write the problem in the Lagrangian form

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\pi}^\top \Phi \lambda - \sum_{i \in [I]} \log \left(\sum_{y \in [Y]} \exp(\Phi\lambda)_{iy} \right) - \gamma \varphi(\lambda) \right\}$$

where γ is the Lagrange multiplier of the constraint $\varphi(\lambda) < t$, which is also the smallest scalar so that the solution λ to this problem satisfies the shrinkage constraint.

When φ is differentiable, the optimality conditions are

$$\Phi^\top (\hat{\pi} - \pi^\lambda) = \gamma \nabla \varphi(\lambda).$$

In general, the optimality conditions are

$$\Phi^\top (\hat{\pi} - \pi^\lambda) \in \gamma \partial \varphi(\lambda),$$

where π_{iy}^λ is the predicted choice probability, and $\partial \varphi(\lambda)$ is the subdifferential of φ at λ.

Optimality conditions, ridge

For the ridge regularization (L2 norm), we get

$$\sum_{iy} \phi_{iyk} (\hat{\pi}_{iy} - \pi^{\lambda}_{iy}) = 2\gamma\lambda_k$$

so the ridge function introduces a discrepancy between theoretical and empirical moments, which is what we need since we want to avoid being too precise with the training data.

That discrepancy is increasing in γ.

Optimality conditions, LASSO

The LASSO estimation problem $\max_{\lambda\in\mathbb{R}^K}\left\{\ell(\lambda)-\gamma\left|\lambda\right|_1\right\}$ has optimality conditions

$$\begin{cases}\lambda_k>0\implies\frac{\partial\ell(\lambda)}{\partial\lambda_k}=\gamma\\ \lambda_k<0\implies\frac{\partial\ell(\lambda)}{\partial\lambda_k}=-\gamma\\ \lambda_k=0\implies\frac{\partial\ell(\lambda)}{\partial\lambda_k}\in[-\gamma,\gamma]\end{cases}$$

Thus the discrepancy between predicted and observed moments do not exceed γ in absolute value. One sees from the optimality conditions that LASSO regularization creates sparsity: it pushes "irrelevant" features out of the model, by setting the corresponding entries of λ to zero.

Proximal gradient descent

Assume for now φ is smooth. We consider two variants of gradient descent:

$$\begin{aligned}\text{explicit scheme} &: \quad \lambda^{t+1} = \lambda^t + \alpha \nabla \ell(\lambda^t) - \alpha\gamma\nabla\varphi(\lambda^t), \\ \text{semi-implicit scheme} &: \quad \lambda^{t+1} = \lambda^t + \alpha \nabla \ell(\lambda^t) - \alpha\gamma\nabla\varphi(\lambda^{t+1}).\end{aligned}$$

Define $\tilde{\lambda}^{t+1} = \lambda^t + \alpha\nabla\ell(\lambda^t)$. The semi-explicit form is equivalent to an optimization problem

$$\lambda^{t+1} = \arg\min_{\lambda} \left\{ \frac{1}{2} \left| \tilde{\lambda}^{t+1} - \lambda \right|^2 + \alpha\gamma\varphi(\lambda) \right\}$$

which remains defined even when φ is nonsmooth.

Soft-thresholding

In the case $\varphi = |.|_1$, the solution can be computed exactly from $\tilde{\lambda}^{t+1}$ by the *soft-thresholding operator*, defined as

$$\lambda_k^{t+1} = \begin{cases} 0 & \text{if } \tilde{\lambda}_k^{t+1} \in [-\alpha\gamma, \alpha\gamma] \\ \tilde{\lambda}_k^{t+1} - \alpha\gamma & \text{if } \tilde{\lambda}_k^{t+1} \geq \alpha\gamma \\ \tilde{\lambda}_k^{t+1} + \alpha\gamma & \text{if } \tilde{\lambda}_k^{t+1} \leq -\alpha\gamma, \end{cases}$$

Unlike gradient descent for LASSO, the proximal gradient algorithm converges:

Proposition 3.7.1

The proximal gradient descent algorithm defined by the semi-implicit scheme converges to the solution to the regularized optimization problem of logistic regression.

Cross-validation

A standard approach to choose γ (model selection) is cross-validation:

- Split the dataset into training and validation subsets.
- Retrieve $\lambda(\gamma)$ for different values of γ based on the training data.
- Use each $\lambda(\gamma)$ to make predictions on the validation data.
- Choose γ^* that minimizes the discrepancy between predictions and observations within the validation dataset.

Section 3.8. Minimax-regret and small-noise limit

Section content

- Scaling the random utility shocks
- Zero-noise limit and minimax-regret estimation
- Small-noise domain and the log-sum-exp trick

Scaling the random utility shocks (1/3)

We consider the logit model with a factor σ to scale the Gumbel random utility:

$$U_{iy} + \sigma \varepsilon_{iy}.$$

As before, we will take a linear parameterization of U and we wish to understand what happens in the estimation problem as we approach the zero-noise limit: $\sigma \to 0$.

Note that in order to have non trivial results, we need to normalize (e.g. to 1) one of the coefficients λ_k. Otherwise, the model with utility $\sum_k \phi_{iyk}\lambda_k + \sigma\varepsilon_{iy}$ is equivalent to a model with utility $\sum_k \phi_{iyk}(\lambda_k/\sigma) + \varepsilon_{iy}$. $\implies$ λ and σ are not simultaneously identified.

We denote by 0 the index of the normalized regressor, so the utility specification becomes

$$\phi_{iy0} + \sum_{k \in [K]} \phi_{iyk}\lambda_k + \sigma\varepsilon_{iy},\ \left(\varepsilon_y\right) \sim \text{i.i.d. Gumbel}.$$

We have assumed that the Y-dimensional random vector (ε_y) is i.i.d. Gumbel in order to remain in the context of the logistic regression. For simplicity, we still view Φ as a $IY \times K$ matrix, so we don't include the reference regressor in the matrix Φ.

The utility term can therefore be written in matrix notation as $\phi_{iy0} + (\Phi\lambda)_{iy} + \sigma\varepsilon_{iy}$.

Using the same arguments as before, the MLE is the solution to

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iy0} + \hat{\pi}^\top (\Phi\lambda) - \sigma \sum_{i \in [I]} \log \left(\sum_{y \in [Y]} \exp \frac{\phi_{iy0} + (\Phi\lambda)_{iy}}{\sigma} \right) \right\}.$$

Defining $\tilde{\lambda} = \lambda/\sigma$, this is equivalent with

$$\max_{\tilde{\lambda} \in \mathbb{R}^K} \{ \hat{\pi}^\top \Phi \tilde{\lambda} - \sum_{i \in [I]} \log(\sum_{y \in [Y]} \exp(\frac{\phi_{iy0}}{\sigma} + (\Phi\tilde{\lambda})_{iy})) \}.$$

Scaling the random utility shocks (3/3)

Again this can be re-written in a GLM form as

$$\max_{\tilde{\lambda}\in\mathbb{R}^K, u\in\mathbb{R}^I} \left\{ \hat{\pi}^\top \left(\Phi\tilde{\lambda} - (\mathbf{I}_I \otimes \mathbf{1}_Y)u\right) - \sum_{iy\in[I\times Y]} \exp\left(\frac{\phi_{iy0}}{\sigma} + (\Phi\tilde{\lambda})_{iy} - u_i\right) \right\}.$$

which is a weighted Poisson regression of $\tilde{\pi}_{iy}$ on regressors Φ, individual fixed effects and weights $w_{iy} = exp(\phi_{iy0}/\sigma)$. The dual to the previous expression is

$$\begin{aligned}
\min_{\pi_{iy}\geq 0} \quad & \sum_{iy\in[I\times Y]} \pi_{iy} \log \pi_{iy} - \sum_{iy\in[I\times Y]} \pi_{iy} \log w_{iy} \\
s.t. \quad & \sum_{iy\in[I\times Y]} \pi_{iy}\phi_{iyk} = \sum_{iy\in[I\times Y]} \hat{\pi}_{iy}\phi_{iyk}, \\
& \sum_{y\in[Y]} \pi_{iy} = 1,
\end{aligned}$$

where λ_k and u_i are the Lagrange multipliers of the first and second sets of constraints.

Small noise domain and the log-sum-exp trick

When σ is small, the terms in ϕ/σ in the exponential may become large positive numbers, leading the exponential to "blow up." We can use the "log-sum-exp trick" and based on the idea that $\log(\exp(a-t)+\exp(b-t)) = \log(\exp(a)+\exp(b)) - t$. Taking $t = \max(a,b)$ prevents a blow up.

Introducing $m_i = \max_y\{(\Phi_0+\Phi\lambda)_{iy}\}$ the following expressions are equivalent but numerically more stable, for the log-likelihood

$$\hat{\pi}^\top(\Phi_0+\Phi\lambda) + \sum_{i\in[I]} m_i - \sigma \sum_{i\in[I]} \log\left(\sum_{y\in[Y]} \exp\frac{(\Phi_0+\Phi\lambda)_{iy}-m_i}{\sigma}\right)$$

and for the predicted choice probability

$$\pi^\lambda_{iy} = \frac{\exp\frac{(\Phi_0+\Phi\lambda)_{iy}-m_i}{\sigma}}{\sum_{z\in[Y]}\exp\frac{(\Phi_0+\Phi\lambda)_{iz}-m_i}{\sigma}}.$$

Zero-noise limit of discrete choice models

When $\sigma \to 0$, the previous primal program converges to (after a change of signs)

$$\min_{\lambda \in \mathbb{R}^K} \sum_{i \in [I]} \max_{y \in [Y]} \{\phi_{iy0} + (\Phi\lambda)_{iy}\} - \sum_{i \in [I]} (\Phi\lambda)_{iy_i}.$$

whose dual is

$$\begin{aligned}
\max_{\pi_{iy} \geq 0} \quad & \sum_{iy \in [I \times Y]} \pi_{iy} \phi_{iy0} \\
s.t. \quad & \sum_{iy \in [I \times Y]} \pi_{iy} \phi_{iyk} = \sum_{iy \in [I \times Y]} \hat{\pi}_{iy} \phi_{iyk} \quad [\lambda_k] \\
& \sum_{y \in [Y]} \pi_{iy} = 1.
\end{aligned}$$

and $\hat{\lambda}^\sigma$ converges to a vector $\hat{\lambda}^0$, solution to the limiting problem above.

Minimax-regret estimator

The *regret* of individual i with respect to option y under a parameter λ is equal to the utility obtained if choosing y minus utility obtained in actual choice y_i, that is

$$\rho_{iy}^{\lambda} = (\phi_{iy0} + (\Phi\lambda)_{iy}) - (\phi_{iy_i0} + (\Phi\lambda)_{iy_i}),$$

and the previous problem can be expressed as a *minimax regret procedure*

$$\min_{\lambda \in \mathbb{R}^K} \sum_{i \in [I]} \max_{y \in [Y]} \rho_{iy}^{\lambda},$$

thus:

Theorem 3.8.1

As $\sigma \to 0$, the maximum likelihood estimator $\hat{\lambda}^{\sigma}$ in the model with scale σ converges to a solution of the minimax regret procedure.

Exercises and Quick Recap Chapter 3

Quick exercises (1/2)

Exercise 3.1: The multinomial logistic regression is sometimes presented under a more restrictive form, which assumes that the utility provided by option y to the decision maker i is: $U_{iy} = \sum_{l \in [L]} X_{il} \theta_{ly}$

or, in matrix form, $U = \mathrm{vec}(\mathbf{X}\Theta)$, where U is a column vector of size IY, $\mathbf{X}$ is an $I \times L$ matrix of full rank, and Θ denotes the $L \times Y$ matrix of terms θ_{ly}. In this specification, the utility associated with each option y is controlled by a different set of parameters, which corresponds to the column y of matrix Θ.

(i) Letting $\lambda = \mathrm{vec}(\Theta)$, which is a column vector of size LY, show that $U = \Xi\lambda$, where $\Xi = \mathbf{X} \otimes I_Y$.

(ii) If R is the $(IY) \times (LY + I)$ matrix given by $R = [\Xi \quad -I_I \otimes \mathbf{1}_Y]$, show that the rank deficiency of R is L, or, in other words, that the rank of R is $LY + I - L$.

Quick exercises (2/2)

Exercise 3.2: Assume that $I = 2$, $Y = 2$, and $K = 2$, and let the regressor vectors be: $\phi_{iy1} = \begin{pmatrix} 0 & 2 & 0 & 3 \end{pmatrix}$ and $\phi_{iy2} = \begin{pmatrix} 4 & 2 & -3 & 0 \end{pmatrix}$.

Does identification hold? Assume the observed choice probabilities are $\hat{\pi} = \begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix}$. Does a logistic estimator exist?

In this chapter, we focused on logistic regression: its estimation, inference, and computation. In Section 3.1, we introduced a RUM where the utility of choosing a specific option is linearly parameterized. This formulation allows for the modeling and estimation of the systematic part of utility via MLE or the method of moments. The former is statistically efficient, while the latter is computationally attractive.

In Section 3.2, we studied the conditional logit model by assuming ϵ_i follows a Gumbel distribution. This forms the basis of logistic regression, and the MLE procedure leads to the multinomial logistic regression estimator. We then explored its connection to GLMs and showed that it can be estimated using a Poisson regression with individual fixed effects.

Quick recap of Chapter 3 (2/2)

Section 3.5 discussed the necessary conditions for the unique identification of the logistic regression parameters, explaining that identification issues arise from model misspecification or specific data configurations. Section 3.6 then derived necessary and sufficient conditions for the existence of a solution.The last part of the chapter addressed overfitting and improving model generalization using regularization techniques, such as LASSO or Ridge. Finally, Section 3.8 examined the zero-noise limit of the logistic regression problem i.e., when the scale of the unobserved heterogeneity (the Gumbel error term) tends to zero.

Bibliography

Chapter 3: Fundamentals of Logistic Regression

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Chapter 4. Characteristics-based models

***Discrete Choice Models*, ch. 4, pp. 87- 121**

Content

As we saw previously, models based on MEV distribution offer a first escape from the limitations of the standard logit model. Recall the apartment demand example, where one nest contains small apartments and another large ones, with individuals refining their choices within each nest.
However, the distinction between nests is often arbitrary and does not allow for modeling continuous trade-offs between size and other amenities. In this chapter, we develop a more flexible framework for modeling random utility: the random coefficient logit model. We show that this model can be conveniently simulated and inverted. In this framework, goods are defined by multidimensional characteristics, and their interaction with consumers' individual attributes generates utility. This also enables us to predict demand for new goods.

We leverage this flexibility and invertibility to perform demand estimation while accounting for price endogeneity. This constitutes the essence of the Berry, Levinsohn, and Pakes (BLP) method.

Section 4.1. The pure characteristics model

Section content

- The Lancasterian framework
- Calculation of the welfare function
- Calculation of the entropy of choice

The Lancasterian framework

The logit model imposes a very rigid structure on random utilities. Nested logit models address some limitations of the standard logit by grouping similar options into nests, but such groupings are arbitrary and restrict modeling trade-offs between attributes. A more flexible alternative is the *characteristics approach*, where utility depends on how consumers value each attribute, as in Lancaster's 1960s framework. This allows for richer predictions, including for new goods—something traditional models fail at.

Let ξ_y be the M-dimensional vector of characteristics of option y, whose entries are ξ_{ym}; η_i is the vector of valuations of the respective characteristics by agent i and has a distribution $\mathbf{P}_\eta$.

An agent $i \in [I]$ draws a utility shock ε_{iy} associated with option y given by

$$\varepsilon_{iy} = \sum_{m \in [M]} \eta_{im} \xi_{ym}.$$

Hence, if η stands for the M-dimensional random vector of distribution $\mathbf{P}_\eta$, and if $\varepsilon \sim \mathcal{P}$ denotes the Y-dimensional random vector of utility shocks associated with each option, then one has

$$\varepsilon = \boldsymbol{\xi} \eta$$

where $\boldsymbol{\xi}$ is the $Y \times M$ matrix of term ξ_{ym}.

Probit example and uniform cubic example

Based on the distribution of η, we have two different examples:

Example 4.1 : the probit model

When $\mathbf{P}_\eta$ is the $\mathcal{N}(0,\Sigma)$ distribution, then the pure characteristics model is called a Probit model; in this case, $\varepsilon \sim \mathcal{N}(0, \boldsymbol{\xi}\Sigma\boldsymbol{\xi}^\top)$.

Example 4.2 : the uniform cubic model

When $\mathbf{P}_\eta = \mathcal{U}([0,1]^M)$, namely when the components η_m of the random vector $\eta \sim \mathbf{P}_\eta$ are independent and all uniformly distributed over the unit interval.

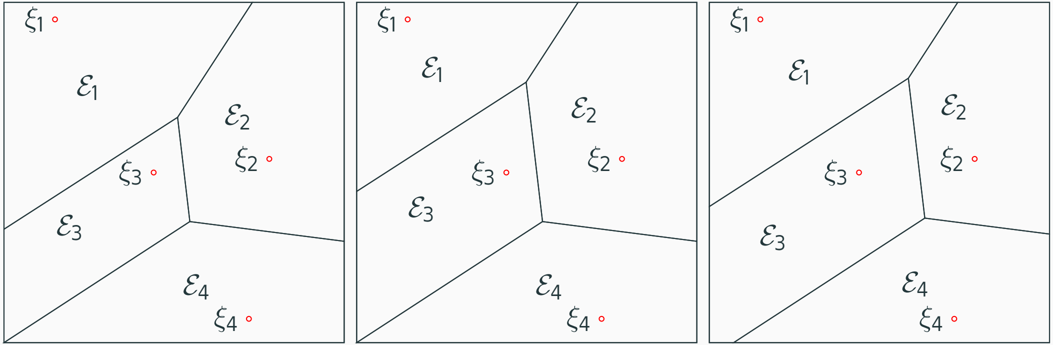
Note on ε distribution

The distribution of ε does not necessarily have full support; consequently, Assumption 1.2 does not always hold. In the probit model, ε possesses full support only if $M \geq Y$ and the matrix ξ is of full rank. In contrast, for the uniform distribution, this condition never holds because the support of ε is compact.

Laguerre diagrams

For $y \in [Y]$, ξ_y is the M-dimensional vector whose mth entry is ξ_{ym}. Given U a vector of systematic utilities, we let $\mathcal{E}_y \subset \mathbb{R}^M$ be the set of vectors $\eta \in \mathbb{R}^M$ that

$$\mathcal{E}_y = \left\{ \eta \in \mathbb{R}^M : U_y + \xi_y^\top \eta \geq U_{y'} + \xi_{y'}^\top \eta, \ \forall y' \in [Y] \right\}.$$



Several remarks

Several remarks on the previous Laguerre diagram:

- If the distribution P_η admits a density, the probability of indifference is zero. The set of strict preferences is simply the interior of E_y, and the distinction between strict and weak inequalities becomes irrelevant almost surely.
- **Geometry of E_y:** The choice region E_y is the intersection of the half-spaces defined by: $(\xi_{y'} - \xi_y)^\top \eta \leq U_y - U_{y'}, \quad \forall y' \neq y$. Since it is the intersection of a finite number of half-spaces, E_y is a **convex polyhedron**.
- **Impact of U_y:** Decreasing U_y (holding other utilities fixed) causes the bounding hyperplanes—orthogonal to $(\xi_y - \xi_{y'})$—to undergo a parallel inward displacement. Consequently, the region E_y shrinks while the adjacent cells expand.

Welfare function

We give the expression of the welfare function G as

$$G(U) = \mathbb{E}\left[\max_{y\in[Y]} \left\{U_y + (\boldsymbol{\xi}\eta)_y\right\}\right] = \sum_{y\in[Y]} \boldsymbol{\pi}_y(U)\left(U_y + \xi_y^\top \bar{\eta}^y\right),$$

where we define $\bar{\eta}^y := \mathbb{E}_{\mathbf{P}_\eta}[\eta \mid \eta \in \mathcal{E}_y]$ to be the barycenter of $\mathcal{E}_y$ with respect to the distribution $\mathbf{P}_\eta$.

In order to compute G, two quantities are needed: first, the probability weight $\boldsymbol{\pi}_y(U)$ that $\mathbf{P}_\eta$ assigns to each Laguerre cell; and second, the barycenter $\bar{\eta}^y$ of each Laguerre cell with respect to $\mathbf{P}_\eta$.

Simulation of the probit model

Letting $\pi_y = \pi_y(U)$, note that π_y is the probability that $\varepsilon_{y'} - \varepsilon_y \leq U_y - U_{y'}$ for all $y' \in [Y]\backslash\{y\}$, and therefore, defining $\epsilon_{y'} = \varepsilon_{y'} - \varepsilon_y$ and $z_{y'} = U_y - U_{y'}$ for $y' \in [Y]\backslash\{y\}$, π_y is equal to $F_\epsilon(z)$, the c.d.f. of the $(Y-1)$-dimensional random vector ϵ evaluated at $z \in \mathbb{R}^{Y-1}$.

Consider the case $Y = 3$, so that $\epsilon = (\epsilon_1, \epsilon_2)$ is a bivariate Gaussian vector with variances σ_1^2 and σ_2^2, and correlation ρ. In that case, we can use the i.i.d. factor representation of ϵ using $(g_1, g_2) \sim \mathcal{N}(0, \mathbf{I}_2)$:

$$\begin{cases} \epsilon_1 = \sigma_1 g_1 \\ \epsilon_2 = \sigma_2 \rho g_1 + \sigma_2 \sqrt{1-\rho^2} g_2. \end{cases}$$

Accept-reject vs. Geweke-Hajivassiliou-Keane simulator

Then, if we define $p = \Pr(\epsilon_1 \leq z_1) = \Phi(z_1/\sigma_1)$ where $\Phi(\cdot)$ is the c.d.f. of the standard normal variable, and $\varphi(x) = \Phi\big((z_2 - \rho\sigma_2\Phi^{-1}(x))/(\sqrt{1-\rho^2}\,\sigma_2)\big)$,

$$F_\epsilon(z) = \mathbb{E}[\Pi_1] \text{ where } \Pi_1 = 1\{X \leq p\}\,\varphi(X) \text{ and } X \sim \mathcal{U}([0,1]),$$

$$\Rightarrow \text{A-R simulator } \frac{1}{I}\sum_{i\in[I]} 1\left\{x_i \leq \Phi\left(\frac{z_1}{\sigma_1}\right)\right\}\Phi\left(\frac{z_2 - \rho\sigma_2\Phi^{-1}(x_i)}{\sqrt{1-\rho^2}\,\sigma_2}\right).$$

But $\mathbb{E}[\Pi_1] = p\mathbb{E}[\varphi(X)1\{X \leq p\}/p] = p\mathbb{E}[\varphi(X) \mid X \leq p] = p\mathbb{E}[\varphi(pX)]$, and thus

$$F_\epsilon(z) = \mathbb{E}[\Pi_2] \text{ where } \Pi_2 = p\varphi(pX) \text{ and } X \sim \mathcal{U}([0,1]),$$

$$\Rightarrow \text{GHK simulator } \frac{1}{I}\sum_{i\in[I]} \Phi\left(\frac{z_1}{\sigma_1}\right)\Phi\left(\frac{z_2 - \rho\sigma_2\Phi^{-1}\left(x_i\Phi\left(\frac{z_1}{\sigma_1}\right)\right)}{\sqrt{1-\rho^2}\,\sigma_2}\right).$$

Importance sampling: the key idea

The key idea is to replace $\mathbb{E}\left[\varphi\left(X\right) \mid X \leq p\right]$ by $\mathbb{E}\left[\varphi\left(pX\right)\right]$. The first one is computed by

$$I^{-1} \sum_i \varphi(X_i) 1\{X_i \leq p\} / p$$

and the second one by

$$I^{-1} \sum_i \varphi(pX_i)$$

Both have the same mean but the variance of the second one is smaller than the first one.

As a result, we have $F_\epsilon(z) = \mathbb{E}[\Pi_1] = \mathbb{E}[\Pi_2]$ and $Var\left(\Pi_1\right) > Var\left(\Pi_2\right)$.

The GHK simulator in the general case

In the general case ($Y \geq 3$), the GHK estimator of $F_\epsilon(z)$ is given by

$$\frac{1}{I}\sum_{i\in[I]}\prod_{j\in[J]}\Phi\left(\frac{z_j - L_{j1}\Phi^{-1}(\hat{x}_{i1}) - \cdots - L_{j,j-1}\Phi^{-1}(\hat{x}_{i(j-1)})}{L_{jj}}\right),$$

where the lower triangular matrix L is the Cholesky factor of Σ_ϵ, the variance-covariance matrix of ϵ, and where the expression of $\hat{x}_{ij}$ is recursively given by

$$\begin{cases} \hat{x}_{i1} = x_{i1}\Phi\left(\frac{z_1}{L_{11}}\right),\ i\in[I] \\ \hat{x}_{ij} = x_{ij}\Phi\left(\frac{z_j - L_{j1}\Phi^{-1}(\hat{x}_{i1}) - \cdots - L_{j,j-1}\Phi^{-1}(\hat{x}_{i(j-1)})}{L_{jj}}\right),\ i\in[I],\ j\geq 2 \end{cases}$$

where in turn the x_{ij} terms are i.i.d. draws from the uniform distribution over $[0,1]$.

Comparing accepting-rejecting with the GHK

We provide demo code from *Discrete Choice Models* in the next slides, where we compare the Accept-Reject simulator with the GHK simulator. We then compute the market share map using the GHK algorithm within the probit model. Readers are strongly encouraged to run this code and experiment with it. Note that dedicated methods exist in the case of the uniform cubic model.

Python code for comparing AR and GHK simulators (1/2)

```
import numpy as np
from statistics import NormalDist
phi = np.vectorize(NormalDist(mu=0, sigma=1).cdf)
phi_inv = np.vectorize(NormalDist().inv_cdf)
np.random.seed(9)
I,B,J=100000,10,2
L = np.array([[np.random.uniform()*(i <= j) for i in range(J)] for j in range(J)]  )
z_j = np.random.normal(size=J)/100
x_b_i_j = np.random.uniform(size=(B,I,J))
# Coding the simulators:
def iar(L,z_j,x_i_j): # integrated accept-reject simulator
  cond_i_j = (phi_inv(x_i_j)[:,None,:] * L[None,:,:]).sum(axis = 2) <= z_j[ None,:]
  argphi_i = (z_j[-1] - (L[-1,:-1][None,:] * phi_inv(x_i_j[:,:-1])).sum(axis =1) ) / L[-1,-1]
  return (( cond_i_j[:,:-1]).all(axis = 1) * phi( argphi_i )).mean()
# continued on the next slide =>>
```

Python code for comparing AR and GHK simulators (2/2)

```
# =>> continued from the previous slide
def ghk(L,z_j,x_i_j): # ghk simulator
    I,J = x_i_j.shape
    xhat_i_j = np.zeros((I,J))
    xhat_i_j[:,0] = x_i_j[:,0] * phi(z_j[0] / L[0,0])
    GHK_i = phi(z_j[0] / L[0,0]) * np.ones(I)
    for j in range(J-1):
        phietc_i = phi((z_j[j+1]-(L[j+1,:(j+1)][None,:]* phi_inv(xhat_i_j[:,:(j+1)])).sum(axis=
              1) ) /L[j+1,j+1])
        xhat_i_j[:,j+1]=x_i_j[:,j+1] * phietc_i
        GHK_i *= phietc_i
    return GHK_i.mean()
# Running simulations:
iarSim_b = np.array([iar(L,z_j,x_b_i_j[b,:,:]) for b in range(B)])
ghkSim_b = np.array([ghk(L,z_j,x_b_i_j[b,:,:]) for b in range(B)])
# Displaying results:
print('IAR, mean=',iarSim_b.mean(),'; std=',iarSim_b.std())
print('GHK, mean=',ghkSim_b.mean(),'; std=',ghkSim_b.std())
```


Python code for probit market share map using GHK (1/2)

```
import numpy as np
np.random.seed(9)
I,Y=100000,5
U_y = np.array([ 1.3, 0.9, -0.7, 0.2, 0.6])
x_i_j = np.random.uniform(size=(I,Y-1))
# Computing:
def ghk(L,z_j,x_i_j): # ghk simulator, as defined above
...
# continued on the next slide =>>
```

Python code for probit market share map using GHK (2/2)

```
# =>> continued from the previous slide
def piprobit_y(U_y,sigma_eta,x_i_j):
  Y=U_y.shape[0]
  I = x_i_j.shape[0]
  pi_y = np.zeros(Y)
  for y in range(Y):
      M = np.array( [[(1. if i==j else 0.) - (1. if j==y else 0.)  for j in range(Y)] for i in
           range(Y) if i != y])
      sigma_eta = M @ sigma_eta @ M.T
      L=  np.linalg.cholesky(sigma_eta)
      z_j  = - M @ U_y
      pi_y[y] = ghk(L,z_j,x_i_j)
  return (pi_y)
#
sigma_eta = np.eye(Y)
pi_y = piprobit_y(U_y,np.eye(Y),x_i_j)
# Displaying results:
print('pi_y=', pi_y,'\nsum(pi_y)=' ,pi_y.sum())
```


Simulation of uniform cubic model

In the simplest form, the algorithms used by the computational geometry libraries typically involve three steps:

- In the **vertex enumeration** step, given the Y vectors of M-dimensional characteristics ξ_y and the systematic utilities U_y, one computes the extreme points of the Laguerre cells $\mathcal{E}_y$.
- In the **triangulation** step, the Laguerre cells (based on their V-representation) are divided into elementary triangles (in dimension 2) or more generally into elementary simplices (in higher dimension), which do not intersect, except on the boundary.
- In the **aggregation** step, the measures of the elementary triangles are summed up and their barycenters are averaged, which obtains the measure and the barycenter of each Laguerre cell.

Calculation of the entropy of choice

We can draw I samples of $\mathbf{P}_\eta$ and if we denote η_{im} the mth dimension of the utility-shifter in draw i, we recall that the random utility is given by $\varepsilon_{iy} = \sum_m \eta_{im}\xi_{ym}$. The sample analog of the entropy of choice is given by the linear programming problem

$$\begin{aligned} -G_I^*(\pi) = \max_{\lambda_{iy}\geq 0} \quad & \sum_{iy\in[I\times Y]} \lambda_{iy}\varepsilon_{iy} \\ s.t. \quad & \sum_{i\in[I]} \lambda_{iy} = \pi_y,\ y \in [Y],\ \sum_{y\in[Y]} \lambda_{iy} = \frac{1}{I},\ i \in [I], \end{aligned}$$

whose dual expression is

$$\begin{aligned} \min_{u_i,U_y} \quad & \frac{1}{I}\sum_{i\in[I]} u_i - \sum_{y\in[Y]} \pi_y U_y \\ s.t. \quad & u_i - U_y \geq \varepsilon_{iy},\ iy \in [I \times Y], \end{aligned}$$

allowing one to recover U_y as a solution.

Continuous extension

Assume that the set $\mathcal{Y}$ of options y is infinite and continuous, and large enough so that the set $\{\xi_y, y \in \mathcal{Y}\}$ should coincide with $\mathbb{R}^M$. In that case one may redefine $U(\xi)$ as the systematic utility U_y associated with the option y such that $\xi = \xi_y$, and the indirect utility of a consumer with heterogeneity η is

$$u(\eta) = \max_{\xi \in \mathbb{R}^M} \{U(\xi) + \xi^\top \eta\}$$

which makes perfect sense, and does not require any renormalization. The function u so defined is convex, and the envelope theorem implies that the quality vector ξ chosen by the consumer with heterogeneity η is $\xi = \nabla u(\eta)$.

Section 4.2. Random coefficient logit specification

Section content

- The random coefficient logit model
- Gaussian-Hermite quadrature

Definition: random coefficient logit utility

The expression of a random coefficient logit utility vector $\varepsilon \sim \mathcal{P}$ is

$$\varepsilon = \xi\eta + \sigma\epsilon,$$

where $\eta \sim \mathbf{P}_\eta$ is a M-dimensional vector of utility-shifters, and $\epsilon \sim \mathcal{G}$ is a random vector of i.i.d. Gumbel terms, independent from η.

Then the welfare function G has expression $G(U) = \mathbb{E}[\max_y\{U_y + (\xi\eta)_y + \sigma\epsilon_y\}]$. By the law of iterated expectations, one can compute that expectation by taking expectations on ϵ first and then on η, which yields

$$G(U) = \mathbb{E}\left[\sigma \log \sum_{y \in [Y]} \exp\left(\frac{U_y + (\xi\eta)_y}{\sigma}\right)\right].$$

The connection with the framework of entropic optimal transport (1/4)

Proposition 4.2.1

The value of the entropy of choice associated with the random coefficient logit model coincides (up to a sign inversion) with the value of the entropic regularized optimal transport problem

$$-G^*(\pi) = \max_{\lambda \in \mathcal{M}(\mathbf{P}_\eta, \mathcal{P}_{\xi,\pi})} \left\{ \mathbb{E}_\lambda \left[\eta^\top \xi \right] - \sigma \mathbb{E}_\lambda \left[\log \lambda \left(\eta, \xi \right) \right] \right\}.$$

As in the pure characteristics model, the sample analogs of the welfare function and the market share map are given by

$$\begin{cases} G_{I,\sigma}(U) = \frac{\sigma}{I} \sum_{i \in [I]} \log \sum_{y \in [Y]} \exp((U_y + \eta_{iy})/\sigma), \\ \left(\pi_{I,\sigma}\right)_y (U) = \frac{1}{I} \sum_{i \in [I]} \frac{\exp((U_y + \eta_{iy})/\sigma)}{\sum_{y' \in [Y]} \exp((U_{y'} + \eta_{iy'})/\sigma)}. \end{cases}$$

The connection with the framework of entropic optimal transport (2/4)

The primal formulation of the entropy of choice in the random coefficient logit model is

$$
\begin{aligned}
-G^{*}_{I,\sigma}(\pi) &= \max_{\lambda_{iy}\geq 0}\left\{\sum_{iy\in[I\times Y]}\lambda_{iy}\eta_{iy}-\sigma\sum_{iy\in[I\times Y]}\lambda_{iy}\log\lambda_{iy}\right\} \\
s.t. \quad & \sum_{i\in[I]}\lambda_{iy}=\pi_y,\ y\in[Y],\ \sum_{y\in[Y]}\lambda_{iy}=\frac{1}{I},\ i\in[I],
\end{aligned}
$$

which has an equivalent dual expression equal to

$$
-G^{*}_{I,\sigma}(\pi)=\min_{u_i,U_y}\left\{\frac{1}{I}\sum_{i\in[I]}u_i-\sum_{y\in[Y]}\pi_y U_y+\sigma\sum_{iy\in[I\times Y]}\exp\left(\frac{\eta_{iy}-u_i+U_y}{\sigma}\right)\right\}.
$$

The connection with the framework of entropic optimal transport (3/4)

The solution $(u_i^\star, U_y^\star)$ is unique up to a constant. We can use the iterated proportional fitting procedure to compute U. We come up with an initial guess of U_y^0 and we iterate

$$\begin{cases} u_i^{t+1} = \sigma \log \left(I \sum_{y \in [Y]} \exp((\eta_{iy} + U_y^t)/\sigma) \right) \\ U_y^{t+1} = -\sigma \log \left(\frac{1}{\pi_y} \sum_{i \in [I]} \exp((\eta_{iy} - u_i^{t+1})/\sigma) \right). \end{cases}$$

Alternatively, we can combine both steps and run the *contraction mapping algorithm*, which consists of iterating

$$U_y^{t+1} = -\sigma \log \left(\frac{1}{I\pi_y} \sum_{i \in [I]} \frac{\exp(\nu_{iy}/\sigma)}{\sum_{y' \in [Y]} \exp((\nu_{iy'} + U_{y'}^t)/\sigma)} \right).$$

If we pick $y^* \in [Y]$ and take the normalization $U_{y^*} = 0$, $\tilde{U}_y^t = U_y^t - U_{y^*}^t$, we can have

$$|\tilde{U}_y^t - U_{y^*}^\star| \leq \sigma d_H(\exp(U_y^t/\sigma), \exp(U_y^\star/\sigma)),$$

and $|\tilde{U}_y^t - U_{y^*}| \leq K^t C$ for $C = \sigma d_H(\exp(U_y^0/\sigma), \exp(U_y^\star/\sigma))$.

Theorem 4.2.1

The solution $(u_i^\star, U_y^\star)$ exists and is unique up to a constant. Further, as $t \to +\infty$, $\tilde{U}_y^t = U_y^t - U_{y^\star}^t$ converges to $U_y^\star$ at rate K^t.

Use of Gauss-Hermite quadrature

Assume for simplicity that $M = 1$ and $\mathbf{P}_\eta$ is the $\mathcal{N}(0,1)$ distribution. We can use formula with $N = I$ and $f(\eta) = \sigma \log \sum_y \exp((U_y + \xi_y \eta)/\sigma)$, and weights w_i given by

$$w_i = \frac{2^{I-1} I!}{I^2 \left(H_{I-1}\left(\eta_i\right)\right)^2},$$

in which case the expression of G can be modified into

$$\begin{cases} G_{I,\sigma}(U) = \sigma \sum_{i \in [I]} w_i \log \sum_{y \in [Y]} \exp((U_y + \xi_y \eta_i)/\sigma), \\ \left(\pi_{I,\sigma}\right)_y(U) = \sum_{i \in [I]} w_i \frac{\exp((U_y + \xi_y \eta_i)/\sigma)}{\sum_{y' \in [Y]} \exp((U_{y'} + \xi_{y'} \eta_i)/\sigma)}, \end{cases}$$

and the entropy of choice is (better) approximated by

$$\begin{aligned} -G^*_{I,\sigma}(\pi) &= \max_{\lambda_{iy} \geq 0} \left\{ \sum_{iy \in [I \times Y]} \lambda_{iy} \xi_y \eta_i - \sigma \sum_{iy \in [I \times Y]} \lambda_{iy} \log \lambda_{iy} \right\} \\ s.t. & \quad \sum_{i \in [I]} \lambda_{iy} = \pi_y, \ y \in [Y], \ \sum_{y \in [Y]} \lambda_{iy} = w_i, \ i \in [I]. \end{aligned}$$

Section 4.3. Demand estimation with endogenous characteristics

Discrete Choice Models, sec. 4.3, pp. 97-102

Section content

- Estimation with endogeneity
- IV-GMM model

The problem

Sometimes some product characteristics are observed by agents, but not by the analyst. *Endogeneity* arises when this unobserved component is correlated with another characteristics included in the regressors, and leads to biased parametric estimates. The structural estimation method of Berry, Levinsohn and Pakes is designed to remedy endogeneity biases.

A motivational example

Assume that the consumer has two options: not purchasing ($y=0$, the default option) or purchasing ($y=1$, the alternative). There are T markets, and the purchasing cost in market $t \in [T]$ is p_t. A consumer i in market t has utility $\lambda_1 + \lambda_2 p_t + \eta_{ti1}$ for purchasing, and utility η_{ti0} for not purchasing, where η_{ti0} and η_{ti1} are i.i.d. Gumbel. The predicted market share is

$$\pi_t^\lambda = \frac{\exp(\lambda_1 + \lambda_2 p_t)}{1 + \exp(\lambda_1 + \lambda_2 p_t)},$$

and $\hat{\lambda}$ is estimated from the observed market share $\hat{\pi}_t$ by the logistic regression

$$\max_{(\lambda_1,\lambda_2)\in\mathbb{R}^2} \sum_{t\in[T]} \hat{\pi}_t(\lambda_1 + \lambda_2 p_t) - \sum_{t\in[T]} \log(1 + \exp(\lambda_1 + \lambda_2 p_t)).$$

Unobservable quality characteristics

We would like to allow for a variation in the quality of the good. Assume that a market-specific quality characteristics κ_t uniformly affects each consumer's valuation for the good in market t, but is not observed by the econometrician.

The utility associated with purchasing becomes $\lambda_1 + \kappa_t + \lambda_2 p_t + \eta_{ti1}$, while the utility of not possessing the good is still η_{ti0}. Assume that κ_t is centered and orthogonal with p_t, so $\mathbb{E}\left[\kappa_t\right] = 0$ and $\mathbb{E}\left[\kappa_t p_t\right] = 0$.

Defining the systematic utility $U_t = \lambda_1 + \lambda_2 p_t + \kappa_t = \ln(\hat{\pi}_t/(1-\hat{\pi}_t))$, we can write the empirical moment conditions

$$\sum_{t\in[T]} (U_t - \lambda_1 - \lambda_2 p_t) = 0 \text{ and } \sum_{t\in[T]} (U_t - \lambda_1 - \lambda_2 p_t) p_t$$

whose solution is $\hat{\lambda}^{OLS}$, the OLS regressor of U_t on the constant and the price p_t.

Dealing with endogeneity

Typically prices are higher when quality is higher. Berry (1994) uses an instrument ζ_t, correlated with p_t, but uncorrelated with κ_t, so that $\mathbb{E}[\zeta_t] = 0$, $\mathbb{E}[\kappa_t \zeta_t] = 0$, and $cov(p_t, \zeta_t) \neq 0$. For instance, cost shifters, such as labor costs or prices of raw materials can be correlated with the price, but independent from the market-specific quality.

In that case, the previous moment conditions are replaced by

$$\sum_{t \in [T]} (U_t - \lambda_1 - \lambda_2 p_t) = 0 \text{ and } \sum_{t \in [T]} (U_t - \lambda_1 - \lambda_2 p_t) \zeta_t = 0,$$

whose solution $\hat{\lambda}^{IV}$ is now the *instrumental variable regressor* of $\hat{\lambda}$, the ordinary least square regressor of $\hat{U}_t = \ln(\hat{\pi}_t / (1 - \hat{\pi}_t))$ on two endogenous regressors, the constant the price p_t, using as instruments the constant and ζ_t.

Berry's instrumental variable approach

Hence the value of $\hat{\lambda}_2^{IV}$ is given by

$$\hat{\lambda}_2^{IV} = \frac{cov_T(\zeta_t, U_t)}{cov_T(\zeta_t, p_t)},$$

where cov_T denotes the sample covariance, and from which $\hat{\lambda}_1$ can easily be deduced.

One can see from the code in the book that the estimator $\hat{\lambda}_2^{OLS}$ is positive: the OLS approach draws an incorrect causal conclusion from the positive correlation between price and the random utility. The instrumental variable regression produces the right conclusion, and leads to an unbiased estimate of λ.

IV-GMM for demand estimation (1/3)

In typical applications, the number of instruments may exceed the number of parameters. Continuing with the same example, suppose we now have three instruments: 1, ζ_t and ζ'_t, for estimating λ_1 and λ_2. The moment conditions indicate the orthogonality

$$\sum_{t\in[T]}(U_t-\lambda_1-\lambda_2 p_t)=\sum_{t\in[T]}(U_t-\lambda_1-\lambda_2 p_t)\zeta_t=\sum_{t\in[T]}(U_t-\lambda_1-\lambda_2 p_t)\zeta'_t=0$$

To handle this problem, Berry, Levinsohn and Pakes have suggested to use IV-GMM. The estimator $\hat{\lambda}$ minimizes the W-norm:

$$\min_{(\lambda_1,\lambda_2)\in\mathbb{R}^2}\|\sum_{t\in[T]}(\hat{U}_t-\lambda_1-\lambda_2 p_t)(1,\zeta_t,\zeta'_t)^\top\|_W^2,$$

where W is a 3×3 symmetric positive definite weighting matrix and $\|.\|_W$ is the quadratic norm associated with W, defined as $\|a\|_W^2=a^\top Wa$.

IV-GMM for demand estimation (2/3)

Letting $\boldsymbol{\zeta} = (1_T, \zeta, \zeta')$ be the $T \times 3$ matrix of instruments stacked column-wise, $\boldsymbol{\Phi} = (1_T, p)$ the $T \times 2$ matrix of the two regressors (constant and prices) stacked column-wise, and $\Pi = \zeta W \zeta^\top$, the problem reformulates as

$$\min_{\lambda \in \mathbb{R}^2} (U - \boldsymbol{\Phi}\lambda)^\top \Pi (U - \boldsymbol{\Phi}\lambda)$$

whose solution is straightforward, and yields the IV-GMM estimator:

$$\hat{\lambda} = \left(\boldsymbol{\Phi}^\top \Pi \boldsymbol{\Phi}\right)^{-1} \left(\boldsymbol{\Phi}^\top \Pi U\right).$$

IV-GMM for demand estimation (3/3)

The efficient GMM estimator is obtained by choosing W as:

$$W = (\mathbb{E}[\kappa^2 (1, \zeta, \zeta')^\top (1, \zeta, \zeta')])^{-1}$$

If κ is independent from ζ and ζ', W can be proportional to

$$\tilde{W} = \mathbb{E}[(1, \zeta, \zeta')^\top (1, \zeta, \zeta')]^{-1},$$

which can be used instead. To conclude, in the heteroskedastic case, one cannot compute W directly, so a two-step GMM is used.

Then, we can capture the interaction between unobserved heterogeneity and observable characteristics by including random coefficients.

Utility shocks with a random coefficient term (1/3)

Assume we may draw I sample points of income from that distribution y_i, and define $\eta_i = -1/y_i$. The utility associated with purchasing the good in market t is

$$U_t + \tau \eta_i p_t + \epsilon_{ti1}, \text{ with } U_t = \lambda_1 + \lambda_2 p_t + \kappa_t,$$

and utility of not purchasing remains ϵ_{ti0}. The interaction term $\tau \eta_i p_t$ captures price sensitivity decreasing in income. We expect $\tau > 0$.

This leads to a random coefficient logit model, and for each market $t \in [T]$, the systematic utility U_t solves

$$\frac{1}{I} \sum_{i \in [I]} \frac{1}{1 + \exp(U_t + \tau \eta_i p_t)} = 1 - \hat{\pi}_t.$$

Let $\hat{U}_t^\tau$ be the solution, which depends on τ.

Utility shocks with a random coefficient term (2/3)

In order to estimate $(\lambda_1, \lambda_2, \tau)$, we introduce instrument ζ''_t to the IV-GMM approach. The GMM objective becomes

$$\min_{(\lambda_1,\lambda_2,\tau)\in\mathbb{R}^2} \| \sum_{t\in[T]} (\hat{U}^{\tau}_t - \lambda_1 - \lambda_2 p_t)(1, \zeta_t, \zeta'_t, \zeta''_t)^\top \|^2_W,$$

where this time W is a 4×4 symmetric positive definite matrix.

As the dependence of $\hat{U}^{\tau}_t - \lambda_1 - \lambda_2 p_t$ is linear in λ, but nonlinear in τ. We define

$$F(\tau) = \min_{(\lambda_1,\lambda_2)\in\mathbb{R}^2} \| \sum_{t\in[T]} (\hat{U}^{\tau}_t - \lambda_1 - \lambda_2 p_t)(1, \zeta_t, \zeta'_t, \zeta''_t)^\top \|^2_W.$$

This inner problem has closed-form solution, yielding

$$F(\tau) = (\hat{U}^{\tau})^\top \Gamma \hat{U}^{\tau},\ \Gamma = \Pi - \Pi\Phi(\Phi^\top \Pi \Phi)^{-1}\Phi^\top \Pi$$

with $\Pi_{tt'} = (1, \zeta_t, \zeta'_t, \zeta''_t) W (1, \zeta_t, \zeta'_t, \zeta''_t)^\top$, and Φ the regressor matrix for 1 and (p_t).

Utility shocks with a random coefficient term (3/3)

It remains to solve the minimization problem over the nonlinear parameter τ, namely minimize F. The optimal $\hat{\tau}$ is found numerically via gradient descent, using

$$F'(\tau) = 2(\hat{U}^{\tau})^{\top} \Gamma d\hat{U}^{\tau} / d\tau,$$

and

$$\sum_{i \in [I]} \frac{\exp(\hat{U}_t^{\tau} + \tau \eta_i p_t)}{(1 + \exp(\hat{U}_t^{\tau} + \tau \eta_i p_t))^2} \left(\frac{d\hat{U}_t^{\tau}}{d\tau} + \eta_i p_t \right) = 0.$$

Finally, plugin $\hat{\tau}$ into the IV-GMM formula to recover $\hat{\lambda}_1$ and $\hat{\lambda}_2$. A second-stage GMM can be used for heteroskedasticity-robust inference.

Section 4.4. A cornerstone in structural estimation: BLP's method

Section content

- The Berry, Levinsohn and Pakes (BLP) method
- Demand-side specification
- Supply-side specification
- The BLP precedure
- BLP Python Practice Notebook

Example 4.3: The original study of BLP

In their seminal 1995 paper, Berry, Levinsohn, and Pakes took on the central question of how to estimate consumer demand and measure market power in the U.S. automobile market, a prime example of a market for differentiated products.

The paper introduced most of the tools we are discussing, including the random coefficients logit model, the treatment of endogeneity, as well as the use of cost shifters, in order to identify how consumer preferences vary across individuals and how firms set prices in response to this heterogeneity.

Demand-side specification

In market $t \in [T]$, we model agent i's utility u_{tiy} associated with alternative $y \in [Y_t]$ as the sum of three terms, one that is systematic in the sense that it does not vary across agents, an other one that results from the interactions between the agents' tastes variations and the products' observed characteristics, and a last one that is purely idiosyncratic and i.i.d. across agent-observation pairs. We thus specify u_{tiy} as

$$\begin{cases} u_{tiy} = U_{ty} + \nu^{\tau}_{tiy} + \epsilon_{tiy} \\ u_{ti0} = \epsilon_{ti0} \end{cases}$$

U_{ty} is a term that may arise from observed product characteristics, denoted ϕ_{tyk}, or from some unobserved residual variations in quality denoted κ_{ty}. We postulate that the systematic U_{ty} can be expressed as

$$U_{ty} = \sum_{k \in [K]} \phi_{tyk} \lambda_k + \kappa_{ty},$$

where $\lambda \in \mathbb{R}^M$ is a parameter vector to be estimated, and the distribution of the term κ_{ty} is left unspecified. A number D of demand instruments ζ_{tyd} for $d \in [D]$ provide the exclusion restrictions needed for identification. These instruments are correlated with the regressors ϕ_{tyk}, but uncorrelated with κ, so that

$$\mathbb{E}[\kappa_{ty} \zeta_{tyd}] = 0.$$

Explanations of the components of the utility (2/2)

ν^{τ}_{tiy} captures various interactions between agent-specific unobserved tastes and observed characteristics. We assume

$$\nu^{\tau}_{tiy} = \sum_{m \in [M]} \eta_{im} \xi_{tym} \tau_m = \sum_{m \in [M]} \nu_{tiy,m} \tau_m,$$

where $(\xi_{tym})_m$ represents various characteristics of alternative y in market t, and $(\eta_{im})_m$ captures agent i's valuation of the characteristic. η_{im} measures the valuation by consumer i of the mth characteristic. $\tau \in \mathbb{R}^M$ controls the relative importance of the valuation of the mth dimension in the overall preference.

ϵ_{tiy} is an idiosyncratic preference shock associated by agent i in market t for option $y \in [0 : Y]$. It is assumed that for each i and t, the $(Y_t + 1)$-dimensional vector $(\eta_{tiy})_y$ has an i.i.d. Gumbel distribution.

Supply-side specification

For the analysis of the supply side, we assume that sellers engage in Bertrand price competition. Each market t is structured into firms, each controlling a subset of options for which they are free to set prices. Each market t is structured into firms, each controls a subset of options free to set prices. Let $\mathbf{H}$ be be the $TY \times TY$ matrix of term $H_{ty,t'y'} = 1$ if y and y' are controlled by the same firm and if $t = t'$, and $H_{ty,t'y'} = 0$ otherwise. The markup on good y is $p_{ty} - c_{ty}$ and the profit of firm producing y in market t is thus

$$\sum_{y' \in [Y]} H_{ty,ty'} \pi_{ty'} (p_{ty'} - c_{ty'}),$$

where the market share π_{ty} depends on the entries of the price vector p through the systematic utility $U_{ty} = \sum_{k \in [K]} \phi_{tyk} \lambda_k + \kappa_{ty}$, but also through the random coefficient part $\nu_{tiy}^{\tau} = \sum_{m \in [M]} \nu_{tiym} \tau_m$.

Market share expression

We recall that

$$\pi_{ty} = \frac{1}{I} \sum_{i \in [I]} \frac{\exp\left(\sum_{k \in [K]} \phi_{tyk} \lambda_k + \kappa_{ty} + \sum_{m \in [M]} \nu_{tiym} \tau_m\right)}{1 + \sum_{y' \in [Y]} \exp\left(\sum_{k \in [K]} \phi_{ty'k} \lambda_k + \kappa_{ty'} + \sum_{m \in [M]} \nu_{tiy'm} \tau_m\right)},$$

or in vectorized terms,

$$\pi = \frac{1}{I} \mathbf{\Sigma}_I f \text{ with } f = \exp\left(V - \mathbf{\Sigma}_Y^\top u\right),$$

where we have defined

$$V = \mathbf{\Sigma}_I^\top \mathbf{\Phi} \lambda + \mathbf{\Sigma}_I^\top \kappa + \boldsymbol{\nu} \tau \text{ and } u = \log\left(\mathbf{1}_{TI} + \mathbf{\Sigma}_Y \exp V\right).$$

As both the regressors of the systematic utility, ϕ, and the components of the random coefficients, ν, may depend on prices p, it follows that the choice probabilities π also depend on p through these channels.

The FOCs for Bertrand profit maximization

The producer of y in market t sets the prices $p_{ty'}$ of the goods y' it controls in order to maximize profit. These goods are the y' such that $H_{ty,ty'} = 1$. The profit of the producer of y in market t is $\sum_{y'} H_{ty,ty'} \pi_{ty'} (p_{ty'} - c_{ty'})$, and by the FOCs, one obtains:

$$\sum_{y' \in [Y]} H_{ty,ty'} \frac{\partial \pi_{ty'}}{\partial p_{ty}} (p_{ty'} - c_{ty'}) + \pi_{ty} = 0.$$

Letting $\boldsymbol{D}$ be the square matrix of size TY with entries $D_{ty,t'y'} = \partial \pi_{t'y'} / \partial p_{ty}$ if y and y' are controlled by the same firm and if $t = t'$, and $D_{ty,t'y'} = 0$ otherwise, the expression can be recast in vectorized terms as

$$\boldsymbol{D}(c - p) = \pi, \text{ where } \boldsymbol{D} = \mathbf{H} \odot \left(\frac{\partial \pi}{\partial p}\right)^{\top},$$

where $\odot$ is the Hadamard (element-wise) product. By matrix inversion, we can have

$$c = p + \boldsymbol{D}^{-1} \pi.$$

BLP: recovering consumer utilities via demand inversion (1/2)

If we fix $\tau \in \mathbb{R}^M$ for now, we can simulate ν^τ_{tiy} using $(\eta_{im})_m \sim \boldsymbol{P}_\eta$. The probability that agent i in market t opts for alternative y is therefore $\pi_{tiy} = \exp(U_{ty} + \nu^\tau_{tiy} - u_{ti})$, where u_{ti} is the expected ex-interim utility of agent i, computed conditionally on the random coefficient ν^τ_{tiy} and by integrating over the logit shock, which is given by $u_{ti} = \log(1 + \sum_y \exp(U_{ty} + \nu^\tau_{tiy}))$, and the market share map is given by $\boldsymbol{\pi}_{ty}(U) = I^{-1} \sum_i \exp(U_{ty} + \nu^\tau_{tiy} - u_{ti})$.

We are interested in the inverse market share map, which is done by solving for U and u in the system

$$\begin{cases} \hat{\pi}_{ty} = \sum_{i \in [I]} \frac{1}{I} \exp(U_{ty} + \nu^\tau_{tiy} - u_{ti}), \\ \frac{1}{I} = \frac{1}{I} \exp(-u_{ti}) + \sum_{y \in [Y]} \frac{1}{I} \exp(U_{ty} + \nu^\tau_{tiy} - u_{ti}). \end{cases}$$

BLP: recovering consumer utilities via demand inversion (2/2)

We denote by $(\hat{U}^\tau, \hat{u}^\tau)$ the solution to that system, where the superscript τ indicates that the market share inversion is done conditional on τ. In vectorized notation, the probability that consumer i in market t chooses alternative y is given by

$$f = \exp(\mathbf{\Sigma}_I^\top U + \boldsymbol{\nu}\tau - \mathbf{\Sigma}_Y^\top u),$$

where we define $\mathbf{\Sigma}_I := I_T \otimes \mathbf{1}_I^\top \otimes I_Y$ and $\mathbf{\Sigma}_Y := I_T \otimes I_I \otimes \mathbf{1}_Y^\top$; and where we denote by $\boldsymbol{\nu}$ the matrix of size $TIY \times M$ whose term is $\nu_{tiy,m} = \eta_{im}\xi_{tym}$. We recall that $\otimes$ is the Kronecker product; that $(\hat{\pi}_{ty}) \in \mathbb{R}^{TY}$ is the vector of market shares; that $(U_{ty}) \in \mathbb{R}^{TY}$ is the vector of systematic utilities; that $(\tau_m) \in \mathbb{R}^M$ is the parameter of the random coefficient; and that $(u_{ti}) \in \mathbb{R}^{TI}$ is the individual welfare. The system above can be reexpressed using vectorized notation as

$$\begin{cases} \hat{\pi} = \frac{1}{I}\mathbf{\Sigma}_I \exp(\mathbf{\Sigma}_I^\top U + \boldsymbol{\nu}\tau - \mathbf{\Sigma}_Y^\top u), \\ \mathbf{1}_{TI} = \exp(-u) + \mathbf{\Sigma}_Y \exp(\mathbf{\Sigma}_I^\top U + \boldsymbol{\nu}\tau - \mathbf{\Sigma}_Y^\top u). \end{cases}$$

BLP: recovering sellers' costs from profit maximization (1/3)

Next, we would like to estimate the vector of costs c_{ty} using expression following from the assumption of profit maximization. This expression involves the market share vector π and the price vector p, both of which are directly observable. However, it also depends on the matrix $\boldsymbol{D}$, which contains the derivatives $\partial\pi/\partial p$ of market shares with respect to prices. These derivatives are not directly observable from the data. Recalling that $\boldsymbol{\Delta}_f$ denotes the diagonal matrix whose diagonal is the vector f, the differential of π with respect to p is given by

$$\frac{\partial \pi}{\partial p} = \frac{1}{I} \boldsymbol{\Sigma}_I \boldsymbol{\Delta}_f \boldsymbol{R}_f \frac{\partial V}{\partial p}, \quad \text{where } \boldsymbol{R}_f = \boldsymbol{I}_{TIY} - (\boldsymbol{I}_{TI} \otimes \mathbf{1}_{Y \times Y}) \boldsymbol{\Delta}_f,$$

and where we recall that $V = \boldsymbol{\Sigma}_I^\top \boldsymbol{\Phi} \lambda + \boldsymbol{\Sigma}_I^\top \kappa + \boldsymbol{\nu} \tau$ is the vector of individual utilities. The differential of V with respect to the price vector p yields

$$\frac{\partial V}{\partial p} = \boldsymbol{\Sigma}_I^\top \frac{\partial \boldsymbol{\Phi}}{\partial p} (\boldsymbol{I}_{TY} \otimes \lambda) + \frac{\partial \boldsymbol{\nu}}{\partial p} (\boldsymbol{I}_{TY} \otimes \tau).$$

BLP: recovering sellers' costs from profit maximization (2/3)

Typically, however, only a few entries of $\mathbf{\Phi}$ are endogenous, while others are exogenous. For that reason, we write

$$\mathbf{\Phi} = \left(\mathbf{\Phi}^e, \mathbf{\Phi}^x\right),$$

where $\mathbf{\Phi}^e$ is a $TY \times K^e$ matrix and $\mathbf{\Phi}^x$ is a $TY \times K^x$ matrix, with $K^e + K^x = K$. We assume that $\partial \mathbf{\Phi}^x / \partial p = 0$, and we write the parameter vector as $\lambda = (\alpha^\top, \beta^\top)^\top$, where α is a vector of size K^e and β is a vector of size K^x. We have

$$\mathbf{\Phi} \lambda = \mathbf{\Phi}^e \alpha + \mathbf{\Phi}^x \beta,$$

and

$$\frac{\partial V}{\partial p} = \mathbf{\Sigma}_I^\top \frac{\partial \mathbf{\Phi}^e}{\partial p} \left(I_{TY} \otimes \alpha\right) + \frac{\partial \boldsymbol{\nu}}{\partial p} \left(I_{TY} \otimes \tau\right).$$

BLP: recovering sellers' costs from profit maximization (3/3)

While $f = \exp(\boldsymbol{\Sigma}_I^\top U + \boldsymbol{\nu}\tau - \boldsymbol{\Sigma}_Y^\top u)$ is not directly observable, it can be estimated by substituting $\hat{U}^\tau$ and $\hat{u}^\tau$, computed in step (a), in place of U and u. Let $\hat{f}$ denote the resulting estimator of f. We can then form $\boldsymbol{R}_{\hat{f}}$, $\boldsymbol{\Delta}_{\hat{f}}$, and obtain an estimator $\hat{\boldsymbol{D}}$ of the matrix $\boldsymbol{D}$. As a result, the cost vector c is estimated by

$$\hat{c}^{\tau,\alpha} = p + \widehat{\boldsymbol{D}}^{-1}\hat{\pi}, \text{ where } \widehat{\boldsymbol{D}} = \boldsymbol{H} \odot \left(\frac{1}{I} \boldsymbol{\Sigma}_I \boldsymbol{\Delta}_{\hat{f}} \boldsymbol{R}_{\hat{f}} \frac{\partial V}{\partial p} \right)^\top .$$

The superscript (τ, α) in the notation $\hat{c}^{\tau,\alpha}$ indicates that the estimator $\widehat{\boldsymbol{D}}$ depends on τ, because the market share inversion is performed conditional on τ, and also on α, since the price elasticities depend on it.

Algorithm 4.4.1 (IPFP algorithm for BLP market share inversion)

Input: v, u^0_{ti}

Parameter: ϵ (tolerance)

while $\max_{ti \in [T \times Y]} ||U^s_{ty} - U^{s-1}_{ty}|| \geq \epsilon$ **do**

$$U^{s+1} = -\log(\Delta_\pi^{-1} I^{-1} \Sigma_I \exp(\nu\tau - \Sigma_Y^T u^s))$$

$$u^{s+1} = \log(1_{TI} + \Sigma_Y \exp(\Sigma_I^T U^{s+1} + \nu\tau))$$

$$s = s + 1$$

end

BLP: IV-GMM estimation (1/4)

The central step of the method consists of running an IV-GMM estimation procedure: regressing $\hat{U}^{\tau}$ on $\mathbf{\Phi}$, and $\hat{c}^{\tau,\alpha}$ on $\mathbf{\Psi}$, in order to estimate the demand parameters $\lambda = (\alpha^{\top}, \beta^{\top})^{\top}$, the supply parameters γ, and the demand parameters τ. The overall objective is to minimize

$$\left\| \begin{pmatrix} \boldsymbol{\zeta}^{\top}(\hat{U}^{\tau} - \mathbf{\Phi}\lambda) \\ \boldsymbol{\chi}^{\top}(\hat{c}^{\tau,\alpha} - \mathbf{\Psi}\gamma) \end{pmatrix} \right\|_{W}^{2} = \left\| \begin{pmatrix} \boldsymbol{\zeta}^{\top}(\hat{U}^{\tau} - \mathbf{\Phi}^{e}\alpha - \mathbf{\Phi}^{x}\beta) \\ \boldsymbol{\chi}^{\top}(\hat{c}^{\tau,\alpha} - \mathbf{\Psi}\gamma) \end{pmatrix} \right\|_{W}^{2}$$

over $(\tau, \lambda, \gamma) \in \mathbb{R}^{M+K+L}$, where $\lambda = (\alpha^{\top}, \beta^{\top})^{\top}$.

We recall that $\|a\|_W^2 = a^{\top} W a$, and W is a $(D+S) \times (D+S)$ matrix that is symmetric positive definite.

We observe that the quantity appearing inside the norm is nonlinear in $\theta = (\tau^\top, \alpha^\top)^\top$ and linear in β and γ.

If we fix θ, we may run a linear IV-GMM procedure with dependent variable $\boldsymbol{Y}^\theta$, defined as the concatenation of $\hat{U}^\tau - \mathbf{\Phi}^e \alpha$ and $\hat{c}^{\tau,\lambda}$, on regressors $\boldsymbol{X}$, given by the aggregation of $\mathbf{\Phi}^x$ and $\mathbf{\Psi}$, and instruments $\boldsymbol{Z}$, given by the aggregation of $\boldsymbol{\zeta}$ and $\boldsymbol{\chi}$.

We define

$$\boldsymbol{Y}^\theta = \begin{pmatrix} \hat{U}^\tau - \mathbf{\Phi}^e \alpha \\ \hat{c}^{\tau,\alpha} \end{pmatrix}, \; \boldsymbol{X} = \begin{pmatrix} \mathbf{\Phi}^x & \mathbf{0}_{TY,L} \\ \mathbf{0}_{TY,K^x} & \mathbf{\Psi} \end{pmatrix} \text{ and } \boldsymbol{Z} = \begin{pmatrix} \boldsymbol{\zeta} & \mathbf{0}_{TY,S} \\ \mathbf{0}_{TY,D} & \boldsymbol{\chi} \end{pmatrix},$$

which are matrices of respective sizes $2TY \times 1$, $2TY \times (K^x + L)$ and $2TY \times (D + S)$.

BLP: IV-GMM estimation (3/4)

The expression of the IV-GMM objective is $\|Z^\top(Y^\theta - X(\beta^\top,\gamma^\top)^\top)\|_W^2$, to be minimized over $\theta = (\tau,\alpha)$, β and γ. If we minimize over β and γ in an inner step, and over θ in an outer step, the inner step boils down to linear IV-GMM. We rewrite the optimization problem as the minimization of $F(\theta)$ over $\theta \in \mathbb{R}^N$, where

$$F(\theta) = \min_{(\beta,\gamma)\in\mathbb{R}^{K^x+L}} \{\|Z^\top(Y^\theta - X(\beta^\top,\gamma^\top)^\top\|_W^2\}/(2TY)^2,$$

so that $F(\theta)$ is the value of the objective function of the linear part of the IV-GMM procedure.

If we denote $\Pi = ZWZ^\top$ and $\Upsilon = \Pi - \Pi^\top X \left(X^\top \Pi X\right)^{-1} X^\top \Pi$, we have an explicit expression for the value of the minimizers β and γ given θ. Denoting by $\beta^\star(\theta)$ and $\gamma^\star(\theta)$ these minimizers, we have the following expressions

$$\begin{pmatrix} \beta^\star(\theta) \\ \gamma^\star(\theta) \end{pmatrix} = \left(X^\top \Pi X\right)^{-1} X^\top \Pi Y^\theta \text{ and } F(\theta) = \left(Y^\theta\right)^\top \Upsilon Y^\theta / (2TY)^2.$$

The function $F(\theta)$ is minimized numerically using gradient descent. It is therefore useful to obtain a formula for the differential of F, and we have

$$\frac{\partial F(\theta)}{\partial \theta} = \frac{1}{2} \frac{1}{(TY)^2} \left(Y^\theta\right)^\top \Upsilon \frac{\partial Y^\theta}{\partial \theta} \text{ and } \frac{\partial Y^\theta}{\partial \theta} = \begin{pmatrix} \frac{\partial \hat{U}^\tau}{\partial \tau} & -\mathbf{\Phi}^e \\ \frac{\partial \hat{c}^{\tau,\alpha}}{\partial \tau} & \frac{\partial \hat{c}^{\tau,\alpha}}{\partial \alpha} \end{pmatrix}.$$

BLP: heteroskedasticity-robust two-step approach

In the first step, we set $W = (Z^\top Z)^{-1}$, and we run the minimization of $F(\theta)$ as described in the previous slide. This yields a first-step estimator $\hat{\theta}$.

We then run a heteroskedasticity-robust second step by running the procedure in the previous slide again but this time using the updated weight matrix:

$$W = \left(Z^\top \operatorname{diag} \left(\left(Y^{\hat{\theta}} - X \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} \right)^2 \right) Z \right)^{-1}.$$

Summary of BLP's method

The following slides provide comprehensive tables detailing the dimensions of the objects, the required data, and the parameters to be estimated in the BLP method. We also outline the necessary matrices to construct before diving into the steps of the BLP algorithm.

Table 4.1: BLP's method - dimensions

Variable	Description
T	Number of markets.
Y	Number of products.
I	Number of simulation points.
M	Number of dimensions of idiosyncratic utility.
K^e	Number of endogenous demand-relevant systematic characteristics.
K^x	Number of exogenous demand-relevant systematic characteristics.
K	Number of demand systematic characteristics, $K = K^e + K^x$.
L	Number of supply-relevant characteristics.
D	Number of demand-side instruments.
S	Number of supply-side instruments.
N	Number of nonlinear parameters, $N = M + K^e$.

Table 4.2: BLP's method - required data (1/2)

Object	Size	Description
$\hat{\pi}$	$(TY, 1)$	$\hat{\pi}_{ty}$ is the market share of product y in market t.
p	$(TY, 1)$	p_{ty} is the price of product y in market t.
$\mathbf{\Phi}^e$	(TY, K^e)	ϕ^e_{ty,k^e} is the k^eth observable demand-relevant endogenous (i.e., price-dependent) characteristics of y in market t entering in the systematic utility U.
$\mathbf{\Phi}^{e\prime}$	(TY, K^e)	$\phi^{e\prime}_{ty,k^e} = \partial \phi^e_{ty,k^e} / \partial p_{ty}$, price-derivative of the previous quantity.
$\mathbf{\Phi}^x$	(TY, K^x)	ϕ^x_{ty,k^x} is the k^xth observable demand-relevant exogenous (i.e., price-independent) characteristics of y in market t entering in the systematic utility U.
η	$(IM, 1)$	η_{im} is the mth entry of the random characteristics of individual i (across markets), that interacts with ξ below to enter in the idiosyncratic utility $\boldsymbol{\nu}\tau$.

Table 4.2: BLP's method - required data (2/2)

Object	Size	Description
ξ	$(TYM, 1)$	ξ_{tym} is the mth entry of the characteristics of product y in market t, that interacts with η above to enter in the idiosyncratic utility $\boldsymbol{\nu}\tau$.
ξ'	$(TYM, 1)$	$\xi'_{tym} = \partial \xi_{tym} / \partial p_y$ is the price-derivative of the previous quantity.
$\boldsymbol{\zeta}$	(TY, D)	$\zeta_{ty,d}$, dth instrument for the demand for product y in market t.
$\boldsymbol{\Psi}$	(TY, L)	$\Psi_{ty,l}$ is the lth supply-relevant characteristics of product y in market t.
$\mathbf{H}$	(TY, TY)	$H_{ty,t'y'}$ is 1 if y and y' are controlled by the same firm and $t = t'$, zero otherwise.
χ	(TY, S)	$\chi_{ty,s}$ is the sth instrument for the demand for product y in market t.

Table 4.3: BLP's method - estimated parameters

Object	Size	Description
α	$(K^e, 1)$	parametrizes the endogenous part of the systematic consumer utility $\mathbf{\Phi}^e\alpha$.
β	$(K^x, 1)$	parametrizes the exogenous part of the systematic consumer utility $\mathbf{\Phi}^x\beta$.
λ	$(K, 1)$	$\lambda = (\alpha^\top, \beta^\top)^\top$ parametrizes the overall systematic utility $U = \mathbf{\Phi}\lambda$, see below.
γ	$(L, 1)$	γ_l parametrizes the seller cost $c = \mathbf{\Psi}\gamma + \sigma$, where σ is a noise term.
τ	$(M, 1)$	τ_m parametrizes the idiosyncratic utility of the consumer $\boldsymbol{\nu}\tau$.
θ	$(N, 1)$	$\theta = (\tau^\top, \alpha^\top)^\top$ nonlinear GMM parameter ($N = M + K^e$).

Table 4.4: BLP's method - matrices

Object	Formula	Size	Comment
$\mathbf{\Sigma}_I$	$\mathbf{\Sigma}_I = I_T \otimes 1_I^\top \otimes I_Y$	(TY, TIY)	summation over $i \in [I]$.
$\mathbf{\Sigma}_Y$	$\mathbf{\Sigma}_Y = I_T \otimes I_I \otimes 1_Y^\top$	(TI, TIY)	summation over $y \in [Y]$.
$\mathbf{\Phi}$	$\mathbf{\Phi} = (\Phi^e, \Phi^x)$	(TY, K)	demand-relevant characteristics of y (systematic part). $K = K^e + K^x$.
$\frac{\partial \mathbf{\Phi}^e}{\partial p}$	$\frac{\partial \mathbf{\Phi}^e}{\partial p} = I_{TY} \circledast \Phi^{e\prime}$	(TY, TYK^e)	price-differential of Φ^e.
$\boldsymbol{\nu}$	$\nu_{tiy,m} = \eta_{im}\xi_{tym}$	(TIY, M)	$\boldsymbol{\nu}_{tiy,m}$ is the mth component of the random utility associated by i with y.
$\boldsymbol{\nu}'$	$\nu'_{tiy,m} = \eta_{im}\frac{\partial \xi_{tym}}{\partial p_y}$	(TIY, M)	$\nu'_{tiy,m} = \partial \nu_{tiy,m} / \partial p_y$ is the price-derivative of the previous quantity.
$\frac{\partial \boldsymbol{\nu}}{\partial p}$	$\frac{\partial \boldsymbol{\nu}}{\partial p} = (I_T \otimes 1_I \otimes I_Y) \circledast \boldsymbol{\nu}'$	(TIY, TYM)	price-differential of $\boldsymbol{\nu}$.

Table 4.5: BLP's method, main procedure (1/5)

Step	Given $\theta = (\tau, \alpha) \in \mathbb{R}^M \times \mathbb{R}^{K^e}$, and W a $(D+S) \times (D+S)$ matrix, compute
1.	Following algorithm 4.4.1, iterate $$\begin{cases} U^{s+1} = -\log(\mathbf{\Delta}_{\hat{\pi}}^{-1} I^{-1} \mathbf{\Sigma}_I \exp(\boldsymbol{\nu}\tau - \mathbf{\Sigma}_Y^\top u^s)) \\ u^{s+1} = \log(\mathbf{1}_{TI} + \mathbf{\Sigma}_Y \exp(\mathbf{\Sigma}_I^\top U^{s+1} + \boldsymbol{\nu}\tau)). \end{cases}$$ over $s \in \mathbb{N}$ until $\max_{ty \in [T \times Y]} \lVert U_{ty}^s - U_{ty}^{s-1} \rVert$ falls below tolerance level.
2.	Set $\hat{f} = \exp(\mathbf{\Sigma}_I^\top \hat{U}^\tau + \boldsymbol{\nu}\tau - \mathbf{\Sigma}_Y^\top \hat{u}^\tau)$.
3.	Set $\mathbf{R}_{\hat{f}} = \mathbf{I}_{TIY} - (\mathbf{I}_{TI} \otimes \mathbf{1}_{Y \times Y})\mathbf{\Delta}_{\hat{f}}$.
4.	Letting $V = \mathbf{\Sigma}_I^\top (\mathbf{\Phi}\lambda + \kappa) + \boldsymbol{\nu}\tau$, set $$\frac{\partial V}{\partial p} = \mathbf{\Sigma}_I^\top \frac{\partial \mathbf{\Phi}^e}{\partial p} (\mathbf{I}_{TY} \otimes \alpha) + \frac{\partial \boldsymbol{\nu}}{\partial p} (\mathbf{I}_{TY} \otimes \tau).$$

Table 4.5: BLP's method - main procedure (2/5)

Step	Compute
5.	Set $$\widehat{\mathbf{D}} = \mathbf{H} \odot \left(\frac{1}{I} \mathbf{\Sigma}_I \mathbf{\Delta}_{\hat{f}} \mathbf{R}_{\hat{f}} \frac{\partial V}{\partial p} \right)^\top .$$
6.	Set $\hat{c}^{\tau,\alpha} = p + \widehat{\mathbf{D}}^{-1} \hat{\pi}$.
7.	Set $$Y^\theta = \begin{pmatrix} \hat{U}^\tau - \mathbf{\Phi}^e \alpha \\ \hat{c}^{\tau,\alpha} \end{pmatrix}, \; X = \begin{pmatrix} \mathbf{\Phi}^x & \mathbf{0}_{TY,L} \\ \mathbf{0}_{TY,K^x} & \mathbf{\Psi} \end{pmatrix}, \; Z = \begin{pmatrix} \zeta & \mathbf{0}_{TY,S} \\ \mathbf{0}_{TY,D} & \chi \end{pmatrix}.$$
8.	Set $\Pi = ZWZ^\top$ and $\Upsilon = \Pi - \Pi^\top X \left(X^\top \Pi X \right)^{-1} X^\top \Pi$.
9.	Set $F(\theta) = \left(Y^\theta \right)^\top \Upsilon Y^\theta / (2TY)^2$.

Table 4.5: BLP's method - main procedure (3/5)

Step	Compute
10.	Set $$\frac{\partial \hat{U}^\tau}{\partial \tau} = \left(-I\boldsymbol{\Delta}_{\hat{\pi}} + \boldsymbol{\Sigma}_I \boldsymbol{\Delta}_{\hat{f}} \boldsymbol{\Sigma}_Y^\top \boldsymbol{\Sigma}_Y \boldsymbol{\Delta}_{\hat{f}} \boldsymbol{\Sigma}_I^\top\right)^{-1} \boldsymbol{\Sigma}_I \boldsymbol{\Delta}_{\hat{f}} \mathbf{R}_{\hat{f}} \boldsymbol{\nu}.$$
11.	Letting $\hat{V} = \boldsymbol{\Sigma}_I^\top U^\tau + \boldsymbol{\nu}\tau$, set $$\frac{\partial \hat{V}}{\partial \theta} = \left(\frac{\partial \hat{V}}{\partial \tau}\ ,\ \mathbf{0}_{TIY,K^e}\right) \text{ where } \frac{\partial \hat{V}}{\partial \tau} = \boldsymbol{\Sigma}_I^\top \frac{\partial \hat{U}^\tau}{\partial \tau} + \boldsymbol{\nu}.$$

Table 4.5: BLP's method - main procedure (4/5)

Step	Compute
12.	Recall that $V = \mathbf{\Sigma}_I^\top (\mathbf{\Phi}\lambda + \kappa) + \boldsymbol{\nu}\tau$ and set $$\left(\frac{\partial^2 V}{\partial\tau\partial p}\right)_{tiy,mt'y'} = \frac{\partial \boldsymbol{\nu}_{tiy,m}}{\partial p_{t'y'}} \text{ and } \left(\frac{\partial^2 V}{\partial\alpha\partial p}\right)_{tiy,k^e t'y'} = \left(\mathbf{\Sigma}_I^\top \frac{\partial \mathbf{\Phi}^e}{\partial p}\right)_{tiy,t'y'm}.$$
13.	Set $\frac{\partial^2 V}{\partial\theta\partial p} = \left(\frac{\partial^2 V}{\partial\tau\partial p}, \frac{\partial^2 V}{\partial\alpha\partial p}\right).$
14.	Set $$\frac{\partial \hat{\mathbf{D}}^\top}{\partial\theta} = \left(\mathbf{1}_N^\top \otimes \mathbf{H}\right) \odot \left(\frac{1}{I}\mathbf{\Sigma}_I \mathbf{\Delta}_{\hat{f}} \mathbf{R}_{\hat{f}} \left(\left(\mathbf{R}_{\hat{f}} \frac{\partial \hat{V}}{\partial\theta} \circledast \mathbf{R}_{\hat{f}} \frac{\partial V}{\partial p}\right) + \frac{\partial^2 V}{\partial\theta\partial p}\right)\right).$$

Table 4.5: BLP's method - main procedure (5/5)

Step	Compute
15.	Set $$\left(\frac{\partial \widehat{\mathbf{D}}}{\partial \theta}\right)_{ty,nt'y'} = \left(\frac{\partial \widehat{\mathbf{D}}^\top}{\partial \theta}\right)_{t'y',nty}.$$
16.	Set $$\frac{\partial \hat{c}^\theta}{\partial \theta} = -\widehat{\mathbf{D}}^{-1} \frac{\partial \widehat{\mathbf{D}}}{\partial \theta} \left(\mathbf{I}_N \otimes \left(\hat{c}^\theta - \mathsf{p}\right)\right).$$
17.	Set $$\frac{\partial F(\theta)}{\partial \theta} = \frac{1}{2} \frac{1}{(TY)^2} \left(Y^{\theta_1}\right)^\top \Upsilon \begin{pmatrix} \frac{\partial \hat{U}^\tau}{\partial \tau} & -\mathbf{\Phi}^e \\ \frac{\partial \hat{c}^{\tau,\alpha}}{\partial \tau} & \frac{\partial \hat{c}^{\tau,\alpha}}{\partial \alpha} \end{pmatrix}.$$

BLP Practice Notebook

We have implemented a BLP Notebook divided into two parts. The first features a manual implementation adapted from the DCM, comparing the results on a synthetic dataset against the PyBLP package. The second part replicates the results of the original BLP automobile paper and serves as an introduction to using PyBLP.

You can find the BLP Notebook at this address.

Exercises and Quick Recap Chapter 4

Short exercise (1/2)

Exercise 4.1: Consider the pure characteristics model of section 4.1 when there is only one characteristic, that is, $K = 1$. Give simplified expressions for the usual objects (welfare function, market share map, entropy of choice, inverse market share).

Short exercises (2/2)

Exercise 4.2: We recall the expression of the market share map associated with the random coefficient logit model: $\pi_y(U) = \frac{1}{I}\sum_{i\in[I]} \frac{\exp(U_y+\nu_{iy})}{\sum_{y'\in[Y]}\exp(U_{y'}+\nu_{iy'})}$.
Show that the Iterative Proportional Fitting Procedure (IPFP) algorithm can equivalently be rewritten as: $U_y^{t+1} = U_y^t - \ln\left(\frac{\pi_y(U^t)}{\hat{\pi}_y}\right)$, or equivalently,
$U_y^{t+1} = U_y^t + \ln(\hat{\pi}_y) - \ln(\pi_y(U^t))$.
This is classically called the '**BLP contraction mapping algorithm**' in the economic literature (Berry, Levinsohn, and Pakes, 1995).

Comment on the resemblance of this update rule to a gradient descent procedure minimizing the distance between observed and predicted shares.

Quick recap of Chapter 4 (1/2)

This chapter dealt with the characteristics-based approach to demand, focusing on models that move beyond the limitations of independent utility shocks for each alternative. In these models, utility is derived from the attributes of the goods, allowing for a more nuanced analysis of consumer preferences. We first covered the Lancasterian framework. We then introduced the mixed logit model, where random coefficients account for preference heterogeneity through both systematic utility and idiosyncratic taste variation. We also detailed its estimation and computation to highlight its potential.

Quick recap of Chapter 4 (2/2)

However, in Section 4.3, we showed that an endogeneity problem arises. We address this by assuming the existence of a product characteristic unobserved by the analyst, which we recover by inverting the market share map and regressing it on other characteristics. To address endogeneity at equilibrium, we employ an IV-GMM approach, defining adequate orthogonal restrictions to allow for parametric estimation. Finally, we introduced the cornerstone of structural estimation, the BLP method presented in full generality, and provided a Notebook to practice its computation.

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Chapter 5. Generalized linear models for allocation and equilibrium problems

Discrete Choice Models, ch. 5, pp. 121-149

Content

In this chapter we will see that the logistic regression framework, and more generally, the addition of fixed effects in generalized linear models, can be used to capture a wide range of econometric models, with a large number of applications covering international trade, labor economics, family economics, and demand estimation, among others.

Section 5.1. Macro logistic regression

Section content

- Observable types
- Formulation of the macro logistic regression

Observable types

We generalize the logistic regression framework described in the previous chapter.

Agents are clustered into observable types $x \in [X]$, and two agents of the same type are indistinguishable by the analyst, but draw different utility shocks.

An individual i's type is denoted x_i and the systematic utility is now

$$\sum_{k \in [K]} \phi_{x_i y k} \lambda_k \text{ instead of } \sum_{k \in [K]} \phi_{iyk} \lambda_k$$

We therefore define:

- $n_x = \sum_i 1\{x_i = x\}$ as the number of agents of type x, and
- $\hat{\mu}_{xy} = \sum_i 1\{x_i = x\} 1\{y_i = y\}$ the number of agents of type x choosing y.
- $m_y = \sum_i 1\{y_i = y\}$ as the number of options y demanded.

We have $n_x = \sum_y \hat{\mu}_{xy}$ and $m_y = \sum_x \hat{\mu}_{xy}$.

The standard logistic regression has $n_x = 1 \; \forall x \in [X]$.

Letting Φ be the matrix of shape $XY \times K$ with term $\phi_{xy,k} = \phi_{xyk}$, and $\hat{\mu}$ be the XY-dimensional vector with term $\hat{\mu}_{xy}$, the (macro) multinomial logistic regression is:

$$\max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\mu}^\top \Phi \lambda - \sum_{x \in [X]} n_x \log \left(\sum_{y \in [Y]} \exp(\Phi\lambda)_{xy} \right) \right\}$$

and as before, we define the predicted choice probabilities

$$\pi^\lambda_{xy} = \frac{\exp(\Phi\lambda)_{xy}}{\sum_{z \in [Y]} \exp(\Phi\lambda)_{xz}}$$

and the first order conditions in the macro logistic regression lead to the usual moment fitting conditions $\sum_{xy} n_x \pi^\lambda_{xy} \phi_{xyk} = \sum_{xy} \hat{\mu}_{xy} \phi_{xyk}$ for all $k \in [K]$.

Without surprise, we can interpret the previous problem as Poisson regression with x-fixed effects. As before, the set of observations is $[\Omega] = [X \times Y]$, the dependent variable is $\hat{\mu}_{xy}$, the regressor matrix $R = \begin{pmatrix} \Phi & -\mathrm{I}_X \otimes 1_Y \end{pmatrix}$, and the problem becomes

$$\max_{\lambda, \tilde{u}} \left\{ \sum_{xy \in [X \times Y]} \hat{\mu}_{xy}((\Phi\lambda)_{xy} - \tilde{u}_x) - \sum_{xy \in [X \times Y]} \exp((\Phi\lambda)_{xy} - \tilde{u}_x) \right\}$$

with $\tilde{u}_x = u_x - \log n_x$. The dual problem is familiar:

$$\begin{aligned} \min_{\mu_{xy} \geq 0} \quad & \sum_{xy \in [X \times Y]} \mu_{xy} \log \mu_{xy} \\ s.t. \quad & \sum_{y \in [Y]} \mu_{xy} = n_x \\ & \sum_{xy \in [X \times Y]} \mu_{xy} \phi_{xyk} = \sum_{xy \in [X \times Y]} \hat{\mu}_{xy} \phi_{xyk}. \end{aligned}$$

Remark on the macro logistic regression

Both the primal and dual forms provided on the previous slide should be familiar, as the macro-level logistic regression is a slight generalization of the "micro" logistic model from Section 3.2, to which it reduces when $n_x = 1$ for all $x \in [X]$.

Section 5.2. Entropic optimal transport

Section content

- Entropy-regularized optimal transport
- Iterated proportional fitting procedure

Choice-specific fixed effect only

Let us consider the case where the regressors are the y-fixed effects. We denote these v_y, so that $\Phi = -\mathbf{1}_X \otimes \mathbf{I}_Y$ (we include a minus in front of the fixed effects for cosmetic reasons). Relabeling $\tilde{u}$ as u, the problem writes:

$$\min_{u,v} \left\{ \sum_{x \in [X]} n_x u_x + \sum_{y \in [Y]} m_y v_y + \sum_{xy \in [X \times Y]} \exp\left(-u_x - v_y\right) \right\}$$

We have $\sum_{xy} \mu_{xy} = I$ and one may assume w.l.o.g. that $I = 1$. The dual problem is:

$$\begin{aligned} \max_{\mu \geq 0} \quad & \left\{ - \sum_{xy \in [X \times Y]} \mu_{xy} \log \mu_{xy} \right\} \\ s.t. \quad & \sum_{y \in [Y]} \mu_{xy} = n_x \text{ and } \sum_{x \in [X]} \mu_{xy} = m_y. \end{aligned}$$

It is easy to see that the solution is the independent coupling: $\mu_{xy} = n_x m_y$.

Regularized optimal transport

Now consider inserting a term in the exponential

$$\min_{u,v} \left\{ \sum_x n_x u_x + \sum_y m_y v_y + \sum_{xy} \exp\left(\phi_{xy} - u_x - v_y\right) \right\}$$

which is a weighted Poisson regression, where observation xy has weight $\exp \phi_{xy}$, and we must take $\hat{\mu}_{xy} = n_x m_y \exp\left(-\phi_{xy}\right)$ as dependent variable. The former problem is dual to

$$\begin{aligned} \max_{\mu \geq 0} \quad & \left\{ \sum_{xy \in [X \times Y]} \mu_{xy}\phi_{xy} - \sum_{xy \in [X \times Y]} \mu_{xy} \log \mu_{xy} \right\} \\ s.t. \quad & \sum_{y \in [Y]} \mu_{xy} = n_x \text{ and } \sum_{x \in [X]} \mu_{xy} = m_y. \end{aligned}$$

which is an optimal transport problem with entropic regularization.

Regularized optimal transport as a Poisson regression

We state:

Proposition 5.2.1

The entropic regularized optimal transport problem can be computed as a weighted Poisson regression of $\hat{\mu}_{xy} = n_x m_y \exp(-\phi_{xy})$ on regressors solely composed of x- and y-fixed effects, with weights $w_{xy} = \exp \phi_{xy}$.

Computation of entropic optimal transport via GLMs

The advantage of recasting the optimal transport problem as a Poisson regression is that it allows us to solve optimal transport problems using standard Generalized Linear Model (GLM) estimation tools (see the Python code on the next slide).

Python code for entropic optimal transport via GLMs

```
import numpy as np, scipy.sparse as sp
from sklearn import linear_model as lm
np.random.seed(7)
X,Y = 5,4
n_x, m_y = np.ones(X)/X, np.ones(Y)/Y
phi_xy = np.random.uniform(size = X*Y)
# Computing:
w_xy = np.exp( phi_xy )
muhat_xy =  np.kron(n_x,m_y) / w_xy
ot_as_glm = lm.PoissonRegressor(fit_intercept=False,alpha=0)
regr = - sp.hstack([ sp.kron(sp.eye(X), np.ones((Y,1))),
                    sp.kron(np.ones((X,1)),sp.eye(Y))])
ot_as_glm.fit( regr, muhat_xy , sample_weight = w_xy )
u_x,v_y  = ot_as_glm.coef_[:X], ot_as_glm.coef_[X:]
# displaying results
print('value=',u_x @ n_x +v_y @ m_y )
```


Computation of entropic optimal transport via IPFP

The previous minimization problem can be solved using gradient descent, but a powerful alternative exists: coordinate descent. The principle is as follows: at iteration t, fix a guess u^t and minimize the objective function with respect to v only. Call this solution v^{t+1}. By inspecting the first-order conditions (FOC), v^{t+1} can be obtained as an explicit function of u^t: $v_y^{t+1} = \log\left(m_y^{-1}\sum_x \exp(\phi_{xy} - u_x^t)\right)$. Similarly, optimizing with respect to u_x only and fixing $v_y = v_y^{t+1}$ yields $u_x^{t+1} = \log\left(n_x^{-1}\sum_y \exp(\phi_{xy} - v_y^{t+1})\right)$. This is known as Sinkhorn's algorithm. Defining $K_{xy} = \exp(\phi_{xy})$, $A_x = \exp(-u_x)$, and $B_y = \exp(-v_y)$ allows us to formulate its standard matrix expression.

The most efficient method to compute the regularized optimal transport problem is the Sinkhorn's algorithm. It consists of coordinate descent on the minimization problem:

Algorithm 5.2.1 iterated proportional fitting procedure (IPFP)

Input: K_{xy}, B_y^0

Parameter: η (tolerance)

while $\max_{y\in[Y]} \|B_y^t - B_y^{t-1}\| \geq \eta$ **do**

$$A_x^t = n_x / \sum_{y\in[Y]} B_y^{t-1} K_{xy}$$

$$B_y^t = m_y / \sum_{x\in[X]} A_x^t K_{xy}$$

Return: A^t, B^t

The following result ensures the convergence of the IPFP. Readers can find convergence rates in Theorem A.1.9 in the Appendix.

Theorem 5.2.1 (Convergence of the IPFP

Consider the sequence (A_x^t, B_y^t) defined in the IPFP algorithm. Then:

- The quantities $(-\log A_x^t, -\log B_y^t)$ converge (up to a constant) towards a solution (u_x, v_y) to the problem.
- The quantity $\mu_{xy}^t = A_x^t B_y^t K_{xy}$ converges toward the unique solution μ_{xy} to the regularized optimal transport problem.

Python code for entropic optimal transport via IPFP

```
import numpy as np, scipy.sparse as sp
np.random.seed(7)
X,Y = 5,4
n_x, m_y = np.ones(X)/X, np.ones(Y)/Y
phi_xy = np.random.uniform(size = X*Y)
tol,maxit = 1e-5,1000
# Computing:
K_x_y = np.exp(phi_xy).reshape((X,Y))
B_y = np.ones(Y)
for i in range(maxit):
  A_x = n_x / (K_x_y @ B_y)
  Bnew_y = m_y / (A_x @ K_x_y)
  if np.abs(Bnew_y-B_y).max() < tol:
    break
  B_y = Bnew_y
u_x,v_y = - np.log(A_x),- np.log(B_y)
# displaying results
print('Value=',u_x @ n_x +v_y @ m_y , 'Steps=',i)
```


Section 5.3. Gravity equation

Section content

- Gravity equation
- The SISTA algorithm

Explaining international trade (1/3)

A central question in international economics is to explain the volume of trade between countries. For this, the gravity equation is a central tool. It connects the size of trades between countries to distance-related variables: geographic distance, common language spoken, cultural similarities, etc.

Example 5.1

Yotov, Piermartini, Monteiro and Larch (2016) provide data on trade flows between pairs of countries. We seek to explain these trade flows using various measures of proximity between countries (e.g., common language, geographic distance, contiguity). For each origin-destination pair $xy \in [X \times Y]$, we observe $\hat{\mu}_{xy}$, the volume of trade from origin country x to destination country y, which is the dependent variable, and we observe a K explanatory variables ϕ_{xyk} which are various measures of distance between country x and country y.

Explaining international trade (3/3)

We consider a macro logistic regression where we include as regressors various measures of distance as well as $y-$fixed effects:

$$\max_{\lambda,u,v}\left\{\sum_{xy\in[X\times Y]}\hat{\mu}_{xy}((\Phi\lambda)_{xy}-u_x-v_y)-\sum_{xy\in[X\times Y]}\exp((\Phi\lambda)_{xy}-u_x-v_y)\right\},$$

and its dual expression becomes

$$\begin{aligned}\min_{\mu_{xy}\geq 0}\quad & \sum_{xy\in[X\times Y]}\mu_{xy}\log\mu_{xy}\\ s.t.\quad & \sum_{y\in[Y]}\mu_{xy}=n_x \text{ and } \sum_{x\in[X]}\mu_{xy}=m_y\\ & \sum_{xy\in[X\times Y]}\mu_{xy}\phi_{xyk}=\sum_{xy\in[X\times Y]}\hat{\mu}_{xy}\phi_{xyk}.\end{aligned}$$

Structural gravity equation

By the FOC, the optimal primal solution (λ, u, v) and the optimal dual solution μ are related by the structural gravity equation:

$$\mu_{xy} = \exp\left((\Phi\lambda)_{xy} - u_x - v_y\right)$$

In the international trade literature $\exp u_x$ and $\exp v_y$ are called **multilateral resistances**, which fit the right flow of exports and imports, that is, denoting $K^{\lambda}_{xy} = \exp\left(\Phi\lambda\right)_{xy}$,

$$\sum_{y \in [Y]} \frac{K^{\lambda}_{xy}}{e^{u_x} e^{v_y}} = n_x \qquad \text{and} \qquad \sum_{x \in [X]} \frac{K^{\lambda}_{xy}}{e^{u_x} e^{v_y}} = m_y.$$

Computation by Poisson regression

The macro-logistic regression can be computed as a Poisson regression with x- and y-fixed effects. Under this correspondence, the set of observations is $\Omega = [X] \times [Y]$. The design matrix R is now composed of three blocks: the main regressors, plus the matrices corresponding to the x- and y-fixed effects typically structured as: $R = \begin{pmatrix} \phi & -I_X \otimes \mathbf{1}_Y & -\mathbf{1}_X \otimes I_Y \end{pmatrix}$ The corresponding code is provided in Appendix B.5.3 of the *DCM*.

Computation by SISTA (1/2)

To solve the problem with x and y fixed effects, we consider the slightly more general LASSO-penalized problem:

$$\max_{\lambda,u,v} \left\{ \sum_{xy\in[X\times Y]} \hat{\mu}_{xy}((\Phi\lambda)_{xy} - u_x - v_y) - \sum_{xy\in[X\times Y]} e^{(\Phi\lambda)_{xy} - u_x - v_y} - \gamma \sum_{k\in[K]} |\lambda_k| \right\},$$

The SISTA (Sinkhorn Iterative Shrinkage-Thresholding Algorithm) procedure consists of iterating over coordinate descent steps with respect to u_x and v_y (as in Sinkhorn's algorithm), and proximal gradient (iterative shirinkage-thresholding) steps with respect to λ_k. Recall that the soft-thresholding operator is defined by $\text{SoftThresholding}_\epsilon(x) = 0$ if $|x| < \epsilon$, $= x - \epsilon$ if $x > \epsilon$, and $= x + \epsilon$ if $x < -\epsilon$.

Computation by SISTA (2/2)

Algorithm 5.3.2 SISTA

Input: λ_k^0, B_y^0

Parameter: η (tolerance), α, γ

while $\max_k |\lambda_k^t - \lambda_k^{t-1}| \geq \eta$ **do**

$$A_x^t = n_x / \sum_{y \in [Y]} B_y^{t-1} \exp(\Phi \lambda^{t-1})_{xy}$$

$$B_y^t = m_y / \sum_{x \in [X]} A_x^t \exp(\Phi \lambda^{t-1})_{xy}$$

$$\mu_{xy}^t = A_x^t B_y^t \exp(\Phi \lambda^{t-1})_{xy}$$

$$\lambda^{t+1} = \text{SoftThresholding}_{\alpha\gamma} \left(\lambda^t + \alpha \Phi^\top (\hat{\mu} - \mu^t) \right)$$

end

Return: A^t, B^t, λ^t

Convergence of SISTA

The SISTA algorithm alternates between coordinate descent steps on the fixed effects u_x and v_y, solvable in closed form ("S" steps for Sinkhorn), and proximal gradient steps on the parameter λ_k ("ISTA" steps). We provide the code in the next slide. The following theorem is due to Carlier, Dupuy, Galichon, and Sun and proves the convergence at a linear rate of SISTA.

Theorem 5.3.1

The parameter vector λ^t in the SISTA algorithm converges to the solution to the gravity equation problem.

We provide the Python code for estimating the gravity equation using **SISTA** on the next slide.

Python code for gravity via SISTA

```
import numpy as np
np.random.seed(7)
X,Y,K = 5,5,3
muhat_x_y = np.random.uniform(size = (X,Y) )
np.fill_diagonal(muhat_x_y,0)
phi_xy_k = np.random.uniform(size = (X*Y,3) )
def soft_thresholding(x, eta):
    return np.sign(x) * np.maximum(np.abs(x) - eta, 0)
# Computing :
n_x,m_y = muhat_x_y.sum(axis =1), muhat_x_y.sum(axis =0)
gamma = 0.1 # set to zero for no l1 penalty
alpha = 1 / np.linalg.norm(phi_xy_k @ phi_xy_k.T)
lambda_k = np.zeros(K)
tol = 1e-5
iter = 1000
B_y = np.ones(Y)
# continued on the next slide =>>
```

Python code for gravity via SISTA (continued)

```
for i in range(iter):
  K_x_y = np.exp(phi_xy_k @ lambda_k).reshape( (X,Y))
  np.fill_diagonal(K_x_y,0)
  A_x = n_x / (K_x_y @ B_y)
  B_y = m_y / (A_x @ K_x_y)
  mu_x_y =  (A_x[:,None] * B_y[None,:] * K_x_y)
  grad_k =  (muhat_x_y - mu_x_y).flatten() @ phi_xy_k
  lambdatilde_k = lambda_k + alpha*grad_k
  new_lambda_k = soft_thresholding(lambdatilde_k , alpha*gamma )
  if np.linalg.norm( lambda_k -new_lambda_k) < alpha*tol:
    iter = i
    break
  lambda_k = new_lambda_k
# displaying results
print('lambda_k=',lambda_k,'Steps=',i)
```


Section 5.4. Empirical models of matching with transferable utility

Section content

- The Becker model
- The Choo and Siow model
- Parametric estimation
- Computation

Two-sided matching models with transferable utility

In this section, we study two-sided matching markets. We consider as an example the case of the heterosexual marriage market.

We assume that a match between two agents generated a joint utility to be split endogenously between partners. This is a model with transferable utility, à-la Becker (1973) with the incorporation of logit random utility, after Choo and Siow (2006). They utilize marriage data from the U.S. Census from the years preceding and following the 1973 Roe v. Wade decision, which established fundamental abortion rights across all states. They formulate a parametric model in which the joint marital surplus depends on the partners' ages and the legal status of abortion in a given state at a given time.

The model setup (1/2)

Men are clustered into X type and women into Y types. Anyone can decide to match or to remain single. When man i of type x matches with woman j of type y, they respectively receive $\alpha_{xy} + \tau_{xy} + \varepsilon_{iy}$ and $\gamma_{xy} - \tau_{xy} + \eta_{jx}$, where:
•**matching affinities** α_{xy} and γ_{xy} are exogenous marital enjoyment terms, whose sum is

$$\phi_{xy} = \alpha_{xy} + \gamma_{xy}$$

which is called the joint systematic surplus, that we are seeking to estimate.
•**transfers:** the quantity τ_{xy} is a utility transfer. It is determined at equilibrium by market-clearing and thus it plays the role of a price.
•**utility shocks**: ε_{iy} and η_{jx} are unobserved random utility shocks that get added to α_{xy} and γ_{xy}, respectively. They capture idiosyncratic tastes for some types of partner.

The model setup (2/2)

If man i remains unmatched, his utility is determined solely by his idiosyncratic shock ε_{i0}. This implies that the systematic component of the payoff from being single is normalized to zero. Analogous reasoning applies to women.

Key assumptions

Large market: there is an infinite number of individuals of each type. We denote by n_x and m_y the shares of men of type x and women of type y in the total population, so

$$\sum_x n_x + \sum_y m_y = 1$$

Logit random utilities: the utility shocks (ε_{iy}) of man i relative to partner type $y \in [0 : Y]$ form a vector of $Y + 1$ i.i.d. draws from a standard Gumbel distribution. Similarly, the utility shocks (η_{jx}) of woman j relative to partner type $x \in [0 : X]$ form a vector of $X + 1$ i.i.d. draws from a standard Gumbel distribution.

Marital demand of men

For a fixed vector of transfers τ_{xy}, man i's problem is

$$u_i = \max_{y \in [Y]} \{U_{xy} + \varepsilon_{iy}, \varepsilon_{i0}\}$$

where $U_{xy} := \alpha_{xy} + \tau_{xy}$. The average utility of type x is

$$u_x = \log(1 + \sum_{y \in [Y]} \exp U_{xy})$$

and the probability that a man of type x picks a marital option $y \in [Y]$ is

$$\exp(U_{xy} - u_x) = \frac{\exp U_{xy}}{1 + \sum_{z \in [Y]} \exp U_{xz}}.$$

The mass of matches xy that are formed at equilibrium is $\mu_{xy} = n_x \exp(U_{xy} - u_x)$.

Marital demand of women

For a fixed vector of transfers τ_{xy}, woman j's problem is

$$v_j = \max_{x \in [X]} \{ V_{xy} + \eta_{jx}, \varepsilon_{j0} \}$$

where $V_{xy} := \gamma_{xy} - \tau_{xy}$. The average utility of type y is

$$v_y = \log(1 + \sum_{x \in [X]} \exp V_{xy})$$

and the probability that a woman of type y picks a marital option $x \in [Y]$ is

$$\exp(V_{xy} - v_y) = \frac{\exp V_{xy}}{1 + \sum_{z \in [X]} \exp V_{zy}}.$$

The mass of matches xy that are formed at equilibrium is $\mu_{xy} = m_y \exp(V_{xy} - v_y)$.

Matching function

Putting this together,

$$\mu_{xy} = n_x \exp(U_{xy} - u_x) \text{ and } \mu_{xy} = m_y \exp(V_{xy} - v_y),$$

and $\mu_{x0} = n_x \exp(-u_x)$ and $\mu_{0y} = m_y \exp(-v_y)$. Taking the geometric mean and using the fact that $U_{xy} + V_{xy} = \phi_{xy}$, we get:

$$\begin{cases} \mu_{xy} = \sqrt{n_x m_y} \exp\left(\frac{\phi_{xy} - u_x - v_y}{2}\right) \\ \mu_{x0} = n_x \exp\left(-u_x\right) \\ \mu_{0y} = m_y \exp\left(-v_y\right). \end{cases}$$

which yields the **Choo and Siow matching function** $\mu_{xy} = \sqrt{\mu_{x0}\mu_{0y}} \exp(\phi_{xy}/2)$.

Equilibrium

- Equilibrium is the determination of two elements:
  (i) who gets what: u_x and v_y, and
  (ii) who matches with whom: μ_{xy}, μ_{x0} and μ_{0y}.

- The demand conditions yield Choo and Siow's matching function

$$\mu_{xy} = \sqrt{\mu_{x0}\mu_{0y}} \exp(\phi_{xy}/2).$$

- The market clearing conditions are

$$\begin{cases} \sum_{y \in [Y]} \mu_{xy} + \mu_{x0} = n_x \\ \sum_{x \in [X]} \mu_{xy} + \mu_{0y} = m_y. \end{cases}$$

Combining the demand conditions with the market-clearing ones yields:

$$\begin{cases} \sum_{y \in [Y]} \sqrt{n_x m_y} \exp\left(\frac{\phi_{xy} - u_x - v_y}{2}\right) + n_x \exp(-u_x) = n_x \\ \sum_{x \in [X]} \sqrt{n_x m_y} \exp\left(\frac{\phi_{xy} - u_x - v_y}{2}\right) + m_y \exp(-v_y) = m_y. \end{cases}$$

which Galichon and Salanié reformulated as the optimality conditions for

$$\min_{u,v} \left\{ \begin{array}{c} \sum_{x \in [X]} n_x u_x + \sum_{y \in [Y]} m_y v_y \\ +2 \sum_{xy \in [X \times Y]} \sqrt{n_x m_y} e^{\frac{\phi_{xy} - u_x - v_y}{2}} + \sum_{x \in [X]} n_x e^{-u_x} + \sum_{y \in [Y]} m_y e^{-v_y} \end{array} \right\}.$$

Strict convexity and coerciveness imply existence and uniqueness of solutions.

Estimation

Now we consider a parametric form of the matching surplus

$$\phi^{\lambda}_{xy} = (\Phi\lambda)_{xy} = \sum_{k\in[K]} \phi_{xyk}\lambda_k$$

and we seek to estimate λ based on the matching probabilities $\hat{\mu}$.

The estimation strategy is based on the method of moments, that is equating empirical and predicted averages of each ϕ_k, that is, finding (λ, u, v) such that

$$\begin{cases} \sum_{y\in[Y]} \sqrt{n_x m_y} \exp\left(\frac{(\Phi\lambda)_{xy} - u_x - v_y}{2}\right) + n_x \exp\left(-u_x\right) = n_x \\ \sum_{x\in[X]} \sqrt{n_x m_y} \exp\left(\frac{(\Phi\lambda)_{xy} - u_x - v_y}{2}\right) + m_y \exp\left(-v_y\right) = m_y \\ \sum_{xy\in[X\times Y]} \mu^{\lambda}_{xy}\phi_{xyk} = \sum_{xy\in[X\times Y]} \hat{\mu}_{xy}\phi_{xyk} \end{cases}$$

Estimation by optimization

As noted by Galichon and Salanié, we see that the previous system can be reformulated as the first order conditions of the following convex optimization problem

$$\min_{\lambda,u,v}\left\{\begin{array}{c}\sum_{x\in[X]} n_x u_x + \sum_{y\in[Y]} m_y v_y - \sum_{xyk\in[X\times Y\times K]} \hat{\mu}_{xy}\phi_{xyk}\lambda_k \\ +2\sum_{xy\in[X\times Y]} \sqrt{n_x m_y} e^{\frac{(\Phi\lambda)_{xy}-u_x-v_y}{2}} + \sum_{x\in[X]} n_x e^{-u_x} + \sum_{y\in[Y]} m_y e^{-v_y}\end{array}\right\},$$

where

- the optimality conditions for u_x provides the margin condition for type x, and
- the condition for v_y provides the margin condition for type y, and
- the condition for λ_k provides the moment condition for the kth moment of ϕ.

Matching estimation as a Poisson regression (1/3)

The observations categories are the types of households: matched of type xy, or unmatched male of type $x0$, or unmatched female of type $0y$. There are thus $B = XY + X + Y$ types of households. Our dataset consists of the number $\hat{\mu}_b$ of households observed in each category $b \in [B]$.

The regressor matrix R includes the determinants of the surplus ϕ plus the dummies $1\{x \in b\}$ for $x \in [X]$ and $1\{y \in b\}$ for $y \in [Y]$, hence we write

$$R = \begin{pmatrix} \Phi & -\mathbf{I}_X \otimes \mathbf{1}_Y & -\mathbf{1}_X \otimes \mathbf{I}_Y \\ \mathbf{0}_{X \times K} & -\mathbf{I}_X & \mathbf{0}_{X \times Y} \\ \mathbf{0}_{Y \times K} & \mathbf{0}_{Y \times X} & -\mathbf{I}_Y \end{pmatrix}.$$

R has shape $B \times P$ where $P = K + X + Y$.

Matching estimation as a Poisson regression (2/3)

Setting $\tilde{u}_x = u_x - \log n_x$ and $\tilde{v}_y = v_y - \log m_y$, the Poisson regression parameter $\theta = (\lambda^\top, \tilde{u}^\top, \tilde{v}^\top)^\top$ is of size P, and the estimating optimization problem rewrites

$$\max_{\theta \in \mathbb{R}^P} \left\{ \hat{\mu}^\top R\theta - \sum_{b \in [B]} s_b \exp \frac{(R\theta)_b}{s_b} \right\}$$

where $s_b = 2$ if b is a pair xy, and $s_b = 1$ if b is a single household $x0$ or $0y$. We state:

Theorem 5.4.1 (Galichon and Salanié)

The parametric matching surplus estimator $\hat{\lambda}$ can be obtained as the regression coefficient of a weighted Poisson regression with dependent variable $(\hat{\mu}_b)$, regressor matrix $\mathbf{\Delta}_s^{-1} R$, and weights (s_b), where $\mathbf{\Delta}_s = diag(s)$.

Theorem 5.4.1 (Galichon and Salanié) (continued)

Letting $B = R^\top \Pi^{-1} R$ and $C = R^\top V_\pi R$, the convergence $\hat{\theta} \to \theta_0$ holds almost surely as $N \to +\infty$, and

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, B^{-1} C B^{-1}\right).$$

Maximum likelihood estimation (1/3)

As an alternative estimation method, we explore the maximum likelihood approach. The total number of households $\sum_{xy} \mu_x y + \sum_x \mu_{x0} + \sum_y \mu_{0y}$ is predicted by $\mathbf{1}_B^\top \exp(S^{-1}R\theta)$, and thus the frequency of households of category $b \in [B]$ is given by

$$f_b^\theta = \frac{\exp((S^{-1}R\theta)_b)}{\mathbf{1}_B^\top \exp\left(S^{-1}R\theta\right)}$$

Denoting $\hat{f}_b = \hat{\mu}_b / I$ the sample frequency of household b, where I is the number of observed households, the maximum likelihood estimator is the solution to

$$\max_\theta \left\{ \hat{f}^\top S^{-1} R\theta - \log\left(\mathbf{1}_B^\top \exp\left(S^{-1}R\theta\right)\right)\right\}$$

which is a logistic regression problem.

Maximum likelihood estimation (2/3)

The optimality conditions of the maximum likelihood estimation problem are:

$$\begin{cases} \sum_{y \in [Y]} \frac{1}{2} f^{\theta}_{xy} + f^{\theta}_{x0} = \sum_{y \in [Y]} \frac{1}{2} \hat{f}_{xy} + \hat{f}_{x0} \\ \sum_{x \in [X]} \frac{1}{2} f^{\theta}_{xy} + f^{\theta}_{0y} = \sum_{x \in [X]} \frac{1}{2} \hat{f}_{xy} + \hat{f}_{0y} \\ \sum_{xy \in [X \times Y]} f^{\theta}_{xy} \phi_{xyk} = \sum_{xy \in [X \times Y]} \hat{f}_{xy} \phi_{xyk}. \end{cases}$$

These equations remind of us the method of moments estimator above, if not for the $1/2$ factor. In maximum likelihood estimation, we match with the average share of the household (individuals of each type). For example, if x is in a household xy, then he only represents half the household, and half the weight. If x is single, then he has all the weight.

We summarize previous remarks into:

Theorem 5.4.2 (MLE esimation in the Choo and Siow model):

The maximum likelihood estimator $\hat{\theta}$, defined by the maximization problem, can be obtained as a multinomial logistic regression with dependent variable $\hat{\mu}$ and regressor matrix $\Pi^{-1}R$. Assuming that the model is well specified (that is, the population π coincides with $\pi_{\hat{\theta}}$ at the true value of the parameter) and letting:

$$F = R^{\top}\Delta_s^{-1}(\Delta_\pi - \pi\pi^{\top})\Delta_s^{-1}R$$

be the Fisher information matrix, the convergence $\hat{\theta} \to \theta_0$ holds almost surely as $N \to +\infty$, and

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, F^{-1}\right).$$

Section 5.5. Coalition formation models

Section content

- Coalition formation
- Equilibrium
- Estimation as a Poisson regression

Coalition formation models

We seek to understand how agents form coalitions based on transferable utility, which generalizes one-to-one matching models just seen. Applications include same marriage, firms...

We still assume transferable utility, that is the total surplus can be split additively between participants. But agents no longer need to belong to distinct groups, and coalition may involve more than two individuals.

Population

Let $[Z]$ be the set of agents' types, for $z \in [Z]$, q_z is the (exogenous) mass of individuals of type z.

A coalition is a group of agents of different types. Within a coalition, there may be several agents of the same type. In a coalition labeled by $b \in [B]$, there are Σ_{zb} agents of type $z \in [Z]$. The size of a coalition is denoted

$$s_b = \sum_{z \in [Z]} \Sigma_{zb} \text{ so } s = \Sigma^\top 1_Z.$$

For a given coalition $b \in \mathbb{N}^Z$, we denote μ_b the (endogeneous) mass of coalitions of type b. The population counting constraint is

$$q_z = \sum_{b \in \mathbb{N}^Z} \mu_b \Sigma_{zb}, \text{ so } q = \Sigma \mu.$$

Surplus

For each coalition b, there is an (exogenous) joint surplus produced ϕ_b called *coalitional surplus* which is split endogenously.

We denote U_{zb} the (endogenous) systematic payoff an agent of type z is able to obtain when participating in coalition b. The sum of these payoffs over the members of the coalition must equate with the joint surplus, so

$$\phi_b = \sum_{z \in [Z]} \Sigma_{zb} U_{zb} \text{ so } \phi = diag(\Sigma^\top U)$$

By convention we set $U_{zb} = -\infty$ if z is not in the coalition (that is, if $\Sigma_{zb} = 0$) and impose $\Sigma_{zb} U_{zb} = 0$ if $\Sigma_{zb} = 0$. Also we assume that the coalition size cannot exceed L, and thus we impose $\phi_b = -\infty$ when $\Sigma_{zb} \geq L$. We set $\phi_b = 0$ for coalitions of a single individual.

Example 1: heterosexual marriage (Choo and Siow)

Example 5.3: We describe the Choo and Siow model as a particular case of coalition formation model. There, $Z = X + Y$ is the number of types of men and women, where $z \in [1 : X]$ will index men's types and $z \in [X + 1 : X + Y]$ will index women's types.

The masses of individuals of each type are $q = (n^\top, m^\top)^\top$.

We set $\phi_b = \phi_{xy}$ when coalition b contains exactly one man of type x and one woman of type y that is when $\Sigma_{xb} = \Sigma_{(y+X)b} = 1$, and $\Sigma_{zb} = 0$ otherwise. We set $\phi_b = -\infty$ in all other cases.

Then $\mu_b = 0$ unless b is a coalition of the form xy, $x0$ or $y0$, and the population counting constraint recovers the usual margin constraints $\sum_y \mu_{xy} + \mu_{x0} = n_x$ and $\sum_x \mu_{xy} + \mu_{0y} = m_y$. If b is the coalition of type xy, U_{zb} is U_{xy} if $z = x$, and V_{xy} if $z = y$, which recovers the familiar condition $U_{xy} + V_{xy} = \phi_{xy}$.

Example 2: same-sex marriage (Ciscato, Galichon and Goussé)

Example 5.4: We consider a homosexual population, and let Z be the number of individuals' types.

We rule out coalitions b unless $s_b \leq 2$:
• If $s_b = 1$, then the surplus is zero.
• If $s_b = 2$, then (i) either there is an index z such that $\Sigma_{zb} = 2$, in which case the coalition is made of two individuals with the same type z; (ii) or the coalition is between two individuals of different types z and z' such that $\Sigma_{zb} = \Sigma_{z'b} = 1$.

The quantity ϕ_b is the joint surplus to be shared between partners.

Example 3: labor market

Example 5.5: Next example is inspired by Corblet (2022). We assume the type of a firm is its industry, and is denoted $z \in [X]$.
Each firm can hire a certain number of executives who all have the same type denoted by $z = X + 1$; a certain number of engineers whose unique type is $z = X + 2$; and a certain number of salespeople whose unique type is $z = X + 3$.

As a result, $Z = X + 3$, and ϕ_b will be finite if and only if $\sum_{z \in [1:X]} \Sigma_{zb} = 1$ (one firm and an arbitrary set of workers) or $\sum_{z \in [1:X]} \Sigma_{zb} = 0$ and $\sum z \in [X + 1 : Z] = 1$ (unassigned worker), and the surplus will depend on the industry of the firm, and the number $\Sigma_{(Z+1)b}$ of executives, $\Sigma_{(Z+2)b}$ of engineers, and $\Sigma_{(Z+3)b}$ of salespeople it employs.

The industry-specific production function may be, for instance, a constant elasticity of substitution (CES) function of the various groups of employees.

Equilibrium (1/2)

Agent i of type z in coalition b, gets utility $U_{zb} + \varepsilon_{ib}$, where ε_{ib} are i.i.d. with a centered Gumbel distribution. Indirect utility:

$$u_i = \max_{b \in [B]} \{U_{zb} + \varepsilon_{ib}\}$$

The average welfare of type z is

$$u_z = \log\Big(\sum_{b \in [B]} \exp U_{zb}\Big),$$

and the choice probability of b by z is

$$\pi_{zb} = \exp(U_{zb} - u_z).$$

Equilibrium (2/2)

At equilibrium, one must have

$$\mu_b \Sigma_{zb} = q_z \exp\left(U_{zb} - u_z\right)$$

Taking the logarithm of the last equation, multiplying by Σ_{zb} and summing over z yields:

$$s_b \log \mu_b = \sum_{z \in [Z]} \Sigma_{zb} \log \frac{q_z}{\Sigma_{zb}} + \phi_b - (\Sigma^\top u)_b = -s_b \ell_b + \phi_b - (\Sigma^\top \tilde{u})_b$$

where we have defined $\ell_b := \left(\sum_z \Sigma_{zb} \log \Sigma_{zb}\right) / s_b$ and $\tilde{u}_z := u_z - \log q_z$. This leads to

$$\mu_b = \exp(-\ell_b) \exp\left(\frac{\phi_b - (\Sigma^\top \tilde{u})_b}{s_b}\right).$$

and the price-like vector $\tilde{u}_z$ is adjusted at equilibrium by the population equation.

Estimation (1/3)

In the estimation problem, we observe the numbers of each type of coalitions $\hat{\mu}_b$, and we wish to estimate the underlying surplus ϕ_b for all b, which we parameterize by $\phi_b^\lambda = (\Phi\lambda)_b = \sum_k \phi_{bk}\lambda_k$, where Φ is a $B \times K$ matrix. Calling $\theta = (\lambda^\top, \tilde{u}^\top)^\top$ the parameter of size $P = K + Z$, the relation becomes

$$\mu_b^\theta = \exp\left(-\ell_b\right) \exp\left(\frac{(\Phi\lambda)_b - (\Sigma^\top \tilde{u})_b}{s_b}\right).$$

We will look for the estimator $\hat{\lambda}$ called the *coalitional surplus estimator* such that the equilibrium vector μ of masses of coalitions predicted by the parameter λ has the same moments as those observed in the sample.

$$\sum_{b \in \mathcal{B}} \mu_b \phi_{bk} = \sum_{b \in \mathcal{B}} \hat{\mu}_b \phi_{bk}.$$

Estimation (2/3)

The parameter θ satisfies the market clearing condition and the moment matching conditions:

$$\left\{ \begin{array}{l} \sum_{b\in[B]} \Sigma_{zb}\mu_b^\theta = \sum_{b\in[B]} \Sigma_{zb}\hat{\mu}_b \\ \sum_{b\in[B]} \phi_{bk}\mu_b^\theta = \sum_{b\in[B]} \phi_{bk}\hat{\mu}_b \end{array} \right.$$

which are the first order conditions to the following optimization problem:

$$\max_{\lambda,\tilde{u}} \left\{ \sum_{b\in[B]} \hat{\mu}_b \left((\Phi\lambda)_b - (\Sigma^\top \tilde{u})_b \right) - \sum_{b\in[B]} s_b e^{-\ell_b} \exp\left(\frac{(\Phi\lambda)_b - (\Sigma^\top \tilde{u})_b}{s_b} \right) \right\}.$$

From the previous problem, we recognize a weighted Poisson regression with dependent variable $\hat{\mu}_b \exp(\ell_b)$, with regressor matrix

$$R = diag(s)^{-1} \left(\Phi, \quad -(\Sigma)^\top \right)$$

and with vector of weights $w_b = s_b \exp(-\ell_b)$. We state:

Theorem 5.5.1

The coalitional surplus estimator $\hat{\lambda}$ defined above can be estimated using a weighted Poisson regression of dependent variable $\hat{\mu}_b \exp(\ell_b)$ on regressor matrix R as defined above, with weights $w_b = s_b \exp(-\ell_b)$.

Coalition formation in same-sex marriage

We provide the Python code for estimating the surplus in same-sex marriages (Example 5.4). Note that this framework can be easily adapted for other applications.

Python code for coalition formation via generalized linear models

```
import numpy as np, scipy.sparse as sp
from sklearn import linear_model as lm
np.random.seed(7)
x_log_x = np.vectorize(lambda x: x * np.log(x) if x > 0 else 0)
Z,K = 4,3
B = Z*(Z+1) // 2
# in this example, same-sex marriage: coalitions of size <= 2
count_b_z = np.array([ list(np.eye(Z)[z1] + np.eye(Z)[z2])
          for z2 in range(Z) for z1 in range(Z) if z1<= z2 ])
muhat_b = np.random.uniform(size=B)
phi_b_k = np.random.uniform(size = (B,K) )
# Computing :
s_b = count_b_z.sum(axis = 1)
R_b_p = np.block([phi_b_k , count_b_z ]) /  s_b[:,None]
l_b =  (x_log_x(count_b_z) ).sum(axis = 1) / s_b
glm = lm.PoissonRegressor(fit_intercept=False,alpha=0)
glm.fit( R_b_p, muhat_b * np.exp(l_b) , sample_weight = s_b * np.exp(-l_b) )
# displaying results
print('lambda_k=',glm.coef_[:K])
```


Section 5.6. Quasilinear hedonic pricing models

Section content

- Hedonic prices
- Hedonic equilibrium

Quasilinear hedonic pricing

In a hedonic model, producers and consumers may produce and consume one unit of a good that comes in several varieties or qualities. There is heterogeneity among producers and among consumers, and as a result, the unitary production cost a producer depend on the interaction of this producer's characteristics, and on the quality of the good. Similarly, the heterogeneity among consumers leads to different unit valuations for the good. Each quality has its own per unit price, called *hedonic price*, which is determined at equilibrium by the adjustment of supply and demand. Both producers and consumers have utilities that are quasilinear in the price.

An example

As an introductory example, consider a market for a car ride service. There is a heterogeneous population of producers (the drivers), with various current location, car size, rating, etc. Similarly, there is a heterogenous population of consumers (the passengers) who also vary according to location, party size, rating, etc. The city is divided into Z areas, where the pick-up area is understood as the quality of the good.

Utilities

There is a mass n_x of drivers with characteristics $x \in [X]$, where x captures all the measurable characteristics about a driver. Each driver i of type x incurs a cost $C_{xz} - \varepsilon_{iz}$ for picking up a passenger at z. If p_z is the price of a ride in area z, the profit of a driver picking up in area xz is then $p_z - C_{xz} + \varepsilon_{iz}$. Participation is forced. We assume that (ε_z) is drawn from an i.i.d. Gumbel distribution.

Passengers have types $y \in [Y]$, each with a mass of m_y. Passengers may walk to other areas to request the service, leading to a decrease in their utility due to walking or waiting. If a passenger j gets picked in area z, they will get utility $U_{yz} - p_z + \eta_{jz}$. All passengers participate, and the total mass of passengers equals the total mass of cars.

Demand and supply

As usual, the average indirect utility of drivers of type $x \in [X]$ and their probability of picking up in area $z \in [Z]$ are respectively

$$u_x = \log\left(\sum_{z \in [Z]} e^{p_z - C_{xz}}\right) \text{ and } \pi_{xz} = \exp(p_z - C_{xz} - u_x)$$

Likewise, the average indirect utility of passengers of type $y \in [Y]$ and their probability of requesting to be picked up in area $z \in [Z]$ are respectively

$$v_y = \log\left(\sum_{z \in [Z]} e^{U_{yz} - p_z}\right) \text{ and } \pi_{yz} = \exp(U_{yz} - p_z - v_y).$$

Equilibrium (1/2)

The market clearing in each area z implies that

$$\sum_{x\in[X]} n_x \exp(p_z - C_{xz} - u_x) = \sum_{y\in[Y]} m_y \exp(U_{yz} - p_z - v_y),$$

When u and v are equilibrium quantities, the logit hedonic pricing formula writes:

$$p_z = \frac{1}{2}\log\frac{\sum_{y\in[Y]} m_y \exp(U_{yz} - v_y)}{\sum_{x\in[X]} n_x \exp(-C_{xz} - u_x)}$$

As before, the equilibrium requirements can be seen as first-order conditions to:

$$\min_{p,\tilde{u},\tilde{v}}\left\{\sum_{x\in[X]} n_x\tilde{u}_x + \sum_{y\in[Y]} m_y\tilde{v}_y + \sum_{xz\in[X\times Z]} e^{p_z - C_{xz} - \tilde{u}_x} + \sum_{yz\in[Y\times Z]} e^{U_{yz} - p_z - \tilde{v}_y}\right\},$$

where $\tilde{u}_x = u_x - \log n_x$ and $\tilde{v}_y = v_y - \log m_y$.

Equilibrium (2/2)

As usual, the problem can be interpreted as a Poisson regression of the vector of dimension $(XZ + YZ)$ obtained by the concatenation of the vector of term $n_x \exp C_{xz}$ for $xz \in [X \times Z]$ and the vector of term $m_y \exp\left(-U_{yz}\right)$ for $yz \in [Y \times Z]$ on the regressor matrix R defined by

$$R = \begin{pmatrix} 1_X \otimes \mathbf{I}_Z & -\mathbf{I}_X \otimes 1_Z & \mathbf{0} \\ -1_Y \otimes \mathbf{I}_Z & \mathbf{0} & -\mathbf{I}_Y \otimes 1_Z \end{pmatrix}$$

where the vector of coefficients $\theta = (p, \tilde{u}, \tilde{v})$ is a vector of weights w that is the concatenation of the vector of term $\exp\left(-C_{xz}\right)$ for $xz \in [X \times Z]$ and the vector of term $\exp U_{yz}$ for $yz \in [Y \times Z]$.

Exercises and Quick Recap Chapter 5

Short exercises (1/3)

Exercise 5.1: Show that the roles played by x and y in the gravity equation are symmetric: that is, the same formulation of the gravity equation would have been obtained by picking x as a categorical dependent variable, and including an x-fixed effect into the macro logistic regression.

Exercise 5.2: Consider a model of one-to-one matching with linear taxes. If a worker of type $x \in [X]$ works with a firm $y \in [Y]$ and they agree on a wage w_{xy}, then their respective utilities are given by:

$$U_{xy} = \alpha_{xy} + (1 - \tau) w_{xy} + \varepsilon_y,$$
$$V_{xy} = \gamma_{xy} - w_{xy} + \eta_x,$$

where:

- α_{xy} is the worker's monetary valuation of the job's amenities;
- $\tau \in (0,1)$ is the (flat) tax rate;
- γ_{xy} is the output of worker x appropriated by the firm, measured in monetary units;
- ε_y and η_x are random utility shocks.

It is assumed that the payoff of an unmatched worker is ε_0, and the payoff of an unmatched firm is η_0. We assume that the random vectors $(\varepsilon_y)_{y\in\{0,\ldots,Y\}}$ and $(\eta_x)_{x\in\{0,\ldots,X\}}$ are independent and identically distributed (i.i.d.) according to a Gumbel distribution, with dimensions $Y+1$ and $X+1$ respectively.

Short exercises (3/3)

(i) Show that the equilibrium in this market maximizes an objective function similar to expression (5.4.12) in Section 5.4. Comment on the differences.

(ii) Do the two quantities

$$A = \sum_{xy \in [X \times Y]} \mu_{xy} \alpha_{xy} - \sum_{xy \in [X \times Y]} \mu_{xy} \log \mu_{xy} - \sum_{x \in [X]} \mu_{x0} \log \mu_{x0},$$

$$\Gamma = \sum_{xy \in [X \times Y]} \mu_{xy} \gamma_{xy} - \sum_{xy \in [X \times Y]} \mu_{xy} \log \mu_{xy} - \sum_{y \in [Y]} \mu_{0y} \log \mu_{0y}$$

increase or decrease when the tax rate τ increases? Same question for their sum $A + \Gamma$. Interpret the result.

Quick recap of Chapter 5 (1/3)

In this chapter, we have demonstrated the versatility of the logistic regression framework from Chapter 3 across a wide range of models, spanning fields such as international trade, labor economics, and family economics.

In Section 5.1, we extended the model to 'macro logistic regression,' which clusters agents into types. This generalization aggregates individuals of the same type, allowing us to capture variations in utility across types without modeling each individual separately. It can be efficiently estimated via Poisson regression.

Section 5.2 presented entropic optimal transport as a specialized instance of this framework, utilizing specific weights and y-fixed effects as the sole regressors. This formulation allows optimal transport problems to be computed conveniently using GLMs.

Quick recap of Chapter 5 (2/3)

In Section 5.3, we introduced the gravity equation by incorporating additional regressors beyond the fixed effects. We used this to model and predict trade flows between entities based on various proximity measures, again estimating the model through Poisson regression with exporter- and importer-fixed effects.

We then applied this framework to heterosexual marriage and labor economics using the Becker model, which assumes matches form to maximize a joint surplus that is subsequently divided among participants.

In Section 5.5, we extended the analysis beyond two-sided markets to coalition formation models. This approach allows for more complex arrangements where the surplus is split among members of a coalition, yet it remains estimable through GLMs.

Quick recap of Chapter 5 (3/3)

We concluded the chapter with hedonic equilibrium pricing models, where supply and demand meet in a quality space. These models can also be reformulated as GLMs with product, seller, and buyer fixed effects.

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Chapter 6. Dynamic discrete choice

Discrete Choice Models, ch. 6, pp. 149-191

Content

Chapter 6: introduction

In this chapter we model discrete choice problems when they are repeated in time. Units are in a certain state at a given time. If a certain choice is made for a given unit, this yields a short-term payoff, but it also implies a probability of transitioning to another state, which depends on the current state and the choice made. Therefore, the decision-maker needs to factor in the effect of the choice on the change of states in the future.

Dynamic programming associates a value of being in each state. The dynamic programming principle relates the value of being in each state at time $t+1$ with the value of being in each state at time t. We will consider versions of the model with finite time horizon and with infinite time horizon (stationarity); with random utility and without random utility; with discounting and without discounting.

Section 6.1. Finite time horizon, no heterogeneity

Section content

- Dynamic programming as a linear programming problem
- Duality and the Bellman equation
- Backward and forward induction

A running example: Harold Zurcher

Example 6.1: Harold Zurcher, a maintenance engineer who is the protagonist in the story told by Rust's 1987 paper, operates a fleet of buses. Each day, he has to decide which buses to send on maintenance and which ones to send to operate.

There is a trade-off between short-term cost of halting the bus for repair, and the long-run benefit of maintaining the bus and thus saving on operation costs. More specifically:

- At each period, for each unit, there are two decisions to be taken: maintain or operate.
- The short-term payoff associated with a decision depends on this decision, but also on the state the bus is in.
- At the end of each period, there is a given probability that the bus's state will transition to another one.

The model in finite horizon and with no heterogeneity

We first consider a model without unobserved heterogeneity (no random utility).

There is a population of individual units (e.g. buses). At each period $t \in [T]$, each unit is in some state $x \in [X]$. Let n_{tx} be the mass of units in state $x \in [X]$ at time $t \in [T]$.

While n_{tx} is endogenous for $t \geq 2$ as it depends on the agent's choices, n_{1x} is exogenous (and set to N_x) – it is the initial condition.

At each period, one choice among a menu of Y possible ones is applied to each unit.

We let μ_{txy} be the mass of units in state x at time t for which choice y is made. The vector μ is called *policy variable*. One has the counting equations

$$\sum_{y \in [Y]} \mu_{txy} = n_{tx}, \quad \forall tx \in [T \times X].$$

Evolution equations

The states of units undergo Markov transitions. Let $P_{x'xy}$ be the probability that the state of a unit that was in state x and for which choice y was made transitions to state x' at the following period. One has

$$n_{tx'} = \sum_{xy \in [X \times Y]} P_{x'xy} \mu_{(t-1)xy}, \quad t \in [2{:}T]$$

and together with the initial conditions for $t = 1$, we get the evolution equations:

$$\begin{cases} n_{1x} = N_x, & t = 1 \\ n_{tx} = \sum_{\tilde{x}y \in [X \times Y]} P_{x\tilde{x}y} \mu_{(t-1)\tilde{x}y}, & t \in [2 : T]. \end{cases}$$

Vectorization

We stack the vector (μ_{txy}) in a single column vector using the row-major order, which means that y varies first, t varies last. Similarly, one vectorizes n still in the row-major order. The counting equations vectorize into

$$(\mathbf{I}_T \otimes \mathbf{I}_X \otimes \mathbf{1}_Y^\top)\mu = n.$$

The evolution equations vectorize into

$$n = \mathbf{e}_T^1 \otimes N + (\mathbf{J}_T \otimes \mathbf{P})\mu$$

where $\mathbf{J}_T$ is the $T \times T$ lower shift matrix with ones on the subdiagonal and zeroes elsewhere, i.e. $(\mathbf{J}_T)_{tt'} = 1\{t = t' + 1\}$, and $\mathbf{P}$ is the $X \times (XY)$ matrix whose (x', xy) term is $P_{x'xy}$, and $\mathbf{e}_T^1$ is the first vector of the canonical basis of $\mathbb{R}^T$.

Combining the counting equations with the evolution equations yields the population dynamics equations,

$$\begin{cases} \sum_{y \in [Y]} \mu_{1xy} = N_x, & x \in [X] \\ \sum_{y \in [Y]} \mu_{txy} = \sum_{\tilde{x}y \in [X \times Y]} P_{x\tilde{x}y} \mu_{(t-1)\tilde{x}y}, & tx \in [(2:T) \times X]. \end{cases}$$

In vectorized form, the population dynamics equations become

$$(\mathbf{I}_T \otimes \mathbf{I}_X \otimes \mathbf{1}_Y^\top - \mathbf{J}_T \otimes \mathbf{P})\mu = \mathbf{e}_T^1 \otimes N.$$

Capacity constraints

There may be other constraints on the policy variables. For instance, there may be a lower or an upper limit to the number of units for which decision y is applied – this will be the case in Harold Zurcher's example if a minimum number of buses are needed for operation, or if the maintenance shop has an upper capacity. We write these constraints $\mathbf{A}\mu \leq c$ where c is a vector of size L, and $\mathbf{A}$ is a matrix of size $L \times (TXY)$.

For instance, if there is a maximum k_y to the total mass of units for which choice y may be applied at each period, then $\mathbf{A} = \mathbf{I}_T \otimes \mathbf{1}_X^\top \otimes \mathbf{I}_Y$ and $c = \mathbf{1}_T \otimes \mathbf{k}$.

Intertemporal objective function

If choice $y \in [Y]$ is applied to a unit in state $x \in [X]$ at time $t \in [T]$, this generates an immediate payoff ϕ_{txy}. ϕ_{txy} is measured in the units of time $t = 1$, so there is no further discounting to do.

As payoffs are measured in the same numéraire across periods, the total payoff is

$$\sum_{txy \in [T \times X \times Y]} \mu_{txy} \phi_{txy}$$

and therefore the decision maker's problem consists of maximizing the latter subject to the evolution equations and the capacity constraints.

The optimal trajectory

The intertemporal decision-maker's problem is therefore

$$
\begin{aligned}
\max_{\mu \geq 0} \quad & \sum_{txy \in [T \times X \times Y]} \mu_{txy} \phi_{txy} \\
s.t. \quad & \sum_{y \in [Y]} \mu_{1xy} = N_x, \quad \forall x \in [X] \\
& \sum_{y \in [Y]} \mu_{txy} = \sum_{\tilde{x} y \in [X \times Y]} P_{x \tilde{x} y} \mu_{(t-1)\tilde{x} y}, \quad \forall tx \in [(2:T) \times X]. \\
& \sum_{txy} A_{ltxy} \mu_{txy} \leq c_l, \quad \forall l \in [L]
\end{aligned}
$$

or in a vectorized form

$$
\begin{aligned}
\max_{\mu \geq 0} \quad & \mu^\top \phi \\
s.t. \quad & (\mathbf{I}_T \otimes \mathbf{I}_X \otimes \mathbf{1}_Y^\top - \mathbf{J}_T \otimes \mathbf{P}) \mu = \mathbf{e}_T^1 \otimes N \\
& \mathbf{A} \mu \leq c.
\end{aligned}
$$

Duality in linear dynamic programming (1/2)

We have:

Proposition 6.1.1

The intertemporal decision-maker's problem has the following dual formulation

$$\begin{aligned} \min_{u \in \mathbb{R}^{TX}, \tau \in \mathbb{R}^L_+} \quad & (\mathbf{e}_T^1 \otimes N)^\top u + c^\top \tau \\ s.t. \quad & (\mathbf{I}_T \otimes \mathbf{I}_X \otimes \mathbf{1}_Y - \mathbf{J}_T^\top \otimes \mathbf{P}^\top) u + \mathbf{A}^\top \tau \geq \phi. \end{aligned}$$

Duality in linear dynamic programming (2/2)

Back to the coordinates notation, the previous dual formulation is expressed as:

$$
\begin{aligned}
\min_{(u_{tx}),(\tau_l)\geq 0} \quad & \sum_{x\in[X]} N_x u_{1x} + \sum_{l\in[L]} c_l \tau_l \\
s.t. \quad & u_{tx} \geq \phi_{txy} - \sum_{l\in[L]} A_{ltxy}\tau_l + \sum_{x'\in[X]} P_{x'xy} u_{(t+1)x'}, \ \forall txy \in [T\times X\times Y] \\
& u_{(T+1)x} = 0, \ \forall x \in [X],
\end{aligned}
$$

where τ is the vector of Lagrange multipliers associated with the capacity constraints, and $\sum_l A_{ltxy}\tau_l$ is interpreted as the shadow cost of option y for a bus in state x at time t.

Bellman equation

The inequality constraints that express that u_{txy} is a *feasible* dual solution can be rewritten

$$u_{tx} \geq \max_{y \in [Y]} \{\phi_{txy} - \sum_{l \in [L]} A_{ltxy} \tau_l + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'}\}, \quad \forall tx \in [T \times X]$$

but if u_{txy} is an *optimal* dual solution, this holds as an equality

$$u_{tx} = \max_{y \in [Y]} \{\phi_{txy} - \sum_{l \in [L]} A_{ltxy} \tau_l + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'}\}, \quad \forall tx \in [T \times X]$$

which is a Bellman equation.

No capacity constraints

Now assume away the capacity constraints $\tau^* = 0$. Then there are no externalities across units, and optimal policies merely aggregate optimal choices at the unit level. In that case, u can be solved for by backward induction, using the Bellman equations

$$\begin{cases} u_{tx} = \max_{y \in [Y]} \{ \phi^\star_{txy} + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'} \}, \ \forall tx \in [T \times X] \\ u_{(T+1)x} = 0, \ \forall x \in [X]. \end{cases}$$

The value function is determined by *backward induction*, and once the value function is obtained, the optimal policy is determined by *forward induction*.

Algorithm 6.1.1 Backward-forward induction

Input: Transition probabilities $P_{x'xy}$, payoff function ϕ^*_{txy}

Output: Value function u_{tx}, policy μ_{txy}

Backward phase (value iteration):

$u_{(T+1)x} = 0 \quad \forall x \in [X]$

for $t \leftarrow T-1$ **to** 1 **by** -1 **do**:

$$u_{tx} \leftarrow \max_{y \in [Y]} \left\{ \phi^\star_{txy} + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'} \right\}.$$

end

Algorithm 6.1.1 Backward-forward induction (continued)

Forward phase (policy iteration):

Set $n_{1x} = N_x \ \forall x \in [X]$

For $t = 1$ **to** T **do**:

forall $x \in [X]$ **do** :

Pick one $y^\circ \in [Y]$ such that $\phi^\star_{txy^\circ} + \sum_{x'} P_{x'xy^\circ} u_{(t+1)x'} = u_{tx}$

Set $\mu_{txy} = n_{tx}$ for $y = y^\circ$, $\mu_{txy} = 0$ otherwise;

end

Compute $n_{(t+1)x} = \sum_{x'y'} P_{xx'y'} \mu_{tx'y'} \quad \forall x \in [X]$.

return μ_{txy}, μ_{tx}

Remark on the model

Remark: In the absence of the capacity constraint $A\mu \leq c$, there are no externalities across the treatment of the units. Consequently, the dynamics of a population with N_x units in state x in the first period coincide with the superposition of N_x independent individual dynamics starting from state x. This configuration is referred to as the **"single-agent dynamic."**

The next slide implements the finite-horizon, no-heterogeneity model using backward-forward induction, by applying the algorithm described previously.

Python code for no heterogeneity, finite horizon via backward induction

```
import numpy as np, gurobipy as grb
T,X,Y = 40,3,2
beta = 0.9
phi_x_y = np.array( [[25,-10], [12,-10 ],[-5,-10]])
phi_t_x_y = np.array([beta**t for t in range(T)])[:,None,None] *  phi_x_y[None,:,:]
n_x = np.array([10,20,5])
P_xprime_x_y = np.array([[[1/3, 1],[0, 1],[0, 1]],[[2/3, 0],[1/3,0],
                        [0, 0]],[[0,0],[2/3, 0],[1, 0]]])
# Computing -- backward:
u_t_x = np.zeros((T,X))
u_t_x[T-1,:] = phi_t_x_y[T-1,:,:].max(axis = 1)
for t in range(T-2,-1,-1):
  u_t_x[t,:] = ( phi_t_x_y[t,:,:] + (P_xprime_x_y * u_t_x[t+1,:][:,None,None]).sum(axis = 0) ).
       max(axis = 1 )
```


Python code for no heterogeneity, finite horizon via backward induction (continued)

```
# Computing -- forward
mu_t_x_y = np.zeros((T,X,Y))
then_x = n_x
for t in range(T-1):
  y_x = ( phi_t_x_y[t,:,:] +
  (P_xprime_x_y * u_t_x[t+1,:][:,None,None]).sum(axis = 0) ).argmax(axis = 1 )
  mu_t_x_y[t,list(range(X)),list(y_x) ]=then_x
  then_x = P_xprime_x_y.reshape((X,X*Y)) @ mu_t_x_y[t,:,:].flatten()
y_x = ( phi_t_x_y[T-1,:,:] ).argmax(axis = 1 )
mu_t_x_y[T-1,list(range(X)),list(y_x) ]= then_x
# Displaying results
print('u_t_x=',u_t_x[0,:])
print('objective=' ,(mu_t_x_y * phi_t_x_y ).sum() )
```

Section 6.2. Adding heterogeneity

Section content

- Logistic Bellman equation
- Duality
- Vectorization

Finite horizon with heterogeneity

We assume that in addition to ϕ_{txy}, there is a random utility shock ε_{ity} associated with choosing option y for unit i at time t – thereby introducing a random utility component.

These shock can be correlated between options for the same unit i, i.e. ε_i and $\varepsilon_{i'}$ are independent but ε_{ity} and $\varepsilon_{ity'}$ can be correlated.
We denote $\mathcal{P}_{tx}$ the distribution of the Y-dimensional random vector $\varepsilon_t = (\varepsilon_{ty})_y$ from which the draws ε_{ity} are made.

Markov assumption

Now, we specify the Markovian structure:

Assumption 6.1

We assume that the sequence $x_t, (x_t, \varepsilon_t), (x_t, y_t), x_{t+1}, \ldots$ constitutes a Markov chain. Conditional on x_t, a Y-dimensional random utility vector ε_t is drawn from a distribution $\mathcal{P}_{tx_t}$; based on x_t and the vector ε_t, the optimal choice y_t is selected; and finally, the next state x_{t+1} is drawn from the transition probability $P_{x'xy}$ given $xy = x_t y_t$.

The units in which the utility shocks are labeled are period $t = 1$ units, so there is no discounting to do.

Markov assumption interpretation

The preceding assumption implies a Markovian structure where, at any stage, only the current state is relevant for predicting the future. Specifically, the distribution of η_t conditional on η_{t-1} is entirely captured by x_t, which effectively rules out "persistence" of the random utility shocks. Consequently, the current action y_t is fully determined by (x_t, ε_t), and the subsequent state x_{t+1} is determined by the state-action pair (x_t, y_t).

We state:

Theorem 6.2.1 (1/2)

In the model with logit heterogeneities, that is when $\mathcal{P}$ is the distribution of i.i.d. standard Gumbel terms, the primal problem has formulation:

$$\begin{aligned}
\max_{\mu_{txy}\geq 0} \quad & \left\{ \sum_{txy\in[T\times X\times Y]} \mu_{txy}\phi_{txy} - \sum_{txy\in[T\times X\times Y]} \mu_{txy} \log \frac{\mu_{txy}}{\sum_{y\in[Y]}\mu_{txy}} \right\} \\
s.t. \quad & \sum_{y\in[Y]} \mu_{1xy} = N_x, \quad x\in[X] \\
& \sum_{y\in[Y]} \mu_{txy} = \sum_{\tilde{x}y\in[X\times Y]} P_{x\tilde{x}y}\mu_{(t-1)\tilde{x}y}, \quad tx\in[(2{:}T)\times X] \\
& \mathbf{A}\mu \leq c
\end{aligned}$$

Intertemporal problem, finite horizon, logit heterogeneity (ctd)

Theorem 6.2.1 (2/2)

and the primal formulation coincides with the dual formulation:

$$\min_{u_{tx},\tau_l} \left\{ \sum_{x\in[X]} N_x u_{1x} + \sum_{l\in[L]} c_l \tau_l \right\}$$

$$s.t. \quad u_{tx} = \log\left(\sum_{y\in[Y]} \exp\left(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + \sum_{x'\in[X]} P_{x'xy} u_{(t+1)x'} \right)\right), \; tx \in [T\times X].$$

Further, if $\mu \geq 0$ is solution to the primal problem and (u,τ) to the dual problem, then defining $n_{tx} = \sum_y \mu_{txy}$ the number of units in state x at time t, the conditional choice probability of choosing option y in state x at time t, denoted $\pi_{txy} := \mu_{txy}/n_{tx}$, is expressed by $\pi_{txy} = \exp\left(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + \sum_{x'\in[X]} P_{x'xy} u_{(t+1)x'} - u_{tx} \right)$.

Bellman equation

The constraint in the dual formulation

$$u_{tx} = \log \left(\sum_{y \in [Y]} \exp \left(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'} \right) \right)$$

is a Bellman equation, which expresses that the average welfare associated with being in state x at time t comes from the optimal decision that maximizes the sum of the short-term payoff, the shadow cost associated with the constraints, and (if t is not the last period) the expected continuation payoff.

Sketch of the duality argument (1/3)

Let u_{1x} be the Lagrange multipliers associated with accounting constraint for $t = 1$ in the primal, u_{tx} for the constraints associated with $t \geq 2$ and τ the L-dimensional vector of Lagrange multipliers associated with the capacity constraints.

The Lagrangian $\mathcal{L}(\mu; u, \tau)$ is

$$
\begin{aligned}
\mathcal{L} \quad (\mu; u, \tau) := & \sum_{txy \in [T \times X \times Y]} \mu_{txy} \phi_{txy} - \sum_{txy \in [T \times X \times Y]} \mu_{txy} \log \frac{\mu_{txy}}{\sum_y \mu_{txy}} \\
& + \sum_{x \in [X]} N_x u_{1x} - \sum_{xy \in [X \times Y]} \mu_{1xy} u_{1x} - \sum_{txy \in [(2:T) \times X \times Y]} \mu_{txy} u_{tx} \\
& + \sum_{tx\tilde{x}y \in [(2:T) \times X \times X \times Y]} P_{x\tilde{x}y} \mu_{(t-1)\tilde{x}y} u_{tx} + \sum_{l \in [L]} c_l \tau_l - \sum_{ltxy \in [L \times T \times X \times Y]} A_{ltxy} \mu_{txy} \tau_l,
\end{aligned}
$$

and the primal formulation is $\max_{\mu \geq 0} \min_{u, \tau} \mathcal{L}(\mu; u, \tau)$.

Sketch of the duality argument (2/3)

The primal formulation coincides with $\min_{u,\tau} \max_{\mu \geq 0} \mathcal{L}(\mu; u, \tau)$. Rearranging yields

$$
\begin{aligned}
\mathcal{L} \quad & (\mu; u, \tau) = \sum_{x \in [X]} N_x u_{1x} + \sum_{l \in [L]} c_l \tau_l \\
& + \sum_{txy \in [(T-1) \times X \times Y]} \mu_{txy} \left(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'} - u_{tx} - \log \frac{\mu_{txy}}{\sum_{y \in [Y]} \mu_{txy}} \right) \\
& + \sum_{txy \in [X \times Y]} \mu_{txy} \left(\phi_{Txy} - (\mathbf{A}^\top \tau)_{Txy} - u_{Tx} - \log \frac{\mu_{Txy}}{\sum_{y \in [Y]} \mu_{Txy}} \right).
\end{aligned}
$$

One can show that we can obtain

$$
\max_{\pi \in \mathbb{R}_+^Y} \left\{ \sum_{y \in [Y]} \pi_y a_y - \sum_{y \in [Y]} \pi_y \log \frac{\pi_y}{\sum_{\tilde{y} \in [Y]} \pi_{\tilde{y}}} \right\}
$$

equals zero if $\sum_y \exp a_y = 1$, and $+\infty$ otherwise.

Sketch of the duality argument (3/3)

As a result, the value of $\max_{\mu \geq 0} \mathcal{L}(\mu; u, \tau)$ is $\sum_x N_x u_{1x} + \sum_l c_l \tau_l$ if each of the following constraints is met:

$$\begin{cases} \sum_{y \in [Y]} \exp(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + \sum_{x' \in [X]} P_{x'xy} u_{(t+1)x'} - u_{tx}) = 1, \; \forall tx \in [(T-1) \times X] \\ \sum_{y \in [Y]} \exp(\phi_{Txy} - (\mathbf{A}^\top \tau)_{Txy} - u_{Tx}) = 1, \; \forall x \in [X] \end{cases}$$

and that value is $+\infty$ otherwise.
Hence the dual of the problem is $\min \sum_x N_x u_{1x} + \sum_l c_l \tau_l$ over the set of vector u that satisfy the constraints.

Vectorized form

The constraint in the dual formulation is vectorized as

$$u = \log\left(\mathbf{\Sigma}_Y \exp\left(\phi - \mathbf{A}^\top \tau + (\mathbf{J}_T^\top \otimes \mathbf{P}^\top) u\right)\right)$$

where $\mathbf{\Sigma}_Y = \mathbf{I}_{TX} \otimes \mathbf{1}_Y^\top$ and similarly, the vectorized expression of the conditional choice probability vector is

$$\pi = \exp(\phi - \mathbf{A}^\top \tau + (\mathbf{J}_T^\top \otimes \mathbf{P}^\top - \mathbf{\Sigma}_Y^\top) u)$$

and the previous constraint (the Bellman equation) can be rewritten as

$$\mathbf{\Sigma}_Y \pi = 1$$

which expresses that π is a *bona fide* probability vector.

Another formulation

Note that we can rewrite the equality constraint of the dual problem in the previous theorem as $u_{tx} \geq \log \sum_{y \in [Y]} \exp(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} - (\mathbf{P}^\top u_{t+1})_{xy})$, which is an inequality constraint. Taking exponentials and dividing by $\exp(u_{tx})$, we can rewrite the dual as:

$$
\begin{aligned}
\min_{u_{tx}, \tau_l} \quad & \sum_{x \in [X]} N_x u_{1x} + \tau^\top c \\
s.t. \quad & \sum_{y \in [Y]} \exp\left(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} - (\mathbf{P}^\top u_{t+1})_{xy} - u_{tx}\right) \leq 1, \ tx \in [T \times X],
\end{aligned}
$$

where we have taken the convention that $u_{(T+1)x} = 0$ for all $x \in [X]$. Denote $n_{tx} \geq 0$ the Lagrange multipliers, and introduce the Lagrangian $\mathcal{S}(u, \tau; n)$, the intertemporal decision problem reformulates as $\min_{u,\tau} \max_n \mathcal{S}(u, \tau; n)$.

We state:

Proposition 6.2.1

The dual problem takes the following minimax formulation

$$\min_{u_{tx},\tau_l} \max_{n_{tx}\geq 0} \left\{ \begin{array}{c} \sum_{x\in[X]} N_x u_{1x} + \tau^\top c \\ +\sum_{txy\in[T\times X\times Y]} n_{tx} e^{\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + (\mathbf{P}^\top u_{t+1})_{xy} - u_{tx}} - \sum_{tx\in[T\times X]} n_{tx} \end{array} \right\},$$

(imposing $u_{(T+1)x} = 0$). Further, (u_{tx}, τ_l, n_{tx}) is a solution to the minimax problem if and only if $\mu_{txy} := n_{tx} \exp(\phi_{txy} - (\mathbf{A}^\top \tau)_{txy} + (\mathbf{P}^\top u_{t+1})_{xy} - u_{tx})$ and (u_{tx}, τ_l) are respectively solutions to the primal and dual problems, and n_{tx} is the mass of units in state x at time t.

Section 6.3. Inference, logit model with finite horizon

Section content

- Representative panel
- Log-sum-exp trick

Estimation (1/3)

We know consider a linear parameterization of the per-period surplus in the dynamic model with logit heterogeneities and finite horizon,

$$(\Phi\lambda)_{xy} = \sum_{k\in[K]} \phi_{txyk}\lambda_k.$$

Assume that we observe a sample $\{(t_i, x_i, y_i),\ i \in [I]\}$, of i.i.d. draws, where in observation i, a choice y_i was made for a unit in state x_i at time t_i. We denote $\hat{\mu}_{txy} = \sum_i 1(t_i x_i y_i = txy)$ the empirical distribution of observations.

Our focus is the estimation of λ using $\hat{\mu}$.

Estimation (2/3)

The conditional likelihood of choice y at time t for a unit in state x is

$$\pi^{\lambda}_{txy} = \frac{\mu^{\lambda}_{txy}}{n_{tx}} = \exp\left((\Phi\lambda)_{txy} + (\boldsymbol{P}^{\top} u_{t+1}) - u_{tx} \right).$$

and setting $\boldsymbol{\Psi} := J_T^{\top} \otimes \boldsymbol{P}^{\top} - \boldsymbol{\Sigma}_Y^{\top}$ and $\boldsymbol{\Sigma}_Y := I_{TX} \otimes \mathbf{1}_Y^{\top}$, and recalling that $\boldsymbol{P}$ is the transition matrix and $(J_T)_{t,t'} = \mathbf{1}(t = t' + 1)$, the log-likelihood of observation txy is

$$\log \pi^{\lambda}_{txy} := (\Phi\lambda)_{txy} + (\boldsymbol{P}^{\top} u_{t+1})_x - u^{\lambda}_{tx} = (\Phi\lambda + \boldsymbol{\Psi} u^{\lambda})_{txy}.$$

where u^{λ} is defined as the solution to the Bellman equation

$$u_{tx} = \log \sum_{y} \exp\left((\Phi\lambda)_{txy} + (\boldsymbol{P}^{\top} u_{t+1})_{xy} \right),\ tx \in [T \times X],$$

which can be rewritten as $\boldsymbol{\Sigma}_Y \pi^{\lambda} = \mathbf{1}_{TX}$, along with final condition $u_{(T+1)x} = 0$.

The sample log-likelihood is:

$$\ell(\lambda) := \sum_{txy} \hat{\mu}_{txy}((\Phi\lambda)_{txy} + (\boldsymbol{P}^\top u_{t+1})_x - u^\lambda_{tx}) = \hat{\mu}^\top(\Phi\lambda + \boldsymbol{\Psi} u^\lambda),$$

and thus:

Proposition 6.3.1

The maximum likelihood in the dynamic discrete choice model with finite horizon and logit heterogeneities is:

$$\begin{aligned} \max_{u,\lambda} \quad & \hat{\mu}^\top (\Phi\lambda + \boldsymbol{\Psi} u) \\ s.t. \quad & \boldsymbol{\Sigma}_Y \exp(\Phi\lambda + \boldsymbol{\Psi} u) = 1_{TX}. \end{aligned}$$

Remark 6.3.1 (Interpretation of optimality): Let η_{tx} be the Lagrange multiplier associated with the preceding constraints, and let $\mu_{txy} = \eta_{tx} \exp((\phi\lambda + \psi u)_{txy})$ be the predicted policy variable. The optimality conditions then yield:

$$\phi^\top \mu = \phi^\top \hat{\mu}$$

This indicates that the predicted moments of the policy variable match the observed ones. Similarly, the optimality conditions yield $\psi^\top \mu = \psi^\top \hat{\mu}$, which gives:

$$\left(\mathbf{J}_T \otimes \mathbf{P} - {}^{\circ}_{Y}\right) \hat{\mu} = \left(\mathbf{J}_T \otimes \mathbf{P} - {}^{\circ}_{Y}\right) \mu$$

This implies that $\sum_y \mu_{1xy} = N_x$.

Remarks on the MLE (2/2)

Remark 6.3.2 (Representative panel): We have a *representative panel* if $\sum_y \hat{\mu}_{(t+1)xy} = n_{(t+1)x} = \sum_{x'y} P_{x,x'y} \hat{\mu}_{tx'y}$. One can then simplify the log-likelihood of the sample into:

$$l(\lambda) = \sum_{txyk} \hat{\mu}_{txy} \phi_{txyk} \lambda_k - \sum_{x \in [X]} N_x u_{1x} = \hat{\mu}^\intercal \phi \lambda - N^\intercal u_1$$

The log-sum-exp trick

Recall the constraint of the problem: $u_{tx} = \log \sum_y \exp((\Phi\lambda)_{txy} + (\boldsymbol{P}^\top u_{t+1})_{xy})$. If the value of the argument of one of the exponentials in the sum over y is relatively large (say, above several dozens), the exponentiation will blow up, causing numerical instability.

The solution is to notice that: $\log \sum_y \exp U_y = c + \log \sum_y \exp(U_y - c)$ for any scalar c. Taking $c = -\max_y U_y$ ensures none of the argument of the exponentials is positive, thus restoring numerical stability.

The value function iteration embeds an intermediate step, as follows:

$$\begin{cases} \bar{u}_{tx} = \max_{y\in[Y]}\{(\Phi\lambda)_{txy} + (\boldsymbol{P}^\top u_{t+1})_{xy}\} \\ u_{tx} = \bar{u}_{tx} + \log \sum_{y\in[Y]} \exp((\Phi\lambda)_{txy} + (\boldsymbol{P}^\top u_{t+1})_{xy} - \bar{u}_{tx}). \end{cases}$$

This method is sometimes called **the log-sum-exp trick.**

Derivatives (1/2)

In order to compute the derivatives of the log-likelihood $\ell(\lambda) = \hat{\mu}^\top(\Phi\lambda + \mathbf{\Psi} u^\lambda)$ with respect to λ, we need to compute those of u. Recall that u^λ is defined by

$$\mathbf{\Sigma}_Y \exp\left(\Phi\lambda + \mathbf{\Psi} u^\lambda\right) = 1_{TX}. \tag{*}$$

First differentiation of (*) yields

$$\mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \left(\Phi + \mathbf{\Psi} du\right) = 0,$$

where $\mathbf{\Delta}_\pi := diag(\pi)$, and thus

$$du = -\left(\mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \mathbf{\Psi}\right)^{-1} \mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \Phi.$$

Derivatives (2/2)

Using $d \log \pi = \Phi + \boldsymbol{\Psi} du$, second differentiation of (*) in the previous slide yields:

$$\boldsymbol{\Sigma}_Y \boldsymbol{\Delta}_\pi \boldsymbol{\Psi} d^2 u + \boldsymbol{\Sigma}_Y \boldsymbol{\Delta}_\pi \mathfrak{M}_{TXY} (d \log \pi \otimes d \log \pi) = 0$$

where the second differential $d^2 u$, is a $TX \times K^2$ matrix and $\mathfrak{M}_A$ for A a positive integer as the $A \times A^2$ matrix given by $\mathfrak{M}_A := \sum_{a \in [A]} \boldsymbol{e}_A^a \otimes (\boldsymbol{e}_A^a)^\top \otimes (\boldsymbol{e}_A^a)^\top$.

Therefore:

$$d^2 u = -\left(\boldsymbol{\Sigma}_Y \boldsymbol{\Delta}_\pi \boldsymbol{\Psi}\right)^{-1} \boldsymbol{\Sigma}_Y \boldsymbol{\Delta}_\pi \mathfrak{M}_{TXY} ((\Phi + \boldsymbol{\Psi} du)^{\otimes 2})$$

where $A^{\otimes 2}$ stands for $A \otimes A$.

Thanks to these expressions, we are compute the first and second derivatives of $\ell(\lambda)$:

$$d\ell(\lambda) = \hat{\mu}^\top(\Phi + \mathbf{\Psi} du) \text{ and } d^2\ell(\lambda) = \hat{\mu}^\top \mathbf{\Psi} d^2 u.$$

As a result, the sample estimator $\hat{\lambda}$ of the true parameter λ is determined by the optimality condition:

$$\hat{\mu}^\top(\Phi + \mathbf{\Psi} du(\hat{\lambda})) = 0$$

.

Let $D^2\ell(\lambda)$ be the Hessian matrix of ℓ, which is the $K \times K$ matrix obtained by reshaping $d^2\ell(\lambda)$ in the row-major order, one has
$\hat{\lambda} - \lambda = -(D^2\ell(\lambda))^{-1}(\Phi^\top + du^\top \mathbf{\Psi}^\top)(\hat{\mu} - \mu) + o_P(\sqrt{I})$. This intuition can be made rigorous using the theory of M-estimators:

We state:

Theorem 6.3.1

Assume that the observations $t_i x_i y_i$ are drawn from the population in an i.i.d. fashion and collected in a sample of size I. Letting $\hat{\mu}_{txy}$ the number of observations of type txy in the sample, the following convergence holds in distribution.

$$\sqrt{I}\left(\hat{\lambda} - \lambda\right) \Rightarrow H^{-1} C H^{-1},$$

where H is estimated by $\hat{H} = D^2 \ell(\lambda)/I$ and $C = BV_\mu B^\top$, where B is estimated by $\hat{B} = (\Phi^\top + Du(\hat{\lambda})^\top \mathbf{\Psi}^\top)$ and V_μ by $\hat{V}_\mu = diag(\hat{\mu})/I - \hat{\mu}\hat{\mu}^\top/I^2$.

Computation with Numpy and PyTorch

The following slides provide the Python code for estimating the finite-horizon logit model, implemented using both NumPy and PyTorch.

Python code for estimation of the logit model with finite horizon in NumPy (1/4)

```
import numpy as np
from scipy import optimize, sparse as sp
np.random.seed(7)
T,X,Y,K = 2,3,2,3
beta = 0.9
phi_x_y_k = np.random.normal(size=(X,Y,K))
muhat_txy = np.random.uniform(size=(T*X*Y))
P_xprime_xy = np.random.uniform(size= (X,X*Y))
P_xprime_xy = P_xprime_xy / P_xprime_xy.sum(axis=0)[None,:]
# Computing:
phi_t_x_y_k = np.array([beta**t for t in range(T)])[:,None,None,None] *  phi_x_y_k[None,:,:,:]
phi_txy_k = phi_t_x_y_k.reshape((-1,K))
SigmaY = sp.kron( sp.eye(T*X),np.ones((1,Y)))
Psi = sp.kron(sp.diags(diagonals=[1]*(T-1), offsets=+1, shape=(T,T)), P_xprime_xy.T) - SigmaY.T
```

Python code for estimation of the logit model with finite horizon in NumPy (2/4)

```
def fu_tx(lambda_k):
  u_t_x = np.zeros((T+1,X))
  for t in range(T-1,-1,-1):
    u_t_x[t,:] = np.log( np.exp( phi_t_x_y_k[t,:,:,:].reshape((-1,K)) @ lambda_k +(u_t_x[t+1,:]
           @ P_xprime_xy)).reshape((X,Y)).sum(axis = 1))
  return u_t_x[:-1,:].flatten()

def fdu_tx(lambda_k):
    u_tx = fu_tx(lambda_k).flatten()
    pi_txy = np.exp( phi_txy_k @ lambda_k + Psi @ u_tx)
    return - sp.linalg.spsolve(SigmaY @ sp.diags(pi_txy) @ Psi , SigmaY @ sp.diags(pi_txy) @
          phi_txy_k)
```


Python code for estimation of the logit model with finite horizon in NumPy (3/4)

```
def neg_F(lambda_k):
    return  - muhat_txy.T @ ( phi_txy_k @ lambda_k + Psi @ fu_tx(lambda_k))
def neg_gradF(lambda_k):
    return muhat_txy.T @ ( - Psi @ fdu_tx(lambda_k) - phi_txy_k)
def neg_hessF(lambda_k):
    u_tx = fu_tx(lambda_k).flatten()
    pi_txy = np.exp( phi_txy_k @ lambda_k + Psi @ u_tx)
    dlogpi = phi_txy_k + Psi @ fdu_tx(lambda_k)
    TXY = T*X*Y
    TTT = sp.csr_matrix( ([1]*TXY,(list(range(TXY)), [a*(TXY+1) for a in range(TXY)])),shape =
          (TXY,TXY*TXY) )
    Tdlogpisquare =  TTT @ sp.kron(dlogpi,dlogpi )
    d2u = - sp.linalg.spsolve( SigmaY @ sp.diags(pi_txy) @ Psi , SigmaY @ sp.diags(pi_txy) @
          Tdlogpisquare)
    return (- muhat_txy.T @ Psi @ d2u).reshape((K,K))

result = optimize.minimize(neg_F, np.zeros(K), jac=neg_gradF, method='BFGS')
```

Python code for estimation of the logit model with finite horizon in NumPy (4/4)

```
# displaying results
if result.success:
    lambdahat_k = result.x
    print("lambdahat:", lambdahat_k)
    I = muhat_txy.sum()
    V= np.diag(muhat_txy) / I - muhat_txy[:,None] @ muhat_txy[None,:] / (I*I)
    HinvB = np.linalg.solve(neg_hessF(lambdahat_k), (phi_txy_k + Psi @ fdu_tx(lambdahat_k)).T)
    print('Cov(lambdahat)=')
    print(HinvB @ V @ HinvB.T)
else:
    print(result.message)
```


Using automatic differentiation

We can bypass the manual computation of the gradient and the Hessian using automatic differentiation, which is handled e.g. using the Torch library.

Python code for estimation of the logit model with finite horizon in Torch (1/3)

```
import torch, numpy as np, scipy.sparse as sp
np.random.seed(7)
T,X,Y,K = 2,3,2,3
beta = 0.9
phi_x_y_k = np.random.normal(size=(X,Y,K))
muhat_txy = np.random.uniform(size=(T*X*Y))
P_xprime_xy = np.random.uniform(size= (X,X*Y))
P_xprime_xy = P_xprime_xy / P_xprime_xy.sum(axis=0)[None,:]
#####
P_xprime_xy = torch.from_numpy(P_xprime_xy).float()
phi_t_x_y_k = np.array([beta**t for t in range(T)])[:,None,None,None] *  phi_x_y_k[None,:,:,:]
phi_t_x_y_k = torch.from_numpy(phi_t_x_y_k).float()
Ehat_phi_k = torch.from_numpy(muhat_txy.T).float() @ phi_t_x_y_k.reshape((-1,K))
SigmaY = sp.kron( sp.eye(T*X),np.ones((1,Y)))
Psi = sp.kron(sp.diags(diagonals=[1]*(T-1), offsets=+1, shape=(T,T)),P_xprime_xy.T) - SigmaY.T
I = muhat_txy.sum()
V= np.diag(muhat_txy) / I - muhat_txy[:,None] @ muhat_txy[None,:] / (I*I)
```


Python code for estimation of the logit model with finite horizon in Torch (2/3)

```
def F(lambda_k):
    u_t_x = torch.zeros((T+1, X))
    for t in range(T-1, -1, -1):
        u_t_x[t,:] = torch.log(torch.exp( (phi_t_x_y_k[t,:,:,:].reshape((-1, K)) @ lambda_k +
             u_t_x[t+1,:] @ P_xprime_xy).reshape((X, Y))).sum(axis=1))
    return Ehat_phi_k @ lambda_k + torch.from_numpy(muhat_txy.T @ Psi).float() @ u_t_x[:-1,:].
         flatten()
lambda_k = torch.ones(K, requires_grad=True)
optimizer = torch.optim.Adam([lambda_k],lr=0.01, maximize=True); num_steps = 1000
for step in range(num_steps):
    optimizer.zero_grad()  # Clear previous gradients
    res = F(lambda_k)      # Compute the function
    res.backward()         # Compute gradients
    optimizer.step()       # Update lambda_k
    if step % 100 == 0:    # Print info every 100 steps
        print(f"Step {step}, Function Value: {res.item()}")
print("lambdahat:", lambda_k) #print("Final Gradient:", lambda_k.grad)
```


Python code for estimation of the logit model with finite horizon in Torch (3/3)

```
lambda_k.grad = None; u_t_x = torch.zeros((T+1, X))
for t in range(T-1, -1, -1):
    u_t_x[t,:] = torch.log(torch.exp((phi_t_x_y_k[t,:,:,:].reshape((-1, K)) @ lambda_k + u_t_x[
         t+1,:] @ P_xprime_xy).reshape((X, Y))).sum(axis=1))
du = torch.zeros((T, X, K)); d2u = torch.zeros((T, X, K, K))
for t in range(T):
    for x in range(X):
          grad1 = torch.autograd.grad(u_t_x[t, x], lambda_k, create_graph=True, retain_graph=
               True, allow_unused=True)[0]
          du[t, x] = grad1.detach()
          for k in range(K):
              grad2 = torch.autograd.grad(grad1[k], lambda_k, retain_graph=True, allow_unused=
                   True)[0]
              d2u[t, x, k] = grad2.detach()
neg_hess =  (- muhat_txy.T @ Psi @ d2u.numpy().reshape((T*X,-1))).reshape((K,K))
HinvB = np.linalg.solve(neg_hess, (phi_t_x_y_k.reshape((-1,K)) + Psi @ du.numpy().reshape((T*X
     ,-1) )).T)
print('Cov(lambdahat)='); print(HinvB @ V @ HinvB.T)
```

Section 6.4. Potential function approach

Section content

- Introducing the function Z

Why introduce the Z function?

This section introduces the potential function approach as an important tool for reformulating previously discussed models and facilitating various extensions.

We study the dynamic model with a finite horizon and logit heterogeneity, assuming no capacity constraints for the moment. We consider the dual program from Section 6.2.1 and focus on how $u_1 = (u_{1x})$, the payoff vector in period 1, depends on $u_2 = (u_{2x})$, the corresponding vector in the subsequent period. We can rewrite u_1 as:

$$\min_{u_{1x}} \max_{n_x \geq 0} \{ \sum_{x \in [X]} N_x u_{1x} + Z(u_1, u_2', n) \}$$

The function Z

We introduce the function Z:

$$Z\left(u,u',n\right) := \sum_{xy\in[X\times Y]} n_x \exp(\phi_{1xy} + (\boldsymbol{P}^\top u')_{xy} - u_x) - \sum_{x\in[X]} n_x.$$

Letting $\mu_{xy} = n_x \exp(\phi_{1xy} + (\boldsymbol{P}^\top u')_{xy} - u_x)$, one has

$$\frac{\partial Z}{\partial n_x} = \sum_{y\in[Y]} \frac{\mu_{xy}}{n_x} - 1, \qquad \frac{\partial Z}{\partial u_x} = -\sum_{y\in[Y]} \mu_{xy}, \qquad \text{and} \quad \frac{\partial Z}{\partial u'_x} = (\boldsymbol{P}\mu)_x,$$

which means that at equilibrium $\partial Z/\partial n_x = 0$, and $\partial Z/\partial u_x + \partial Z/\partial u'_x = 0$.

Let's start with the simplest application.

Example 6.2 : static logit model

Static logit discrete choice problem with N_x agents of type x:
$u_x = \mathbb{E}[\max_{y \in [Y]} \{\phi_{xy} + \varepsilon_y\}]$ is solution to

$$\min_u \left\{ \sum_{x \in [X]} N_x u_x + Z\left(u, 0, N\right) \right\},$$

with the expression for the function Z given by (6.4.2). Indeed, by first-order conditions, one has $N_x = \sum_y N_x \exp(\phi_{xy} - u_x)$, and thus $u_x = \log \sum_y \exp \phi_{xy}$.

We move on to a parametric version.

Example 6.3 : static logistic regression

Parametric version of the above model, where $(\Phi\lambda)_{xy} = \sum_k \phi_{kxy}\lambda_k$. $\hat{\mu}_{xy}$ is the number of agents x who have chosen option y, and $N_x = \sum_{xy} \hat{\mu}_{xy}$. Logistic regression is

$$\min_{u,\lambda} \left\{ \sum_{xy \in [X \times Y]} \hat{\mu}_{xy}(u_x - (\Phi\lambda)_{xy}) + Z(\lambda, u, 0, N) \right\},$$

where the function Z is given by

$$Z\left(\lambda, u, u', n\right) := \sum_{xy \in [X \times Y]} n_x \exp\left((\Phi\lambda)_{xy} + (\boldsymbol{P}^\top u')_{xy} - u_x\right) - \sum_{x \in [X]} n_x.$$

Next, matching models.

Example 6.4: matching estimation

Surplus estimation in the static Choo–Siow model of section 5.4 is obtained by

$$\min_{u,v,\lambda}\max_{n,m}\left\{n^\top u+m^\top v-\sum_{xy\in[X\times Y]}\hat{\mu}_{xy}(\Phi\lambda)_{xy}+Z\left(\lambda,u,v,n,m\right)\right\},$$

where the function Z is given by

$$Z\quad(\lambda,u,v,n,m)=2\sum_{xy\in[X\times Y]}\sqrt{n_x m_y}\exp\left(\frac{(\Phi\lambda)_{xy}-u_x-v_y}{2}\right)$$
$$+\sum_{x\in[X]}n_x\exp(-u_x)+\sum_{y\in[Y]}m_y\exp(-v_y)-\sum_{x\in[X]}n_x-\sum_{y\in[Y]}m_y.$$

Lastly, the dynamic case.

Example 6.6: dynamic discrete choice

The finite-horizon dynamic discrete choice estimation problem expresses as

$$\max_{u,\lambda} \min_{n\geq 0} \left\{ \hat{\mu}^\top \Phi \lambda + N^\top u - \sum_{t\in[T]} Z_t \left(\lambda, u_t, u_{t+1}, n_t \right) \right\},$$

where

$$Z\left(\lambda, u, u', n\right) := \sum_{txy\in[T\times X\times Y]} n_{tx} \exp\left((\Phi\lambda)_{txy} + (\boldsymbol{P}^\top u')_{xy} - u_{tx} \right) - \sum_{tx\in[T\times X]} n_{tx}.$$

Why introduce the function Z?

The five examples just presented are simple reformulations of models we previously introduced and solved without using the function Z. The reader may question the usefulness of introducing Z if it serves only to paraphrase existing results and methods. However, as we shall see later, the function Z will be crucial for extending our framework to dynamic discrete choice problems with an infinite horizon, as well as to more general dynamic models of coalition formation.

Section 6.5. The model with infinite horizon

Section content

- Infinite-horizon dynamic discrete choice
- Stationary solution
- Nested fixed point estimation algorithm
- MPEC estimation algorithm

We now consider the problem with an infinite time horizon, so $T = +\infty$, and constant geometric discounting $\beta \in [0,1]$, so $\phi_{xy} = \beta^{t-1}\tilde{\phi}_{xy}$.

Choosing option y for a unit in state x at time t generates payoff $\beta^{t-1}\tilde{\phi}_{xy} + \beta^{t-1}\varepsilon_y$, where the entries of the vector ε are i.i.d. draws from a Gumbel distribution. (The payoffs are measured in time $t = 1$ units).

If the policy variable is μ, the total intertemporal payoff is

$$\sum_{txy \in [\infty \times X \times Y]} \mu_{txy} \beta^{t-1} \tilde{\phi}_{xy} - \sum_{txy \in [\infty \times X \times Y]} \beta^{t-1} \mu_{txy} \log\left(\frac{\mu_{txy}}{\sum_{y \in [Y]} \mu_{txy}}\right)$$

The capacity constraints are assumed to be time-independent:

$$\sum_{xy \in [X \times Y]} \tilde{A}_{l,xy} \mu_{txy} \leq \tilde{c}_l \text{ for each } tl \in [\infty \times L].$$

The dual problem in infinite horizon

Denote τ_{tl} the Lagrange multiplier of the capacity constraint at (t,l). The dual is the minimization of

$$\sum_{x\in[X]} N_x u_{1x} + \sum_{tl\in[\infty\times L]} \tau_{tl}\tilde{c}_l$$

subject to $u_{tx} = \mathbb{E}[\max_y\{\beta^{t-1}\tilde{\phi}_{xy} - (\tilde{\boldsymbol{A}}\tau_t)_{xy} + (\boldsymbol{P}^\top u_{t+1})_{xy} + \beta^{t-1}\varepsilon_y\}]$ (the Bellman equation). Letting $\tilde{u}_{xt} = u_{tx}/\beta^{t-1}$ and $\tilde{\tau}_t = \tau_t/\beta^{t-1}$, this becomes:

$$\begin{aligned} \min \quad & \sum_x N_x \tilde{u}_{1x} + \sum_{tl} \beta^{t-1}\tilde{\tau}_{tl}\tilde{c}_l \\ s.t. \quad & \tilde{u}_{tx} = \log \sum_y \exp(\tilde{\phi}_{xy} - (\tilde{\boldsymbol{A}}^\top \tilde{\tau}_t)_{xy} + \beta(\boldsymbol{P}^\top \tilde{u}_{t+1})_{xy}), \end{aligned}$$

and we shall drop the tildes for better readability.

We state:

Theorem 6.5.1

The intertemporal decision program of the decision-maker has primal formulation:

$$
\begin{aligned}
&\max_{\mu_{txy} \geq 0} \{ \sum_{txy \in [\infty \times X \times Y]} \mu_{txy} \beta^{t-1} \phi_{xy} - \sum_{txy \in [\infty \times X \times Y]} \beta^{t-1} \mu_{txy} \log(\frac{\mu_{txy}}{\sum_{y \in [Y]} \mu_{txy}}) \} \\
&s.t. \\
&\sum_{y \in [Y]} \mu_{1xy} = N_x, \ x \in [X] \\
&\sum_{x'y \in [X \times Y]} P_{x,x'y} \mu_{tx'y} = \sum_{y \in [Y]} \mu_{(t+1)xy}, \ tx \in [\infty \times X] \\
&\sum_{xy \in [X \times Y]} A_{l,xy} \mu_{txy} \leq c_l, \ tl \in [\infty \times L],
\end{aligned}
$$

Duality theorem (2/2)

Theorem 6.5.1 (continued)

and the dual formulation is:

$$\min_{u,\tau} \sum_{x\in[X]} N_x u_{1x} + \sum_{tl\in[\infty,L]} \beta^{t-1}\tau_{tl} c_l$$

$$s.t.\ u_{tx} = \log \sum_{y\in[Y]} \exp(\phi_{xy} - (\boldsymbol{A}^\top \tau_t)_{xy} + \beta(\boldsymbol{P}^\top u_{t+1})_{xy})$$

Dynamic equations (1/2)

Assume no capacity constraints , then:

$$Z\left(u, u', n\right) = \sum_{xy \in [X \times Y]} n_x \exp(\phi_{xy} + \beta(\boldsymbol{P}^\top u')_{xy} - u_x) - \sum_{x \in [X]} n_x$$

Let μ_{txy} and u_{tx} be optimal primal and dual solutions, and set $n_{tx} = \sum_y \mu_{txy}$ and $u_{tx} = \log \sum_{y \in [Y]} \exp(\phi_{xy} + \beta(\mathbf{P}^\top u_{t+1})_{xy})$, we can express optimality as:

- **market clearing:**
  $n_{tx} = \sum_y \mu_{txy} = -\partial Z(u_t, u_{t+1}, n_t)/\partial u_x$;
- **value iteration:**
  $\log \sum_{x \in [X]} \exp(\phi_{xy} + \beta(\boldsymbol{P}^\top u_{t+1})_{xy}) - u_{tx} = \partial Z(u_t, u_{t+1}, n_t)/\partial n_x = 0$;
- **Markov transitions:**
  $n_{(t+1)x} = \sum_{x'y} P_{x|x'y} \mu_{tx'y} = \beta^{-1} \partial Z(u_t, u_{t+1}, n_t)/\partial u'_x$.

Definition 6.5.1

The dynamic equations of the model with logit heterogeneity and infinite are given using the expression of Z by

$$\begin{cases} n_{1x} = N_x \ \forall x \in [X] \\ 0 = \frac{1}{\beta} \frac{\partial Z}{\partial u'_x} \left(u_t, u_{t+1}, n_t \right) + \frac{\partial Z}{\partial u_x} \left(u_{t+1}, u_{t+2}, n_{t+1} \right), \ \forall tx \in [\infty \times X], \\ 0 = \frac{\partial Z}{\partial n_x} \left(u_t, u_{t+1}, n_t \right) \ \forall tx \in [\infty \times X], \end{cases}$$

Remark. We can take into exogenous arrivals/departures by replacing zero by e_{tx}, the net exiting flow, on the right hand-side of the second equation.

Remark on various generalizations on the infinite horizon model

Remark 6.5.1 (Various generalizations): These equations can be modified to incorporate various features. If we need to deal with capacity constraints $A\mu \leq c$, we change the expression for Z, and previous dynamic equations apply.

In another generalization, we can account for births and deaths. If we have an arrival flow b_{tx} of units in state x at time t, and a flow d_{tx} of units leaving the system from state x at time t, we can denote the net exit flow as $e_{tx} = d_{tx} - b_{tx}$. The previous equations are then modified to:

$$\begin{cases} n_{1x} = N_x, \forall x \in [X], \\ 0 = \dfrac{1}{\beta} \dfrac{\partial Z}{\partial u'_x} (u_t, u_{t+1}, n_t) + \dfrac{\partial Z}{\partial u_x} (u_{t+1}, u_{t+2}, n_{t+1}), \ \forall tx \in [\infty \times X], \\ 0 = \dfrac{\partial Z}{\partial n_x} (u_t, u_{t+1}, n_t), \forall (t, x) \in [\infty \times X]. \end{cases}$$

If we return to the case without capacity constraint, then we have:

Proposition 6.5.1

Consider the dynamic discrete choice problem with logit heterogeneity, infinite horizon and no capacity constraint. Then, the primal and dual solutions (μ_{txy}) and (u_{tx}) are such that the dual vector of solutions (u_{tx}) does not depend on t, and the primal solutions μ_{txy} are given by $\mu_{txy} = n_{tx} \exp(\phi_{xy} + \beta(\boldsymbol{P}^\top u)_{xy} - u_x)$, where the quantities n_t, u, τ are the solution to:

$$\begin{cases} n_{1x} = N_x \ \forall x \in [X] \\ 0 = \frac{1}{\beta}\frac{\partial Z}{\partial u'_x}\left(u, u, n_t\right) + \frac{\partial Z}{\partial u_x}\left(u, u, n_{t+1}\right) \ \forall tx \in [\infty \times X], \ \forall tx \in [\infty \times X] \\ 0 = \frac{\partial Z}{\partial n_x}\left(u, u, n_t\right) \ \forall tx \in [\infty \times X], \end{cases}$$

and the function Z is given above.

Idea of the proof

The proof of Theorem 6.5.1 relies on the fact that u_{tx} remains bounded. Specifically, its absolute value cannot exceed a constant K, which equals $(1-\beta)^{-1}$ times the maximum short-term, per-unit expected payoff (in the logit case, this maximum is bounded by $\|\phi\|_\infty + \log Y$).

From Theorem A.3.1, the value iteration operator is a contraction mapping modulus β. It implies that given two bounded solutions u_{tx} and u'_{tx} to the value iteration equation, we have for any integer $s \geq 0$:

$$|u_t - u'_t|_\infty \leq \beta^s |u_{t+s} - u'_{t+s}|_\infty \leq 2\beta^s K,$$

which, letting $s \to +\infty$, implies that in the infinite horizon setting there is a unique bounded solution to the value iteration equation. Taking $u'_{t+1} = u_t$, we see that this solution is a fixed point of the value iteration operator.

The stationary solution (1/2)

Note that although the primal solution μ_{txy} is time-dependent, the choice probabilities, which are given by $\pi_{txy} = \exp(\phi_{xy} + \beta(\boldsymbol{P}^\top u)_{xy} - u_x)$, are stationary.

One may write down the equation associated with a fully stationary solution. Because the total size of the population $\sum_x n_{tx}$ is conserved, we need to impose it.

Definition 6.5.2

A stationary solution is such that $\mu_{xy} = n_x \exp(\Phi_{xy} + \beta(\boldsymbol{P}^\top u)_{xy} - u_x)$, where, taking Z as in (6.5.7), n and u solve

$$\begin{cases} \sum_{x\in[X]} n_x = 1 \\ 0 = \frac{1}{\beta}\frac{\partial Z}{\partial u'_x}(u,u,n) + \frac{\partial Z}{\partial u_x}(u,u,n) \;\; \forall x \in [X] \\ 0 = \frac{\partial Z}{\partial n_x}(u,u,n) \;\; \forall x \in [X]. \end{cases}$$

The stationary solution (2/2)

Theorem 6.5.2

Consider the dynamic discrete choice problem with logit heterogeneity, infinite horizon and no capacity constraint.

Assume that for every pair of states x and x' in $[X]$, there is an action $y \in [Y]$ such that $\boldsymbol{P}_{x',xy} > 0$.

Then, a stationary solution $n^\star$ exists, and it is the limit as $t \to +\infty$ of n_t solution to:

$$\begin{cases} n_{1x} = N_x \; \forall x \in [X] \\ 0 = \frac{1}{\beta} \frac{\partial Z}{\partial u'_x} (u, u, n_t) + \frac{\partial Z}{\partial u_x} (u, u, n_t) \; \forall tx \in [\infty \times X] \\ 0 = \frac{\partial Z}{\partial n_x} (u, u, n_t) \; \forall tx \in [\infty \times X], \end{cases}$$

irrespective of the starting point $n_1 = N$.

Proof of the existence of the stationary solution n^*

We consider $Q_{x'x} = \sum_y P_{x'|xy} \pi_{xy}$, where $\pi_{xy} = \exp(\phi_{xy} + \beta(P^\top u)_{xy} - u_x)$. Since Q is strictly positive and $1_X^\top Q = 1_X^\top$, the Perron–Frobenius theorem (A.3.4) establishes that its dominant eigenvalue is 1.

Letting n be the corresponding right eigenvector ($Qn = n$), we have:

$$n_{x'} = \sum_{x,y \in X \times Y} n_x \pi_{xy} P_{x'|xy}.$$

By setting $\mu_{xy} = n_x \pi_{xy}$, we obtain $P\mu = n$, which ensures a stationary population ($n_{t+1} = n_t$).

Finally, as the discount factor $\beta \to 1$ (infinite patience), these conditions can be reformulated as the solution to a minimax problem, as developed below:

Unit discount rate

When $\beta = 1$, which is the limit of unit discount rate, and still with Z given by expression (6.5.7), the equations, the equations associated with the fully stationary solution become:

$$\begin{cases} \sum_{x \in [X]} n_x = 1 \\ 0 = \frac{\partial Z}{\partial u'_x}(u, u, n) + \frac{\partial Z}{\partial u_x}(u, u, n) \ \ \forall x \in [X] \\ 0 = \frac{\partial Z}{\partial n_x}(u, u, n) \ \ \forall x \in [X]. \end{cases}$$

which are the first order conditions associated with the saddle-point problem

$$\min_{u \in \mathbb{R}^X} \max_{n \in \Delta_X} Z(u, u, n)$$

where $\Delta_X = \{n \in \mathbb{R}_+^X : \sum_x n_x = 1\}$.

Section 6.6. Inference, logit model with infinite horizon

Section content

- MPEC formulation
- The nested fixed-point algorithm
- Augmented Lagrangian algorithm for dynamic choice estimation

Introduction

Let $\beta \in [0,1)$ and assume that we have a sample of size I of observations of the form $x_i y_i$, $i \in [I]$. Assume that the observations are independently sampled. The conditional log-likelihood of observation i is therefore $\log \pi^{\lambda}_{x_i y_i}$, where π^{λ}_{xy} is given by

$$\pi^{\lambda}_{xy} = \exp\left((\Phi\lambda + \beta \boldsymbol{P}^{\top} u^{\lambda})_{xy} - u^{\lambda}_{x} \right),$$

where u^{λ}_{x} is the unique solution to the *nested fixed-point equation,*

$$u^{\lambda}_{x} = \log \sum_{y \in [Y]} \exp(\Phi\lambda + \beta \boldsymbol{P}^{\top} u^{\lambda})_{xy},$$

which can be written in a vectorized manner as

$$\boldsymbol{\Sigma}_Y \exp\left(\Phi\lambda + \boldsymbol{\Psi} u^{\lambda} \right) = 1_X,$$

where we have let $\boldsymbol{\Sigma}_Y := \boldsymbol{I}_X \otimes 1_Y^{\top}$ and $\boldsymbol{\Psi} := \beta \boldsymbol{P}^{\top} - \boldsymbol{\Sigma}_Y^{\top}$.

The maximum likelihood estimator

As the log-likelihood of the sample is the sum of the log-likelihood terms associated with the observations, we have $\ell_l(\lambda) = \sum_i (\Phi\lambda)_{x_i y_i} + \beta \sum_i (\boldsymbol{P}^\top u^\lambda)_{x_i y_i} - \sum_i u^\lambda_{x_i}$, which, letting $\hat{\mu}_{xy} = \sum_i 1\{x_i = x, y_i = y\}$, can be written as $\ell(\lambda) = \hat{\mu}^\top(\Phi\lambda + (\beta \boldsymbol{P}^\top - \boldsymbol{I}_X \otimes 1_Y)u^\lambda)$. Therefore the maximum likelihood estimator of λ is given by

$$\max_{\lambda \in \mathbb{R}^K} \ell_l(\lambda) = \max_{\lambda \in \mathbb{R}^K} \left\{ \hat{\mu}^\top \left(\Phi\lambda + \boldsymbol{\Psi} u^\lambda \right) \right\}.$$

This problem has the following equivalent MPEC formulation, where the acronym stands for "mathematical program under equilibrium constraint":

$$\begin{aligned} \max_{\lambda \in \mathbb{R}^K, u \in \mathbb{R}^X} \quad & \left\{ \hat{\mu}^\top (\Phi\lambda + \boldsymbol{\Psi} u) \right\} \\ s.t. \quad & \boldsymbol{\Sigma}_Y \exp(\Phi\lambda + \boldsymbol{\Psi} u) = 1_X. \end{aligned}$$

First and second differentials of the value function

Denoting du the differential of u with respect to λ, which is

$$du = -(\mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \mathbf{\Psi})^{-1} \mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \Phi.$$

Turning to the second differential, we obtain, using indexed notation,

$$0 = \sum_y \pi_{xy} \left(\sum_{x'} \mathbf{\Psi}_{xy,x'} \frac{\partial^2 u_{x'}}{\partial \lambda_k \partial \lambda_l} \right) + \sum_y \pi_{xy} \frac{\partial \log \pi_{xy}}{\partial \lambda_l} \left(\phi_{xyk} + \sum_{x'} \mathbf{\Psi}_{xy,x'} \frac{\partial u_{x'}}{\partial \lambda_k} \right).$$

Therefore, as $d \log \pi = \Phi + \mathbf{\Psi} du$, the equation can be rewritten in a vectorized form as $0 = \mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \mathbf{\Psi} d^2 u + \mathbf{\Sigma}_Y \mathbf{\Delta}_\pi ((\Phi + \mathbf{\Psi} du)^{\circledast 2})$, where the notation $M^{\circledast 2}$ stands for $M \circledast M$. As a result, we obtain the following expression for the second differential of u:

$$d^2 u = -(\mathbf{\Sigma}_Y \mathbf{\Delta}_\pi \mathbf{\Psi})^{-1} \mathbf{\Sigma}_Y \mathbf{\Delta}_\pi ((\Phi + \mathbf{\Psi} du)^{\circledast 2}).$$

The nested fixed-point algorithm

As the log-likelihood of the sample is $\ell(\lambda) = \hat{\mu}^{\top}\left(\Phi\lambda + \mathbf{\Psi} u^{\lambda}\right)$, its differential is

$$d\ell(\lambda) = \hat{\mu}^{\top}\left(\mathbf{I}_{XY} - {}^{-}\left({}^{\bullet}{}_{Y}{}^{\prime}{}_{\pi}{}^{-}\right)^{-1}{}^{\bullet}{}_{Y}{}^{\prime}{}_{\pi}\right)\Phi,$$

and from the formula for the second differential of u, one can readily deduce the expression of $d^2\ell(\lambda) = \hat{\mu}^{\top}\mathbf{\Psi} d^2 u$.

The essential step of the method consists of updating the parameter value λ by an increment that is proportional to the gradient of $\ell(\lambda)$. The computation of that gradient requires the differential of the value function du; however, the computation of du in turns requires the computation of the value function u itself, which is performed by the nested fixed-point algorithm.

The nested fixed-point algorithm

Algorithm 6.6.2

Input: λ^0 (initial guess)

Parameters: β (discount factor), ϵ (tolerance), α (learning rate), S (max iterations)

Output: Estimated parameters $\hat{\lambda}$ (estimator)

Initialize: $s \leftarrow 0$, $\lambda^s \leftarrow \lambda^0$

cont $\leftarrow$ **True**

Algorithm 6.6.2 (continued)

Optimization Loop:

While *cont* = **True do**:

- Compute u_x with $\lambda = \lambda^s$ using fixed-point iteration (6.6.2);
- Compute $\nabla(\lambda^s)$ using (6.6.8) and the value of u_x;
- Update $\lambda^{s+1} = \lambda^s - \alpha\nabla(\lambda^s)$;
- $s \leftarrow s + 1$;
- **If** $\|\lambda^{s+1} - \lambda^s\| < \epsilon$ **or** $s = S$ **then**
  - *cont* $\leftarrow$ **False**;

Return: $\hat{\lambda} = \lambda^s$

Python code for the nested fixed point algorithm

The following slides provide the Python code for estimating the infinite-horizon model using the nested fixed-point algorithm.

Python code for estimation in infinite horizon, nested fixed point (1/5)

```
import numpy as np
from scipy import optimize, sparse as sp
np.random.seed(7)
X,Y,K = 10,2,3
beta = 0.9
rel_tol = 1e-5
epsilon = 0.05
max_it = 10000
phi_x_y_k = np.random.normal(size=(X,Y,K))/10
muhat_xy = np.random.uniform(size=(X*Y))
P_xprime_xy = np.random.uniform(size= (X,X*Y))
P_xprime_xy = P_xprime_xy / P_xprime_xy.sum(axis=0)[None,:]
# Computing:
phi_xy_k = phi_x_y_k.reshape((-1,K))
SigmaY = sp.kron( sp.eye(X),np.ones((1,Y)))
Psi = beta* P_xprime_xy.T - np.kron( np.eye(X),np.ones((Y,1)))
tol_lambda = rel_tol* np.abs(phi_x_y_k).max()*muhat_xy.sum()
# continued on the next slide =>>
```

Python code for estimation in infinite horizon, nested fixed point (2/5)

```
# =>> continued from the previous slide
def fu_x(lambda_k, tol_u=1e-5, max_it = 10000):
    u_x = np.zeros(X)
    phi_x_y = (phi_xy_k @ lambda_k).reshape((X,Y))
    M = 1+ (np.abs(phi_x_y).max() + np.log(Y)) / (1-beta)
    S = int(max ( (np.log(tol_u) - 2* M) / np.log(beta) ,10) )
    for s in range(min(S,max_it)):
        m_x = (phi_x_y + beta * (P_xprime_xy.T @ u_x).reshape((X,Y))).max(axis = 1)
        u_x = m_x + np.log( np.exp(phi_x_y + beta * (P_xprime_xy.T @ u_x).reshape((X,Y)) -m_x
             [:,None] ).sum(axis = 1) )
    return(u_x)
# continued on the next slide =>>
```

Python code for estimation in infinite horizon, nested fixed point (3/5)

```
# =>> continued from the previous slide
def fdu_x(lambda_k):
    u_x = fu_x(lambda_k).flatten()
    pi_xy = np.exp( phi_xy_k @ lambda_k + Psi @ u_x)
    return - sp.linalg.spsolve(SigmaY @ sp.diags(pi_xy) @ Psi , SigmaY @ sp.diags(pi_xy) @
         phi_xy_k)

def neg_F(lambda_k):
    return  - muhat_xy.T @ ( phi_xy_k @ lambda_k + Psi @ fu_x(lambda_k))

def neg_gradF(lambda_k):
    return muhat_xy.T @ ( - Psi @ fdu_x(lambda_k) - phi_xy_k)
# continued on the next slide =>>
```

Python code for estimation in infinite horizon, nested fixed point (4/5)

```
# =>> continued from the previous slide
def neg_hessF(lambda_k):
    u_x = fu_x(lambda_k).flatten()
    pi_xy = np.exp( phi_xy_k @ lambda_k + Psi @ u_x)
    dlogpi = phi_xy_k + Psi @ fdu_x(lambda_k)
    XY = X*Y
    TTT = sp.csr_matrix( ([1]*XY,(list(range(XY)), [a*(XY+1) for a in range(XY)])),shape = (XY,
         XY*XY) )
    Tdlogpisquare =  TTT @ sp.kron(dlogpi,dlogpi )
    d2u = - sp.linalg.spsolve( SigmaY @ sp.diags(pi_xy) @ Psi , SigmaY @ sp.diags(pi_xy) @
         Tdlogpisquare)
    return (- muhat_xy.T @ Psi @ d2u).reshape((K,K))

result = optimize.minimize(neg_F, np.zeros(K), jac=neg_gradF, method='BFGS')
# continued on the next slide =>>
```

Python code for estimation in infinite horizon, nested fixed point (5/5)

```
# =>> continued from the previous slide
# displaying results
if result.success:
    lambdahat_k = result.x
    print("lambdahat:", lambdahat_k)
    I = muhat_xy.sum()
    V = np.diag(muhat_xy) / I - muhat_xy[:,None] @ muhat_xy[None,:] / (I*I)
    HinvB = np.linalg.solve(neg_hessF(lambdahat_k), (phi_xy_k + Psi @ fdu_x(lambdahat_k)).T)
    print('Cov(lambdahat)=')
    print(HinvB @ V @ HinvB.T)
else:
    print(result.message)
```


The MPEC problem can be rewritten as $\max_{\lambda,u} \min_n \mathcal{L}(\lambda, u; n)$, where the Lagrangian $\mathcal{L}$ is defined by

$$\mathcal{L}(\lambda, u; n) := \sum_{xy} \hat{\mu}_{xy}((\Phi\lambda + \beta \boldsymbol{P}^\top u)_{xy} - u_x) + \sum_x n_x - \sum_{xy} n_x \exp(\Phi\lambda + \boldsymbol{\Psi} u)_{xy},$$

that is, $\mathcal{L}(\lambda, u; n) = \hat{\mu}^\top(\Phi\lambda + \boldsymbol{\Psi} u) - Z(\lambda, u, n)$, where Z is the penalty function defined by $Z(\lambda, u, n) := \sum_{xy} n_x \exp((\Phi\lambda + \beta \boldsymbol{P}^\top u)_{xy} - u_x) - \sum_x n_x$. Equivalently, this problem can be formulated as $\max_{\lambda,u} \min_n \mathcal{L}(\lambda, u; n)$. Letting $\theta = (\lambda, u, n)$, and denoting

$$\mu^\theta_{xy} := \sum_{xy} n_x \exp((\Phi\lambda + \beta \boldsymbol{P}^\top u)_{xy} - u_x).$$

Minimax formulation (2/2)

One can express the derivatives of Z from (6.5.7) with respect to λ, u, and n as:

$$\begin{cases} \frac{\partial Z}{\partial \lambda_k}(\lambda, u, n) = \sum_{xy \in [X \times Y]} \mu^{\theta}_{xy} \phi_{xyk}, \\ \frac{\partial Z}{\partial u_x}(\lambda, u, n) = \beta \sum_{x'y \in [X \times Y]} \boldsymbol{P}_{x,x'y} \mu^{\theta}_{x'y} - \sum_{y \in [Y]} \mu^{\theta}_{xy}, \\ \frac{\partial Z}{\partial n_x}(\lambda, u, n) = \sum_{y \in [Y]} \frac{\mu^{\theta}_{xy}}{n_x} - 1. \end{cases}$$

These quantities are relatively straightforward to interpret. $\partial Z / \partial \lambda_x$ is the kth moment of ϕ_k predicted by μ^{θ}_{xy}. $\partial Z / \partial u_x$ is β times the number of units that are transitioning to state x from the previous period minus the number of units currently in state x; in a stationary solution, this is equal to $\beta - 1$ times the number of units in state x. $\partial Z / \partial n_x$ is the ratio of the number of units that are in state x divided by n_x; at equilibrium, this should be equal to 1.

Interpretation of the optimality conditions (1/2)

By first-order conditions, one has the following relations:

- Optimality with respect to λ_k implies

$$\sum_{xy \in [X \times Y]} \mu^{\theta}_{xy} \phi_{xyk} = \sum_{xy \in [X \times Y]} \hat{\mu}_{xy} \phi_{xyk} \quad \text{for all } k \in [K],$$

  which means that the K predicted moments that Φ associates with the predicted policy vector μ^{θ}_{xy} match with the observed corresponding moments.

Interpretation of the optimality conditions (2/2)

- Optimality with respect to the value u_x yields

$$\beta \sum_{x'y \in [X \times Y]} \boldsymbol{P}_{x,x'y} \mu^{\theta}_{x'y} - \sum_{y \in [Y]} \mu^{\theta}_{xy} = \beta \sum_{x'y \in [X \times Y]} \boldsymbol{P}_{x,x'y} \hat{\mu}_{x'y} - \sum_{y \in [Y]} \hat{\mu}_{xy} \quad \forall x \in [X],$$

which expresses that the predictor μ^{θ}_{xy} is such that β times the total number of units transitioning to state x from the previous period minus the total number of units in state x matches its observed analog.

- Optimality with respect to the marginal number n_x gives us

$$\sum_{y \in [Y]} \frac{\mu^{\theta}_{xy}}{n_x} = 1 \quad \text{for all } x \in [X],$$

which expresses that $n_x = \sum_y \mu^{\theta}_{xy}$, allowing one to interpret n_x.

Computation using the augmented Lagrangian method

We can use the augmented Lagrangian algorithm to compute the maximum likelihood estimator using the MPEC formulation. Given the expression of the Lagrangian, we can formulate the augmented Lagrangian as:

$$\mathcal{L}_A(\lambda, u; n; \gamma) = \hat{\mu}^\top (\Phi\lambda + \Psi u) - Z(\lambda, u, n) - \gamma \|\nabla_n Z(\lambda, u, n)\|^2,$$

Algorithm 6.6.3 (Augmented Lagrangian algorithm)

Input: Input values λ^0, u^0, n^0

Parameter: Step size ϵ, tolerance η, maximum iterations S, sequence $\gamma^t \to +\infty$

Initialize iteration counter s=0 While $||Z(\lambda^s, u^s)||_{+\infty} \geq \eta$ **and** $s < S$ **do**:

$$(\lambda^{s+1}, u^{s+1}) = \arg\max_{\lambda,u} \mathcal{L}_A(\lambda, u; n^s; \gamma^s)$$

$$n_x^{s+1} = n_x^s + \gamma^s(1 - \sum_y \exp((\Phi\lambda^{s+1} + \beta \mathbf{P}^\top u^{s+1})_{xy} - u_x^{s+1})$$

$$s = s + 1$$

end Return: λ^s, u^s, n^s

Asymptotic distribution of the estimator

One can obtain the Hessian matrix $D^2\ell(\lambda)$ by reshaping the entries of the K^2-dimensional vector $d^2\ell(\lambda)$ into a $K \times K$ matrix, and

Theorem 6.6.1

Assume that the observations $x_i y_i$ are drawn from the population in an i.i.d. fashion and collected in a sample of size I. Letting $\hat{\mu}_{xy}$ be the number of observations of type xy in the sample, the following convergence holds in distribution:

$$\sqrt{I}(\hat{\lambda} - \lambda) \xrightarrow{d} \mathcal{N}(0, H^{-1} C H^{-1}),$$

where H is estimated by $\hat{H} = D^2\ell(\lambda)/I$ and $C = BVB^\top$, where B is estimated by $\hat{B} = \Phi^\top + Du(\hat{\lambda})^{\top -\top}$ and V by $\hat{V} = diag(\hat{\mu})/I - \hat{\mu}\hat{\mu}^\top/I^2$.

Section 6.7. Nonparametric identification, infinite horizon

Section content

- Nonparametric identification

Set estimators (1/2)

Given the conditional choice probabilities $\hat{\pi}_{xy}$ computed from the observations $\hat{\mu}_{xy}$, and a discounting rate $\beta \in [0,1]$, the set estimator of ϕ is the set of those vectors $\hat{\phi}$ that lead to predicting the conditional choice probabilities $\hat{\pi}_{xy}$. Recall that

$$\log \pi = \phi + (\beta \boldsymbol{P}^\top - \boldsymbol{\Sigma}_Y^\top) u,$$

where the expression of $\boldsymbol{\Sigma}_Y^\top$ was given by $\boldsymbol{\Sigma}_Y^\top = \boldsymbol{I}_X \otimes \mathbf{1}_Y$, and where π is subject to the restriction $\boldsymbol{\Sigma}_Y \pi = \mathbf{1}_X$. As a result, the set estimator of ϕ is given by

$$\{\log \hat{\pi} + (\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top) u : u \in \mathbb{R}^X\}.$$

Set estimators (2/2)

In the static case ($\beta = 0$), the set estimator of ϕ is simply the set of vectors $\hat{\phi}$ such that $\hat{\phi}_{xy} = \log \hat{\pi}_{xy} + u_x$. This means that $\log \hat{\pi}$ belongs in that set, as well as any quantity obtained by adding any x-fixed effects to it.

In the general dynamic case, the set estimator of ϕ is the set of vectors $\hat{\phi}$ such that $\hat{\phi}_{xy} = \log \hat{\pi}_{xy} + u_x - \beta \sum_{x'} P_{x',xy} u_{x'}$, for any vector $u \in \mathbb{R}^X$. This means that for $\beta > 0$, $\log \hat{\pi}$ still belongs in the set estimator; but in general, $\log \hat{\pi}$ plus an arbitrary x-fixed effect will not belong in the set estimator.

Letting $F : \mathbb{R}^{XY} \to \mathbb{R}^P$, we say that F is an identified statistic if $\hat{\phi} \mapsto F(\hat{\phi})$ is constant on the set estimator of ϕ. In the current setting, this means that if we let for instance $\hat{\phi} := \log \hat{\pi}$, then for any vector $u \in \mathbb{R}^X$, we have $F(\hat{\phi} + (\mathbf{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top)u) = F(\log \hat{\pi})$. When F is differentiable, F is an identified statistic if and only if

$$dF(\hat{\phi})(\mathbf{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top) = 0$$

holds for all $\hat{\phi}$ in the set estimator of ϕ. For instance, in the static case, that is, when $\beta = 0$, $F(\hat{\phi}) = (\boldsymbol{I}_{XY} - \mathbf{\Sigma}_Y^\top \mathbf{\Sigma}_Y / Y)\hat{\phi}$ is an identified statistic since $(\boldsymbol{I}_{XY} - \mathbf{\Sigma}_Y^\top \mathbf{\Sigma}_Y / Y)\mathbf{\Sigma}_Y^\top = 0$. This corresponds to taking $F_{xy}(\hat{\phi}) = \log \hat{\pi}_{xy} - Y^{-1} \sum_y \log \hat{\pi}_{xy}$, which is the deaveraging operation consisting of removing the conditional average of $\log \hat{\pi}_{xy}$ given x.

Identifying restrictions

Letting $\rho : \mathbb{R}^{XY} \to \mathbb{R}^{X}$, we say that the restrictions $\rho(\hat{\phi}) = 0$ are identifying if there is only one vector $\hat{\phi}$ in the set estimator of ϕ such that $\rho(\hat{\phi}) = 0$. In our setting, this means that if we set $\hat{\phi} := \log \hat{\pi} + (\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top) u$ and if we assume that $\rho(\hat{\phi}) = 0$, then the equation $\rho(\hat{\phi} + (\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top) r) = 0$ has no other solution than $r = 0_X$. We have:

Lemma 6.7.1

If $0 \leq \beta < 1$, the matrix $(\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top)$ is of full rank.

For instance, requiring $\phi_{x1} = 0$ for all x is an identifying restriction. In that case, $\rho(\phi) = (\boldsymbol{I}_X \otimes (\boldsymbol{e}_Y^1)^\top)\phi$, and we have $d\rho(\phi)(\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top) = (\boldsymbol{I}_X \otimes (\boldsymbol{e}_Y^1)^\top)(\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top)$ $= \boldsymbol{I}_X - \beta \tilde{\boldsymbol{Q}}^\top$, where the matrix $\tilde{\boldsymbol{Q}}_{x',x}$ is defined by $\tilde{\boldsymbol{Q}}_{x',x} = \boldsymbol{P}_{x',x1}$, and a similar argument to the one given in lemma 6.7.1 implies that $\boldsymbol{I}_X - \beta \tilde{\boldsymbol{Q}}^\top$ is invertible.

Proof of Lemma 6.7.1

We want to show that $(\Sigma_Y^\top - \beta P^\top)u = 0$ implies $u = 0$. First, we assume $\Sigma_Y^\top u - \beta P^\top u = 0$ and let $Q = P\Sigma_Y^\top / Y$, we left-multiply the equation by Σ_Y / Y to obtain:

$$\frac{\Sigma_Y \Sigma_Y^\top}{Y} u - \beta Q^\top u = 0.$$

Since $\Sigma_Y \Sigma_Y^\top / Y = I_X$, this simplifies to $u = \beta Q^\top u$. Iterating this relation yields $u = \beta^k (Q^\top)^k u$ for all integers $k \geq 1$. As $k \to \infty$, the discounting factor drives this to zero, which implies that $u = 0$.

Section 6.8. The conditional choice probability approach

Section content

- Hotz and Miller's conditional choice probability (CCP) approach

Nonparametric identification (1/2)

The identifying restrictions $\phi_{x1} = 0$ for all $x \in [X]$ allow us to assume $\phi_{x1} = 0$ without loss of generality. This corresponds to setting $\rho(\phi) = \boldsymbol{\delta}^1 \phi$, where $\boldsymbol{\delta}^1 := \boldsymbol{I}_X \otimes (\boldsymbol{e}_Y^1)^\top$. The identifying equation $\log \pi_{xy} = (\phi + (\beta \boldsymbol{P}^\top - \boldsymbol{\Sigma}_Y^\top) u)_{xy}$ can be rewritten for $y = 1$ as

$$\log \pi_{x1} = \beta(\boldsymbol{P}^\top u)_{x1} - u_x, \ \forall x \in [X],$$

which, as $\boldsymbol{\delta}^1 \log \pi$ is the vector $(\log \pi_{x1})$ and $\boldsymbol{\delta}^1 \phi = 0$, we can rewrite in vectorized terms as $\boldsymbol{\delta}^1 \log \pi = \boldsymbol{\delta}^1 (\beta \boldsymbol{P}^\top - \boldsymbol{\Sigma}_Y^\top) u$, and therefore u is identified by Hotz and Miller's formula for the *conditional choice probability estimator*, which is

$$u = \left(\boldsymbol{\delta}^1 (\beta \boldsymbol{P}^\top - \boldsymbol{\Sigma}_Y^\top) \right)^{-1} \boldsymbol{\delta}^1 \log \pi,$$

and $\phi = \log \pi + (\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top) u$ can readily be deduced.

Theorem 6.8.1

Given the identifying restrictions $\phi_{x1} = 0$ for all $x \in [X]$, ϕ is identified from the data by $\phi = \boldsymbol{A} \log \pi$, where $\boldsymbol{A} = \boldsymbol{I}_{XY} + \left(\boldsymbol{\Sigma}_Y^\top - \beta \boldsymbol{P}^\top\right) \left(\boldsymbol{\delta}^1(\beta \boldsymbol{P}^\top - \boldsymbol{\Sigma}_Y^\top)\right)^{-1} \boldsymbol{\delta}^1$.
If $\hat{\pi}$ is a consistent estimator of π that is asymptotically normal, so that the convergence $I^{1/2}\hat{\pi} \xrightarrow{d} \mathcal{N}(0, V_\pi)$ holds in distribution as $I \to \infty$, we let $\hat{\phi} = \boldsymbol{A} \log \hat{\pi}$, and we have

$$I^{1/2}\hat{\phi} \xrightarrow{d} \mathcal{N}\left(0, \boldsymbol{A}\boldsymbol{\Delta}_\pi^{-1} V_\pi \boldsymbol{\Delta}_\pi^{-1} \boldsymbol{A}^\top\right),$$

where $\boldsymbol{\Delta}_\pi$ is the diagonal square matrix of size XY with the vector π on the diagonal.

Parametric estimation using generalized least squares

We now consider a parametric version of the conditional choice probability estimator. Assume as before that ϕ has a linear parameterization $\phi^{\lambda} = \Phi\lambda$, where, as before, Φ is a $XY \times K$ matrix whose (xy, k) entry is ϕ_{xyk}. We rewrite the equation in previous section to get

$$\log \pi = \Phi\lambda + (\beta \boldsymbol{P}^{\top} - \boldsymbol{\Sigma}_{Y}^{\top})u,$$

and letting $\boldsymbol{W}$ be a square matrix of weights of size XY that is symmetric, positive definite, we may estimate (λ, u) by generalized least squares using the following problem:

$$\min_{\lambda,u} \left| \log \hat{\pi} - \Phi\lambda - (\beta \boldsymbol{P}^{\top} - \boldsymbol{\Sigma}_{Y}^{\top})u \right|_{\boldsymbol{W}},$$

where $|z|^2_{\boldsymbol{W}} = z^{\top}\boldsymbol{W}z$. Methods exist to pick a matrix $\boldsymbol{W}$ with good statistical efficiency properties.

Section 6.9. Dynamic matching models

Section content

- Dynamic model of two-sided matching with transferable utility

Basic settings (1/2)

We assume a matching game is played in each period. The systematic part of the current-period value created by a match of type xy is $\phi_{xy} = (\Phi\lambda)xy = \sum_k \phi xyk\lambda_k$, where λ is a K-dimensional parameter vector to be estimated.

If x or y remain unmatched, their systematic utilities are $\phi_{x0} = (\Phi\lambda)x0$ and $\phi 0y = (\Phi\lambda)_{0y}$.

Conditional on being matched with firm y, a type-x worker transitions to x' with probability $P_{x',xy}$. Similarly, conditional on being matched with worker x, a type-y firm transitions to y' with probability $Q_{y',xy}$.

Let $\boldsymbol{P}$ be the $X \times X(Y+1)$ matrix of $P_{x',xy}$ and $\boldsymbol{Q}$ the $Y \times (X+1)Y$ matrix of $Q_{y',xy}$.

Let u'_x and v'_y be the expected continuation values for types x and y, respectively.

Basic settings (2/2)

The intertemporal joint surplus is given by

$$(\Phi\lambda)_{xy} + \beta(\boldsymbol{P}^\top u')_{xy} + \beta(\boldsymbol{Q}^\top v')_{xy} + \varepsilon_y + \eta_x,$$

and similarly, if a worker of type x is unmatched, then this worker will get payoff equal to $\phi_{x0} + \beta(\boldsymbol{P}^\top u')_{x0} + \varepsilon_0$, and if a firm of type y is unmatched, then its payoff will be $\phi_{0y} + \beta(\boldsymbol{Q}^\top v')_{0y} + \eta_0$.

Interpretation of the settings

- the systematic joint short-term payoff $(\Phi\lambda)_{xy}$ (for a matched pair) or the systematic reservation payoff $(\Phi\lambda)_{x0}$ or $(\Phi\lambda)_{0y}$ (for unmatched agents);
- the discounted expected future utility on the worker side $\beta(\boldsymbol{P}^\top u')_{xy}$, which is the average of the next-period utility associated to the possible next-period type, weighted by the probability of the transitions, and similarly, the discounted expected future utility on the firm side, $\beta(\boldsymbol{Q}^\top v')_{xy}$;
- the random preference shock (ε_y) that workers associate with a firm of type y; and, similarly, the random production shock (η_x) generated by firms with various workers types.

The problem in current period

In the current period, agents solve a Choo and Siow matching model where the intertemporal payoff $(\Phi\lambda)_{xy} + \beta(\boldsymbol{P}^\top u')_{xy} + \beta(\boldsymbol{Q}^\top v')_{xy}$ replaces the single-period payoff $(\Phi\lambda)_{xy}$. Given the current numbers of agents of each type, indicated by n_x and m_y, and given the next-period utilities u'_x and v'_y that agents of each type anticipate receiving in the next period, one seeks to determine:

- u_x and v_y, the current-period average payoff associated with each type;
- n'_x and m'_y, the next-period numbers of agents in each type.

Data setup

We consider a sample of size I that consists of observations $x_i y_i$ that are either a pair of matched individuals, in which case $x_i \in [X]$ and $y_i \in [Y]$, or unmatched individuals, in which case either $x_i = 0$ or $y_i = 0$. We define $\hat{\mu}_{xy} = \sum_i \mathbf{1}_{x_i = x, y_i = y}$ to be the number of observations of each type, and $\hat{f}_{xy} = \hat{\mu}_{xy} / I$ to be the empirical frequencies associated with each observation. Our goal is to estimate the parameter λ of the matching surplus $\Phi\lambda$ based on our data.

Choo and Siow equations in the dynamic case

Recall that the expression of μ is given by Choo and Siow's equations, which in the present context become

$$\begin{cases} \mu_{xy} = \sqrt{n_x m_y} \exp\left(\frac{(\Phi\lambda)_{xy} + \beta(\boldsymbol{P}^\top u')_{xy} + \beta(\boldsymbol{Q}^\top v')_{xy} - u_x - v_y}{2}\right), \\ \mu_{x0} = n_x \exp\left((\Phi\lambda)_{x0} + \beta(\boldsymbol{P}^\top u')_{x0} - u_x\right), \\ \mu_{0y} = m_y \exp\left((\Phi\lambda)_{0y} + \beta(\boldsymbol{Q}^\top v')_{0y} - v_y\right). \end{cases}$$

Definition and differentiation of function Z

The function Z is

$$Z = 2 \sum_{xy \in [X \times Y]} \mu_{xy} + \sum_{x \in [X]} \mu_{x0} + \sum_{y \in [Y]} \mu_{0y} - \sum_{x \in [X]} n_x - \sum_{y} m_y,$$

where μ is a function of the parameters λ, u, v, u', v', n, so that the partial derivatives of Z with respect to its arguments are

$$\frac{\partial Z}{\partial u_x} = -\left(\sum_{y \in [Y]} \mu_{xy} + \mu_{x0} \right) \quad \text{and} \quad \frac{\partial Z}{\partial v_y} = -\left(\sum_{x \in [X]} \mu_{xy} + \mu_{0y} \right),$$

$$\frac{\partial Z}{\partial u'_x} = \beta \sum_{x'y \in [X \times 0:Y]} P_{x,x'y} \mu_{x'y} \quad \text{and} \quad \frac{\partial Z}{\partial v'_y} = \beta \sum_{xy' \in [0:X \times Y]} Q_{y,xy'} \mu_{xy'},$$

$$\frac{\partial Z}{\partial n_x} = \sum_{y \in [Y]} \frac{\mu_{xy}}{n_x} + \frac{\mu_{x0}}{n_x} - 1 \quad \text{and} \quad \frac{\partial Z}{\partial m_y} = \sum_{x \in [X]} \frac{\mu_{xy}}{m_y} + \frac{\mu_{0y}}{m_y} - 1,$$

$$\text{and} \quad \frac{\partial Z}{\partial \lambda_k} = \sum_{xy \in [X \times Y]} \mu_{xy} \phi_{xyk} + \sum_{x \in [X]} \mu_{x0} \phi_{x0k} + \sum_{y \in [Y]} \mu_{0y} \phi_{0yk}.$$

Stationarity conditions and equilibrium equations

For each value of λ, we can therefore write the equilibrium equations of the model that express stationarity as $u = u'$, $v = v'$, $n = n'$, and $m = m'$, and we have the following equations on (u, v, n, m):

$$\sum_{y \in [Y]} \mu_{xy} + \mu_{x0} = n_x \quad \text{and} \quad \sum_{x \in [X]} \mu_{xy} + \mu_{0y} = m_y,$$

$$\sum_{x'y \in [X \times 0:Y]} P_{x,x'y} \mu_{x'y} = n_x \quad \text{and} \quad \sum_{xy' \in [0:X \times Y]} Q_{y,xy'} \mu_{xy'} = m_y,$$

where the expression of μ is a function of (u, v, n, m), setting $u' = u$ and $v' = v$.

Vectorization (1/2)

We vectorize these equations in order to permit tractable computations. Set $p^\top = (u^\top, v^\top)$,

$$\bullet^\top = \begin{pmatrix} \mathbf{I}_X \otimes \mathbf{1}_Y & \mathbf{1}_X \otimes \mathbf{I}_Y \\ \mathbf{I}_X & \mathbf{0}_{X \times Y} \\ \mathbf{0}_{Y \times X} & \mathbf{I}_Y \end{pmatrix} \text{ and } \mathbf{R}^\top = \begin{pmatrix} (\mathbf{P}^\top)_{xy,x'} & (\mathbf{Q}^\top)_{xy,y'} \\ (\mathbf{P}^\top)_{x0,x'} & \mathbf{0}_{X \times Y} \\ \mathbf{0}_{Y \times X} & (\mathbf{Q}^\top)_{0y,y'} \end{pmatrix}$$

and $\nu^\top = (\log n^\top, \log m^\top)$. Let $B = XY + X + Y$ be the number of entries of $\mu = \left(\mu_{xy}, \mu_{x0}, \mu_{0y}\right)$. For $b \in [B]$, let s_b be the size of match b, that is, $s_b = 2$ if $b \leq XY$, in which case b is one of the xy for $x \in [X]$ and $y \in [Y]$, and $s_b = 1$ if $b > XY$, in which case b stands for either $x0$ for $x \in [X]$ or $0y$ for $y \in [Y]$. Let $\mathbf{\Delta}_s$ be the diagonal matrix of size B whose diagonal is s. Let $\mathbf{\Psi} = \beta \boldsymbol{R}^\top + \mathbf{\Sigma}^\top$. Define $\theta^\top = (p^\top, \nu^\top)$, $D_1 = \mathbf{\Delta}_s^{-1} \Phi$, $D_2 = \left(\beta \mathbf{\Delta}_s^{-1} \boldsymbol{R}^\top - \mathbf{\Delta}_s^{-1} \mathbf{\Sigma}^\top, \quad \mathbf{\Delta}_s^{-1} \mathbf{\Sigma}^\top\right)$, $E_1 = \left(\mathbf{\Sigma}^\top, \quad \boldsymbol{R}^\top\right)$, $E_2 = \left(\boldsymbol{I}_{X+Y}, \quad \boldsymbol{I}_{X+Y}\right)$, and $D_3 = \left(\mathbf{0}_{(X+Y)\times(X+Y)}, \quad \boldsymbol{I}_{(X+Y)}\right)$.

Vectorization (2/2)

The probability of observing $b \in [B]$ is equal to f_b, where f has the following vectorized equation:

$$f := \frac{\mu}{\mathbf{1}_B^\top \mu} \quad \text{where } \mu = \exp\left(D_1\lambda + D_2\theta^\lambda\right),$$

where given the value of λ, the vector θ is implicitly determined, which can be rewritten as

$$E_1^\top \exp\left(D_1\lambda + D_2\theta\right) = E_2^\top \exp\left(D_3\theta\right).$$

Derivatives of θ with respect to λ

By differentiation of equation above, denoting $\mathbf{\Delta}_{\mu} = diag(\mu)$, and $\mathbf{\Delta}_{\exp(\nu)} = diag(\exp(\nu))$ where $\nu = D_3\theta$, we have that the differential of θ with respect to λ satisfies

$$E_1^{\top}\mathbf{\Delta}_{\mu}\left(D_1 + D_2 d\theta\right) = E_2^{\top}\mathbf{\Delta}_{\exp(\nu)} D_3 d\theta.$$

Letting $K := E_2^{\top}\mathbf{\Delta}_{\exp(\nu)} D_3 - E_1^{\top}\mathbf{\Delta}_{\mu} D_2$, a calculation shows that the first and second differentials of θ with respect to λ are given by

$$\begin{cases} d\theta = K^{-1} E_1^{\top}\mathbf{\Delta}_{\mu} D_1, \\ \\ d^2\theta = K^{-1}\left(E_2^{\top}\mathbf{\Delta}_{\exp(\nu)}\left(D_3 d\theta\right)^{\circledast 2} - E_1^{\top}\mathbf{\Delta}_{\mu}\left(D_1 + D_2 d\theta\right)^{\circledast 2}\right), \end{cases}$$

where the notation $M^{\circledast 2}$ stands for $M \circledast M$.

Likelihood function and its derivatives (1/2)

Recall that I is the sample size, and that $\hat{f} := \hat{\mu}/I$ is the vector of the empirical frequencies of the observations. Letting $M(\lambda) := \log f = D_1\lambda + D_2\theta - \log(1_B^\top \exp(D_1\lambda + D_2\theta))$, where θ is implicitly a function of λ, the expression for the log-likelihood of the sample (rescaled by I) is $\ell_I(\lambda) := \hat{f}^\top M(\lambda)$, and the maximum likelihood estimator $\hat{\lambda}$ then maximizes $\ell_I(\lambda)$. Recalling the expression of f , we can deduce from the differential expressions that

$$\begin{cases} d\ell_I(\lambda) = \hat{f}^\top (D_1 + D_2 d\theta) - f^\top (D_1 + D_2 d\theta), \\ \\ d^2\ell_I(\lambda) = \hat{f}^\top D_2 d^2\theta - f^\top D_2 d^2\theta, \end{cases}$$

and we can deduce the Hessian of ℓ_I by reshaping $d^2\ell$ into an $I \times I$ matrix.

Theorem 6.9 (Corblet, Fox, and Galichon)

Assume that the observations $x_i y_i \in B$ are drawn from the population in an i.i.d. fashion from a data-generating process consistent with the model described in this section, and are collected in a sample of size I. Letting $\hat{\mu}_{xy}$ be the number of observations of type xy in the sample, the following convergence holds in distribution:

$$\sqrt{I}(\hat{\lambda} - \lambda) \xrightarrow{d} \mathcal{N}(0, H^{-1}),$$

where $H = D^2 \left(f^\top M \right) (\lambda)$

Exercises and Quick Recap Chapter 6

Short Exercises (1/2)

Exercise 6.4: Write down the equations that govern the evolution of (ntx,mty) in the model introduced in section 6.9. Show that the quantities $\sum_x n_{tx}$ and $\sum_y m_{ty}$ are independent of t.

Exercise 6.2 (Dynamic matching): The following code generates dynamic matching data. Estimate the dynamic matching model introduced in Section 6.9 based on that dataset.

```
# Data generation for the estimation of dynamic matching models
import numpy as np
from scipy import optimize, sparse as sp
from scipy.sparse import kron
```


Short Exercises (3/3)

```
np.random.seed(7)
X, Y, K = 15, 12, 3
B = X * Y + X + Y
beta, rel_tol, epsilon, max_it = 0.9, 1e-5, 0.05, 10000

phi_b_k = np.random.normal(size=(B, K))
muhat_b = np.random.uniform(size=B)

P_xprime_b = np.random.uniform(size=(X, B))
P_xprime_b = P_xprime_b / P_xprime_b.sum(axis=0)[None, :]

Q_yprime_b = np.random.uniform(size=(Y, B))
Q_yprime_b = Q_yprime_b / Q_yprime_b.sum(axis=0)[None, :]
```


Quick recap of Chapter 6 (1/2)

This chapter expanded upon previously studied models by incorporating a dynamic perspective, where decision-makers face sequential choices that influence both current payoffs and future opportunities. These models have a wide range of applications, highlighting the trade-offs between myopic short-term decisions and choices optimized for long-term benefits.

We began by analyzing a finite-horizon model with no heterogeneity, framed as a linear programming problem, to establish the foundational mechanics of the dynamic choice process. In Section 6.2, by introducing i.i.d. Gumbel shocks, we transformed the model to exhibit a logit structure, significantly facilitating estimation and analysis. We employed a fully vectorized formulation to streamline the computational and statistical aspects.

Quick recap of Chapter 6 (2/2)

We subsequently introduced a potential function, Z, demonstrating how the governing equations of the dynamic model emerge directly from its partial derivatives. This approach reveals that the model can be estimated using GLM techniques, highlighting its adaptability across various frameworks.

Sections 6.5 and 6.6 extended the analysis to infinite-horizon models, focusing on equilibrium properties and statistical inference. We discussed estimation strategies, including the nested fixed-point algorithm and MPEC, along with the asymptotic distribution of the MLE.

Section 6.8 addressed identification issues and presented an alternative estimation method based on the inversion of Conditional Choice Probabilities. Finally, we considered dynamic matching problems, demonstrating the flexibility of the framework in modeling situations where agents form pairs or coalitions over time.

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Chapter 6: Dynamic Discrete Choice

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Chapter 7. Constrained choice

***Discrete Choice Models*, ch. 7, pp. 191-217**

Content

Introduction to constrained choice

In this chapter, we explore constrained choice problems, where the limited availability of certain options necessitates rationing. Many familiar markets operate under these conditions—such as those for popular concert tickets or walk-in-only restaurants. In these settings, prices do not adjust to perfectly balance supply and demand; instead, other variables (such as wait times or allocation probabilities) must adjust. The primary distinction between this framework and the one studied in Chapter 5 is the absence of money: while Chapter 5 focused on equilibrium models with transferable utility (price adjustments), this chapter investigates markets with nontransferable utility.

Section 7.1. Basics

Section content

- Shadow price
- Constrained market share
- Excess capacity

Introduction to constrained choice problem

Agents pick their hospital $y \in [Y]$ or opt out ($y = 0$) base on the utility $\max_y \{U_y + \varepsilon_y, \varepsilon_0\}$. The demand for hospital y, $\boldsymbol{\pi}_y(U)$, may be more than the capacity constraint $\bar{\pi}_y$ of the hospital.

If hospital y is overdemanded, one may predict that there will be congestion. We shall assume that the utility will decrease by $\tau_y \geq 0$.

$$U_y \to U_y - \tau_y,$$

where the quantity τ_y is interpreted as the "shadow price" of the capacity constraint.

The market share of hospital y becomes $\pi_y(U - \tau)$, and the presence of τ ensures that the constraints $\pi_y(U - \tau) \leq \bar{\pi}_y$ are satisfied for each hospital $y \in [Y]$.

Problem system

These considerations lead us to investigate the solution $\tau \in \mathbb{R}^Y$ of the system

$$\begin{cases} \tau_y \geq 0 & \forall y \in [Y] \\ \bar{\pi}_y - \pi_y(U - \tau) \geq 0 & \forall y \in [Y] \\ \tau_y \left(\bar{\pi}_y - \pi_y(U - \tau) \right) = 0 & \forall y \in [Y]. \end{cases}$$

Theorem 7.1.1

Under assumptions 1.1 (continuity) and 1.2 (full support), the system shown above has a unique solution.

For all $y \in [Y]$, the shadow price τ_y and the overcapacity in option y, defined by

$$\rho_y := \bar{\pi}_y - \pi_y (U - \tau)$$

are non-negative and cannot be strictly positive at the same time. We introduce the Y-dimensional vector θ whose element θ_y has positive part equal to τ_y, and negative part equal to ρ_y. Formally,

$$\theta = \tau - \rho \quad \text{and} \quad \tau = \theta^+, \rho = \theta^-,$$

where for any vector $a \in \mathbb{R}^Y$, $a_y^+ = \max(a_y, 0)$, and $a_y^- = \max(-a_y, 0)$. Therefor, if τ is the solution to the system, we have $\pi (U - \theta^+) + \theta^- = \bar{\pi}$. Conversely, if θ satisfies the latter condition, then $\tau = \theta^+$ solves the system.

Proposition 7.1.1

Assume the distribution of ε satisfies assumptions 1.1 and 1.2. Let $\bar{\pi} \in \mathbb{R}_+^Y$ such that $\bar{\pi}_y > 0$ for all $y \in [Y]$. Then the solution τ to the system is given by $\tau = \theta^+$, where $\theta \in \mathbb{R}^Y$ is the solution to the set of equations

$$\boldsymbol{\pi}\left(U - \theta^+\right) + \theta^- = \bar{\pi}.$$

Proposition 7.1.2

Under assumptions 1.1 and 1.2, the map $\mathbf{q}$ defined by

$$\mathbf{q}\left(\theta\right) = \bar{\pi} - \theta^{-} - \boldsymbol{\pi}(U - \theta^{+})$$

is a M-function.

As $\pi = \nabla G$ and G^{*} is submodular as a result of Proposition 1.2.1, then π is an M-function, and, q is a Z-map.

Proof: We show that it is strongly non reversing. We assume $\theta \leq \theta'$ and $q(\theta) \geq q(\theta')$. Then $\theta^{-} \geq \theta'^{-}$, and $\bar{\pi} - \theta^{-} - \pi(U - \theta^{+}) \geq \bar{\pi} - \theta'^{-} - \pi(U - \theta'^{+})$.

Summing, we have $\pi(U - \theta^+) \leq \pi(U - \theta'^+)$, since π is an M-function, it implies $U - \theta^+ \leq U - \theta'^+$. Resulting, we have $\theta'^+ \leq \theta^+$, since $\theta^+ \leq \theta'^+$ by assumption, we deduce that $\theta^+ = \theta'^+$.

The inequality $q(\theta) \geq q(\theta')$ thus simplifies to $\theta^- \leq \theta'^-$. Hence, $\theta^- = \theta'^-$ as well, and therefore $\theta = \theta'$.

Proposition 7.1.3

Under assumptions 1.1 and 1.2, a solution θ to equations in proposition 7.1.1 exists and is unique.

Proof of Theorem 7.1.1 (Existence and Uniqueness of the System's Solution)

Proof of theorem 7.1.1: Combining the preceding results, Proposition 7.1.2 establishes that τ is a solution to the system of equations only if $\tau = \theta^+$, where θ is the solution to the equation presented in Section 7.1.1. Furthermore, Proposition 7.1.3 guarantees that this solution for θ exists and is unique. Consequently, we deduce that there is a unique solution to the overall system.
It leads to a few important definitions:

Definition 7.1.1

Under assumptions 1.1 and 1.2, let τ be the unique solution. Based on that vector τ, we define the following notions:

(i) the *shadow price map* $\boldsymbol{\tau}(U, \bar{\pi})$ which is τ itself $\boldsymbol{\tau}(U, \bar{\pi}) := \tau$,

(ii) the *constrained market share map* $\tilde{\boldsymbol{\pi}}(U, \bar{\pi})$ is defined as

$$\tilde{\boldsymbol{\pi}}(U, \bar{\pi}) := \boldsymbol{\pi}\left(U - \tau\right),$$

(iii) the *rejection map*, or *excess capacity map* $\boldsymbol{\rho}(U, \bar{\pi})$, defined as the remaining capacity

$$\boldsymbol{\rho}(U, \bar{\pi}) = \bar{\pi} - \boldsymbol{\pi}\left(U - \tau\right).$$

Example (continued): logit model

We normalize the utility of the default option to $U_0 = 0$, and we assume that there is no constraint on it, so $\bar{\pi}_0 = 1$. We have $\exp(U_y - \tau_y - u) \leq \bar{\pi}_y$, and $\tau_y > 0$ implies $\exp(U_y - \tau_y - u) = \bar{\pi}_y$, that is $\tau_y = U_y - u - \log \bar{\pi}_y$. Now $\tau_y = 0$ implies $\exp(U_y - u) \leq \bar{\pi}_y$, that is $U_y - u - \log \bar{\pi}_y \leq 0$. We have

$$\tilde{\boldsymbol{\pi}}_y(U, \bar{\pi}) = \min\left(e^{U_y - u}, \bar{\pi}_y\right),$$

where $u \in \mathbb{R}$ is obtained by the scalar equation

$$e^{-u} + \sum_{y \in [Y]} \min\left(e^{U_y - u}, \bar{\pi}_y\right) = 1.$$

Section 7.2. Constrained welfare and entropy

Section content

- Constrained welfare function

Introduction of constrained welfare function

We introduce $\tilde{G}(U, \bar{\pi})$ the constrained welfare function

$$\begin{aligned} \tilde{G}(U, \bar{\pi}) = \max_{\pi \in \mathbb{R}_+^Y} \quad & \left\{ \sum_{y \in [Y]} \pi_y U_y - G^*(\pi) \right\} \\ s.t. \quad & \pi \leq \bar{\pi}. \end{aligned}$$

The constrained market share map $\tilde{\boldsymbol{\pi}}(U, \bar{\pi})$ is the solution, and that $\boldsymbol{\tau}(U, \bar{\pi})$ arises as the vector of Lagrange multipliers. The Lagrangian problem is $\mathcal{L}(\tau, \pi) = \sum_y \bar{\pi}_y \tau_y + \sum_y \pi_y (U_y - \tau_y) - G^*(\pi)$, and an easy calculation shows that

$$\tilde{G}(U, \bar{\pi}) = \min_{\tau \geq 0} \left\{ \sum_{y \in [Y]} \bar{\pi}_y \tau_y + G(U - \tau) \right\}.$$

Introduction of constrained welfare function

Theorem 7.2.1

Under assumptions 1.1 and 1.2, we have:

$$\tilde{\boldsymbol{\pi}}(U, \bar{\pi}) = \nabla_U \tilde{G}(U, \bar{\pi}) \text{and} \boldsymbol{\tau}(U, \bar{\pi}) = \nabla_{\bar{\pi}} \tilde{G}(U, \bar{\pi}).$$

Example 7.1

If we simulate a sample ε_{iy} such that for each $i \in [I]$, $(\varepsilon_{iy})_y$ is randomly drawn from distribution $\mathcal{P}$, we recall that $-G_I^*(\pi)$ is $\max_{\lambda_{iy} \geq 0} \sum_{iy} \lambda_{iy} \varepsilon_{iy}$ subject to constraints $\sum_y \lambda_{iy} = 1/I$ and $\sum_i \lambda_{iy} = \pi_y$, and then

$$
\begin{aligned}
\tilde{G}_I(U, \bar{\pi}) = \max_{\pi \in \mathbb{R}_+^Y, \lambda \in \mathbb{R}_+^{I \times Y}} \quad & \sum_{y \in [Y]} \pi_y U_y + \sum_{iy \in [I \times Y]} \lambda_{iy} \varepsilon_{iy} \\
s.t. \quad & \sum_{y \in [Y]} \lambda_{iy} = 1/I \\
& \sum_{i \in [I]} \lambda_{iy} = \pi_y \\
& \pi \leq \bar{\pi},
\end{aligned}
$$

Example 7.1 (continued)

which we can slightly simplify as

$$\tilde{G}_I(U,\bar{\pi}) = \max_{\lambda\in\mathbb{R}_+^{I\times Y}} \sum_{iy\in[I\times Y]} \lambda_{iy}\left(U_y+\varepsilon_{iy}\right)$$

$$s.t. \quad \sum_{y\in[Y]} \lambda_{iy} = 1/I \quad [u_i]; \quad \sum_{i\in[I]} \lambda_{iy} \leq \bar{\pi} \quad [\tau_y \geq 0],$$

whose dual formulation is

$$\tilde{G}_I(U,\bar{\pi}) = \min_{u\in\mathbb{R}^I,\tau\in\mathbb{R}_+^Y} \sum_{i\in[I]} \frac{1}{I}u_i + \sum_{y\in[Y]} \bar{\pi}_y\tau_y$$

$$s.t. \quad u_i \geq U_y - \tau_y + \varepsilon_{iy} \quad [\lambda_{iy}\geq 0].$$

Section 7.3. Comparative statics

Section content

- Monotone comparative statics

Intuitively, we expect that when $\bar{\pi}$ decreases, the constraints are tighter, and therefore the shadow prices increase. The vector of shadow prices τ is given by $\tau = \theta^+$, where $\theta \in \mathbb{R}^Y$ is the unique solution to

$$\boldsymbol{\pi}(U - \theta^+) + \theta^- = \bar{\pi}.$$

Assume that $\bar{\pi} \leq \bar{\pi}'$ are such that $\boldsymbol{\pi}(U - \theta^+) + \theta^- = \bar{\pi}$ and $\boldsymbol{\pi}(U - \theta'^+) + \theta'^- = \bar{\pi}'$. By proposition 7.1.2 the map $\theta \mapsto \bar{\pi} - \boldsymbol{\pi}(U - \theta^+) - \theta^-$ follows that $\theta \geq \theta'$. As a result,

Theorem 7.3.1

Under assumptions 1.1 and 1.2:

(i) the map $\bar{\pi} \mapsto \boldsymbol{\tau}(U, \bar{\pi})$ is antitone, and

(ii) the map $\bar{\pi} \mapsto \boldsymbol{\rho}(U, \bar{\pi})$ is isotone.

Explanations

Theorem 7.3.1 is a manifestation of substitutability. In discrete choice models, agents form demand for only one option, and therefore, they must view options as substitutable.

The first part of theorem 7.3.1 shows that tighter constraints lead to at least higher shadow prices increase. This comes from substitutability: reduced supply of one option shifts demand to others, pushing them toward their capacity limits and increasing their shadow prices.

The second statement of theorem 7.3.1 says that rejected offers under tighter constraints remain rejected as constraints loosen because chosen offers continue to dominate them. Again, this depends on substitutability.

Section 7.4. A dynamic model of waiting lines

Section content

- A dynamic model of waiting lines

Equilibrium with waiting times (1/2)

We assume a flow $\bar{\pi}_y \geq 0$ of arrivals of options of type $y \in [Y]$ per period. If demand exceeds supply, individuals who arrive at time t queue, incurring a waiting time $\tau_y^t \geq 0$, which reduces utility to $U_y - \tau_y^t$. The market share will then be $\boldsymbol{\pi}_y \left(U - \tau^t\right)$. Letting N_y^t be the mass of the stock of units of options y at the start of period t, we have

$$N_y^{t+1} = N_y^t + \bar{\pi}_y - \boldsymbol{\pi}_y \left(U - \tau^t\right).$$

A stationary solution has $N_y^t = N_y \in \mathbb{R} \cup \{\infty\}$ and $\tau_y^t = \tau_y$ for all t, therefore

$$\begin{cases} N_y = +\infty \implies \boldsymbol{\pi}_y \left(U - \tau\right) \leq \bar{\pi}_y \\ N_y < +\infty \implies \boldsymbol{\pi}_y \left(U - \tau\right) = \bar{\pi}_y. \end{cases}$$

Equilibrium with waiting times (2/2)

If $\tau_y > 0$, it implies that there is a waiting line for option y, leading to $\pi_y(U - \tau) = \bar{\pi}_y$. Thus, τ solves the system and the stationary solution to the model is such that the stationary length of the waiting line τ_y for y is:

$$\tau = \boldsymbol{\tau}\left(U, \bar{\pi}\right).$$

Section 7.5. An empirical model of matching without prices

Section content

- Aggregate welfare functions
- Equilibrium matching without prices

Introduction to matching without prices

We consider a matching market with fixed wages.

A firm i of type x has utility $\alpha_{xy} + \varepsilon_{iy}$ from hiring a worker of type y, and ε_{i0} if unmatched, where α_{xy} is deterministic and exogenous, and $(\varepsilon_{iy})_y \sim \mathbf{P}_x$ is the vector drawn by individual firm i and includes the wage paid to the worker.

Similarly, a worker j of type y has utility $\gamma_{xy} + \eta_{jx}$ from working with a firm of type x, and η_{j0} if unmatched, where symmetrically, γ_{xy} is deterministic and exogenous, and $(\eta_{jx})_x \sim \mathbf{Q}_y$ is the vector drawn by the individual worker j and includes the wage receiving from the firm.

Our goal will be to investigate a notion of equilibrium matching when prices are absent.

Aggregate welfare functions (1/2)

Let U_{xy} and V_{xy} denote the systematic utilities of firms and workers, then the unconstrained welfare functions of firms and workers are defined as $G(U)$ and $H(V)$,

$$
\begin{aligned}
G(U) &= \sum_{x \in [X]} n_x \mathbb{E}_{\mathcal{P}_x} \left[\max_{y \in [Y]} \left\{ U_{xy} + \varepsilon_y, \varepsilon_0 \right\} \right] \\
H(V) &= \sum_{y \in [Y]} m_y \mathbb{E}_{\mathcal{Q}_y} \left[\max_{x \in [X]} \left\{ V_{xy} + \eta_x, \eta_0 \right\} \right]
\end{aligned}
$$

and the corresponding functions of entropy of choice as their convex conjugates are

$$
\begin{aligned}
G^*(\mu) &= \max_{U \in \mathbb{R}^{X \times Y}} \left\{ \sum_{xy \in [X \times Y]} \mu_{xy} U_{xy} - G(U) \right\} \\
H^*(\mu) &= \max_{V \in \mathbb{R}^{X \times Y}} \left\{ \sum_{xy \in [X \times Y]} \mu_{xy} V_{xy} - H(V) \right\}.
\end{aligned}
$$

Aggregate welfare functions (2/2)

Finally, we define the constrained welfare functions of firms and workers $\tilde{G}(U, \bar{\mu})$ and $\tilde{H}(V, \bar{\mu})$ under capacity constraints $\mu_{xy} \leq \bar{\mu}_{xy}$ as

$$\begin{aligned}
\tilde{G}(U, \bar{\mu}) &= \max_{\mu \in \mathbb{R}_+^{X \times Y}} \left\{ \sum_{xy \in [X \times Y]} \mu_{xy} U_{xy} - G^*(\mu) \quad s.t.\ \mu_{xy} \leq \bar{\mu}_{xy}\ \forall xy \right\} \\
\tilde{H}(V, \bar{\mu}) &= \max_{\mu \in \mathbb{R}_+^{X \times Y}} \left\{ \sum_{xy \in [X \times Y]} \mu_{xy} V_{xy} - H^*(\mu) \quad s.t.\ \mu_{xy} \leq \bar{\mu}_{xy}\ \forall xy \right\},
\end{aligned}$$

and we define:

$$\begin{aligned}
\tilde{\boldsymbol{\mu}}^\alpha(\bar{\mu}) = \frac{\partial \tilde{G}}{\partial U}(\alpha, \bar{\mu}) \quad &\text{and} \quad \tilde{\boldsymbol{\mu}}^\gamma(\bar{\mu}) = \frac{\partial \tilde{H}}{\partial V}(\gamma, \bar{\mu}), \\
\boldsymbol{\tau}^\alpha(\bar{\mu}) = \frac{\partial \tilde{G}}{\partial \bar{\mu}}(\alpha, \bar{\mu}) \quad &\text{and} \quad \boldsymbol{\tau}^\gamma(\bar{\mu}) = \frac{\partial \tilde{H}}{\partial \bar{\mu}}(\gamma, \bar{\mu}), \\
\boldsymbol{\rho}^\alpha(\bar{\mu}) = \bar{\mu} - \tilde{\boldsymbol{\mu}}^\alpha(\bar{\mu}) \quad &\text{and} \quad \boldsymbol{\rho}^\gamma(\bar{\mu}) = \bar{\mu} - \tilde{\boldsymbol{\mu}}^\gamma(\bar{\mu}).
\end{aligned}$$

Equilibrium matching without prices (1/3)

In the absence of market-clearing prices, some categories x or y of firms or workers will be overdemanded, leading to negative effects.

We model this via shadow prices $\tau_{xy}^{\alpha} \geq 0$ and $\tau_{xy}^{\gamma} \geq 0$, which reduce the utility from matches: firm i receives utility $\alpha_{xy} - \tau_{xy}^{\alpha} + \varepsilon_{iy}$, and worker j receives utility $\gamma_{xy} - \tau_{xy}^{\gamma} + \eta_{jx}$.

The inequality $\tau_{xy}^{\alpha} > 0$ means that some firms of type x compete to attract workers of type y; this means that x is the "long side" in the xy relation. This implies that y is the "short side," and therefore that $\tau_{xy}^{\gamma} = 0$. As a result, we have $\min\{\tau_{xy}^{\alpha}, \tau_{xy}^{\gamma}\} = 0$.

An equilibrium matching without prices is a vector (μ_{xy}) of matching patterns that are demanded by both sides, which means that $\mu = \nabla G(\alpha - \tau^{\alpha}) = \nabla H(\gamma - \tau^{\gamma})$.

Definition 7.5.1

A vector $\mu \in \mathbb{R}_{+}^{X \times Y}$ is an equilibrium matching without prices if there exist two vectors τ^{α} and τ^{γ} in $\mathbb{R}_{+}^{X \times Y}$ such that

$$\begin{cases} \min \left\{ \tau_{xy}^{\alpha}, \tau_{xy}^{\gamma} \right\} = 0 \quad \forall xy \in [X \times Y] \\ \mu = \nabla G \left(\alpha - \tau^{\alpha} \right) = \nabla H \left(\gamma - \tau^{\gamma} \right). \end{cases}$$

Theorem 7.5.1

Assume that the distribution of random utility shocks ε and η satisfies assumptions 1.1 and 1.2. Then an equilibrium matching without prices exists and is unique.

Sketch of proof: The proof of uniqueness is based on the fact that the map from $\mathbb{R}^{X \times Y}$ to itself, defined by $\tau \mapsto \nabla G(\alpha - \tau^{-}) - \nabla H(\gamma - \tau^{+})$, is an M-function. The proof of existence is constructive; one way to establish it is to demonstrate that the DA algorithm, described later, converges to an equilibrium matching without prices.

Example 7.2

The usual relations between the choice ratios and the systematic utilities are

$$\frac{\mu_{xy}}{\mu_{x0}} = \exp\left(\alpha_{xy} - \tau^{\alpha}_{xy}\right) \quad \text{and} \quad \frac{\mu_{xy}}{\mu_{0y}} = \exp\left(\alpha_{xy} - \tau^{\gamma}_{xy}\right),$$

where $\min\left\{\tau^{\alpha}_{xy}, \tau^{\gamma}_{xy}\right\} = 0$, which yields

$$\mu_{xy} = \min\left\{\mu_{x0}\exp\alpha_{xy}, \mu_{0y}\exp\gamma_{xy}\right\},$$

and therefore μ_{x0} and μ_{0y} are solutions to

$$\begin{cases} \mu_{x0} + \sum_{y\in[Y]} \min\left\{\mu_{x0}\exp\alpha_{xy}, \mu_{0y}\exp\gamma_{xy}\right\} = n_x \\ \mu_{0y} + \sum_{x\in[X]} \min\left\{\mu_{x0}\exp\alpha_{xy}, \mu_{0y}\exp\gamma_{xy}\right\} = m_y. \end{cases}$$

Example 7.2 (continued)

One can appreciate the constrast between this model of equilibrium matching without prices and the model of equilibrium matching with prices where the equilibrium relation is replaced by Choo and Siow's formula which, as we recall, is

$$\mu_{xy} = \sqrt{\mu_{x0}\mu_{0y} \exp\left(\alpha_{xy} + \gamma_{xy}\right)}.$$

Section 7.6. Deferred acceptance

Section content

- Deferred acceptance algorithm

Introduction

This section investigates Gale and Shapley's DA algorithm from a non-standard perspective. We follow Galichon, Hsieh, and Sylvestre by recasting the DA algorithm within the context of aggregate discrete choice analysis. This approach allows us to formulate a notion of aggregate stable matching with nontransferable utility. We assume that the distribution of ε and η should satisfy continuity and full support assumptions.

Algorithm description (1/2)

As there are n_x individual of type x and m_y individual of type y, the number of offers initially available $\mu_{xy}^{A,0} = \min(n_x, m_y)$.

If firms were to pick workers in an unconstrained manner, firm i would pick the worker y maximizing the firm's utility, which would make $\mu^{P,1} = \nabla G(\alpha)$.

If some workers types receive too many offers $\sum_x \mu_{xy}^{P,1} > m_y$, workers select their preferred ones based on welfare $\tilde{H}(\gamma, \mu^{P,1})$. The number of offers from firms of type x kept by workers of type y is

$$\mu_{xy}^{K,1} = \tilde{\boldsymbol{\mu}}^{\gamma}(\mu^{P,1})$$

and the rejected offers are removed, updating availability,

$$\mu_{xy}^{A,1} = \mu_{xy}^{A,0} - (\mu_{xy}^{P,1} - \mu_{xy}^{K,1}).$$

Algorithm description (2/2)

The second period of the algorithm resembles the first one, with the difference that the offers that were rejected in the first period can no longer be made at the second one. Therefore, in the second one, the number of offers $\mu_{xy}^{P,2}$ that can be made by firms of type x to workers of type y is constrained by the number of available offers $\mu_{xy}^{A,1}$.

The provisional welfare of workers at the second period is therefore $\tilde{G}(\alpha, \mu_{xy}^{A,1})$, and the number of offers made by type x firms to type y workers is

$$\mu_{xy}^{P,2} = \tilde{\mu}^{\alpha}(\mu^{A,1}),$$

to which workers will again respond by picking their favorite offers among those offered, just as in the previous step.

Algorithm 7.6.1 Deferred acceptance algorithm

Input: Systematic utilities α_{xy} and γ_{yx}, tolerance threshold $\epsilon > 0$ **Output**: Matching patterns (μ^K_{xy}) **Initialize**: $t = 0$ and $\mu^{A,0}_{xy} = \min(n_x, m_y)$ **repeat**

$$\begin{aligned} \text{Proposal phase:} \quad & \mu^{P,t+1}_{xy} = \tilde{\mu}^{\alpha}(\mu^{A,t}) \\ \text{Disposal phase:} \quad & \mu^{K,t+1}_{xy} = \tilde{\mu}^{\gamma}(\mu^{P,t+1}) \\ \text{Update phase:} \quad & \mu^{A,t+1}_{xy} = \mu^{A,t}_{xy} - (\mu^{P,t+1}_{xy} - \mu^{K,t+1}_{xy}) \quad \text{and} \quad t = t+1 \end{aligned}$$

until : $\max_{x,y}\{\mu^{P,t+1}_{xy} - \mu^{K,t+1}_{xy} < \epsilon$; **return**: $(\mu^{K,t+1}_{xy})$

The following slides provide a generic Python implementation of the DA algorithm, which requires us to specify the constrained choice functions $\tilde{\mu}^{\alpha}$ and $\tilde{\mu}^{\gamma}$.

Python code for the deferred acceptance algorithm (1/2)

```
# Before running, need to define the class ChoiceProblem and
# its child classes as above

import numpy as np, scipy.optimize as opt, gurobipy as grb
np.random.seed(7)
X,Y = 50,45
n_x = np.ones(X)
m_y = np.ones(Y)
alpha_x_y = np.random.uniform(size=(X,Y) )
gamma_x_y = np.random.uniform(size=(X,Y) )
muA_x_y = np.minimum(n_x[:,None],m_y[None,:] )
muP_x_y = np.zeros((X,Y))
muK_x_y = np.zeros((X,Y))
proposers = LogitChoiceProblem(n_x,alpha_x_y)
responders = LogitChoiceProblem(m_y,gamma_x_y.T)
# continued on the next slide =>>
```


Python code for the deferred acceptance algorithm (2/2)

```
# proposers = IndividualChoiceProblem(n_x,alpha_x_y)
# responders = IndividualChoiceProblem(m_y,gamma_x_y.T)
# eps_i_y = np.random.normal(size=(10,Y+1)); eta_i_x = np.random.normal(size=(10,X+1))
# proposers = SimulatedChoiceProblem(n_x, alpha_x_y,eps_i_y)
# responders = SimulatedChoiceProblem(m_y,gamma_x_y.T,eta_i_x)
maxiter=int(1e4); rel_tol = 1e-3
# Main loop:
for i in range(maxiter):
  for x in range(X):
    muP_x_y =  proposers.choice(constraint = muA_x_y )
  for y in range(Y):
    muK_x_y = responders.choice( constraint = muP_x_y.T).T
  muR_x_y = muP_x_y - muK_x_y
  if muR_x_y.sum() < rel_tol:
    break
  muA_x_y = muA_x_y - muR_x_y
print('Converged in '+str(i)+' step.')
print ('Number of matched individuals='+str(muK_x_y.sum()))
```

Theorem 7.6.1

Under assumptions 1.1 and 1.2, the quantities $\mu^{P,t}$ and $\mu^{K,t}$ defined in the deferred acceptance algorithm both converge to an equilibrium matching without prices.

Proof of the theorem

The proof of theorem 7.6.1 is decomposed in several claims. We let $\tau_{xy}^{\alpha,t} = \tau_{xy}^{\alpha}\left(\mu^{A,t}\right)$ and $\tau_{xy}^{\gamma,t} = \tau_{xy}^{\gamma}\left(\mu^{P,t}\right)$.

<u>Claim 1.</u> There cannot be less offers proposed at time $t+1$ than offers proposed and kept at time t, i.e. $\mu^{P,t+1} \geq \mu^{K,t}$.

<u>Claim 2.</u> $\tau_{xy}^{\alpha,t}$ is nondecreasing in t and $\tau_{xy}^{\gamma,t}$ is nonincreasing in t.

<u>Claim 3.</u> One has $\min\{\tau_{xy}^{\alpha,t}, \tau_{xy}^{\gamma,t}\} = 0$ for all txy.

<u>Claim 4.</u> Each of the sequences $\mu_{xy}^{A,t}$, $\mu_{xy}^{P,t}$, $\mu_{xy}^{K,t}$, $\tau_{xy}^{\alpha,t}$ and $\tau_{xy}^{\gamma,t}$ converges as $t \to +\infty$ for each xy.

<u>Claim 5.</u> Both $\mu^{P,t}$ and $\mu^{K,t}$ converge as $t \to +\infty$ to an equilibrium matching without prices.

Section 7.7. Reinterpretation à la Hatfield–Milgrom

Section content

- An reinterpretation of deferred acceptance algorithms

Period 1

We go through the steps of algorithm provided in the previous section and revisit them under the light of characterization provided by proposition 7.1.1.

Period 1, proposal: all the matches $\mu_{xy}^{A,0} = M_{xy} := \min\{n_x, m_y\}$ are initially available. As $\nabla G(\alpha) \leq M$, all firms make their best offers without constraints and the solution is

$$\tau^{\alpha,1} = 0 \quad \text{and} \quad \nabla G(\alpha) + R^{\alpha,1} = M.$$

Period 1, disposal: workers respond to firm proposals $\nabla G(\alpha)$. Solve for $\tau^{\gamma,1}$ and $\rho^{\gamma,1}$:

$$\min\{\tau^{\gamma,1}, \rho^{\gamma,1}\} = 0 \quad \text{and} \quad \nabla H\left(\gamma - \tau^{\gamma,1}\right) + \rho^{\gamma,1} = M - R^{\alpha,1}.$$

Period 1, update: update remaining offer availability after rejections to $\mu^{A,1} = M - \rho^{\gamma,1}$.

Period 2 to 3

Period 2, proposal: under capacity constraint determined by $\mu^{A,1} = M - \rho^{\gamma,1}$, we solve for a pair $(\tau^{\alpha,2}, R^{\alpha,2})$ that

$$\min\{\tau^{\alpha,2}, R^{\alpha,2}\} = 0 \quad \text{and} \quad \nabla G\left(\alpha - \tau^{\alpha,2}\right) + R^{\alpha,2} = M - \rho^{\gamma,1}.$$

Period 2, disposal: workers respond to constrained offers. Let $R^{\gamma,2} := \rho^{\gamma,1} + \rho^{\gamma,2}$ denote cumulative rejected offers. We solve for

$$\min\{\tau^{\gamma,2}, R^{\gamma,2}\} = 0 \quad \text{and} \quad \nabla H\left(\gamma - \tau^{\gamma,2}\right) + R^{\gamma,2} = M - R^{\alpha,2}.$$

Period 3, proposal: use updated availability, $\mu^{A,2} = M - R^{\gamma,2}$, to solve:

$$\min\{\tau^{\alpha,3}, R^{\alpha,3}\} = 0 \quad \text{and} \quad \nabla G\left(\alpha - \tau^{\alpha,3}\right) + R^{\alpha,3} = M - R^{\gamma,2}.$$

Period t, the general evolution

For the general evolution of the algorithm, we initially set $R^{\alpha} = M$ and $R\gamma = 0$.

Period t, proposal: look for $(\tau^{\alpha,t}, R^{\alpha,t})$ such that $\min\{\tau^{\alpha,t}, R^{\alpha,t}\} = 0$ and which is solution to

$$\nabla G\left(\alpha - \tau^{\alpha,t}\right) = M - R^{\alpha,t} - R^{\gamma,t-1},$$

Period t, disposal: look for $(\tau^{\gamma,t}, R^{\gamma,t})$ such that $\min\{\tau^{\gamma,t}, R^{\gamma,t}\} = 0$ and which is solution to

$$\nabla H\left(\gamma - \tau^{\gamma,t}\right) = M - R^{\alpha,t} - R^{\gamma,t}.$$

A further simplification can be rewritten as

$$\begin{cases} R^{\alpha,t} = \boldsymbol{\rho}^{\alpha}\left(M - R^{\gamma,t-1}\right), \\ R^{\gamma,t} = \boldsymbol{\rho}^{\gamma}\left(M - R^{\alpha,t}\right). \end{cases}$$

Theorem 7.7.1

Under assumptions 1.1 and 1.2, the quantities

$$\tilde{\boldsymbol{\mu}}^{\alpha}(M - R^{\gamma,t}) \quad \text{and} \quad \tilde{\boldsymbol{\mu}}^{\gamma}(M - R^{\alpha,t+1})$$

where $R^{\alpha,t}$ and $R^{\gamma,t+1}$ are defined in this algorithm respectively coincide with $\mu^{P,t+1}$ and $\mu^{K,t+1}$, the number of offers proposed and kept that are defined in last algorithm.

First, this algorithm consists of the repeated application of

$$F(R^{\alpha}, R^{\gamma}) = (\boldsymbol{\rho}^{\alpha}(M - R^{\gamma}), \boldsymbol{\rho}^{\gamma}(M - R^{\alpha}))$$

where $\boldsymbol{\rho}^{\alpha}$ and $\boldsymbol{\rho}^{\gamma}$ are isotone, therefore F is antitone and $F^2 = F \circ F$ is isotone. One may view the quantity R^{α} as a fixed point of $R^{\alpha} \mapsto \boldsymbol{\rho}^{\alpha}(M - \boldsymbol{\rho}^{\gamma}(M - R^{\alpha}))$, which is an isotone map.

Second, the equilibrium matching μ given by $\mu = \tilde{\boldsymbol{\mu}}^{\alpha}(M - R^{\gamma}) = \tilde{\boldsymbol{\mu}}^{\gamma}(M - R^{\alpha})$ satisfies

$$M_{xy} - \mu_{xy} = R^{\alpha}_{xy} + R^{\gamma}_{xy}.$$

Exercises and Quick Recap Chapter 7

Quick Exercises (1/3)

Exercise 7.1 (Equivalence with the classical notion of stability): This exercise aims to show that the notion of equilibrium matching without prices defined in Definition 7.5.1, when $\varepsilon_y = 0$, $\eta_x = 0$, $n_x = 1$, and $m_y = 1$ for all $x \in [X]$ and $y \in [Y]$ is a direct generalization of the notion of stable matching with nontransferable utility defined in Gale and Shapley's (1962) paper.

We assume that $\alpha_{x0} = 0$ and $\gamma_{0y} = 0$, and that strict preferences hold: $\alpha_{xy} = \alpha_{xy'}$ implies $y = y'$, and $\gamma_{xy} = \gamma_{x'y}$ implies $x = x'$. Let μ_{xy} for $(x, y) \in [X \times Y] \cup ([X] \times \{0\}) \cup (\{0\} \times [Y])$ be a vector with nonnegative entries. We say that μ is a stable matching in the sense of Gale and Shapley if:

Quick Exercises (2/3)

(i) $\mu_{xy} \in \{0,1\}$, with $\sum_{y\in[Y]} \mu_{xy} \leq 1$ and $\sum_{x\in[X]} \mu_{xy} \leq 1$ for all $x \in [X]$ and $y \in [Y]$.

(ii) There is no blocking pair. That is, defining $u_x^\mu = \sum_y \mu_{xy}\alpha_{xy}$ and $v_y^\mu = \sum_x \mu_{xy}\gamma_{xy}$ as the utilities that x and y obtain under matching μ, we have:

$u_x^\mu \leq \alpha_{xy}$ and $v_y^\mu \leq \gamma_{xy}$ jointly imply $\mu_{xy} = 1$,
$u_x^\mu \leq 0$ implies $\mu_{x0} = 0$,
$v_y^\mu \leq 0$ implies $\mu_{0y} = 0$.

Quick Exercises (3/3)

(i) Argue that when $\varepsilon_y = 0$ and $\eta_x = 0$ (and therefore when Assumptions 1.1 and 1.2 are not satisfied), the natural generalization of an equilibrium matching without prices in Definition 7.5.1 consists of the determination of vectors τ^{α}_{xy} and τ^{γ}_{xy} such that $\min\{\tau^{\alpha}_{xy}, \tau^{\gamma}_{xy}\} = 0$ for all $(x, y) \in [X \times Y]$, and such that:

$$\begin{cases} \mu_{xy} > 0 \implies y \in \arg\max_{y' \in [0:Y]}\{\alpha_{xy'} - \tau^{\alpha}_{xy'}\}, & \forall (x, y) \in [X \times (0 : Y)], \\ \mu_{xy} > 0 \implies x \in \arg\max_{x' \in [0:X]}\{\gamma_{x'y} - \tau^{\gamma}_{x'y}\}, & \forall (x, y) \in [(0 : X) \times Y] \end{cases} \tag{6}$$

(for convenience, one has set $\tau^{\alpha}_{x0} = 0$ and $\tau^{\gamma}_{0y} = 0$).

(ii) Let μ be a stable matching in the sense of Gale and Shapley. Show that one can find τ^{α}_{xy} and τ^{γ}_{xy} such that conditions (6) hold.

(iii) Conversely, suppose conditions (6) hold and $\mu_{xy} \in \{0, 1\}$, with $\sum_{y \in [Y]} \mu_{xy} \leq 1$ and $\sum_{x \in [X]} \mu_{xy} \leq 1$ for all $x \in [X]$ and $y \in [Y]$. Show that μ is a stable matching in the sense of Gale and Shapley.

Quick recap of Chapter 7 (1/2)

This chapter covered constrained choice models—discrete choice models governed by capacity constraints that limit the availability of certain options. We first introduced the foundational concepts and derived the relevant quantities for the logit model. The subsequent section explored these models from a welfare perspective, introducing the notion of constrained welfare. We demonstrated that, outside the logit framework, it is often necessary to resort to simulation methods, such as linear programming.

We also presented two key results in monotone comparative statics: the antitonicity of the shadow price map and the isotonicity of the rejection map, both with respect to the capacity vector. We explained how these properties serve as a manifestation of substitutability. In Section 7.4, waiting line models provided a concrete economic interpretation of these shadow prices.

Quick recap of Chapter 7 (2/2)

Section 7.5 adapted the concept of equilibrium matching to a nontransferable utility context, formalizing the notion of equilibrium matching without prices. We then translated the Gale and Shapley Deferred Acceptance (DA) algorithm into this discrete choice setting (Algorithm 7.6.1) and proved its convergence to a price-free equilibrium matching. Finally, in Section 7.7, we reinterpreted this algorithm as the iteration of an isotone fixed-point operator, following the approach of Hatfield and Milgrom.

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Chapter 7: Constrained Choice

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Chapter 8. Models with general heterogeneities

***Discrete Choice Models*, ch. 8, pp. 217-239**

Content

In the preceding chapters, we presented the general theory of random utility models without imposing specific distributional assumptions on the error term ε (aside from continuity and full support). Chapter 7 maintained this level of generality while introducing new constrained objects. Furthermore, Chapters 3 through 6 relied primarily on Gumbel heterogeneity and the logistic framework. The objective of this current chapter is to bridge this gap by extending the models of choice, allocation, and equilibrium in both static and dynamic settings to accommodate general unobserved heterogeneities.

Section 8.1 One-sided choice and minimax regret

Section content

- Choice regret
- Z function

Description of the problem

We consider Y options and a decision-maker of observable type $x \in [X]$, where the random utility for option y is $U_{xy} + \varepsilon_y$, with $\varepsilon \sim \mathcal{P}_x$ that satisfies assumptions 1.1 and 1.2. The systematic utility takes a parametric form $U_{xy} = \sum_k \phi_{xyk} \lambda_k$, whose vector form is $U = \Phi\lambda$. The welfare function is $G_x(U) = \mathbb{E}_{\mathcal{P}_x}[\max_y \{U_y + \varepsilon_y\}]$, and the corresponding entropy of choice is $G_x^*(\pi) = \max_U \{\pi^\top U - G_x(U)\}$.

Given sample $\hat{\mu}_{xy}$ (counts of type x choosing y), the natural generalization of the logistic regression is

$$\max_{\lambda \in \mathbb{R}^K} \{\hat{\mu}^\top \Phi\lambda - \sum_{x \in [X]} n_x G_x(\Phi\lambda)\}.$$

The first-order conditions are

$$\Phi^\top \hat{\mu} = \Phi^\top \mu^\lambda,$$

where $\mu_{xy}^\lambda = n_x \partial G_x / \partial U_y (\Phi\lambda)$ is the predicted number.

Interpretation as a minimax regret procedure (1/2)

For an observed choice $\tilde{y}$ by type x, the regret relative to alternative $y \in [Y]$ is

$$R^{\lambda}_{xy\tilde{y}}(\varepsilon) := (\Phi\lambda)_{xy} + \varepsilon_y - (\Phi\lambda)_{x\tilde{y}} - \varepsilon_{x\tilde{y}}.$$

The realization of the random vector ε and its joint distribution with $\tilde{Y}$ are unknown.

The maximum regret with respect to an alternative is defined as

$$\max_y R^{\lambda}_{xy\tilde{y}}(\varepsilon) = \max_y \{(\Phi\lambda)_{xy} + \varepsilon_{xy}\} - (\Phi\lambda)_{x\tilde{y}} - \varepsilon_{x\tilde{y}}$$

and the integrated maximum regret is given by

$$\sum_{x\in[X]} n_x \mathbb{E}_{\mathcal{P}_x}[\max_y R^{\lambda}_{xy\tilde{y}}(\varepsilon)] = \sum_{x\in[X]} n_x G_x(\Phi\lambda) - \hat{\mu}^\top \Phi\lambda - \sum_{x\in[X]} n_x \mathbb{E}_{\mathcal{P}_x}[\varepsilon_{x\tilde{y}}]. \tag{7}$$

.

By the virtues of theorem 1.3.3, this largest value of $\mathbb{E}_{\mathcal{P}_x}[\varepsilon_{x\tilde{y}}]$ is equal to $-G_x^*(\mu/n_x)$, and therefore we have the definition:

Definition 8.1.1 (Maximum choice regret)

The maximum choice regret associated with the parameter vector $\lambda \in \mathbb{R}^K$ is

$$R(\lambda) = \sum_{x \in [X]} n_x G_x(\Phi\lambda) - \hat{\mu}^\top \Phi\lambda + \sum_{x \in [X]} n_x G_x^* \left(\frac{\hat{\mu}}{n}\right),$$

and minimax-regret estimator $\hat{\lambda}$ is the parameter vector λ that achieves the minimum of $R(\lambda)$.

A reformulation (1/2)

Introducing the margining-out matrix $\mathbf{\Sigma}_Y = \mathbf{I}_X \otimes \mathbf{1}_Y^\top$, we recall that $n = \mathbf{\Sigma}_Y \mu$ is the vectorized form of $n_x = \sum_y \mu_{xy}$.

Letting the value of each state x be $u_x = G_x(\Phi\lambda)$, we recall that $\mathbf{\Sigma}_Y^\top u$ is the doubly-indexed vector of size XY whose xy term is u_x, and $u_x = G_x(\Phi\lambda)$ can be equivalently reexpressed as $G_x(\Phi\lambda - \mathbf{\Sigma}_Y^\top u) = 0$.

Then the original problem can therefore be re-expressed as

$$\begin{aligned} \max_{\lambda \in \mathbb{R}^K, u \in \mathbb{R}^X} \quad & \left\{ \hat{\mu}^\top \left(\Phi\lambda - \mathbf{\Sigma}_Y^\top u \right) \right\} \\ s.t. \quad & G_x(\Phi\lambda - \mathbf{\Sigma}_Y^\top u) = 0 \; \forall x \in [X]. \end{aligned}$$

Letting $\tilde{n}_x$ be the Lagrange multiplier, the saddlepoint formulation of the problem becomes

$$\max_{\lambda\in\mathbb{R}^K, u\in\mathbb{R}^X} \min_{\tilde{n}\in\mathbb{R}^X} \{\hat{\mu}^\top \left(\Phi\lambda - \boldsymbol{\Sigma}_Y^\top u\right) - G\left(\Phi\lambda - \boldsymbol{\Sigma}_Y^\top u; \tilde{n}\right)\} \text{ where } G\left(U;\tilde{n}\right) := \sum_{x\in[X]} \tilde{n}_x G_x(U)$$

We have by Karush-Kuhn-Tucker conditions with respect to u that

$$\left(\boldsymbol{\Sigma}_Y \hat{\mu}\right)_x = \tilde{n}_x \sum_{y\in[Y]} \partial G_x / \partial U_y (\Phi\lambda - u_x) = \tilde{n}_x,$$

and therefore at the optimum, $\tilde{n} = n$ and the problem boils down to

$$\begin{aligned} \max_{\lambda\in\mathbb{R}^K, u\in\mathbb{R}^X} \quad & \{\hat{\mu}^\top \left(\Phi\lambda - \boldsymbol{\Sigma}_Y^\top u\right)\} \\ s.t. \quad & G(\Phi\lambda - \boldsymbol{\Sigma}_Y^\top u; n) = 0. \end{aligned}$$

Introduction of the Z function

Another way to express the constraint $u_x = G_x(\Phi\lambda)$ is $1 = \exp(-u_x + G_x(\Phi\lambda))$, and letting N_x the corresponding Lagrange multiplier, the original problem can rewrite in a saddlepoint formulation as

$$\max_{\lambda \in \mathbb{R}^K, u \in \mathbb{R}^X} \min_{N \in \mathbb{R}^X} \{\hat{\mu}^\top \left(\Phi\lambda - \mathbf{\Sigma}_Y^\top u \right) + \sum_{x \in [X]} Z_x \left(\lambda, u_x; N_x \right),$$

$$\text{where } Z_x \left(\lambda, u_x; N_x \right) := N_x - N_x \exp(-u_x + G_x(\Phi\lambda))\}.$$

While the interest of the latter formulation is questionable in general, considering the particular case of the logit random utility reveals its usefulness in that case. Indeed, the logit structure implies $\exp(G_x(\Phi\lambda)) = \sum_y \exp((\Phi\lambda)_{xy})$, and therefore

$$Z_x \left(\lambda, u; N \right) = N_x - N_x \sum_{y \in [Y]} \exp((\Phi\lambda)_{xy} - u_x)\}.$$

Section 8.2 Empirical matching models

Section content

- The iterated proportional fitting procedure (IPFP)
- CCP approach

Basic settings

For a heterosexual marriage market, we assume that if a man of type $x \in [X]$ and a woman of type $y \in [Y]$ match, they get respective surpluses $U_{xy} + \varepsilon_y$ and $V_{xy} + \eta_x$, where the vector $\varepsilon \sim \mathcal{P}_x$ and the vector $\eta \sim \mathcal{Q}_y$, and $U_{xy} + V_{xy} = \phi_{xy}$. It is assumed that $\mathcal{P}_x$ and $\mathcal{Q}_y$ satisfy assumptions 1.1 and 1.2 for all x and y.

The expected indirect utility of a man of type x is

$$u_x = \mathbb{E}[\max_y \{U_{xy} + \varepsilon_y, \varepsilon_0\}] = G_x(U),$$

while the expected indirect utility of a woman of type y is

$$v_y = \mathbb{E}[\max_x \{V_{xy} + \eta_x, \eta_0\}] = H_y(V).$$

We let $G(U) = \sum_x n_x G_x(U)$ and $H(V) = \sum_y m_y H_y(V)$ be the total welfare of men and women, respectively, when their systematic utilities associated with matches of type xy are respectively U_{xy} and V_{xy}.

Equilibrium characterization (1/3)

It follows from the Daly-Zachary-Williams theorem that the number of men of type x who prefer women of type y is given by

$$\mu_{xy} = \partial G(U)/\partial U_{xy},$$

and the number of women of type y who prefer men of type x is given by

$$\mu_{xy} = \partial H(V)/\partial V_{xy}.$$

As $U + V = \phi$, equilibrium is therefore given by

$$\frac{\partial G}{\partial U_{xy}}(U) = \frac{\partial H}{\partial V_{xy}}(V).$$

Theorem 8.2.1

If $\mathcal{P}_x$ and $\mathcal{Q}_y$ satisfy assumptions 1.1 and 1.2, the equilibrium matching μ satisfies

$$\max_{\mu \in \mathbb{R}_+^{X \times Y}} \left\{ \sum_{xy \in [X \times Y]} \mu_{xy} \phi_{xy} - \mathcal{E}(\mu) \right\}$$

where $\mathcal{E}(\mu) := G^*(\mu) + H^*(\mu)$ defines the entropy of matching associated with the random utility structure, and the systematic utilities U_{xy} and V_{xy} arise as the solution to

$$\begin{aligned} \min_{U,V \in \mathbb{R}^{X \times Y}} \quad & \{G(U) + H(V)\} \\ s.t. \quad & U_{xy} + V_{xy} = \phi_{xy} \; \forall xy \in [X \times Y] . \end{aligned}$$

Proof: The equilibrium conditions highlighted on Slide 501 are the optimality conditions associated with the convex minimization problem in Theorem 8.2.1. The Lagrangian associated with this problem is: $\mathcal{L}(U, V; \mu) = \mu^\top \phi - \mu^\top U + G(U) - \mu^\top V + H(V)$. Therefore, the dual of problem from Slide 501 is given by: $\max_\mu \min_{U,V} \mathcal{L}(U, V; \mu)$. Evaluating the inner minimization yields: $\min_{U,V} \mathcal{L}(U, V; \mu) = \mu^\top \phi - G^*(\mu) - H^*(\mu)$, from which formulation of the maximization problem directly follows.

Example 1.2 (continued): logit model

In the case when $\mathcal{P}_x$ and $\mathcal{Q}_y$ are the distributions of i.i.d Gumbel random vectors, one has

$$\begin{cases} G^*(\mu) = \sum_{x\in[X]} \left(\sum_{y\in[Y]} \mu_{xy} \log \frac{\mu_{xy}}{n_x} + \mu_{x0} \log \frac{\mu_{x0}}{n_x} \right) \\ H^*(\mu) = \sum_{y\in[Y]} \left(\sum_{x\in[X]} \mu_{xy} \log \frac{\mu_{xy}}{m_y} + \mu_{0y} \log \frac{\mu_{0y}}{m_y} \right) \end{cases},$$

where μ_{x0} is defined from the other entries of (μ_{xy}) by $\mu_{x0} = n_x - \sum_{y\in[Y]} \mu_{xy}$, and similarly μ_{0y} is defined as $\mu_{0y} = m_y - \sum_{x\in[X]} \mu_{xy}$, which implies

$$\mathcal{E}(\mu) = \sum_{xy\in[X\times Y]} \mu_{xy} \ln \frac{\mu_{xy}^2}{n_x m_y} + \sum_{x\in[X]} \mu_{x0} \log \frac{\mu_{x0}}{n_x} + \sum_{y\in[Y]} \mu_{0y} \log \frac{\mu_{0y}}{m_y}$$

Example 1.2 (continued): logit model

so that the problem becomes

$$\max_{\mu \in \mathbb{R}_+^{X \times Y}} \left\{ \begin{array}{l} \sum_{xy \in [X \times Y]} \mu_{xy} \phi_{xy} - \sum_{xy \in [X \times Y]} \mu_{xy} \ln \frac{\mu_{xy}^2}{n_x m_y} \\ - \sum_{x \in [X]} (n_x - \sum_{y \in [Y]} \mu_{xy}) \log \left(1 - \sum_{y \in [Y]} \frac{\mu_{xy}}{n_x} \right) \\ - \sum_{y \in [Y]} (m_x - \sum_{x \in [X]} \mu_{xy}) \log \left(1 - \sum_{x \in [X]} \frac{\mu_{xy}}{m_y} \right) \end{array} \right\}.$$

We know from section 6.4 that there is a much simpler formulation of the TU-logit matching model, which the following discussion will show how to obtain.

Define unmatched masses as $\mu_{x0} = n_x - \sum_{y\in[Y]} \mu_{xy}$ and $\mu_{0y} = m_y - \sum_{x\in[X]} \mu_{xy}$, and introduce a convex function E defined on $\mathbb{R}_+^{XY+X+Y}$ and whose argument is the vector $(\mu_{xy}, \mu_{x0}, \mu_{0y})$ which is such that

$$\mathcal{E}(\mu) = E((\mu_{xy}), (n_x - \sum_{y\in[Y]} \mu_{xy}), (m_y - \sum_{x\in[X]} \mu_{xy})).$$

We shall now regard μ as the vector obtained by the concatenation of $vec_R(\mu_{xy})$, (μ_{x0}), and (μ_{0y}), Φ as the vector obtained by the concatenation of ϕ_{xy} and the zero vector of size $X + Y$ (given that $\phi_{x0} = \phi_{0y} = 0$), and let

$$\begin{cases} \mathbf{\Sigma}_X := \begin{pmatrix} \mathbf{I}_X \otimes \mathbf{1}_Y^\top & \mathbf{I}_X & \mathbf{0}_{X\times Y} \end{pmatrix} \\ \mathbf{\Sigma}_Y := \begin{pmatrix} \mathbf{1}_X^\top \otimes \mathbf{I}_Y & \mathbf{0}_{Y\times X} & \mathbf{I}_Y \end{pmatrix}, \end{cases}$$

so that the x-margin constraints $\sum_{y\in[Y]} \mu_{xy} + \mu_{x0} = n_x$ read $\mathbf{\Sigma}_X \mu = n$ and the y-margin constraints and $\sum_{x\in[X]} \mu_{xy} + \mu_{0y} = m_y$ read $\mathbf{\Sigma}_Y \mu = m$.

The equivalent reformulation (2/3)

We rewrite the problem as

$$\begin{aligned}\max_{\mu \in \mathbb{R}_+^{XY+X+Y}} \quad & \left\{\mu^\top \Phi - E\left(\mu\right)\right\} \\ s.t. \quad & \mathbf{\Sigma}_X \mu = n \\ & \mathbf{\Sigma}_Y \mu = m\end{aligned}$$

and we now need to introduce u_x and v_y as the Lagrange multipliers of the constraints, the Lagrangian formulation of the latter is

$$\max_{\mu \in \mathbb{R}_+^{XY+X+Y}} \min_{u \in \mathbb{R}^X, v \in \mathbb{R}^Y} \left\{n^\top u + m^\top v + \mu^\top \left(\Phi - \mathbf{\Sigma}_X^\top u - \mathbf{\Sigma}_X^\top v\right) - E\left(\mu\right)\right\}.$$

By strong duality, we have the following result:

Theorem 8.2.2

Under the assumptions of theorem 8.2.1, an equivalent formulation of problem is

$$\min_{(u,v)\in\mathbb{R}^{X+Y}} \{n^\top u + m^\top v + E^* \left(\Phi - \mathbf{\Sigma}_X^\top u - \mathbf{\Sigma}_X^\top v\right)\},$$

where E^*, defined on $\mathbb{R}^{XY+X+Y}$ is the Legendre-Fenchel transform of E, expressed as $E^*(a) = \max_\mu\{\mu^\top a - E(\mu) : \mu \in \mathbb{R}_+^{XY+X+Y}\}$.

Example 1.2 (continued): logit model

We can take $E(\mu) = \sum_{xy} \mu_{xy} \ln(\mu_{xy}^2/(n_x m_y)) + \sum_x \mu_{x0} \ln(\mu_{x0}/n_x) + \sum_y \mu_{0y} \ln(\mu_{0y}/m_y) - 2\sum_{xy} \mu_{xy} - \sum_x \mu_{x0} - \sum_y \mu_{0y} + \sum_x n_x + \sum_y m_y$, and

$$E^*(a) = 2 \sum_{xy \in [X \times Y]} \sqrt{n_x m_y} e^{\frac{a_{xy}}{2}} + \sum_{x \in [X]} n_x e^{a_{x0}} + \sum_{y \in [Y]} m_y e^{a_{0y}} - \sum_{x \in [X]} n_x - \sum_{y \in [Y]} m_y,$$

so that $E^*\left(\Phi - \mathbf{\Sigma}_X^\top u - \mathbf{\Sigma}_X^\top v\right)$ is precisely the function Z introduced in section 6.4. This leads to the expression obtained in chapter 5, that is

$$\min_{(u,v) \in \mathbb{R}^{X+Y}} \left\{ \begin{array}{c} \sum_{x \in [X]} n_x u_x + 2\sum_{xy \in [X \times Y]} \sqrt{n_x m_y} \exp\left(\frac{\phi_{xy} - u_x - v_y}{2}\right) \\ + \sum_{y \in [Y]} m_y v_y + \sum_{x \in [X]} n_x \exp(-u_x) + \sum_{y \in [Y]} m_y \exp(-v_y) \end{array} \right\}.$$

Equilibrium computation (1/2)

We define

$$F(u,v) = n^\top u + m^\top v + E^* \left(\Phi - \mathbf{\Sigma}_X^\top u - \mathbf{\Sigma}_X^\top v \right)$$

and describe a generalized IPFP algorithm that alternates maximization over u and v.

The coordinate gradient descent may not have computational benefits in the general case. Therefore, one may prefer using gradient descent in order to compute the optimal (u,v) pair that minimizes F, which would lead to iterations of the form

$$\begin{cases} \mu^t = \nabla E \left(\Phi - \mathbf{\Sigma}_X^\top u - \mathbf{\Sigma}_Y^\top v \right), \\ u_x^{t+1} = u_x^t - \epsilon \left(n - \left(\mathbf{\Sigma}_X^\top \mu^t \right) \right)_x, \; x \in [X] \\ v_y^{t+1} = v_y^t - \epsilon \left(m - \left(\mathbf{\Sigma}_Y^\top \mu^t \right) \right)_y . \; y \in [Y] \end{cases}$$

Equilibrium computation (2/2)

The computation of E itself may be complex in some cases, and the cases when E is available in closed-form are relatively rares. We can base our computational approach directly on the initial formulation of the dual shown previously in this section, and we can use the following gradient descent method

$$U_{xy}^{t+1} = U_{xy}^{t} - \epsilon \Big(\frac{\partial G}{\partial U_{xy}} \left(U^t \right) - \frac{\partial H}{\partial V_{xy}} \left(\Phi - U^t \right) \Big), \; xy \in [X \times Y], \tag{8}$$

which can be interpreted as a tâtonnement procedure.

Algorithm 8.2.1: Generalized IPFP algorithm

Input: F, v_y^0

Parameter: η (tolerance)

while $\max_{y\in[Y]} \|v_y^t - v_y^{t-1}\| \geq \eta$ **do**

$$u_x^t = \arg\max_u F(u, v^{t-1})$$

$$v_y^t = \arg\max_v F(u^t, v)$$

end

Return: u^t, v^t

The following slides provide a Python implementation of this gradient descent procedure.

Python code for probit matching model via gradient descent (1/2)

```
import numpy as np
from statistics import NormalDist
phi = np.vectorize(NormalDist(mu=0, sigma=1).cdf)
phi_inv = np.vectorize(NormalDist().inv_cdf)
def ghk(L,z_j,x_i_j):
    ... # see above
def piprobit_y(U_y,sigma_eps,x_i_j):
    ... # see above
X,Y = 5,3
np.random.seed(7)
phi_x_y = np.random.uniform(size=(X,Y))
n_x = np.ones(X)/X # no unassigned individuals, total masses conicide
m_y = np.ones(Y)/Y
I=J= 1000; maxiter= 1000
tol = 1e-3
eps = 0.1
unif1_i_j = np.random.uniform(size=(I,Y-1))
unif2_i_j = np.random.uniform(size=(J,X-1))
# continued on the next slide =>>
```


Python code for probit matching model via gradient descent (2/2)

```
# =>> continued from the previous slide
# Computing:
def muprobit (U_x_y, n_x,Sigma ,unif_i_j):
    mu_x_y = np.zeros(U_x_y.shape)
    for x in range(U_x_y.shape[0]):
        mu_x_y[x,:] = n_x[x]* piprobit_y(U_x_y[x,:],Sigma,unif_i_j)
    return mu_x_y
nablaG = lambda U_x_y:  muprobit(U_x_y,n_x,np.eye(Y),unif1_i_j)
nablaH = lambda V_x_y:  muprobit(V_x_y.T,m_y, np.eye(X), unif2_i_j).T
#
w_x_y = np.zeros((X,Y))
for step  in range(maxiter):
    delta_x_y = nablaG(w_x_y) - nablaH(phi_x_y - w_x_y)
    if np.abs(delta_x_y).max()<tol:
        break
    w_x_y -= eps * delta_x_y
# Displaying results
print('Converged in ',step,' steps')
print('w_x_y=',w_x_y)
```


We assume the surplus function $\Phi_{xy}^{\lambda} = \sum_k \phi_{xyk}\lambda_k$, whose vectorized manner is $\Phi^{\lambda} = \mathbf{\Phi}\lambda$. Letting $\hat{\mu}_{xy}$ be the number of matches of type xy, we can base an estimator of λ on the following convex optimization problem

$$\begin{array}{ll} \max\limits_{\lambda,U,V} & \left\{ \hat{\mu}^{\top}\mathbf{\Phi}\lambda - G(U) - H(V) \right\} \\ s.t. & U + V = \mathbf{\Phi}\lambda. \end{array}$$

Denoting μ the vector of Lagrange multipliers, we have:

$$\nabla G(U) = \mu = \nabla H(V),$$

while the optimality conditions with respect to λ yield

$$\mathbf{\Phi}^{\top}\mu = \mathbf{\Phi}^{\top}\hat{\mu},$$

or more explicitly, $\sum_{xy} \mu_{xy}\phi_{xyk} = \sum_{xy} \hat{\mu}_{xy}\phi_{xyk}$.

Letting N be the total number of households, and $\hat{f}_{xy} = \hat{\mu}_{xy}/N$ be the empirical frequency of a match of type xy for $xy \in [X \times Y + X \times 0 + 0 \times Y]$, we can rescale the matching estimation problem by a factor N to obtain that the estimator $\hat{\lambda}$ of λ is given by

$$\max_{\lambda \in \mathbb{R}^K} \{\hat{f}^\top \mathbf{\Phi} \lambda - W(\lambda, \hat{f})\},$$

where

$$W(\lambda, \mu) := \min_{U \in \mathbb{R}^{X \times Y}} \{G(U; \mathbf{\Sigma}_X \mu) + H(\mathbf{\Phi}\lambda - U; \mathbf{\Sigma}_Y \mu)\}$$

and the notation $G(U; n)$ indicates the dependence in n, that is $G(U; n) = \sum_x n_x G_x(U)$, and similarly for $H(V; m) = \sum_y m_y H_y(V)$.

Inference (2/3)

Assuming that the true value of λ is $\lambda^{\star}$, and that in the population, the frequency of a match of type xy is $f^{\star}_{xy}$, we have $\varphi(\hat{\lambda}, \hat{f}) = 0$ and $\varphi\left(\lambda^{\star}, f^{\star}\right)$ where $\varphi(\lambda, f) = \nabla_{\lambda} W\left(\lambda, f\right) - \mathbf{\Phi}^{\top} f$. Assuming that the observations are drawn from the population in an i.i.d. manner, we have the convergence in distribution of the empirical frequency toward the population frequency

$$\sqrt{N}\left(\hat{f} - f^{\star}\right) \xrightarrow{d} N\left(0, V\right),$$

where $V = diag\left(f\right) - ff^{\top}$. By linearization, we have therefore the first-order approximation $D_{\lambda}\varphi(\lambda^{\star}, f^{\star})(\hat{\lambda} - \lambda^{\star}) + D_{f}\varphi(\lambda^{\star}, f^{\star})(\hat{f} - f^{\star}) + o_{P}(1/\sqrt{N})$.

Theorem 8.2.3

Under assumptions 1.1 and 1.2, the estimator $\hat{\lambda}$ that solves the estimation problem is consistent, and the following convergence holds in distribution

$$\sqrt{N}\left(\hat{\lambda}-\lambda^{\star}\right)\xrightarrow{d}N\left(0,H^{-1}BVBH^{-1}\right),$$

where $V=diag\left(f\right)-ff^{\top}$, $B=D_{f}\varphi\left(\lambda^{\star},f^{\star}\right)=D_{\lambda f}^{2}W(\lambda^{\star},f^{\star})-\mathbf{\Phi}$, and $H=D_{\lambda}\varphi(\lambda^{\star},f^{\star})=D_{\lambda\lambda}^{2}W(\lambda^{\star},f^{\star})$. These quantities can be consistently estimated themselves by plugging in $\hat{\lambda}$ and $\hat{f}$ instead of $\lambda^{\star}$ and $f^{\star}$.

Simulation (1/3)

When G and H cannot be obtained in closed-form, we can resort to simulation. Let I be the number of draws ε_{ixy} and η_{ixy} from the distributions $\mathcal{P}_x$ and $\mathcal{Q}_y$ of the random utilities on both sides of the market for each cell. The idea is to replace $G(U) = \sum_x n_x G_x(U)$ by $G_I(U)$ defined as

$$G_I(U) = \sum_{x \in [X]} \frac{n_x}{I} \sum_{i \in [I]} \max_y \left\{ U_{xy} + \varepsilon_{ixy}, \varepsilon_{ix0} \right\}$$

and replace $H(V)$ by $H_I(V)$ that has a similar expression.

Simulation (2/3)

The estimation problem then becomes the following linear programming problem

$$
\begin{aligned}
\max_{\lambda,u,v,U,V} \quad & \left\{ \hat{\mu}^\top \mathbf{\Phi} \lambda - \sum_{ix \in [I \times X]} \frac{n_x}{I} u_{ix} - \sum_{iy \in [I \times Y]} \frac{m_y}{I} v_{iy} \right\} \\
s.t. \quad & U_{xy} + V_{xy} \geq (\mathbf{\Phi}\lambda)_{xy} \ \forall xy \in [X \times Y] \\
& u_{ix} \geq U_{xy} + \varepsilon_{ixy} \ \forall ixy \in [I \times X \times Y] \\
& u_{ix} \geq \varepsilon_{ix0} \ \forall ix \in [I \times X] \\
& v_{iy} \geq V_{xy} + \eta_{ixy} \ \forall ixy \in [I \times X \times Y] \\
& v_{iy} \geq \eta_{i0y} \ \forall iy \in [I \times Y],
\end{aligned}
$$

which vectorizes into

$$
\max_{\theta} \ b^\top \theta \quad s.t. \ C\theta \geq d.
$$

Simulation (3/3)

The optimization variables are concatenated into $\theta = (\lambda^\top, u^\top, v^\top, U^\top, V^\top)^\top$, the objective function is determined by $b = (\hat{\mu}^\top \mathbf{\Phi}, -\mathbf{1}_I^\top \otimes n^\top / I, -\mathbf{1}_I^\top \otimes m^\top / I, \mathbf{0}_{2XY}^\top)^\top$, the right hand-side of the constraint is given by $d = (\mathbf{0}_{XY}^\top, \varepsilon^\top, \varepsilon_0^\top, \eta^\top, \eta_0^\top)^\top$, and the constraint matrix by

$$C = \begin{pmatrix} -\mathbf{\Phi} & . & . & \mathbf{I}_{XY} & \mathbf{I}_{XY} \\ . & \mathbf{I}_{IX} \otimes \mathbf{1}_Y & . & -\mathbf{1}_I \otimes \mathbf{I}_{XY} & . \\ . & \mathbf{I}_{IX} & . & . & . \\ . & . & \mathbf{I}_I \otimes \mathbf{1}_X \otimes \mathbf{I}_Y & . & -\mathbf{1}_I \otimes \mathbf{I}_{XY} \\ . & . & \mathbf{I}_{IY} & . & . \end{pmatrix},$$

where the dots stand for zero block matrices.

CCP approach

Given the observation of $\hat{\mu}_{xy}$ for $xy \in [X \times Y] \cup [X \times 0] \cup [0 \times Y]$, we may compute

$$\hat{\phi}_{xy} = \frac{\partial G_x^*}{\partial \pi_y} \left((\frac{\hat{\mu}_{xy}}{n_x})_{y \in [Y]} \right) + \frac{\partial H_y^*}{\partial \pi_x} \left((\frac{\hat{\mu}_{xy}}{m_y})_{x \in [X]} \right),$$

and next, run generalized least squares of $\hat{\phi}$ so obtained on the basis functions that are given by the rows of $\mathbf{\Phi}$, looking for the parameter vector λ providing the minimal distance between the predicted surplus $\mathbf{\Phi}\lambda$ and the inferred surplus $\hat{\phi}$. We get

$$\min_{\lambda} \left| \hat{\phi} - \mathbf{\Phi}\lambda \right|_{\mathbf{W}},$$

where as before, $|z|^2_{\mathbf{W}} = z^\top \mathbf{W} z$.

Section 8.3 Coalition formation

Section content

- Coalitional equilibrium

Description of the problem

Individuals belong to types $z \in [Z]$, with a mass q_z for each type. A coalition indexed by $b \in [B]$ contains Σ_{zb} agents of type $z \in [Z]$, where Σ_{zb} is a nonnegative integer. The set of allowed coalitions and assume B is finite. We recall that the size of the coalition b is $s_b = (\Sigma^\top 1_Z)_b.s_b = 1_Z^\top b$. For $b \in B$, the quantity $\mu_b \in \mathbb{R}_+$ is the mass of coalitions of type b that are formed, and the market clearing condition is

$$q_z = \sum_{b \in [B]} \Sigma_{zb} \mu_b = (\Sigma \mu)_z, \tag{9}$$

An individual i of type z in coalition b derives utility $U_{zb} + \varepsilon_{ib}$, where the random vector (ε_{ib}) follows a distribution $\mathcal{P}_z$, and the total amount of utility is

$$\sum_{z \in [Z]} b_z U_{zb} = \phi_b.$$

Equilibrium definition (1/2)

We let $G_z(U) = \mathbb{E}_{\mathcal{P}_z}[\max_b\{U_{zb} + \varepsilon_{ib}\}]$ and $G(U) = \sum_z q_z G_z(U)$ be the welfare of agents of type z and aggregate welfare respectively. G_z^*, denoting the Legendre-Fenchel transform of G_z, is the entropy of choice associated with G, and G^* is given by $G^*(\mu) = \sum_z q_z G_z^*(\mu/q_z)$. The mass of agents of type z who choose coalition b is given by $\partial G(U)/\partial U_{zb}$, which should coincide with $\Sigma_{zb}\mu_b$.

Definition 8.3.1 (Coalitional equilibrium)

A coalitional equilibrium outcome is a vector of numbers $(\mu_b, U_{zb}) \in \mathbb{R}^B \times \mathbb{R}^{ZB}$ such that

$$\begin{cases} \sum_{z\in[Z]} \Sigma_{zb} U_{zb} = \phi_b, \ \forall b \in [B], \\ \Sigma_{zb}\mu_b = \frac{\partial G(U)}{\partial U_{zb}}, \ \forall zb \in [Z \times B] \,. \end{cases}$$

Note: We do not explicitly impose the market-clearing condition in our definition of coalitional equilibrium, as it is implicitly satisfied by the relation: $\sum_b b_z \mu_b = \sum_b \frac{\partial G(U)}{\partial U_{zb}} = q_z$.

Theorem 8.3.1

A vector $(\mu_b, U_{zb}) \in \mathbb{R}^B \times \mathbb{R}^{ZB}$ is a coalitional equilibrium if and only if the vector of mass (μ_b) is a solution to the primal problem

$$\max_{\mu_b \in \mathbb{R}^B_+} \left\{ \sum_{b \in [B]} \mu_b \phi_b - G^* \left(\left(\Sigma_{zb} \mu_b \right)_{zb \in [Z \times B]} \right) \right\} \tag{10}$$

and the payoff vector U is solution to the corresponding dual

$$\begin{aligned} \min_{U \in \mathbb{R}^{ZB}} \quad & G(U) \\ s.t. \quad & \sum_{z \in [Z]} \Sigma_{zb} U_{zb} = \phi_b. \end{aligned}$$

Equilibrium computation (2/2)

Theorem 8.3.1 (continued)

Moreover, the value of the primal and the dual coincide, and the solution vector to the primal problem is the vector of Lagrange multipliers for the dual.

Note that if μ_b is the Lagrange multiplier in the dual problem constraint, the optimality condition with respect to U_{zb} yield the equilibrium relation

$$\frac{\partial G(U)}{\partial U_{zb}} = \Sigma_{zb}\mu_b.$$

We assume that our data consists of a vector $\hat{\mu}_b$ that counts the number of times a coalition of type b is formed. The surplus function is parameterized linearly by $\lambda \in \mathbb{R}^K$, with $\phi_b^\lambda = (\mathbf{\Phi}\lambda)_b$. Based on earlier results, the estimator $\hat{\lambda}$ is the solution of

$$\max_{\lambda \in \mathbf{R}^K, U \in \mathbb{R}_+^{\mathcal{B}}} \quad \left\{ \hat{\mu}^\top \breve{} \lambda - G(U) \right\} \tag{11}$$
$$s.t. \quad \sum_{z \in [Z]} b_z U_{zb} = (\breve{}\lambda)_b.$$

Denoting μ_b the Lagrange multiplier, the optimality conditions with respect to U yield

$$\frac{\partial G(U)}{\partial U_{zb}} = \Sigma_{zb} \mu_b,$$

and the optimality conditions with respect to λ give

$$\hat{\mu}^\top \mathbf{\Phi} = \mu^\top \mathbf{\Phi}.$$

Theorem 8.3.2

The solution of the estimation problem provides a coalitional equilibrium matching that has the same K moments of the basis function provided by the rows of the matrix $\mathbf{\Phi}$ than the observed matching $\hat{\mu}$.

Theorem 8.2.3 readily extends to the general case of coalitional equilibria to provide the consistency and the asymptotic normality of the estimator $\hat{\lambda}$.

Section 8.4 Dynamic discrete choice

Section content

- Finite horizon
- Infinite horizon
- CCP inversion approach

Basic settings

We extend the framework to random utilities that are not necessarily logit. Define $\mathcal{P}_{tx}$ as the Y-dimensional random vector of the utility shocks ε_{ity} that an agent i of type x associates to various options y at time t is Gumbel and i.i.d. across y.

In chapter 6 starting from section 6.2, we assumed that $\mathcal{P}_{tx}$ was the distribution of i.i.d. Gumbel random vectors for all t and x; however, in this section, we shall relax that assumption and solely assume that $\mathcal{P}_{tx}$ satisfies assumptions 1.1 and 1.2 for all t and x. Throughout the section, G_{tx} shall stand for $G_{\mathcal{P}_{tx}}$.

Finite horizon, computation

We reconsider the model from section 6.2. Assume that the policy variable μ is subject to capacity constrained that can be expressed as $A\mu \leq c$.

The Bellman equation is $u_{tx} = G_{tx}((\phi - A^\top \tau)_{txy} + \sum_{x'} \mathbf{P}_{x',xy} u_{(t+1)x'})_y)$. Recalling the definition of $\mathbf{J}_T$ as the $T \times T$ lower shift matrix with ones on the subdiagonal and zeroes elsewhere, i.e. $(\mathbf{J}_T)_{tt'} = 1\{t = t' + 1\}$, this constraint can be written in a vectorized way as $u_{tx} = G_{tx}((\phi - A^\top \tau + \mathbf{J}_T^\top \otimes \mathbf{P}^\top u)_{txy, y \in [Y]})$, and then

$$\min_{u \in \mathbb{R}^{TX}, \tau \in \mathbb{R}^L} \quad \sum_{x \in [X]} N_x u_x^1 + \sum_{l \in [L]} c_l \tau_l$$
$$s.t. \quad G_{tx}\left(\left(\phi - A^\top \tau + \left(\mathbf{J}_T^\top \otimes \mathbf{P}^\top - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y^\top\right) u\right)_{txy, y \in [Y]}\right) = 0 \;\forall tx \in [T \times X],$$

where we recall the usual definition $\mathbf{\Sigma}_Y = \mathbf{I}_X^\top \otimes 1_Y$.

We assume no capacity constraint, and use a linear surplus parameterization $\phi = \Phi\lambda$, where λ is K-dimensional.

$$\max_{u\in\mathbb{R}^{TX},\lambda\in\mathbb{R}^K} \quad \hat{\mu}^\top \left(\Phi\lambda + \left(\mathbf{J}_T^\top \otimes \mathbf{P}^\top - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y^\top\right) u\right)$$

$$s.t. \quad G_{tx}\left(\left(\Phi\lambda + \left(\mathbf{J}_T^\top \otimes \mathbf{P}^\top - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y^\top\right) u\right)_{txy,y\in[Y]}\right) = 0 \; \forall tx \in [T \times X] \, .$$

Let n^*_{tx} be the Lagrange multiplier of the constraint; let $\pi^*_{txy} = \partial G_{tx}/\partial U_y$; and let $\mu^*_{txy} = n^*_{tx}\pi^*_{txy}$. We have by optimality with respect to λ that

$$\hat{\mu}^\top \Phi = \mu^{*\top}\Phi,$$

and the optimality conditions with respect to u yield

$$\left(\mathbf{J}_T \otimes \mathbf{P} - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y\right)\hat{\mu} = \left(\mathbf{J}_T \otimes \mathbf{P} - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y\right)\mu^*,$$

which has the same interpretation in remark 6.3.1.

Finite horizon, simulation

We can simulate this model and estimate it using linear programming by

$$
\begin{aligned}
\max_{\lambda_k, u_{tx}, \tilde{u}_{itx}} \quad & \left\{ \hat{\mu}^\top \left(\Phi\lambda + \left(\mathbf{J}_T^\top \otimes \mathbf{P}^\top - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y^\top \right) u \right) \right\} \\
s.t. \quad & u_{tx} = \frac{1}{I} \sum_{i \in [I]} \tilde{u}_{itx} \; \forall x \in [T \times X], \\
& \tilde{u}_{itx} \geq \left(\Phi\lambda + \mathbf{P}^\top u_{t+1} \right)_{xy} + \varepsilon_{itxy} \; \forall itxy \in [I \times T \times X \times Y],
\end{aligned}
$$

which leads to the following vectorized formulation

$$
\begin{array}{l}
\max\limits_{\lambda \in \mathbb{R}^K, u \in \mathbb{R}^{TX}, \tilde{u} \in \mathbb{R}^{ITX}} \hat{\mu}^\top \Phi \lambda + \hat{\mu}^\top \left(\mathbf{J}_T^\top \otimes \mathbf{P}^\top - \mathbf{I}_T \otimes \mathbf{\Sigma}_Y^\top \right) u \\
s.t. \\
\begin{array}{rcccccl}
\mathbf{0}_{TX \times K} \lambda & + & & \mathbf{I}_{TX} u & - & \frac{1}{I} \left(\mathbf{1}_I^\top \otimes \mathbf{I}_{TX} \right) \tilde{u} & = & \mathbf{0}_{TX}, \\
-\left(\mathbf{1}_I \otimes \Phi \right) \lambda & - & \left(\mathbf{1}_I \otimes \mathbf{J}_T^\top \otimes \mathbf{P}^\top \right) u & & + & \left(\mathbf{I}_{ITX} \otimes \mathbf{1}_Y \right) \tilde{u} & \geq & \varepsilon.
\end{array}
\end{array}
$$

Based on data consisting of $\hat{\mu}_{xy}$, the number of times decision y was made for a unit in state x, we seek to estimate the surplus ϕ_{xy} among a parametric family $\phi^{\lambda}_{xy} = \sum_k \phi_{xyk}\lambda_k$, or $\phi^{\lambda} = \Phi\lambda$ where Φ is the matrix of size $XY \times K$ of term ϕ_{xyk}. The stationary Bellman equation is

$$u_x = G_x((\phi + \beta \mathbf{P}^{\top} u)_{xy, y\in[Y]}),$$

which rewrites in a vectorized way as $G_x((\phi + (\beta\mathbf{P}^{\top} - \Sigma_Y^{\top})u)_{xy,y\in[Y]}) = 0$, and then

$$\begin{aligned} \max_{u\in\mathbb{R}^X, \lambda\in\mathbb{R}^K} \quad & \hat{\mu}^{\top}\left(\Phi\lambda + \left(\beta\mathbf{P}^{\top} - \mathbf{\Sigma}_Y^{\top}\right)u\right) \\ s.t. \quad & G_x\left(\left(\Phi\lambda + \left(\beta\mathbf{P}^{\top} - \mathbf{\Sigma}_Y^{\top}\right)u\right)_{xy,y\in[Y]}\right) = 0 \ \forall x \in [X]\,. \end{aligned} \tag{12}$$

Infinite horizon, estimation (2/2)

If we let (n_x^*) be the vector of Lagrange multipliers associated with the constraints, $\pi_{xy}^* = \partial G_x / \partial U_y$ be the vector of conditional choice probabilities, and if we let $\mu_{xy}^* = n_x^* \pi_{xy}^*$ be the predicted policy variable, the optimality conditions with respect to λ in the estimation problem lead to

$$\hat{\mu}^\top \Phi = \mu^{*\top} \Phi,$$

which has the usual interpretation that the predicted moments of the K basis functions of Φ match the observed ones, and the optimality conditions with respect to u yield

$$\left(\beta \mathbf{P} - {}^{\bullet}{}_{Y}\right) \hat{\mu} = \left(\beta \mathbf{P} - {}^{\bullet}{}_{Y}\right) \mu^*,$$

which again means that the predicted policy variable μ^* assigns the same change of mass in units in each state as the one observed in the sample.

Infinite horizon, computation

Again, we can simulate the random utilities ε_y to approximate the estimation problem.

$$\begin{aligned}
\max_{\lambda_k, u_x, \tilde{u}_{ix}} \quad & \left\{ \hat{\mu}^\top \left(\Phi\lambda + \left(\beta \mathbf{P}^\top - \mathbf{\Sigma}_Y^\top \right) u \right) \right\} \\
s.t. \quad & u_x = \frac{1}{I} \sum_{i \in [I]} \tilde{u}_{ix} \; \forall x \in [X], \\
& \tilde{u}_{ix} \geq \left(\Phi\lambda + \beta \mathbf{P}^\top u \right)_{xy} + \varepsilon_{ixy} \; \forall ixy \in [I \times X \times Y],
\end{aligned}$$

which vectorizes into

$$\begin{aligned}
& \max_{\lambda \in \mathbb{R}^K, u \in \mathbb{R}^X, \tilde{u} \in \mathbb{R}^{IX}} \hat{\mu}^\top \Phi \lambda + \left(\beta \hat{\mu}^\top \mathbf{P}^\top - \hat{\mu}^\top \mathbf{\Sigma}_Y^\top \right) u \\
& s.t. \\
& \begin{array}{rcrcrl}
\mathbf{0}_{X \times K} \lambda & + & \mathbf{I}_X u & - & \frac{1}{I} \left(\mathbf{1}_I^\top \otimes \mathbf{I}_X \right) \tilde{u} & = \mathbf{0}_X \\
- \left(\mathbf{1}_I \otimes \Phi \right) \lambda & - & \beta \left(\mathbf{1}_I \otimes \mathbf{P}^\top \right) u & + & \left(\mathbf{I}_{IX} \otimes \mathbf{1}_Y \right) \tilde{u} & \geq \varepsilon.
\end{array}
\end{aligned}$$

CCP inversion approach

As in section 6.8, we are in the infinite-horizon setting, where we seek to estimate the per-period surplus ϕ. To this end, we take a linear parameterization $\phi^{\lambda} = \Phi\lambda$, where as before, Φ is a $XY \times K$ matrix whose (xy, k) entry is ϕ_{xyk}. The Bellman equation in the infinite horizon then inverts into

$$\left(\nabla G_x^* \left(\pi\right)\right)_{xy \in [X \times Y]} = \Phi\lambda + \left(\beta \mathbf{P}^\top - \mathbf{\Sigma}_Y^\top\right) u,$$

and given $\mathbf{W}$ a symmetric, positive definite square matrix of weights of size XY, we estimate (λ, u) by generalized least square using the following program

$$\min_{\lambda, u} \left|\left(\nabla G_x^* \left(\hat{\pi}\right)\right)_{xy \in [X \times Y]} - \Phi\lambda - \left(\beta \mathbf{P}^\top - \mathbf{\Sigma}_Y^\top\right) u\right|_{\mathbf{W}},$$

where we recall $|z|_{\mathbf{W}}^2 = z^\top \mathbf{W} z$.

Exercises and Quick Recap Chapter 8

Quick exercises

Exercise 8.1: What becomes of the model described in section (8.2) when $\sum_x n_x = \sum_y m_y$ and when agents are forced to match ?
Exercise 8.2: Making use of lemma 2.3.1, show that the nested logit model (with one nest) can be recast as a (two-period) finite horizon dynamic discrete choice model.

Quick recap of Chapter 8 (1/2)

We extended the models of choice, allocation, and equilibrium from the logistic frameworks to general structures of random utilities. We first introduced a one-sided choice model, recalling our previous work and leading to the introduction of a broader multivariate generalized linear model framework for choice beyond simple logistic regression.

We then explored one-to-one matching models with general heterogeneities, describing them and providing computational methods, such as gradient descent, for finding the equilibria. This was followed by the exploration of coalition formation models, where different types of agents form coalitions to generalize matching models. We discussed coalitional equilibrium, characterization, and estimation methods.

Quick recap of Chapter 8 (2/2)

Section 8.4 extended dynamic discrete choice models to general random utility structures. For a finite horizon, this was achieved through the Bellman equation and estimation via linear programming. For the infinite horizon, we provided an estimation procedure and simulation methods. Finally, the CCP inversion approach was explored.

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Chapter 8: Models with General Heterogeneities

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Mathematical Appendices

The aim of these appendices is to provide the reader with the necessary mathematical tools to understand the chapters of the book. A solid college-level mathematical background is required to fully grasp the material.

Appendix A.1: Optimization

***Discrete Choice Models*, appendix A.1, pp. 239-256**

Content

Appendix A.1.1. Convex analysis

Section content

- Convex function
- Legendre-Fenchel transform
- Coercivity
- Subdifferential
- Gradient, Hessian and Differentiability of a convex function

Basic notions (1/2)

We define a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ to be convex if (i) f is not identically equal to $+\infty$; (ii) f is *lower semi-continuous*, meaning that its sublevel sets $\{x \in \mathbb{R}^n : f(x) \leq t\}$ are closed for all $t \in \mathbb{R}$; and (iii) the following convexity inequality

$$\lambda f(x) + (1-\lambda) f(y) \geq f(\lambda x + (1-\lambda) y)$$

holds for all $\lambda \in [0,1]$ and for all $x, y \in \mathbb{R}^n$. A function f is *concave* if $-f$ is convex. We define f to be *strictly convex* if in addition, the convexity inequality holds strictly whenever $\lambda \in (0,1)$ and $x \neq y$. The domain of a convex function f, denoted *dom* (f), is the set of $x \in \mathbb{R}^n$ such that $f(x) \in \mathbb{R}$.

Compared with the usual definition of convex functions, we impose the additional requirements (i) of nonempty domain and (ii) of lower semi-continuity.

Proposition A.1.1

If $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ satisfies the convexity inequality, then it is continuous on the interior of its domain.

Proposition A.1.2

If $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a convex function and if the set of its minimizers is non-empty, then the latter is a convex set.

Proposition A.1.3

If f is twice differentiable, then f is convex if and only if $dom(f)$ is convex and $D^2 f(x) \succeq_{\mathrm{psd}} 0$, $\forall x \in dom(f)$.

Definition (Legendre-Fenchel transform)

Given a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ not identically equal to $+\infty$, one may define its Legendre–Fenchel transform, also known as the *convex conjugate* of f, as

$$f^*(y) = \sup_{x \in \mathbb{R}^n} \left\{ x^\top y - f(x) \right\}.$$

Proposition A.1.4

For any x and y in $\mathbb{R}^n$, one has

$$f(x) + f^*(y) \geq x^\top y.$$

Legendre-Fenchel transform (2/3)

The first proposition says that the Legendre–Fenchel transform reverts the order between functions.

Proposition A.1.5

$f(x) \leq g(x)$ for all x implies $f^*(y) \geq g^*(y)$ for all y.

The second proposition investigates how the Legendre–Fenchel transform responds under a location-scale transform.

Proposition A.1.6

If $g(x) = \lambda f(x) + \gamma$ for $f : \mathbb{R}^n \to \mathbb{R}$, $\lambda > 0$ and $\gamma \in \mathbb{R}$, then $g^*(y) = \lambda f^*(y/\lambda) - \gamma$.

Legendre-Fenchel transform (3/3)

Proposition A.1.7

Given a convex function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$:

(i) if f is strictly convex, then f^* is differentiable on the interior of its domain;

(ii) if f is differentiable on the interior of its domain, and if $\|\nabla f(x)\| \to +\infty$ as x tends to a point on the frontier of the domain of f, then f^* is strictly convex on its domain.

Note that the Legendre-Fenchel transform applies to a function that is not necessarily convex. It is easy to verify that f^* is always a convex function unless it is identically equal to $+\infty$. This leads to the following definition.

Definition A.1.2

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Then f is *coercive* if $f(x) \to +\infty$ as $\|x\| \to +\infty$.

Proposition A.1.8

Consider $f : \mathbb{R}^n \to \mathbb{R}$ a convex and coercive function. Then f has a minimizer on $\mathbb{R}^n$.

Definition A.1.3 (Subdifferential)

Given a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ not identically equal to $+\infty$, one defines the subdifferential of f as

$$\partial f(x) = \left\{ y \in \mathbb{R}^n : f(z) \geq f(x) + y^\top (z - x) \; \forall z \in \mathbb{R}^n \right\}.$$

Proposition A.1.9

If f is convex, one has

$$\partial f(x) = \arg\max_{y \in \mathbb{R}^n} \left\{ x^\top y - f^*(y) \right\} = \left\{ y \in \mathbb{R}^n : f(x) + f^*(y) = x^\top y \right\}.$$

Proposition A.1.10

Consider a convex function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$. Then the following statements are equivalent:

(i) $y \in \partial f(x)$

(ii) $x \in \partial f^*(y)$

(iii) $f(x) + f^*(y) = x^\top y$.

Proposition A.1.11

A convex function f is differentiable at $x \in dom(f)$ if and only if $\partial f(x)$ has a single element. In that case, $\partial f(x) = \{\nabla f(x)\}$.

Proposition A.1.12

Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ that is convex and differentiable on an open set X. Then f is C^1 on X.

Proposition A.1.13

If f is strictly convex and C^1, then $\nabla f^* = (\nabla f)^{-1}$, where the inverse is understood in the sense of the composition of functions.

Proposition A.1.14

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. If f and f^* are twice differentiable respectively at x and $y = \nabla f(x)$, then $D^2 f^*(y) = (D^2 f)^{-1}(x)$, where the inverse is understood in the sense of matrix algebra.

Proposition A.1.15

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Then:

(i) the set $\{x \in \mathbb{R}^n : f \text{ is not differentiable at } x\}$ is of zero Lebesgue measure,

(ii) the set $\{x \in \mathbb{R}^n : f \text{ is not twice differentiable at } x\}$ is of zero Lebesgue measure.

Appendix A.1.2. Optimization by gradient descent

Section content

- Gradient descent
- β-smooth function

Description

Consider the following minimization problem

$$\min_{x \in \mathbb{R}^d} f(x),$$

where f is a convex function. In this appendix, we provide conditions on f that imply the convergence of the gradient descent algorithm, defined by $x^0 = x_0 \in \mathbb{R}^d$, and

$$x^{t+1} = x^t - \alpha \nabla f(x^t),$$

where $\alpha > 0$ is a scalar which does not depend on time, and which is called the *learning rate*. Equivalently, we could consider the maximization of a concave function; the results of this section apply after a simple change of the sign of f, so the gradient descent for maximizing a concave function f is changed into $x^{t+1} = x^t + \alpha \nabla f(x^t)$.

Definition A.1.4

A convex function f is β-*smooth* for $\beta > 0$ if for all $x, x' \in dom(f)$ and $t \in [0,1]$, one has $f(tx + (1-t)x') \geq tf(x) + (1-t)f(x') - \frac{\beta t(1-t)}{2}|x - x'|^2$.
A concave function f is β-smooth if $-f$ is β-smooth in the sense of convex functions defined above.

Proposition A.1.16

When a (convex or concave) function f is C^2, it is β-smooth if and only if for all $x \in dom(f)$, the largest absolute value of the eigenvalues of $D^2 f(x)$ is less or equal than β.

Theorem A.1.1

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a C^1, convex, β-smooth function. Assume that f has a minimizer x^* on $\mathbb{R}^n$. Consider the sequence x^t obtained by the gradient descent with constant learning rate $\alpha = \beta^{-1}$. Then

$$\left|\nabla f\left(x^t\right)\right| \leq \frac{\beta^2 \left|x^0 - x^*\right|}{\sqrt{(1+t)\,t}} \text{ for } t \geq 1,$$

and

$$0 \leq f\left(x^t\right) - f\left(x^*\right) \leq \frac{\beta \left|x^0 - x^*\right|^2}{2\,(t+1)} \text{ for } t \geq 0.$$

Appendix A.1.3. Proximal gradient descent

Section content

- Proximal gradient

Proximal gradient descent applies to settings where one seeks to minimize $f(x) = g(x) + \varphi(x)$, where g is smooth but φ is not necessarily smooth. Standard gradient descent would lead to

$$x^{t+1} = x^t - \alpha \left(\nabla g\left(x^t\right) + \nabla \varphi\left(x^t\right)\right),$$

but instead, we shall apply a semi-implicit scheme, where we replace $\nabla \varphi(x^t)$ by $\nabla \varphi\left(x^{t+1}\right)$ and we seek to have

$$x^{t+1} = x^t - \alpha \left(\nabla g\left(x^t\right) + \nabla \varphi\left(x^{t+1}\right)\right).$$

We can rewrite this as looking for x^{t+1} such that $x^{t+1} + \alpha \nabla \varphi \left(x^{t+1} \right) = \tilde{x}^{t+1}$, where $\tilde{x}^{t+1} := x^t - \alpha \nabla g \left(x^t \right)$. This equation arises as the first-order condition of the convex minimization problem:

$$x^{t+1} = \arg\min_{x \in \mathbb{R}^d} \left\{ \frac{1}{2} |x|^2 + \alpha \varphi \left(x \right) - x^\top \tilde{x}^{t+1} \right\},$$

or equivalently $x^{t+1} = \mathit{prox}_{\alpha\varphi} \left(\tilde{x}^{t+1} \right)$, where the *proximal operator* associated with function φ is defined as

$$\mathit{prox}_{\varphi} \left(\tilde{x} \right) = \arg\min_{x \in \mathbb{R}^d} \left\{ \frac{1}{2} \left| x - \tilde{x} \right|^2 + \varphi \left(x \right) \right\}.$$

Example A.1

The following are commonly used proximal operators:

(a) When $\varphi(x) = 0$, $prox_{\varphi}(\tilde{x}) = \tilde{x}$.

(b) When $\varphi(x) = x^{\top}\Sigma x/2$, $prox_{\varphi}(\tilde{x}) = (I + \Sigma)^{-1}\tilde{x}$.

(c) When $\varphi(x) = \lambda|x|_1$. Then we have $(prox_{\varphi}(\tilde{x}))_i = S_{\lambda}(x_i)$, where $S_{\lambda}(u) := (u-\lambda)^{+} - (u+\lambda)^{-}$ is called the *soft thresholding function*.

Algorithm A.1.1 Proximal Descent Algorithm

Input: The smooth part g to be minimized, rules for picking the descent direction $d(g,x)$, the step size $\alpha(g,x,d)$, and the proximal operator $\mathrm{prox}_{\alpha\varphi}$.

Parameter: The starting point x^0, the tolerance ξ.

Set $t=0$, $x^t = x^0$.

repeat

$d^t = d(g,x^t), \quad \alpha^t = \alpha(g,x^t,d^t)$

$\tilde{x}^{t+1} := x^t - \alpha^t d^t$

$x^{t+1} := \mathrm{prox}_{\alpha\varphi}(\tilde{x}^{t+1})$

$t = t+1$

until $\|x^{t+1} - x^t\| \leq \xi$

Return: x^t

Appendix A.1.4. Minimax theory

Section content

- Duality
- Saddlepoint

Duality

Duality: Let $\mathcal{X}$ and $\mathcal{Y}$ be two subsets of $\mathbb{R}^n$, and consider $L : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. For any $x' \in \mathcal{X}$ and $y' \in \mathcal{Y}$, we have $\inf_{x \in \mathcal{X}} L(x, y') \leq L(x', y') \leq \sup_{y \in \mathcal{Y}} L(x', y)$, thus

$$\varphi(y') := \inf_{x \in \mathcal{X}} L(x, y') \leq \sup_{y \in \mathcal{Y}} L(x', y) =: \psi(x'),$$

thus $\sup \varphi \leq \inf \psi$, and therefore $\sup_{y \in \mathcal{Y}} \inf_{x \in \mathcal{X}} L(x, y) \leq \inf_{x \in \mathcal{X}} \sup_{y \in \mathcal{Y}} L(x, y)$.

Lemma A.1.1 (Weak duality)

We have the *weak duality inequality*

$$V_D := \sup_{y \in \mathcal{Y}} \inf_{x \in \mathcal{X}} L(x, y) \leq \inf_{x \in \mathcal{X}} \sup_{y \in \mathcal{Y}} L(x, y) =: V_P.$$

When the duality gap $V_P - V_D \geq 0$ is zero, one says that *strong duality* holds.

Saddlepoint

A *saddlepoint* for L is a pair $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$ such that $L(x^*, y) \leq L(x^*, y^*) \leq L(x, y^*)$ holds for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$.

Lemma A.1.2

The following are equivalent:

(i) $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$ is a saddlepoint, and

(ii) x^* is optimal for the primal problem, y^* is optimal for the dual problem, and there is no duality gap, that is

$$x^* \in \arg\min_{x \in \mathcal{X}} \sup_{y \in \mathcal{Y}} L(x, y), \; y^* \in \arg\max_{y \in \mathcal{Y}} \inf_{x \in \mathcal{X}} L(x, y),$$
$$\sup_{y \in \mathcal{Y}} \inf_{x \in \mathcal{X}} L(x, y) = \inf_{x \in \mathcal{X}} \sup_{y \in \mathcal{Y}} L(x, y).$$

Appendix A.1.5. Linear programming

Section content

- Standard form
- Feasible and optimal solutions
- Application of duality
- Complementary slackness

The standard problem form

A linear programming problem consists in optimizing (maximizing or minimizing) a linear objective function under constraints that are all linear. Any nontrivial linear programming problem can be reformulated as a problem *in standard form*, which is

$$V_P = \max_{x \in \mathbb{R}^n_+} c^\top x \quad \text{s.t. } Ax = d, \tag{A.1.8}$$

where A is an $I \times J$ matrix of full rank and such that $I \leq J$. Thus I is the number of constraints and J is the number of variables.

Feasible and optimal solutions

Problem (A.1.8) is called the *primal program*. If $x \in \mathbb{R}^J$ satisfies the constraints $x \geq 0$ and $Ax = d$, it is called a *feasible solution*; if it is a solution to problem (A.1.8), it is called an *optimal solution*.

Note that the set of feasible solutions may be empty; if $A = 1$ and $d = -1$ for instance, then there is no feasible solution. In this case, because a maximum over an empty set should have a smaller value than any finite number, we assign value $V_P = -\infty$ to problem (A.1.8).

Even when the set of feasible solutions is nonempty, the set of optimal solutions may still be empty because the objective function may not be bounded above on the feasible set; this will be the case for example when $A = 0$ and $d = 1$ and $c = 1$, in which case the value of problem (A.1.8) is $V_P = +\infty$.

Dual program

One can rewrite the primal problem as

$$V_P = \max_{(x_j)\geq 0} \min_{(y_i)} \left\{ c^\top x + (d - Ax)^\top y \right\}.$$

Indeed, the solution to the inner minimization problem will be $-\infty$ as soon as one of the primal constraints is not satisfied. By the minimax inequality,

$$V_P \leq V_D := \min_{(y_i)} \max_{(x_j)\geq 0} \left\{ c^\top x + (d - Ax)^\top y \right\} = \min_{(y_i)} \left\{ d^\top y + \max_{(x_j)\geq 0} \left\{ x^\top (c - A^\top y) \right\} \right\},$$

and we see that $\max_{(x_j)\geq 0} \left\{ x^\top w \right\}$ is equal to 0 if all the w_j are nonpositive, and to $+\infty$ if some w_j is positive. Thus V_D is the value of the *dual program*, defined by

$$V_D := \min_{(y_i)} d^\top y \tag{13}$$

$$\text{s.t. } A^\top y \geq c.$$

Proposition A.1.17

The following are the only possible cases regarding the values of the primal and the dual:

- $V_P = V_D = +\infty$: the primal is unbounded and the dual in unfeasible;
- $V_P = V_D \in \mathbb{R}$: both the primal and the dual are feasible and bounded;
- $V_P = V_D = -\infty$: the primal is unfeasible and the dual in unbounded;
- $V_P = -\infty < V_D = +\infty$: both the primal and the dual are unfeasible.

Note that the case when both the primal and dual programs are unfeasible is the only case when strong duality does not hold.

Features of dual problem (2/2)

Complementary slackness. If $(x_j^*) \geq 0$ is an optimal primal solution, and (y_i^*) dual solution, complementary slackness holds, that is, $(x_j^*) > 0$ implies $(A^\intercal y^*)_j = c_j$, which is summarized in the next statement.

Theorem A.1.2 (Finite-dimensional linear programming)

In the setting described above:

(i) The weak duality inequality holds:

$$V_P \leq V_D.$$

(ii) As soon as the primal or the dual program has an optimal solution, then both programs have an optimal solution, and strong duality holds:

$$V_P = V_D.$$

Theorem A.1.2 (continued)

(iii) If $(x_j^\star) \geq 0$ is an optimal primal solution, and $(y_i^\star)$ is an optimal dual solution, then the complementary slackness conditions hold:

$$x_j^\star > 0 \text{ implies } (A^\top y^\star)_j = c_j.$$

(iv) Conversely, if $(x_j^\star) \geq 0$ is a feasible primal solution, and $(y_i^\star)$ is a feasible dual solution such that the complementary slackness conditions hold, then $(x_j^\star) \geq 0$ is an optimal primal solution, and $(y_i^\star)$ is an optimal dual solution.

Appendix A.1.6. Nonlinear programming

Section content

- Karush–Kuhn–Tucker conditions
- Slater's theorem

Consider a problem of the form

$$V_P := \min_{x \in \mathbb{R}^{\mathcal{I}}} f(x) \qquad \text{(A.1.11)}$$
$$\text{s.t. } g_j(x) \leq 0, \ \forall j \in \mathcal{J},$$

where f and the g_j are convex functions. Problem (A.1.11) is then called a "convex" program.

Theorem A.1.3 (weak duality)

One has $V_P \geq V_D$, where

$$V_D := \sup_{y \in \mathbb{R}_+^{\mathcal{J}}} \varphi(y), \qquad \text{(A.1.12)}$$

where $\varphi(y) := \inf_{x \in \mathbb{R}^{\mathcal{I}}} \left\{ f(x) + y^\top g(x) \right\}$ is a concave function of y.

Theorem A.1.4

Assume the problem (A.1.11) is convex. Then, given $(x^*, y^*) \in \mathbb{R}^{\mathcal{I}} \times \mathbb{R}^{\mathcal{J}}$, the following two statements are equivalent:

(i) (x^*, y^*) is a saddlepoint for L on $\mathbb{R}^{\mathcal{I}} \times \mathbb{R}_+^{\mathcal{J}}$, and

(ii) (x^*, y^*) satisfies the Karush–Kuhn–Tucker (KKT) conditions:

(a) *primal feasibility:* $g_j(x^*) \leq 0, \ \forall j \in \mathcal{J}$,

(b) *dual feasibility:* $y_j^* \geq 0, \ \forall j \in \mathcal{J}$,

(c) *complementary slackness:* $y_j^* > 0 \implies g_j(x^*) = 0, \ \forall j \in \mathcal{J}$,

(d) *stationarity:* $0_{\mathcal{I}} \in \partial_x L(x^*, y^*)$.

We let $\mathcal{J}_A \subseteq \mathcal{J}$ such that $g_j(x) = a_j^\top x - b_j$ is an affine function for $j \in \mathcal{J}_A$. Hence, we may assume that g_j is not affine for $j \notin \mathcal{J}_A$, and we are able to state Slater's condition:

Assumption A.1 (Slater's condition)

There exists $x^* \in \mathbb{R}^{\mathcal{J}}$ such that

$$g_j(x^*) < 0, \ \forall j \notin \mathcal{J}_A.$$

Theorem A.1.5 (Slater's theorem)

If problem (A.1.11) is convex and Slater's condition is met, then strong duality holds: the value of the primal problem (A.1.11) coincides with the value of the dual problem (A.1.12).

Appendix A.1.7. Augmented Lagrangian method

Section content

- Augmented Lagrangian method

Preparation

Consider a variant of problem (A.1.11) with equality constraint:

$$V_P := \min_{x \in \mathbb{R}^{\mathcal{I}}} f(x) \quad \text{s.t. } g(x) = 0, \tag{A.1.14}$$

where $f : \mathbb{R}^{\mathcal{I}} \to \mathbb{R}$ and $g : \mathbb{R}^{\mathcal{I}} \to \mathbb{R}^{\mathcal{J}}$ are C^2 functions. The (classical) Lagrangian for this constrained minimization problem is:

$$L(x, y) = f(x) + y^\top g(x).$$

Augmented Lagrangian method

The augmented Lagrangian method, also known as the method of multipliers, adds a penalty term that helps stabilize the convergence of the method when dealing with constraint violations. We define the augmented Lagrangian as

$$L_A(x, y, \gamma) = f(x) + y^\top g(x) + \frac{\gamma}{2} \|g(x)\|^2,$$

where $\gamma > 0$ increases the cost of violating the constraint.

Theorem A.1.6

Assume f and g are continuous functions. Assume (γ^t) is an increasing real-valued sequence that tends to $+\infty$. Let (y^t) be a bounded sequence of elements of $\mathbb{R}^{\mathcal{J}}$, and consider x^t a global minimum of $L(x, y^t, \gamma^t)$ over $x \in \mathbb{R}^{\mathcal{I}}$. Then every limit point of (x^t) is a solution to (A.1.14).

Algorithm A.1.2: Augmented Lagrangian method

Input: Objective function $f(x)$, constraint function $g(x)$

Parameter: Initial variables (x^0, y^0), increasing sequence $\gamma^t \to +\infty$, tolerance ξ

Output: Approximate primal solution x^*

Set $t = 0$

repeat

$x^{t+1} = \arg\min_x L_A(x, y^t, \gamma^t)$

$y^{t+1} = y^t + \gamma^t g(x^{t+1})$

$t = t + 1$

until $||g(x^t)|| \leq \xi$;

Appendix A.1.8. Optimal transport

Section content

- Monge–Kantorovich duality
- Entropic regularization
- Iterated proportional fitting procedure (IPFP)

Basic setting

Assume that $\mathcal{X}$ and $\mathcal{Y}$ are closed subsets of $\mathbb{R}^d$ and $\mathbb{R}^{d'}$, respectively. Let n and m be two probability measures of $\mathcal{X}$ and $\mathcal{Y}$ respectively, and let $\mathcal{M}(n, m)$ be the set of joint distributions μ over $\mathcal{X} \times \mathcal{Y}$ such that if $(X, Y) \sim \mu$, then $X \sim n$ and $Y \sim m$. Let $\Phi : \mathcal{X} \times \mathcal{Y} \to \mathbb{R} \cup \{-\infty\}$ and consider the so-called *primal problem*

$$\sup_{\mu \in \mathcal{M}(n,m)} \mathbb{E}_\mu[\Phi(X, Y)],$$

which is an infinite-dimensional linear programming problem whose dual expresses as

$$\inf_{u(x)+v(y) \geq \Phi(x,y)} \mathbb{E}_n[u(X)] + \mathbb{E}_m[v(Y)],$$

where it is understood that the infimum is taken over all n- and m-integrable functions u and v.

Theorem A.1.7

Assume that Φ is upper semicontinuous and that there are $\overline{a} \in L^1(n)$ and $\overline{b} \in L^1(m)$ such that $\Phi(x, y) \leq \overline{a}(x) + \overline{b}(y)$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. Then:

(i) The values of the primal and the dual problems coincide, i.e.

$$\sup_{\mu \in \mathcal{M}(n,m)} \mathbb{E}_\mu[\Phi(X, Y)] = \inf_{u(X)+v(Y) \geq \Phi(X,Y)} \mathbb{E}_n[u(X)] + \mathbb{E}_m[v(Y)]. \quad \text{(A.1.19)}$$

(ii) The primal problem on the left-hand side of (A.1.19) has an optimal solution μ.

(iii) Assume further that there are $\underline{a} \in L^1(n)$ and $\underline{b} \in L^1(m)$ such that $\Phi(x, y) \geq \underline{a}(x) + \underline{b}(y)$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. Then the dual problem on the right-hand side of (A.1.19) has an optimal solution (u, v) as well.

Discrete setting (1/2)

In the discrete setting when $\mathcal{X}$ and $\mathcal{Y}$ are finite sets, n and m are two nonnegative vectors of $\mathbb{R}^{\mathcal{X}}$ and $\mathbb{R}^{\mathcal{Y}}$ respectively such that $\sum_x n_x = \sum_y m_y = 1$, and the set of transport plans is given as

$$\mathcal{M}(n,m) := \left\{ \mu \in \mathbb{R}_+^{\mathcal{X}\times\mathcal{Y}} : \begin{array}{l} \sum_{y\in\mathcal{Y}} \mu_{xy} = n_x, \ \forall x \in \mathcal{X} \\ \sum_{x\in\mathcal{X}} \mu_{xy} = m_y, \ \forall y \in \mathcal{Y} \end{array} \right\}.$$

Discrete setting (2/2)

The primal problem is then obtained as the maximization of the surplus $\sum_{xy} \mu_{xy} \Phi_{xy}$ over $\mu \in \mathcal{M}(n, m)$, that is

$$\max_{\mu_{xy} \geq 0} \sum_{xy \in \mathcal{X} \times \mathcal{Y}} \mu_{xy} \Phi_{xy} \quad \text{(A.1.21)}$$
$$\text{s.t. } \mu \in \mathcal{M}(n, m),$$

and its dual is

$$\min_{u_x, v_y} \sum_{x \in \mathcal{X}} n_x u_x + \sum_{y \in \mathcal{Y}} m_y v_y \quad \text{(A.1.22)}$$
$$\text{s.t. } u_x + v_y \geq \Phi_{xy}.$$

Both the primal and dual problems being feasible, strong duality holds and the value of the program is finite.

Proposition A.1.18

Let $\mu \geq 0$ be primal feasible and (u, v) be dual feasible. Then the following two assertions are equivalent:

(i) μ is optimal for the primal problem (A.1.21) and (u, v) is optimal for the dual problem (A.1.22).

(ii) The complementarity relation holds:

$$\forall xy \in \mathcal{X} \times \mathcal{Y},\ \mu_{xy} > 0 \implies u_x + v_y = \Phi_{xy}. \tag{A.1.23}$$

Entropic regularization (1/2)

It simply consists of adding an entropic penalty term to the objective function in the primal problem (A.1.21), which becomes

$$\max_{\mu \in \mathcal{M}(n,m)} \sum_{xy \in \mathcal{X} \times \mathcal{Y}} \mu_{xy} \Phi_{xy} - T \sum_{xy \in \mathcal{X} \times \mathcal{Y}} \mu_{xy} \log \mu_{xy}, \tag{A.1.24}$$

where $T > 0$ is a temperature parameter that controls how much we regularize. Calculation shows that the dual to problem (A.1.24) is

$$\min_{u,v} \sum_{x \in \mathcal{X}} n_x u_x + \sum_{y \in \mathcal{Y}} m_y v_y + T \sum_{xy \in \mathcal{X} \times \mathcal{Y}} \exp\left(\frac{\Phi_{xy} - u_x - v_y}{T} \right), \tag{A.1.25}$$

whose FOCs determine u and v by what is called a *Bernstein–Schrödinger system* of equations:

$$\begin{cases} n_x = \sum_{y \in \mathcal{Y}} \exp\left((\Phi_{xy} - u_x - v_y)/T \right) \\ m_y = \sum_{x \in \mathcal{X}} \exp\left((\Phi_{xy} - u_x - v_y)/T \right). \end{cases} \tag{A.1.26}$$

Theorem A.1.8

Consider the Bernstein–Schrödinger system (A.1.26). Then:

(i) The system has a unique solution (u, v) up to a constant – that is, if (u, v) and (u', v') are solutions, then there is a constant $c \in \mathbb{R}$ such that $u'_x = u_x + c$ and $v'_y = v_y - c$.

(ii) The set of solutions to the system coincides with the set of minimizers of the dual problem (A.1.25).

(iii) If (u, v) is a solution to the system, then μ_{xy}, defined by $\mu_{xy} := \exp\left(\frac{\Phi_{xy} - u_x - v_y}{T}\right)$, is a solution to the primal regularized problem (A.1.24). Conversely, every primal solution arises in that way.

IPFP

Let $a_x = \exp(u_x/T)$ and $b_y = \exp(v_y/T)$. Given b_y^0, we construct iterative estimates (a^t, b^t) of the solutions (a, b) as follows:

$$a_x^{t+1} = \frac{1}{n_x} \sum_{y \in \mathcal{Y}} \exp\left(\frac{\Phi_{xy}}{T}\right) \frac{1}{b_y^t},$$

and next, in the same loop, we look for v_y^{t+1} to satisfy the second set of equations of system (A.1.26), so that

$$b_y^{t+1} = \frac{1}{m_y} \sum_{x \in \mathcal{X}} \exp\left(\frac{\Phi_{xy}}{T}\right) \frac{1}{a_x^{t+1}}.$$

Let A be the $|\mathcal{X}| \times |\mathcal{Y}|$ matrix of term $A_{xy} = \exp(\Phi_{xy}/T)/n_x$ and let B be the $|\mathcal{Y}| \times |\mathcal{X}|$ matrix of term $B_{yx} = \exp(\Phi_{xy}/T)/m_y$. Let $K_A = \kappa(A)$ and $K_B = \kappa(B)$, and let $K = K_A K_B$, which is smaller than one. We have

Theorem A.1.9 (convergence of the IPFP)

The solution $(u^\star, v^\star)$ to system (A.1.26) exists and is unique up to a constant. Then, letting $a_x^\star = \exp(u_x^\star/T)$ and $b_y^\star = \exp(v_y^\star/T)$, we have for any $t \geq 0$, $d_H(a^t, a^\star) \leq K^t d_H(a^0, a^\star)$, and similarly, $d_H(b^t, b^\star) \leq K^t d_H(b^0, b^\star)$.

Convergence of the IPFP (2/2)

Proof: Define the mappings F and G such that $F_x(a) = 1/a_x$ and $G_y(b) = 1/b_y$, noting that these mappings preserve the Hilbert projective metric. The iterative update for a^{t+1} can be rewritten in vector form as $a^{t+1} = AG(b^t)$, and the update for b^{t+1} as $b^{t+1} = BF(a^t)$. Combining these two equations directly yields the desired result. ■

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Appendix A.2: Probability and statistics

Discrete Choice Models, appendix A.2, pp. 256-284

Content

Appendix A.2: Probability and statistics

Discrete Choice Models, appendix A.2, pp. 256-284 (continued)

Content

Appendix A.2.1. Geometry of the simplex

Section content

- Simplical subdivision
- Parallel subsimplex
- Dirichlet distribution

Description

A probability vector $\pi \in \mathbb{R}^J$ can be thought of as an element of the $(J-1)$-dimensional *standard simplex* $\{\pi \in \mathbb{R}^J_+ : \sum_j \pi_j = 1\}$, whose J extreme points are $\boldsymbol{e}^j_J$, the elements of the canonical basis of $\mathbb{R}^J$.

More generally, a $(J-1)$-dimensional simplex is the convex hull of J points $(a_1, \ldots, a_J)$ in $\mathbb{R}^J$. The simplex is said to be "generated" by the vectors a_j. A 1-dimensional simplex $(J=2)$ is a segment and a 2-dimensional simplex $(J=3)$ is a triangle.

Any point a of the simplex $\Delta = conv(a_1, \ldots, a_J)$ can be written as a barycenter of the points that generate this simplex. In particular, a point a_0 in the interior of Δ has expression $a_0 = \sum_j t_j a_j$ with $t_j > 0$ and $\sum_j t_j = 1$. For $j \in [J]$, we define the *simplical subdivision* Δ_j with respect to a_0 by $\Delta_j = conv(\{a_k : k \neq j\} \cup \{a_0\})$.

We claim:

Proposition A.2.1

Let $b \in \Delta$ so that b writes $b = \sum_j \theta_j a_j$ with $\theta_j \geq 0$ and $\sum_j \theta_j = 1$. Then $b \in \Delta_j$ if and only if $\theta_j t_k \leq t_j \theta_k$ for all $k \in [J]$.

The $(J-1)$-dimensional volume of $\Delta = conv(a_1, ..., a_J)$ is given by

$$vol(\Delta) = \frac{1}{(J-1)!}\sqrt{G(\{a_j - a_J, j \in [J-1]\})}$$

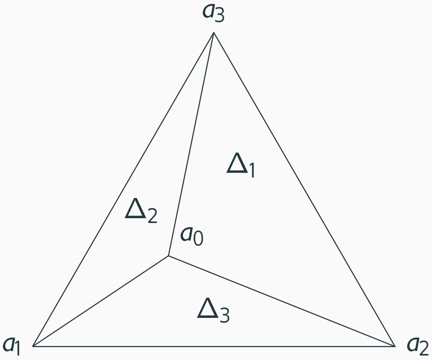
where $G(\{v_j, j \in [J-1]\})$ is the Gram determinant of the vectors (v_j) for $j \in [J-1]$, which is the determinant of the *Gram matrix* of these vectors, itself defined as the $(J-1)$ by $(J-1)$ matrix whose (j,j')th term for $1 \leq j, j' \leq J-1$ is the scalar product $(v_j)(v_{j'})$. If we substitute $a_0 = \sum_j t_j a_j$ to a_J, and more generally to any a_j, we get an expression for the area of the simplical subdivision Δ_j, namely:

Proposition A.2.2

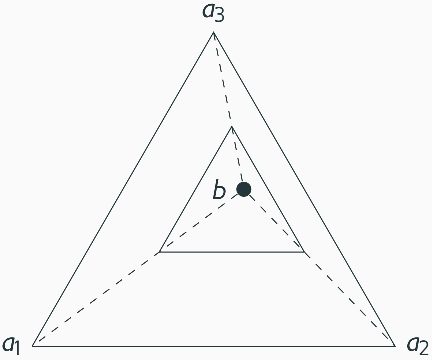
We have

$$vol(\Delta_j) = t_j \; vol(\Delta).$$

Figure of simplex



(a) Simplical subdivisions of a simplex.



(b) Parallel subsimplex of a simplex.

Figure 3: Left: simplical subdivisions. Right: parallel subsimplex.

Parallel subsimplex (1/2)

Consider $\Delta = conv(a_1, \ldots, a_J)$, and for $\tau \in \mathbb{R}^J$, let the *parallel subsimplex* $\Delta(\tau)$ be the subset of Δ consisting of vectors $a = \sum_j t_j a_j$ with $t_j \geq \tau_j \geq 0$ and $\sum_j t_j = 1$.

One has $\Delta(0) = \Delta$. Consider $\tau \in \mathbb{R}^J_+$. We have:

- If $\sum_j \tau_j > 1$, then $\Delta(\tau)$ is empty.
- If $\sum_j \tau_j = 1$, then $\Delta(\tau) = \{b\}$ where $b = \sum_j \tau_j a_j$.
- If $\sum_j \tau_j < 1$, then we let $k = \sum_j \tau_j$ and we define $\theta_j = \tau_j / k$, and $b = \sum_j \theta_j a_j$. We can show that $\Delta(\tau)$ is obtained from Δ by an homothety of ratio $1 - k$ and of center b.

Proposition A.2.3

We have

$$vol\big(\Delta(\tau)\big) = \left(\left(1 - \sum_{j \in [J]} \tau_j\right)^+\right)^{J-1} vol(\Delta). \tag{14}$$

Dirichlet distribution (1/2)

We call *flat Dirichlet distribution* of order J the uniform distribution on the $J-1$ subsimplex $\{a \in \mathbb{R}^J : a_j \geq 0, \sum_j a_j = 1\}$.

This distribution is denoted $\mathcal{D}_J$. The flat Dirichlet distribution on the 1-dimensional standard simplex is thus the distribution of (D_1, D_2) given by $D_1 = U$, $D_2 = 1 - U$ for $U \sim \mathcal{U}$, a uniform random variable over $[0, 1]$. More generally, the Dirichlet distribution of order I is characterized by the fact that if $D \sim \mathcal{D}_J$, then

$$\Pr\left(D_j \geq x_j,\ \forall j \in [J]\right) = \left(\left(1 - \sum_{j \in [J]} x_j\right)^+\right)^{J-1}.$$

If $D \sim \mathcal{D}_J$, then for any $j \in [J]$, D_j has a c.d.f. at $z \in [0, 1]$ equal to $1 - (1 - z)^{J-1}$. As a result, we can set $x_j = b_j / z$, and we have:

Dirichlet distribution (2/2)

Proposition A.2.4

If $D \sim \mathcal{D}_J$ has a flat Dirichlet distribution of order J, then

$$\max_{j \in [J]} \frac{b_j}{D_j} \stackrel{d}{=} \frac{\sum_{j \in [J]} b_j}{D_0},$$

where D_0 has the same distribution as each of the components D_j of the random vector D.

Proposition A.2.5

If $X_1, \ldots, X_I$ are i.i.d. exponential random variables, then $D_i = X_i / (\sum_i X_i)$ is flat Dirichlet of order I.

Appendix A.2.2. Positive stable distributions

Section content

- Positive (add-)stable

A probability distribution with support contained in $\mathbb{R}_+$ is *positive (add-)stable* if for $Z_1, \ldots, Z_l$ i.i.d. variables with that distribution, there exists a constant $c_l \in \mathbb{R}_+$ such that one has

$$Z_1 + \cdots + Z_l \overset{d}{=} c_l Z, \text{ where } Z \overset{d}{=} Z_i \; \forall i \in [l]. \tag{A.2.6}$$

Equation (A.2.6) holds if and only if the Laplace transform of the distribution, denoted $L(t) := \mathbb{E}[\exp(-tZ)]$, satisfies $L^l(t) = L(c_l t)$ for all $t \geq 0$ and $l \in \mathbb{N}$. We can thus apply the following result to $\varphi(t) = -\log L(t)$:

Lemma A.2.1

Let $\varphi : \mathbb{R}_+ \to \mathbb{R}_+$ be a continuous function such that for each l there exists some $c_l \in \mathbb{R}_+$ such that $l\varphi(t) = \varphi(c_l t)$ holds for all $t \in \mathbb{R}_+$. Then $\varphi(t) = t^\alpha$ for some $\alpha \in \mathbb{R}$.

Theorem A.2.1

A probability distribution is positive add-stable if and only if there is a scalar parameter $\alpha \in (0,1]$, called the *stability parameter,* such that the Laplace transform of the distribution is equal to

$$\mathbb{E}\left[e^{-tZ}\right] = \exp\left(-t^{\alpha}\right),$$

in which case the distribution is called the *positive stable distribution of parameter* α, denoted $\mathcal{PS}_{\alpha}$, and $c_l = l^{1/\alpha}$ in (A.2.6), and thus the latter expression becomes

$$Z_1 + \cdots + Z_l \stackrel{d}{=} l^{1/\alpha} Z.$$

A compact probabilistic representation

For $\alpha < 1$, the domain of $\mathcal{PS}_\alpha$ is $\mathbb{R}_+$, and its mean is $+\infty$. If U and V are independent $\mathcal{U}([0,1])$ random variables, then

$$\left(\frac{\sin\left((1-\alpha)\pi U\right)}{-\log V}\right)^{\frac{1-\alpha}{\alpha}} \frac{\sin\left(\alpha\pi U\right)}{\left(\sin\left(\pi U\right)\right)^{1/\alpha}} \sim \mathcal{PS}_\alpha.$$

Proposition A.2.6

If $Z_1 \sim \mathcal{PS}_{\alpha_1}$ and $Z_2 \sim \mathcal{PS}_{\alpha_2}$ are independent random variables, then $Z_1 Z_2^{1/\alpha_1} \sim \mathcal{PS}_{\alpha_1\alpha_2}$.

Relation between the positive add-stable distributions (1/2)

One can see that if $X_1, X_2, \ldots, X_n$ are independent variables that all have the same distribution as $\alpha \log Z$ with $Z \sim \mathcal{PS}_\alpha$, then

$$\operatorname{softmax}^{\alpha}_{i\in[I]} \{X_i + x_i\} \stackrel{d}{=} \log\Big(\sum_{i\in[I]} \exp x_i\Big) + X,$$

where X has the same distribution as the X_i's, and the *soft-maximum operator* is defined by $\operatorname{softmax}^{\alpha}_{i}\{z_i\} = \alpha \log \sum_i \exp(z_i/\alpha)$. Since $\operatorname{softmax}^{\alpha}_{i}\{z_i\} \to \max_i\{z_i\}$ as $\alpha \to 0$, we may expect for $\alpha \log Z$ to converge in some sense to the standard Gumbel distribution, which is max-stable.

Theorem A.2.2

Let $\mathcal{S}_\alpha$ be the distribution of $\alpha \log Z$, where $Z \sim \mathcal{PS}_\alpha$. Then

$$W_2\left(\mathcal{S}_\alpha, \mathcal{G}\right) \leq \alpha \sqrt{\gamma^2 + \frac{\pi^2}{6}},$$

where $W_2\left(\mathcal{S}_\alpha, \mathcal{G}\right)$ is the 2-Wasserstein distance, and $\mathcal{G}$ is the standard Gumbel distribution.

Appendix A.2.3. Rosenblatt quantiles and importance sampling

Section content

- Rosenblatt quantile
- Importance sampling

Rosenblatt quantile map (1/2)

Definition A.2.1

The Rosenblatt quantile map associated with the distribution $\mathcal{Q}$ is the map $T : \mathbb{R}^J \to \mathbb{R}^J$ defined by

$$\begin{cases} T_1(x) = F_{Z_1}^{-1}(x_1) \\ T_2(x) = F_{Z_2|Z_1}^{-1}(x_2 \mid Z_1 = T_1(x_1)) \\ \quad \vdots \\ T_J(x) = F_{Z_J|Z_1,\dots,Z_{J-1}}^{-1}\left(x_J \mid Z_j = T_j(x) \;\; \forall j \in [J-1]\right). \end{cases}$$

T_1 depends only on x_1, T_2 depends on x_1 and x_2, etc. As a result, the Jacobian of T is lower triangular with positive diagonal terms.

Rosenblatt quantile map (2/2)

Lemma A.2.2

If T is the Rosenblatt quantile associated with distribution $\mathcal{Q}$, then $X \sim \mathcal{U}([0,1]^J)$ implies $T(X) \sim \mathcal{Q}$.

Example A.2

When $\mathcal{Q} = \mathcal{N}(0, \Sigma)$, the Rosenblatt quantile of $\mathcal{Q}$ is given by $T(x) = L\Phi^{-1}(x)$, where L is the Choleski factorization of Σ, i.e. the unique lower triangular matrix with a positive diagonal such that $LL^\top = \Sigma$, and Φ is the c.d.f. of the standard normal distribution, with Φ^{-1} its inverse, the quantile map. Here Φ^{-1} applies component-wise on the entries of $x \in \mathbb{R}^J$, so that $\left(\Phi^{-1}(x)\right)_j = \Phi^{-1}(x_j)$.

Importance sampling: basic setting

Assume we want to compute

$$\pi = \Pr\left(Z_1 \leq z_1, Z_2 \leq z_2\right),$$

and suppose we know the Rosenblatt quantile for the distribution $\mathcal{Q}$ of Z, which is $T(x) = (T_1(x_1), T_2(x_1, x_2))$. We denote $T_1^{-1}(z_1)$ the inverse map of T_1, and $T_2^{-1}(x_1, z_2)$ the inverse map of $x_2 \mapsto T_2(x_1, x_2)$, holding x_1 fixed.

We assume that both T_1^{-1} and T_2^{-1} are either known in closed form, or can easily be obtained by numerical computation.

Two alternative simulators

For $ij \in [I \times J]$ ($J = 2$), we draw x_{ij}, an i.i.d. sample of uniform random variables over $[0,1]$. One may compute π using the *naive accept-reject simulator*

$$\hat{\pi}_I^{NAR} = \frac{1}{I} \sum_{i \in [I]} \mathbf{1}\left\{T_1\left(x_{i1}\right) \leq z_1\right\} \mathbf{1}\left\{T_2\left(x_{i1}, x_{i2}\right) \leq z_2\right\}$$

whose variance is $\pi\left(1-\pi\right)/I$.

However, note that one does not have to sample two dimensions of (X_1, X_2); indeed, $\pi = \mathbb{E}[\mathbf{1}\{X_1 \leq T_1^{-1}(z_1)\} T_2^{-1}(X_1, z_2)]$, and therefore one may draw I samples x_i from the $\mathcal{U}([0,1])$ distribution and define the *integrated accept-reject simulator*

$$\hat{\pi}_I^{IAR} = \frac{1}{I} \sum_{i \in [I]} \mathbf{1}\{x_i \leq T_1^{-1}(z_1)\} T_2^{-1}(x_i, z_2).$$

Importance sampling simulator (1/2)

One may argue that $\pi = \pi_1 \pi_{2|1}$ where $\pi_1 = \Pr\left(Z_1 \leq z_1\right)$, and $\pi_{2|1} = \Pr\left(Z_2 \leq z_2 \mid Z_1 \leq z_1\right)$. Letting $X \sim \mathcal{U}([0,1]^2)$, and denoting $Z_1 = T_1(X_1)$ and $Z_2 = T_2(X_1, X_2)$, we have

$$\pi_1 = T_1^{-1}(z_1)$$

as well as $\Pr\left(Z_2 \leq z_2 \mid Z_1 \leq z_1\right) = \Pr\left(T_2\left(X_1, X_2\right) \leq z_2 \mid X_1 \leq T_1^{-1}(z_1)\right)$. However, the distribution of $X_1 \sim \mathcal{U}([0,1])$ conditional on $X_1 \leq T_1^{-1}(z_1)$ is simply the uniform distribution over $[0, T_1^{-1}(z_1)]$, which is the unconditional distribution of $\tilde{X}_1 T_1^{-1}(z_1)$ for $\tilde{X}_1 \sim \mathcal{U}([0,1])$. Therefore $\Pr(Z_2 \leq z_2 \mid Z_1 \leq z_1) = \Pr(T_2(\tilde{X}_1 T_1^{-1}(z_1), \tilde{X}_2) \leq z_2)$, where $(\tilde{X}_1, \tilde{X}_2) \sim \mathcal{U}([0,1]^2)$, which is equal to $\Pr(\tilde{X}_2 \leq T_2^{-1}(\tilde{X}_1 T_1^{-1}(z_1), z_2))$, and therefore

$$\pi_{2|1} = \mathbb{E}\left[T_2^{-1}(\tilde{X}_1 T_1^{-1}(z_1), z_2))\right]$$

Importance sampling simulator (2/2)

Still using the i.i.d. sample of I draws $\tilde{x}_i$, $i \in [I]$ from the uniform distribution over $[0,1]$, we get the *importance sampling simulator*

$$\hat{\pi}_I^{IS} = \frac{\pi_1}{I} \sum_{i \in [I]} T_2^{-1}(\tilde{x}_i T_1^{-1}(z_1), z_2)).$$

More generally, for $J \geq 2$, we denote $T_j^{-1}\left(x_1, \ldots, x_{j-1}, z_j\right)$ the inverse of $x_j \mapsto T_j\left(x_1, \ldots, x_{j-1}, x_j\right)$ at z_j. In the Gaussian case, when $\mathcal{Q} = \mathcal{N}\left(0, \Sigma\right)$ and L is the Cholesky lower triangular matrix such that $LL^\top = \Sigma$, we have

$$\begin{cases} T_j\left(x\right) = L_{j1}\Phi^{-1}(x_1) + \cdots + L_{j,j-1}\Phi^{-1}(x_{j-1}) + L_{jj}\Phi^{-1}(x_j) \\ T_j^{-1}\left(x_1, \ldots, x_{j-1}, z_j\right) = \Phi(-\frac{L_{j1}}{L_{jj}}\Phi^{-1}(x_1) - \cdots - \frac{L_{j,j-1}}{L_{jj}}\Phi^{-1}(x_{j-1}) + \frac{z_j}{L_{jj}}). \end{cases}$$

Lemma A.2.3

We have

$$\Pr(Z_1 \leq z_1, \ldots, Z_J \leq z_J) = \mathbb{E}[\prod_{j \in [J]} T_j^{-1}\left(\hat{X}_1, \hat{X}_2, \ldots, \hat{X}_{j-1}, z_j\right)].$$

where, given $\tilde{X} \sim \mathcal{U}([0,1]^J)$, the J-dimensional random vector $\hat{X}$ is defined recursively by

$$\begin{cases} \hat{X}_1 = \tilde{X}_1 \; T_1^{-1}(z_1) \\ \hat{X}_2 = \tilde{X}_2 \; T_2^{-1}\left(\hat{X}_1, z_2\right) \\ \vdots \\ \hat{X}_J = \tilde{X}_J \; T_J^{-1}\left(\hat{X}_1, \ldots, \hat{X}_{J-1}, z_J\right). \end{cases}$$

Conclusion of importance sampling (2/2)

This lemma is the basis of the GHK (Geweke–Hajivassiliou–Keane) algorithm, described in example 4.1. Numerical studies show that the variance of the importance sampling simulator (that is, in the context of discrete choice, the GHK simulator) is typically much lower than the variance of the accept-reject simulator.

Appendix A.2.4. Numerical integration by Gauss-Hermite quadrature

Section content

- The Hermite polynomials
- Gauss-Hermite quadrature

Hermite polynomial

The (physicist's) Hermite polynomials are the polynomials H_N such that

$$H_N\left(x\right) = \left(-1\right)^N e^{x^2} \frac{d^N}{dx^N} e^{-x^2}$$

from which we can deduce the recurrence relation

$$H_{N+1}\left(x\right) = 2xH_N\left(x\right) - 2NH_{N-1}\left(x\right),$$

and one can show that $H_N(x)$ is a polynomial of degree N with 2^N as its leading coefficient. We have

$$\int H_N\left(x\right) H_M\left(x\right) \frac{e^{-x^2}}{\sqrt{\pi}} dx = 2^N N! \text{ if } N = M \text{ and } = 0 \text{ otherwise.}$$

Thus the Hermite polynomials $H_0, H_1, ..., H_N$ form an orthogonal basis of the space of polynomials of degree N.

Gauss-Hermite quadrature

If x_n, $n \in [N]$ are the roots of H_N, then a good approximation of the integral of $f(x)\exp(-x^2)$ over the real line is provided by

$$\int_{-\infty}^{+\infty} f(x)\, e^{-x^2} dx \approx \sum_{n \in [N]} w_n f(x_n) \text{ where } w_n = \frac{2^{N-1} N!}{N^2 \left(H_{N-1}(x_n)\right)^2},$$

and the approximation is exact if f is a polynomial of degree at most $2N-1$. As a result, if $X \sim \mathcal{N}(0,1)$, the expectation $\mathbb{E}[f(X)]$ is approximated by

$$\int_{-\infty}^{+\infty} f(x) \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = \int_{-\infty}^{+\infty} f\left(\sqrt{2}x\right) \frac{e^{-x^2}}{\sqrt{\pi}} dx \approx \frac{1}{\sqrt{\pi}} \sum_{n \in [N]} w_n f\left(\sqrt{2}x_n\right).$$

Appendix A.2.5. M-estimation

Section content

- M-estimation

M-estimation is a general class of estimation methods that includes, as special cases, maximum likelihood estimation, least squares, and many others. An M-estimator of a parameter vector $\theta \in \mathbb{R}^K$ is defined as the solution to the following optimization problem:

$$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}^K} \sum_{i \in [I]} m(Z_i, \theta),$$

where $\{Z_i\}_{i \in [I]}$, which are vectors of $\mathbb{R}^P$, are observed data, and $m(z, \theta)$ is a real-valued function often referred to as the objective function or criterion function.

Several assumptions

(A1) The data $\{Z_i\}_{i\in[I]}$ are independently and identically distributed from a probability distribution $\mathcal{P}$.

(A2) The function $\theta \mapsto m(z,\theta)$ is twice continuously differentiable with respect to θ for every $z \in \mathbb{R}^P$.

(A3) The function $z \mapsto m(z,\theta)$ is Borel measurable for every $\theta \in \mathbb{R}^K$.

(A4) One has $\mathbb{E}_{\mathcal{P}}[\sup_{\theta\in\Theta} m(Z,\theta)] < +\infty$ for every compact subset Θ of $\mathbb{R}^K$.

(A5) The true parameter value $\theta_0 \in \mathbb{R}^K$ is the only minimizer of the population analog of problem (A.2.20), namely $\theta \mapsto \mathbb{E}_{\mathcal{P}}[m(Z,\theta)]$.

(A6) One has $\mathbb{E}_{\mathcal{P}}\left[\sup_{\theta\in\Theta} |\partial^2 m(Z,\theta)/\partial\theta_k\partial\theta_l|\right] < +\infty$ for every compact subset Θ of $\mathbb{R}^K$ and every k and l in $[K]$.

(A7) Each random variable $\partial m(Z,\theta_0)/\partial\theta_k$ has finite second moments for $k \in [K]$.

Theorem A.2.3

Let $\hat{\theta}$ be an M-estimator of θ based on a sample of size I. Then:

(i) Under assumptions (A1) to (A4), the uniform law of large numbers holds: the sample criterion function converges uniformly in probability to its expectation, i.e., $\sup_{\theta \in \Theta} \left| \frac{1}{I} \sum_{i \in [I]} m(Z_i, \theta) - \mathbb{E}_{\mathcal{P}}[m(Z, \theta)] \right| \xrightarrow{p} 0$ holds as $I \to \infty$ for every compact subset Θ of $\mathbb{R}^K$.

(ii) Under assumptions (A1) to (A5), consistency holds: $\hat{\theta} \xrightarrow{p} \theta_0$ as $I \to \infty$.

(iii) Under assumptions (A1) to (A7), we have the asymptotic normality of $\hat{\theta}$, that is $\sqrt{I}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, H^{-1} C H^{-1})$, where the matrices H and C are given by $H_{kl} = \mathbb{E}_{\mathcal{P}} \left[\frac{\partial^2 m(Z, \theta_0)}{\partial \theta_k \partial \theta_l} \right]$ and $C_{kl} = \mathbb{E}_{\mathcal{P}} \left[\frac{\partial m(Z, \theta_0)}{\partial \theta_k} \frac{\partial m(Z, \theta_0)}{\partial \theta_l} \right]$.

Appendix A.2.6. Generalized method of moments

Section content

- The GMM estimator
- Large sample properties
- Efficient weighting

Description

Generalized method of moments (GMM) consists of estimating $\theta^{\star} \in \mathbb{R}^K$ based on the L moment restrictions

$$\mathbb{E}\left[m_{l\tilde{\omega}}(\theta^{\star})\right] = 0, \forall l \in [L],$$

for a data-generating process $\tilde{\omega} \sim \mathcal{P}$. Our data is assumed to be an i.i.d. sample $\{\omega_i, i \in [I]\}$ drawn from $\mathcal{P}$. Throughout we shall denote $\tilde{m}(\theta^{\star})$ the L-dimensional random vector $(m_{l\tilde{\omega}})_l$ where $\tilde{\omega} \sim \mathcal{P}$.

The GMM estimator (1/2)

Assume that the support of $\mathcal{P}$ has cardinality $\Omega \in \mathbb{N}$ and can be described by $[\Omega] = \{1, \ldots, \Omega\}$. Then for $\omega \in [\Omega]$, we define $\hat{\pi}_\omega = I^{-1} \sum_i 1\{\omega_i = \omega\}$ the empirical frequency of outcome ω. Letting $\pi_\omega = \Pr_{\mathcal{P}}(\tilde{\omega} = \omega)$ be the population analog of the latter quantity, which forms a column vector π of size $\Omega \times 1$, and denoting $M(\theta)$ the $L \times \Omega$ matrix of term $m_{l\omega}(\theta)$,

$$M(\theta^\star)\pi = 0,$$

and given a symmetric positive definite matrix W of size $L \times L$,

$$\theta^\star = \arg\min_{\theta \in \mathbb{R}^K} F(\theta, \pi)$$

where for $\theta \in \mathbb{R}^K$ and π a Ω-dimensional probability column vector, F is defined by

$$F(\theta, \pi) = \frac{1}{2} \pi^\top M(\theta)^\top W M(\theta) \pi,$$

The GMM estimator (2/2)

allowing us to define the GMM estimator of $\theta^{\star}$ as

$$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}^K} F\left(\theta, \hat{\pi}\right),$$

whose first order conditions are $\partial F(\hat{\theta}, \hat{\pi}) / \partial \theta_k = 0$ for $k \in [K]$, where

$$\frac{\partial F(\theta, \pi)}{\partial \theta_k} = \pi^{\top} M(\theta)^{\top} W \frac{\partial M}{\partial \theta_k}(\theta) \pi.$$

Large sample properties (1/2)

Assume from now on that $M(\theta)$ is linear in θ, and let $\boldsymbol{G}$ be the $L \times K$ matrix of term $G_{lk} = \sum_{\omega} \frac{\partial M_{l\omega}}{\partial \theta_k} \pi_{\omega}$, which therefore does not depend on θ, and which is such that $\mathbb{E}[\tilde{m}(\theta)] = \boldsymbol{G}\theta$. Letting $\boldsymbol{M}$ be the $L \times \Omega$ matrix of term $M_{l\omega} = M_{l\omega}(\theta^{\star})$,

$$D^2_{\theta\pi} F(\theta^{\star}, \pi) = \boldsymbol{G}^{\top} W \boldsymbol{M} \text{ and } D^2_{\theta\theta} F(\theta^{\star}, \pi) = \boldsymbol{G}^{\top} W \boldsymbol{G}$$

and a first-order expansion of $\partial_{\theta} F(\theta^{\star}, \pi) = 0$ around the true θ and the population π yields

$$\begin{aligned} \hat{\theta} - \theta^{\star} &= -\left(\partial^2_{\theta\theta} F(\theta^{\star}, \pi)\right)^{-1} \partial^2_{\theta\pi} F(\theta^{\star}, \pi)(\hat{\pi} - \pi) + o_P(I^{-1/2}) \\ &= -\left(\boldsymbol{G}^{\top} W \boldsymbol{G}\right)^{-1} \left(\boldsymbol{G}^{\top} W \boldsymbol{M}\right)(\hat{\pi} - \pi) + o_P(I^{-1/2}). \end{aligned}$$

Large sample properties (1/2)

Now with i.i.d. data we have that the variance-covariance matrix of $\hat{\pi}$ is V_π / I where $V_\pi := (\Delta_\pi - \pi\pi^\top)$, and therefore the variance-covariance matrix of $\boldsymbol{M}\hat{\pi} = I^{-1} \sum_i m_{l\omega_i}(\theta^\star)$ is Σ / I, where the $L \times L$ matrix Σ is defined by

$$\Sigma = \boldsymbol{M} V_\pi \boldsymbol{M}^\top,$$

which is the variance-covariance matrix of $\tilde{m}(\theta^\star)$, the L-dimensional random vector $(m_{l\tilde{\omega}})_l$. Thus

$$\sqrt{I}(\hat{\theta} - \theta^\star) \xrightarrow{d} \mathcal{N}\left(0, \left(\boldsymbol{G}^\top W \boldsymbol{G}\right)^{-1} \boldsymbol{G}^\top W \Sigma W^\top \boldsymbol{G} \left(\boldsymbol{G}^\top W \boldsymbol{G}\right)^{-1}\right).$$

Efficient weighting (1/2)

Letting $F = \Sigma^{-1/2} G$ which is a $L \times K$ matrix of rank $K \leq L$, we have $I_L \geq F \left(F^\top F\right)^{-1} F^\top$ and therefore for any symmetric definite positive matrix S, one has $F^\top S^2 F \geq \left(F^\top SF\right)\left(F^\top F\right)^{-1}\left(F^\top SF\right)$, and therefore $\left(F^\top SF\right)^{-1} F^\top S^2 F \left(F^\top SF\right)^{-1} \geq \left(F^\top F\right)^{-1}$. Taking $S = \Sigma^{1/2} W \Sigma^{1/2}$ and recalling the expression of F yields

$$\left(G^\top WG\right)^{-1} G^\top W \Sigma W G \left(G^\top WG\right)^{-1} \geq \left(G^\top \Sigma^{-1} G\right)^{-1},$$

which implies the estimator $\hat{\theta}$ achieves minimum variance for $W = \Sigma^{-1}$, in which case it is called *efficient GMM estimator*.

Theorem A.2.4

The efficient GMM estimator is obtained for $W = \Sigma^{-1}$, in which case

$$\sqrt{I}(\hat{\theta} - \theta^{\star}) \xrightarrow{d} \mathcal{N}\left(0, \left(\boldsymbol{G}^{\top}\Sigma^{-1}\boldsymbol{G}\right)^{-1}\right). \tag{A.2.31}$$

Note that Σ is the variance-covariance matrix of $\tilde{m}(\theta^{\star})$, which requires knowledge of the true value of the parameter, and thus efficient GMM estimation is therefore not directly feasible. In practice, a first consistent estimator of θ is obtained using a reasonable generic choice of the matrix W, and that parameter value is then plugged in to compute a consistent estimator of Σ.

Appendix A.2.7. Instrumental variables

Section content

- Just identified
- Overidentified
- Two-stage-least-square estimator
- GMM estimation

Description

Our dataset consists in for each observation $i \in [I]$, $y_i \in \mathbb{R}$ the dependent variable, $x_i \in \mathbb{R}^K$ the vector of (potentially endogenous) regressor, and $z_i \in \mathbb{R}^L$ the vector of instruments. We assume that the data-generating process is $\tilde{y} = \tilde{x}^\top \beta^\star + \tilde{\epsilon}$ where the error $\tilde{\epsilon}$ is assumed to be orthogonal to the vector of instruments, that is $\mathbb{E}[\tilde{\epsilon}\tilde{z}] = 0$, or in other words, $\beta^\star$ satisfies the moment restriction

$$\mathbb{E}[(\tilde{y} - \tilde{x}^\top \beta)\tilde{z}] = 0. \tag{A.2.32}$$

Denoting Y the column vector of size I of entries y_i, X the $I \times K$ matrix whose rows are the x_i's, and Z be the $I \times L$ matrix whose rows are the z_i's, we obtain the sample analog of (A.2.32), namely

$$Z^\top (Y - X\beta) = 0. \tag{A.2.33}$$

Just identified case

Standard instrumental variable regression assumes that there are as many instruments as parameters: $L = K$, which is the just identified case. Then there is sufficient dependence between the instruments and the regressors so that the $K \times K$ matrix $\mathbb{E}[\tilde{z}\tilde{x}^\top]$ be of full rank. In that case, restriction (A.2.33) yields the standard IV estimator, which is given by

$$\hat{\beta}^{IV} = (Z^\top X)^{-1} Z^\top Y.$$

Overidentified case

Often however, there are more restrictions than parameters, so $L > K$. In that case, β provided by (A.2.33) is overidentified, and one needs to essentially pick K among L instrumental restrictions. This is done by left-multiplying (A.2.33) by some $K \times L$ selection matrix P so that the $K \times K$ matrix $PZ^\top$ is invertible, and $PZ^\top (Y - X\beta) = 0$ yields $\beta = (PZ^\top X)^{-1} PZ^\top Y$.

Two-stage-least-square estimator

The well-known two-stage-least-square estimator provides a first rationale for obtaining the selection matrix P. It consists of forming the best linear predictor of X from Z, call it $\hat{X}$, and regressing Y on $\hat{X}$. The best linear predictor is obtained by regressing X on Z: $X = Z\delta + \eta$, which gives $\delta = \left(Z^\top Z\right)^{-1} Z^\top X$, which yields $\hat{X} = Z\delta = Z\left(Z^\top Z\right)^{-1} Z^\top X$. Note that the $K \times K$ matrix $Z\left(Z^\top Z\right)^{-1} Z^\top$ is the orthogonal projection matrix on the linear space spanned by the row of Z, and our selection matrix P introduced above has expression $P = X^\top Z\left(Z^\top Z\right)^{-1}$, which leads us to the two-stage-least-square estimator

$$\hat{\beta}^{2SLS} = (X^\top Z\left(Z^\top Z\right)^{-1} Z^\top X)^{-1} X^\top Z\left(Z^\top Z\right)^{-1} Z^\top Y.$$

GMM estimation (1/3)

The parameter denoted θ is simply the regression parameter β, so that $\theta = \beta$, and we look for the value of β that minimizes

$$\mathbb{E}[\epsilon(\beta)z]^\top W \mathbb{E}[\epsilon(\beta)z] \tag{A.2.34}$$

where the weighting matrix W is the symmetric positive definite matrix of size $L \times L$, and where $\epsilon(\beta) = y - x^\top \beta$. The sample analog of (A.2.34) is

$$\min_{\beta \in \mathbb{R}^K} (Y - X\beta)^\top ZWZ^\top (Y - X\beta),$$

and the objective function F is $F(\beta) = (Y - X\beta)^\top ZWZ^\top (Y - X\beta)/I^2$. Optimality conditions yield

$$\beta^\star = \left(X^\top \Pi X\right)^{-1} \left(X^\top \Pi Y\right) \text{ and } F(\beta^\star) = Y^\top \Upsilon Y / I^2$$

where $\Pi = ZWZ^\top$, and $\Upsilon = \Pi - \Pi^\top X \left(X^\top \Pi X\right)^{-1} X^\top \Pi$.

The $L \times \Omega$ matrix-valued function M becomes

$$M_{l\omega}(\beta) = z_{l\omega}\left(y_\omega - \sum_{k\in[K]} x_{k\omega}\beta_k\right),$$

and the sample analog of $M(\beta)\mu$ is $M(\beta)\hat{\mu} = Z^\top(Y - X\beta)/I$. It is straightforward to verify that $G = \mathbb{E}[\tilde{z}\tilde{x}^\top]$, whose sample analog is

$$\hat{G} = Z^\top X/I,$$

and that $\Sigma = \mathbb{V}\left(\tilde{z}\tilde{\epsilon}\right)$ is the variance-covariance matrix of the random vector $\tilde{z}\tilde{\epsilon}$, whose sample analog is

$$\Sigma = \hat{M}(\beta^\star)(\frac{\boldsymbol{I}_I}{I} - \frac{\mathbf{1}_I\mathbf{1}_I^\top}{I^2})\hat{M}(\beta^\star)^\top \tag{A.2.37}$$

where $\hat{M}(\beta)$ is the $L \times I$ matrix whose (l,i)-term is $\hat{M}_{li}(\beta) = z_{li}y_i - z_{li}\sum_{k\in[K]} x_{k\omega}\beta_k$.

GMM estimation (3/3)

Expression (A.2.37) is not feasible, as it requires the knowledge of $\beta^{\star}$, but one can use instead a consistent estimator provided for example by the two-stage-least square estimator, and thus

$$\hat{\Sigma} = \hat{M}(\hat{\beta}^{2SLS})(\frac{\mathbf{I}_I}{I} - \frac{\mathbf{1}_I \mathbf{1}_I^{\top}}{I^2})\hat{M}(\hat{\beta}^{2SLS})^{\top}.$$

The first order conditions associated with the generalized method of moments estimator is $(Y - X\beta)^{\top} ZWG = 0$, which, given the expression of G, yields

$$\hat{\beta}^{GMM}(W) = (X^{\top} ZWZ^{\top} X)^{-1} X^{\top} ZWZ^{\top} Y.$$

Conclusion

The generalized method of moments consists of taking the selection matrix $P = X^\top ZW$. The two-stage-least square estimator, which has $P = X^\top Z\left(Z^\top Z\right)^{-1}$, is obtained as a GMM estimator with weight matrix $W = \left(Z^\top Z\right)^{-1}/I$.

$$\begin{cases} W_1 = (\frac{Z^\top Z}{I})^{-1} \\ \hat{\beta}^{2SLS} = \hat{\beta}^{GMM}(W_1) \\ \hat{\Sigma} = \hat{M}(\hat{\beta}^{2SLS})(\frac{I_I}{I} - \frac{1_I 1_I^\top}{I^2})\hat{M}(\hat{\beta}^{2SLS})^\top \\ W_2 = \hat{\Sigma}^{-1} \\ \hat{\beta}^{IV-GMM} = \hat{\beta}^{GMM}(W_2). \end{cases}$$

It follows that

$$\sqrt{I}(\hat{\beta}^{IV-GMM} - \beta^\star) \xrightarrow{d} \mathcal{N}\left(0, \left(\left(\frac{X^\top Z}{I}\right)\hat{\Sigma}^{-1}\left(\frac{Z^\top X}{I}\right)\right)^{-1}\right).$$

Appendix A.2.8. Generalized linear models

Section content

- GLM specification
- GLM estimator
- Weighted GLMs
- Coercivity and existence of a GLM estimator

GLM specification (1/2)

Generalized linear models (GLMs) extend linear regressions in the sense that if $\tilde{\mu}$ is the dependent variable and $\tilde{\rho}$ the P-dimensional column vector of explanatory variables (regressors), the linear specification $\mathbb{E}[\tilde{\mu} \mid \tilde{\rho}] = \tilde{\rho}^\top \theta$ is replaced by $f(\mathbb{E}[\tilde{\mu} \mid \tilde{\rho}]) = \tilde{\rho}^\top \theta$, where the index $\tilde{\rho}^\top \theta$ is often called the *score* associated with covariates $\tilde{\rho}$. The function f, which is called the *link function*, is increasing and continuous from an open (non necessarily bounded) interval $(\mu_l, \mu_u) \subseteq \mathbb{R}$ onto $\mathbb{R}$; it is therefore invertible,

Definition A.2.2

The generalized linear model with link function f, with dependent variable $\tilde{\mu}$ and with vector of explanatory variables $\tilde{\rho}$ is given by the relationship

$$\mathbb{E}[\tilde{\mu} \mid \tilde{\rho}] = f^{-1}\left(\tilde{\rho}^\top \theta\right). \tag{A.2.39}$$

GLM specification (2/2)

Much of the appeal of generalized linear models comes from the possibility of computing them as a convex optimization problem. Indeed, letting F be the primitive of f, it follows from convex analysis that the primitive of f^{-1} is F^*, the Legendre–Fenchel conjugate of F. As a result, the relation (A.2.39) recasts as $\mathbb{E}\left[\tilde{\mu} \mid \tilde{\rho}\right] = \left(F^*\right)' \left(\tilde{\rho}^\top \theta\right)$, from which it follows that the true parameter θ is the solution to

$$\max_{\theta \in \mathbb{R}^P} \mathbb{E}\left[\tilde{\mu}^\top \tilde{\rho}^\top \theta - F^*\left(\tilde{\rho}^\top \theta\right)\right]. \tag{A.2.40}$$

Theorem A.2.5

The GLM estimator $\hat{\theta}$ is obtained as the solution to

$$\max_{\theta \in \mathbb{R}^P} \left\{ \hat{\mu}^\top R\theta - \sum_{\omega \in [\Omega]} F^* \left((R\theta)_\omega \right) \right\}, \tag{A.2.41}$$

whose dual expression is

$$\min_{\mu \geq 0} \sum_{\omega \in [\Omega]} F \left(\mu_\omega \right) \quad \text{s.t. } \mu^\top R = \hat{\mu}^\top R.$$

The estimator $\hat{\theta}$ solution satisfies $\sum_{\omega \in [\Omega]} \left(f^{-1}((R\hat{\theta})_\omega) - \hat{\mu}_\omega \right) R_{\omega p} = 0$, for all $p \in [P]$, and the solution $\bar{\mu}_\omega$ satisfies $\bar{\mu}_\omega = f^{-1}((R\hat{\theta})_\omega)$.

Weighted GLMs

We shall sometimes need to place different weights w_ω to different observations ω. Letting $\mathbf{\Delta}_w = diag\,(w)$, the *weighted GLM estimator* now becomes

$$\max_{\theta \in \mathbb{R}^P} \left\{ \hat{\mu}^\top \mathbf{\Delta}_w R\theta - w^\top F^* \left(R\theta \right) \right\},$$

whose first-order conditions are $\sum_\omega w_\omega \left(\hat{\mu}_\omega - f^{-1}\left((R\theta)_\omega \right) \right) R_{\omega p} = 0$, and whose dual formulation consists of minimizing $\sum_\omega w_\omega F(\mu_\omega)$ subject to $\mu^\top \mathbf{\Delta}_w R = \hat{\mu}^\top \mathbf{\Delta}_w R$. We obtain

$$\min_{\mu \geq 0} \sum_{\omega \in [\Omega]} w_\omega F(\mu_\omega) \tag{15}$$

$$\text{s.t. } \mu^\top \acute{}_w R = \hat{\mu}^\top \acute{}_w R.$$

Coercivity and existence of a GLM estimator

Take for instance $F^*(z) = \exp(z)$ with a single observation and a single parameter dimension, and assume $R = 1$. Then:

- if $\mu = -1$, the maximum in (A.2.41) is infinite, as the limit of the objective function when $\theta \to -\infty$ is $+\infty$;
- likewise, if $\mu = 0$, the value of problem (A.2.41) is zero;
- finally, if $\mu = 1$, the problem has a maximizer which is $\theta = 0$ (and hence the problem has a finite value).

Existence of a GLM estimator

Theorem A.2.6

Assume that $f^{-1}(w) > 0$ for all $w \in \mathbb{R}$, and that $F^{*}(w) \to 0$ as $w \to -\infty$ and $F^{*}(w)/w \to +\infty$ as $w \to +\infty$, and consider problem (A.2.41). Then:

(i) the value of the problem is finite if and only if there is a vector $\bar{\mu}$ such that $\bar{\mu}_{\omega} \geq 0$ for all $\omega \in [\Omega]$ and $\bar{\mu}^{\top} R = \hat{\mu}^{\top} R$.

(ii) the problem has a maximizer if and only if there is a vector $\bar{\mu}$ such that $\bar{\mu}_{\omega} > 0$ for all $\omega \in [\Omega]$ and $\bar{\mu}^{\top} R = \hat{\mu}^{\top} R$.

Coercivity detector (1/2)

The following criterion, based on linear programming, allows for a simple test of conditions (i) and (ii) in theorem above. Consider the following linear programming problem, called *coercivity detector*

$$V = \max_{t \geq 0, \bar{\mu}_\omega \geq 0} t \qquad \text{subject to } t\mathbf{1}_\Omega - \bar{\mu} \leq 0 \text{ and } \bar{\mu}^\top R = \hat{\mu}^\top R, \tag{A.2.45}$$

which is always feasible, as $t = 0$, $\bar{\mu} = \mu$ satisfies the constraints. By strong duality, the value of the coercivity detector coincides with the value of its dual expression

$$V = \min_{u_\omega \geq 0, \lambda_k} \lambda^\top Z\mu \qquad \text{subject to } \mathbf{1}_\Omega^\top u \geq 1 \text{ and } u \leq R\lambda, \tag{A.2.46}$$

and inspection of the dual problem reveals that it is not always feasible: for instance, when $R = 0$, there is no vector $u_\omega \geq 0$ such that $u \leq R\lambda$ and $\sum_\omega u_\omega \geq 1$. Hence V can take values in $\mathbb{R}_+ \cup \{+\infty\}$.

Theorem A.2.7

Consider V, the value of both the primal problem (A.2.45) and its dual (A.2.46). Then:

(i) $V = +\infty$ if and only if problem (A.2.41) is infinite.

(ii) $0 < V < +\infty$ if and only if problem (A.2.41) has a finite value and a maximizer.

(iii) $V = 0$ if and only if problem (A.2.41) has a finite value but no maximizer.

Appendix A.2.9. Poisson regression

Section content

- Poisson dependence
- Weighted Poisson regression
- Asymptotics

Poisson distribution

The random variable $\tilde{\mu}$ valued in $\mathbb{N}$ follows a *Poisson distribution* of parameter m if

$$\Pr\left(\tilde{\mu} = \mu\right) = e^{-m} \frac{m^{\mu}}{\mu!}.$$

The Poisson distribution has mean m and variance m. If $\rho \in \mathbb{R}^{P}$ is a vector of covariates, we assume that the conditional distribution of $\tilde{\mu}$ given $\tilde{\rho} = \rho$ is a Poisson distribution of parameter $m\left(\rho\right) = \exp\left(\rho^{\top}\theta\right)$.

Poisson dependence

Definition A.2.3 (Poisson dependence)

Assume that the conditional distribution of μ given ρ is given by

$$\Pr\left(\tilde{\mu} = \mu \mid \tilde{\rho} = \rho\right) = \exp(-\exp(\rho^\top \theta)) \frac{\exp(\mu \rho^\top \theta)}{\mu!}.$$

It follows from the Poisson dependence relation that both the conditional expectation and the variance of $\tilde{\mu}$ given $\tilde{\rho} = \rho$ are equal to the parameter $m\left(\rho\right) = \exp\left(\rho^\top \theta\right)$, therefore

$$\mathbb{E}[\tilde{\mu} = \mu \mid \tilde{\rho} = \rho] = \exp\left(\rho^\top \theta\right) \quad \text{and} \quad \mathbb{V}\left[\tilde{\mu} = \mu \mid \tilde{\rho} = \rho\right] = \exp\left(\rho^\top \theta\right).$$

If we observe a sample of observations $(\hat{\mu}_\omega, \rho_\omega)_{\omega \in [\Omega]}$ indexed by $\omega \in [\Omega]$, the log-likelihood of observation ω is given by $\mu_\omega \rho_\omega^\top \theta - \exp\left(\rho_\omega^\top \theta\right)$, and therefore

$$\ell(\theta) = \hat{\mu}^\top R\theta - \sum_{\omega \in [\Omega]} \exp\left(\rho_\omega^\top \theta\right), \tag{A.2.49}$$

where the design matrix R is the $\Omega \times P$ matrix whose rows are the ρ_ω's.

Definition A.2.4

The Poisson regression consists of the maximization of the log-likelihood function (A.2.49). The Poisson regression estimator $\hat{\theta}$ is thus the parameter vector θ solution to

$$\max_{\theta \in \mathbb{R}^P} \left\{ \hat{\mu}^\top R\theta - \sum_{\omega \in [\Omega]} \exp\left((R\theta)_\omega\right) \right\}. \tag{A.2.50}$$

Poisson regression (2/2)

A Poisson regression can be recast as a *generalized linear model* (GLM) with the logarithm as link function. Then $f^{-1} = \exp$, and thus $F^{*} = \exp$ and expression (A.2.41) boils down to the Poisson regression. A primitive F is $F\left(m\right) = m \log m$, and by the dual of the Poisson regression thus formulates as

$$\begin{array}{ll} \min_{\mu \geq 0} & \sum_{\omega \in [\Omega]} \mu_{\omega} \log \mu_{\omega} \\ \text{s.t.} & \mu^{\top} \Phi = \hat{\mu}^{\top} \Phi. \end{array}$$

Weighted Poisson regression

For the weighted Poisson regression with weight w_ω placed on observation ω, the primal formulation (A.2.50) becomes

$$\max_{\theta \in \mathbb{R}^P} \left\{ \hat{\mu}^\top \mathbf{\Delta}_w R\theta - w^\top \exp\left(R\theta\right) \right\} \tag{A.2.52}$$

where $\mathbf{\Delta}_w = diag(w)$, and the dual formulation becomes

$$\min_{\mu \geq 0} \sum_{\omega \in [\Omega]} \mu_\omega \log \frac{\mu_\omega}{w_\omega}$$

$$\text{s.t. } \mu^\top \Phi = \hat{\mu}^\top \Phi.$$

Asymptotics (1/3)

It will be useful to investigate the asymptotic distribution of $\hat{\theta}$ under a particular setting, namely when Ω remains fixed, but when $N = \sum_{\omega} \hat{\mu}_{\omega}$ gets large. In that case, we need to introduce the concept of *homogeneous* weighted Poisson regression, as follows. We shall assume that

$$\hat{\pi} := \frac{\hat{\mu}}{N}$$

converges almost surely to π, and that the following convergence in distribution holds

$$\sqrt{N} \left(\hat{\pi} - \pi \right) \xrightarrow{d} \mathcal{N} \left(0, V_{\pi} \right).$$

Definition A.2.5

A weighted Poisson regression with design matrix R and vector of weights $w \in \mathbb{R}_{+}^{\Omega}$ is called homogeneous if $1_{\Omega} \in \mathrm{Im} R$, which implies that there is a unique vector $v \in \mathbb{R}^{P}$ such that $Rv = 1_{\Omega}$.

Asymptotics (2/3)

In that case, we consider the Poisson regression of $\hat{\mu}$ on R with weights w, which formulates as problem (A.2.52) above. Letting $\mu^{\theta} = \exp(R\theta)$, the optimality conditions that characterize $\hat{\theta}$ are

$$R^{\top}\mathbf{\Delta}_{w}\mu^{\hat{\theta}} = R^{\top}\mathbf{\Delta}_{w}\hat{\mu}.$$

Now assume that we rescale the dependent variable, and we define $\hat{\mu}_K = \hat{\mu}/K$. Let $\hat{\theta}_K$ the resulting estimator, one has $\mu^{\hat{\theta}-(\ln K)v} = \mu^{\hat{\theta}}/K$, and thus $\hat{\theta}_K = \hat{\theta} - (\ln K)v$.

Assuming the homogeneous case, we replace $\hat{\mu}$ with $\hat{\pi} = \hat{\mu}/N$ in the Poisson regression. This changes the estimator by a $(\ln N)v$ term. We therefore regress $\hat{\pi}$ on R, and denote the resulting estimator by $\hat{\theta}$, defined via $R^\top \mathbf{\Delta}_w \pi^{\hat{\theta}} = R^\top \mathbf{\Delta}_w \hat{\pi}$, where $\pi^\theta = \exp(R\theta)$. Let θ_0 be the limit of $\hat{\theta}$ as $\hat{\pi} \to \pi$. We now recall the asymptotic behavior of $\hat{\pi}$. We have $R^\top \mathbf{\Delta}_w \exp(R\hat{\theta}) = R^\top \mathbf{\Delta}_w \hat{\pi}$, which, letting $\delta\hat{\theta} = \hat{\theta} - \theta_0$ and $\delta\hat{\pi} = \hat{\pi} - \pi$, gets linearized into $R^\top \mathbf{\Delta}_w \mathbf{\Delta}_{\pi^{\theta_0}} R\delta\hat{\theta} = R^\top \mathbf{\Delta}_w \delta\hat{\pi} + o_P(\sqrt{N})$ and therefore:

Theorem A.2.8

Under the assumptions made above, and letting $B = R^\top \mathbf{\Delta}_w \mathbf{\Delta}_{\pi^{\theta_0}} R$ and $C = R^\top \mathbf{\Delta}_w \mathbf{V}_\pi \mathbf{\Delta}_w R$, the convergence $\hat{\theta} \to \theta_0$ holds almost surely as $N \to +\infty$, and

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, B^{-1}CB^{-1}\right).$$

Appendix A.2.10. Exponential families

Section content

- Exponential family
- Latent data
- A Bayesian point of view

Consider a parametric family $f(z;\theta)$ with support $\mathcal{Z} \subseteq \mathbb{R}^P$ and parameter $\theta \in \mathbb{R}^P$. Let $\Lambda : \mathbb{R}^P \to \mathbb{R}$ be a convex function, and let $\Lambda^* : \mathbb{R}^P \to \mathbb{R}$ be its Legendre transform, defined by $\Lambda^*(\theta) = \max_\zeta \{\zeta^\top \theta - \Lambda(\zeta)\}$. If we observe i.i.d. data $\{z_\omega : \omega \in [\Omega]\}$ sampled from $f(z;\theta^\star)$, we may define $\bar{z} = \Omega^{-1} \sum_\omega z_\omega$, and the problem

$$\arg\max_{\theta \in \mathbb{R}^P} \left\{ \frac{1}{\Omega} \sum_{\omega \in \Omega} z_\omega^\top \theta - \Lambda^*(\theta) \right\} \tag{A.2.58}$$

which yields $\nabla \Lambda^*(\hat{\theta}) = \bar{z}$ for $\hat{\theta}$, the optimal value of the parameter θ in (A.2.58). In that case the likelihood associated with observation z, $\log f(z;\theta)$ coincides with $z^\top \theta - \Lambda^*(\theta)$ up to an irrelevant function of z $a(z)$, and thus

$$f(z;\theta) = \exp(z^\top \theta - \Lambda^*(\theta) - a(z)).$$

Description (2/2)

Further, as Λ is the Legendre-Fenchel transform of Λ^*, the value of the likelihood at the optimal value of the parameter is equal to $\Lambda(\bar{z}) - a(\bar{z})$. Obviously, imposing that $f(.;\theta)$ should be a density requires that is sums to one when integrating over $z \in \mathcal{Z}$, and thus requires that

$$\Lambda^* (\theta) = \log \int_{\mathcal{Z}} \exp(z^\top \theta - a(z)) dz. \tag{A.2.60}$$

Two remarks

First, note that while problem (A.2.58) bears close resemblance with a generalized linear model, it is not one in a strict sense as in our exposition of generalized linear models, we have requested that the link function should be univariate; while here, the analog of the link function, which is $\nabla\Lambda^*$ is a function from $\mathbb{R}^p$ to $\mathbb{R}^p$. Another distinction is that generalized linear models do not require Λ^* to be of the form (A.2.60), which means that generalized linear models do not necessarily fall within the maximum likelihood framework.

The last point brings us to a second remark: any function $a(z)$ on $\mathcal{Z}$ certainly defines a function Λ^* by (A.2.60) that is convex on its domain, but conversely, not all convex functions Λ^* can be represented as (A.2.60). For instance, in dimension one and when $\mathcal{Z} = \mathbb{R}_-$, the Bernstein theorem imposes that the signs of the higher-order derivatives of $\theta \mapsto \exp(\Lambda^*(\theta))$ alternate in order for $a(z)$ to exist.

Example of exponential families

The following classical parametric families of distributions can be reformulated (often, after a nonlinear transformation of the variables) as exponential families:

(i) Gaussian distributions: In the univariate case, the p.d.f. of a Gaussian distribution with mean μ and variance σ^2 is proportional to $\exp(-(x-\mu)^2/(2\sigma^2))$, so this is (up to a transformation) an exponential family with $P=2$, $z_1 = x$, $z_2 = x^2$, $\theta_1 = \mu/\sigma^2$, and $\theta_2 = -1/(2\sigma^2)$, $a(z) = 0$ and $\mathcal{Z} = \{z \in \mathbb{R}^2 : z_2 = z_1^2\}$. The extension to the multivariate case is straightforward.

Example of exponential families

(ii) Poisson distribution: A Poisson distribution with parameter λ assigns a probability mass proportional to $\exp(-\lambda)\lambda^{x} = \exp(x \ln \lambda - \lambda)$ to each $x \in \mathbb{N}$, so this is an exponential family with $P = 1$, $\mathcal{Z} = \mathbb{N}$, $z = x$, $\theta = \ln \lambda$ and $\Lambda^{*}(\theta) = \exp \theta$.

(iii) Exponential distribution: An exponential distribution with parameter $\lambda \geq 0$ has p.d.f. $\lambda \exp(-\lambda x)$ for $x \geq 0$, which is an exponential family with $P = 1$, $\mathcal{Z} = \mathbb{R}_{+}$, $\theta = -\lambda$ and $\Lambda^{*}(\theta) = -\ln(-\lambda)$.

A number of other examples exist.

We denote by $\mathbb{E}_\theta$ and $\mathbb{V}_\theta$ the expectation and variance-covariance matrix with respect to the distribution $f\left(.|\theta\right)$, and we have:

Proposition A.2.7

If $\tilde{z} \sim f(.;\theta)$, then

$$\nabla\Lambda^*\left(\theta\right) = \mathbb{E}_\theta\left[\tilde{z}\right] \text{ and } D^2\Lambda^*\left(\theta\right) = \mathbb{V}_\theta\left[\tilde{z}\right],$$

and if $\{z_\omega : \omega \in [\Omega]\}$ is an i.i.d. sample of random variables drawn from distribution $f\left(z;\theta^\star\right)$ the maximum likelihood estimator $\hat{\theta}$ of $\theta^\star$ is such that $\nabla\Lambda^*(\hat{\theta}) = \bar{z}$, where $\bar{z} = \Omega^{-1}\sum_\omega z_\omega$ has been defined as the sample average of the observations.

Exponential families allow for a relatively simple way to handle missing data. Indeed, if $\mathcal{Z}' \subseteq \mathcal{Z}$ then it is easy to see that the distribution of $\tilde{z} \sim f(.;\theta)$ given $\tilde{z} \in \mathcal{Z}'$ is $f(z|\mathcal{Z}';\theta) = \exp(z^\top \theta - \Lambda^*_{\mathcal{Z}'}(\theta))$, where $\Lambda^*_{\mathcal{Z}'}(\theta) = \log \int_{\mathcal{Z}'} \exp(z^\top \theta) dz$, and one has

$$\nabla \Lambda^*_{\mathcal{Z}'}(\theta) = \mathbb{E}_\theta \left[\tilde{z} | \tilde{z} \in \mathcal{Z}' \right] \text{ and } \Pr\left(z \in \mathcal{Z}' \right) = \exp(\Lambda^*_{\mathcal{Z}'}(\theta) - \Lambda^*_{\mathcal{Z}}(\theta)).$$

Proposition A.2.8

The maximum likelihood estimator $\hat{\theta}$ based on $\{w_1, ..., w_\Omega\}$ where $w = \varphi(z)$ is characterized by

$$\frac{1}{\Omega} \sum_{\omega \in [\Omega]} \mathbb{E}_{\hat{\theta}} \left[\tilde{z} | \varphi(\tilde{z}) = w_\omega \right] = \mathbb{E}_{\hat{\theta}} \left[\tilde{z} \right].$$

A Bayesian point of view

If we put a prior distribution $\pi_0(\theta)$ on the parameter, it is easy to see that the posterior distribution of θ conditional on observing one observation $\hat{z}$ is

$$\pi(\theta|\hat{z}) = \frac{\pi_0(\theta)\exp(\hat{z}^\top \theta - \Lambda^*(\theta))}{\int \pi_0(\theta')\exp(\hat{z}^\top \theta' - \Lambda^*(\theta'))d\theta'}.$$

If one assumes a parametric form for the distribution π of θ – which is to say, if one parameterizes the distribution of the parameter, and if one lets $(\zeta, \nu) \in \mathbb{R}^P \times \mathbb{R}$ be that hyper-parameter, we may consider

$$\pi(\theta; \zeta, \nu) \propto \exp(\zeta^\top \theta - \nu \Lambda^*(\theta))$$

where the $\propto$ sign indicates that one should divide by a normalization constant. Assuming that $\pi_0(.) = \pi(.; \zeta_0, \nu_0)$, we can show that $\pi_1(\theta) = \pi(.; \zeta_1, \nu_1)$, where $\zeta_1 = \zeta_0 + \hat{z}$ and $\nu_1 = \nu_0 + 1$.

Appendix A.2.11. The EM algorithm

Section content

- EM algorithm
- Large simulated pool
- Interpretation as coordinate descent

Overview

The Expectation–Maximization (EM) algorithm is a powerful iterative method used for finding maximum likelihood estimates in statistical models where the data is incomplete or has missing values.

The core idea of the EM algorithm is to iteratively improve parameter estimates for models with latent variables or incomplete data by alternating between two steps: the Expectation (E) step and the Maximization (M) step.

In the E-step, the algorithm estimates the missing data based on the current parameters of the model, while in the M-step, it updates the model parameters to maximize the likelihood given the estimated data from the E-step.

Setup with latent variables

Consider a latent variable Z whose distribution belongs to a parametric family $f_Z(z;\theta)$. However, Z is not fully observed but only partially, in the sense that only $Y = \varphi(Z)$ is, where the function φ is not invertible.

One observes an i.i.d. sample $z_\omega = \varphi(z_\omega)$ where the latent variables z_ω are drawn from the distribution $f_Z(\cdot;\theta^\star)$. The most straightforward approach consists of denoting $f_Y(y;\theta)$ the distribution of $Y = \varphi(Z)$ if $Z \sim f_Z(\cdot;\theta)$, and carrying maximum likelihood estimation, where the log-likelihood of the sample is $\sum_\omega \log f_Y(y_\omega;\theta)$.

In the case of censored data, $\varphi(z) = \min\{Z, z_0\}$ and the log-likelihood of the sample is

$$\max_\theta \left\{ \sum_{\omega\in[\Omega]} 1\{y_\omega < z_0\} \log f_Z(y_\omega;\theta) + \sum_{\omega\in[\Omega]} 1\{y_\omega = z_0\} \log \int_{z_0}^{+\infty} f_Z(z;\theta)dz \right\}.$$

Reason to use EM algorithm (1/2)

Maximizing a quantity may however not be easy in practice due to difficulties in evaluating $\int_{z_0}^{+\infty} f_Z(z;\theta)dz$, which may involve an integration in high dimension. A first way out of this problem consists of replacing the objective function by

$$\sum_{\omega\in[\Omega]} 1\{y_\omega < z_0\} \log f_Z(y_\omega;\theta) + \sum_{\omega\in[\Omega]} 1\{y_\omega = z_0\} \log(B^{-1} \sum_{b\in[B]} 1\{z_b \geq z_0\}),$$

where the z_b's are B additional samples of simulated data drawn from $f_Z(\cdot;\theta)$, and maximizing over θ.

However, this may not be a satisfactory solution. First, a maximization problem typically requires the gradient of the objective function; second, deeper problem is that it may be difficult to sample draws z_b from $f_Z(\cdot;\theta)$.

Reason to use EM algorithm (2/2)

Let us assume that for each observation ω, we may draw a sample of simulated points $z_{b\omega}$ drawn from $f_{Z|Y}(\cdot|y_\omega;\theta)$, so that by definition, we have $\varphi(z_{b\omega}) = y_\omega$.

The distribution of each $z_{b\omega}$ is $f_{Z|Y}(\cdot|y_\omega;\theta)\, f_Y(y_\omega;\theta^\star)$, which differs from $f_Z(\cdot;\theta^\star)$ unless $\theta = \theta^\star$. If we knew $\theta^\star$, we could use it in order to draw points $z_{b\omega}$ from $f_{Z|Y}(\cdot|y_\omega;\theta^\star)$, and thus the distribution of $z_{b\omega}$ would be $f_Z(\cdot;\theta^\star)$.

Conversely, if the distribution of $z_{b\omega}$ were $f_Z(\cdot;\theta^\star)$, then maximum likelihood estimation would recover the parameter $\theta^\star$.

The EM algorithm procedure

At period t, given the current parameter estimate θ^t, one draws in the E-step (expectation step) for each $\omega \in [\Omega]$, B simulated points $z_{b\omega}$ ($b \in [B]$) sampled in an i.i.d. fashion (conditionally on y_ω) from the distribution $f_{Z|Y}(\cdot|y_\omega; \theta^t)$. Each $z_{b\omega}$ has therefore distribution $f_{Z|Y}(\cdot|y_\omega; \theta^t) f_Y(y_\omega; \theta^\star)$.

In the M-step (maximization step), we maximize the likelihood of the sample thus generated, and we update the parameter estimate into θ^{t+1}.

Large simulated pool

We shall limit $B \to +\infty$ to make a claim about the algorithm. In that case, combining the E-step and the M-step, the algorithm becomes

$$\theta^{t+1} = \arg\max_{\theta \in \mathbb{R}^p} \sum_{\omega \in [\Omega]} \mathbb{E}_{f_{Z|Y}(\cdot|y_\omega, \theta^t)} \left[\log f_Z(Z; \theta) \right]$$

where the random term Z in the expectation has distribution $f_{Z|Y}(\cdot|y_\omega, \theta^t)$, which depends on y_ω.

Example A.4

Assume that $f_Z(\cdot; \theta)$ is an exponential family as described in section A.2.10. Because of proposition A.2.8, the EM algorithm consists of updating θ^{k+1} such that

$$\mathbb{E}_{\theta^{k+1}} [Z] = \mathbb{E}_{\theta^k} [Z | y_\omega].$$

Interpretation as coordinate descent (1/2)

Letting

$$\mathcal{L}(g,\theta) = \sum_{\omega \in [\Omega]} \mathbb{E}_{g(\cdot|y_\omega)} \left[\log \frac{f_Z(Z,\theta)}{g(Z|y_\omega)} \right],$$

we note that the expression $\mathcal{L}(g,\theta)$ can be rewritten as

$$\mathcal{L}(g,\theta) = \sum_{\omega \in [\Omega]} \mathbb{E}_{g(\cdot|y_\omega)} \left[\log \frac{f_{Z|Y}(Z|y_\omega;\theta)}{g(Z|y_\omega)} \right] + \sum_{\omega \in [\Omega]} \log f_Y(y_\omega;\theta)$$

$$\text{and } \mathcal{L}(g,\theta) = \sum_{\omega \in [\Omega]} \mathbb{E}_{g(\cdot|y_\omega)} \big[\log f_Z(Z,\theta) \big] - \sum_{\omega \in [\Omega]} \mathbb{E}_{g(\cdot|y_\omega)} \big[\log g(Z|y_\omega) \big],$$

Interpretation as coordinate descent (2/2)

Proposition A.2.9

One has

$$g^t = \arg\max_g \ \mathcal{L}(g, \theta^t) \quad \text{and} \quad \theta^{t+1} = \arg\max_{\theta \in \mathbb{R}^P} \ \mathcal{L}(g^t, \theta).$$

As a result, we have $\mathcal{L}(g^{t+1}, \theta^{t+1}) \geq \mathcal{L}(g^{t+1}, \theta^t) \geq \mathcal{L}(g^t, \theta^t)$, but $\mathcal{L}(g^t, \theta^t) = \max_g \mathcal{L}(g, \theta^t)$, and $\max_g \mathcal{L}(g, \theta^t)$ is obtained for $g = g^t = f_{Z|Y}(\cdot|y_\omega; \theta)$ and has therefore $\mathcal{L}(g^t, \theta^t) = \sum_{\omega \in [\Omega]} \log f_Y(y_\omega; \theta^t)$.

Theorem A.2.9

The value of the likelihood $\sum_\omega \log f_Y(y_\omega; \theta^t)$ is nondecreasing with respect to t.

Algorithm A.2.3: EM algorithm

Input: The observed sample $\{y_\omega\}$; the likelihood of the latent variable $f_Z(.;\theta)$; a generator of $f_{Z|Y}(.|y;\theta)$

Parameter: the starting point θ^0, the tolerance ξ

Set $t = 0, \theta^t = \theta^0$

repeat

Algorithm A.2.3: EM algorithm

E-step : for each $\omega \in [\Omega]$ draw B i.i.d. points from the conditional distribution of Z given by $Y = y_\omega$ at the current estimate of θ :

$$z_{b\omega} \sim f_{Z|Y}(\cdot|y_\omega;\theta^t), \quad b \in [B]$$

M-step : update the estimate of θ by maximum likelihood estimation on the sample thus generated:

$$\theta^{t+1} = \arg\max_{\theta \in \mathbb{R}^p} \frac{1}{B} \sum_{b\omega \in [B \times \Omega]} \log f_Z(z_{b\omega};\theta)$$

$t = t + 1$

until $||\theta^{t+1} - \theta^t|| \leq \xi$;

Return: θ^t

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Appendix A.2 : Probability and Statistics

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Appendix A.3: Numerical analysis

***Discrete Choice Models*, appendix A.3, pp. 284-307**

Content

Appendix A.3.1. Vectors and tensors

Section content

- Tensors
- Hadamard product
- (Row-wise) Kronechker product
- Vectorization of matrix products
- Diagonal selection matrix
- Permutation matrix
- Commutation matrix

Introduction

If we have data $T_{x,y,z}$ for $x \in [X]$, $y \in [Y]$, and $z \in [Z]$, we may naturally view T as a three-dimensional tensor of shape (X, Y, Z);

But we may also lump the x and the y indices into $xy \in [X \times Y]$ and view T as a two-dimensional tensor (matrix) of shape (XY, Z);

Alternatively, we may also view the data as a matrix of shape (X, YZ); or we may stack all the data in a single vector and view T as a one-dimensional tensor of shape (XYZ).

Vectorization

Regardless of its shape, any tensor is represented as a sequence of numbers – the *vectorized* form – in the memory of the computer.

- Stacking the lines of matrices ("row-major order") is used by `C`, and is the default convention for `NumPy`, `PyTorch`, and `TensorFlow`. For example, a 2×2 matrix A becomes $vec_R(A) = (A_{11}, A_{12}, A_{21}, A_{22})$.
- Stacking the columns of matrices ("column-major order") is used by `Fortran`, `Matlab`, `R`, and most underlying core linear algebra libraries as `BLAS`. The matrix described above becomes $vec_C(A) = (A_{11}, A_{21}, A_{12}, A_{22})$.

Recall the standard notation $[N] = \{1, \ldots, N\}$. Given P integers $N_1, N_2, \ldots, N_P$, $[N_1 \times N_2 \times \cdots \times N_P]$ denotes the list of P-tuples $n_1 n_2 \ldots n_P$ where $n_p \in [N_p]$ for $p \in [P]$, and the P-tuples are ordered in the lexicographic increasing order.

Hadamard product

Given A and B two matrices (or more generally tensors) of the same size, the Hadamard product $A \odot B$ is the tensor whose entries are the element-wise product of the entries of A and B.

For instance, if A and B are two matrices of size $I \times J$, $(AB)_{ij} = A_{ij} B_{ij}$. In `NumPy`, the Hadamard product is implemented by the standard multiplication operator `*`.

Kronecker product

Given A a matrix of size $M \times N$ and B a matrix of size $P \times Q$, the *Kronecker product* $A \otimes B$ of A and B is the $MP \times NQ$ matrix defined by blocks as

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1N}B \\ \vdots & \ddots & \vdots \\ a_{M1}B & \cdots & a_{MN}B \end{bmatrix}.$$

The Kronecker product is intimately connected with vectorization. We have

$$\text{for all } mp \in [M \times P] \text{ and } nq \in [N \times Q],\ (A \otimes B)_{mp,nq} = a_{m,n} b_{p,q}.$$

If $u \in \mathbb{R}^I$ and $v \in \mathbb{R}^J$ are both column vectors, then $u \otimes v^\top$ is a rank-one matrix of size $I \times J$, and $vec_R(u \otimes v^\top) = u \otimes v$. We have the important identity

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

Vectorization of matrix products

If A is a matrix of size $M \times N$, if X is a matrix of size $N \times P$, and if B is a matrix of size $P \times Q$, we expect the vector of entries of the matrix product AXB to be a linear transform of the entries of the vector of entries of X. That is, there must be a relation of the form $vec_R(AXB) = K\, vec_R(X)$, where K is a matrix to be determined. We have

$$(AXB)_{m,q} = \sum_{n \in [N]} \sum_{p \in [P]} A_{m,n} X_{n,p} B_{p,q}$$

which we can rewrite in a vectorized way as $\sum_{np \in [N \times P]} (A_{m,n} B_{p,q}) X_{np}$. We note that $(A_{m,n} B_{p,q}) = (A_{m,n} B^\top_{q,p}) = (A \otimes B^\top)_{mq,np}$, which implies

$$(AXB)_{m,q} = \sum_{np \in [N \times P]} (A \otimes B^\top)_{mq,np} X_{np}.$$

As a result, the desired relation between the entries of AXB and those of X is

$$vec_R(AXB) = \left(A \otimes B^\top\right) vec_R(X).$$

Reduction and expansion

Consider a tensor $(U_{x,y,z})$ of shape (X, Y, Z). Its (sum) *reduction* with respect to the y is the tensor S of shape (X, Z) whose (X, Y) entry is $S_{xz} = \sum_{y \in [Y]} U_{xyz}$. The vector of entries of the reduced tensor S is given by

$$vec_R(S) = \left(I_X \otimes \mathbf{1}_Y^\top \otimes I_Z\right) vec_R(U).$$

Conversely, given a tensor $(V_{x,z})$ of shape (X, Z), its *expansion* of size Y between the first and second dimensions is the tensor T of shape (X, Y, Z) whose (x, y, z) entry is $T_{x,y,z} = V_{x,z}$. The vector of entries of the expanded tensor T is given by

$$vec_R(T) = (I_X \otimes \mathbf{1}_Y \otimes I_Z)\, vec_R(V).$$

In `NumPy` for instance, the tensor reduction operation would be implemented using `S = U.sum(axis=1)`, and the tensor expansion operation would be implemented using `T = V[:,None,:]`.

Row-wise Kronecker product

We define the row-wise Kronecker product (also called Khatri-Rao product) $\circledast$ as an operator acting on two matrices with the name number of rows, say a matrix A of size $I \times J$ and a matrix B of size $I \times K$. The row-wise Kronecker product is defined as the $I \times JK$ matrix whose ith row is the Kronecker product of the ith row of A and the ith row of B. That is, for $i \in [I]$ and $jk \in [J \times K]$,

$$(A \circledast B)_{i,jk} = A_{ij} B_{ik}.$$

so that if a and b are column vectors, then $a \circledast b = a \otimes b$ and $a^\top \circledast b^\top = a^\top \odot b^\top$. Hence, the row-wise Kronecker product acts on columns vectors as the standard Kronecker product, and on row vectors as the Hadamard product.

Diagonal selection matrix (1/2)

For a positive integer I, we introduce the *diagonal selection matrix* $\mathfrak{D}_I$, a matrix of size $I \times I^2$ defined by

$$\mathfrak{D}_I := \sum_{i \in [I]} \boldsymbol{e}_I^i \otimes (\boldsymbol{e}_I^i)^\top \otimes (\boldsymbol{e}_I^i)^\top,$$

where we recall that e_I^i is the i^{th} vector of the canonical basis of $\mathbb{R}^I$. This tensor has the important property that if A is a matrix of size $I^2 \times J$ whose rows are doubly indexed by $ii' \in [I^2]$ and whose columns are indexed by $j \in [J]$, so that the term of A is $(A_{ii',j})$, then $\mathfrak{D}_I A$ is a matrix of size $I \times J$ such that

$$(\mathfrak{D}_I A)_{ij} = A_{ii,j}.$$

Diagonal selection matrix (2/2)

In particular, if P is a $K \times K$ square matrix, then $vec_R(P)$ is a column vector of size K^2, and $\mathfrak{D}_K vec_R(P)$ is a column vector of size K, and we have $(\mathfrak{D}_K vec_R(P))_k = P_{k,k}$, and thus, calling $diagonal(P)$ the diagonal vector of the square matrix P

$$\mathfrak{D}_K vec_R(P) = diagonal(P).$$

Conversely, if v is a column vector of size K, then, calling $\mathbf{\Delta}_v$ the $K \times K$ diagonal matrix whose diagonal is the vector v, we have $\mathfrak{D}_K v = vec_R(\mathbf{\Delta}_v)$. We have the identity

$$A \circledast B = \mathfrak{D}_I(A \otimes B).$$

If $f : \mathbb{R}^I \to \mathbb{R}^K$, one defines the first differential df and the second differential $d^2 f$ at $x \in \mathbb{R}^I$ as the matrices of respective size $K \times I$ and $K \times I^2$ such that

$$df\left(x\right) = \sum_{ik \in [I \times K]} \frac{\partial f_k\left(x\right)}{\partial x_i} \mathbf{e}_K^k \otimes \left(\mathbf{e}_I^i\right)^\top \text{ and } d^2 f\left(x\right) = \sum_{ijk \in [I \times I \times K]} \frac{\partial^2 f_k\left(x\right)}{\partial x_i \partial x_j} \mathbf{e}_K^k \otimes \left(\mathbf{e}_I^i\right)^\top \otimes \left(\mathbf{e}_I^j\right)^\top,$$

so that df is the usual Jacobian of f. More generally, if $f(x)$ is valued in the space of matrices of size $K \times L$, then the differential $df(x)$ is a matrix of size $K \times IL$ given by

$$df(x) = \sum_{ikl} \frac{\partial f_{kl}(x)}{\partial x_i} \mathbf{e}_K^k \otimes \left(\mathbf{e}_I^i\right)^\top \otimes \left(\mathbf{e}_L^l\right)^\top$$

so that, when $K = 1$ $d^2 f = d(df)$.

If a function $f : \mathbb{R}^I \to \mathbb{R}^J$ is given by $f(x) = g(h(x))$ where $h : \mathbb{R}^I \to \mathbb{R}^J$ and $g : \mathbb{R} \to \mathbb{R}$ with the convention that if $y \in \mathbb{R}^J$, $g(y)$ is the vector of $\mathbb{R}^J$ whose j^{th} entry is $g(y_j)$, and if g and h are twice differentiable, we have

$$df(x) = \mathbf{\Delta}_{g'(h(x))} dh(x)$$

where for any $y \in \mathbb{R}^J$, $\mathbf{\Delta}_y$ denotes the $J \times J$ diagonal matrix with y on the diagonal, and

$$d^2 f(x) = \mathbf{\Delta}_{g'(h(x))} d^2 h(x) + \mathbf{\Delta}_{g''(h(x))} dh(x)^{\circledast 2},$$

where $dh(x)^{\circledast 2}$ denotes $dh(x) \circledast dh(x)$. As $dh(x)$ is a $J \times I$ matrix, $dh(x)^{\circledast 2}$ is a $J \times I^2$ matrix, the row-wise Kronecker product of $dh(x)$ by itself.

Differentiation (3/3)

If $f : \mathbb{R}^P \to \mathbb{R}^I$, and if $\mathbf{\Delta}_f : \mathbb{R}^P \to \mathbb{R}^{I \times I}$ is the function that sends x to the diagonal matrix with $f(x)$ on its diagonal, then

$$d\mathbf{\Delta}_f(x) = df(x) \circledast \boldsymbol{I}_I.$$

If $f(x)$ and $g(x)$ are two matrix-valued functions of a parameter $x \in \mathbb{R}^P$ with respective shapes $I \times J$, is $J \times K$, the differential of the matrix product fg is given by

$$d\left(fg\right)(x) = df(x)\left(\boldsymbol{I}_P \otimes g(x)\right) + f(x)dg(x).$$

If $f(x)$ and $g(x)$ are two matrix-valued functions of a parameter $x \in \mathbb{R}^P$ with the same shape $I \times J$, the differential of the Hadamard product is given by

$$d\left(f \odot g\right)(x) = df(x) \odot \left(1_P^\top \otimes g\right) + \left(1_P^\top \otimes f\right) \odot dg(x).$$

Appendix A.3.2. Schur complement

Section content

- Schur complement

Description of Schur complement

Consider a block matrix M written in the form:

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix},$$

where A, B, C, and D are submatrices of appropriate dimensions. The Schur complement provides a way to reduce or simplify the inversion of M under certain conditions. The Schur complement of the block A in the matrix M is defined as:

$$S = D - CA^{-1}B,$$

provided that A is invertible. Similarly, the Schur complement of the block D is:

$$T = A - BD^{-1}C,$$

provided that D is invertible.

Properties of the Schur complement

(i) If M is symmetric and positive definite, then both Schur complements S and T are also symmetric and positive definite.

(ii) The determinant of M can be expressed using the respective Schur complements as:

$$\det(M) = \det(A)\det(S) = \det(T)\det(D),$$

when A (and, respectively D) is invertible

(iii) The inverse of M can be written as:

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{bmatrix},$$

when both A and S are invertible, and

$$M^{-1} = \begin{bmatrix} T^{-1} & -T^{-1}BD^{-1} \\ -D^{-1}CT^{-1} & D^{-1} + D^{-1}CT^{-1}BD^{-1} \end{bmatrix},$$

Appendix A.3.3. Automatic differentiation

Section content

- Automatic differentiation
- The `torch` library

Automatic differentiation

Automatic differentiation (AD) leverages the fact that every computer program, no matter how complex, executes a sequence of arithmetic operations and elementary functions.

- The **forward step** computes the values of each intermediate variable based on the input variables as it progresses through the computational graph from inputs to outputs.
- The **backward step** computes the derivative of the output variable with respect to each intermediate variable, working backward from the output to the inputs. This mode is particularly efficient for functions with a large number of inputs and a small number of outputs, making it popular in machine learning.

Example

Consider a function $f(x, y) = x^2 + xy + \sin(y)$. To differentiate this function with respect to x and y, one would construct a computational graph representing the function, and then apply the automatic differentiation algorithm to compute the gradients. The derivatives of f with respect to x and y can be computed as follows:

$$\frac{\partial f}{\partial x}(x, y) = 2x + y, \qquad \frac{\partial f}{\partial y}(x, y) = x + \cos(y),$$

where these derivatives can be obtained by applying the chain rule to the computational graph of f.

Demonstration figure

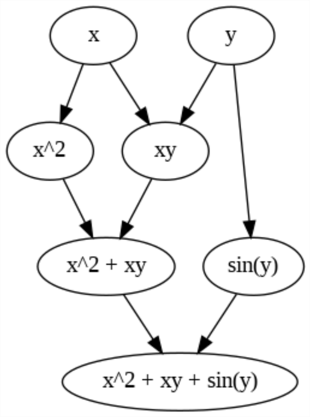

Figure 4: The graph shows the forward pass where the computation of the function $f(x, y)$ takes place, and the backward pass, where gradients are computed using the chain rule.

The `torch` library

The `torch.autograd` package in `PyTorch` automatically computes the gradients of tensors. Consider the above example, and the goal is to compute the gradient of $f(x, y)$ with respect to x and y.

```
import torch
x = torch.tensor([1.0], requires_grad=True)
y = torch.tensor([2.0], requires_grad=True)
f = x * x + x * y + torch.sin(y)
f.backward()
print(x.grad,y.grad)
```

Automatic differentiation in `PyTorch` is implemented using a computational graph. This graph is a directed acyclic graph (DAG) where nodes represent operations or variables, and edges represent the dependencies between these operations.

Appendix A.3.4. Perron–Frobenius theory

Section content

- Abstract Shapley operators
- Nonexpansiveness of order-preserving operators
- Hilbert's projective metric
- The Perron–Frobenius theorem

Abstract Shapley operators

In the context of repeated zero-sum games, given a discounting factor $\beta \in [0,1]$, the *Shapley operator* is a function $S^\beta : \mathbb{R}^X \to \mathbb{R}^X$ defined by

$$S_x^\beta(u) = \max_{y \in [Y]} \min_{z \in [Z]} \left\{ \Phi_{xyz} + \beta \sum_{x' \in [X]} P_{x'|xyz} u_{x'} \right\},$$

where $x \in [X]$ is the state of the game, $y \in [Y]$ is the action of player 1, $z \in [Z]$ is the action of the opponent, and Φ_{xyz} is the short-term payoff that player 1 receives from player 2 if the game is in state x, if player 1 plays action y and player 2 plays action z. $P_{x'|xyz}$ is the probability of a transition from x to x' if the players' actions are y and z.

Some properties

For $u \in \mathbb{R}^X$, let $|u|_\infty = \max_x |u_x|$ denote the *infinity norm*. Recall that the partial order on $\mathbb{R}^X$ is such that $u \leq v$ if and only if $u_x \leq v_x$ for all $x \in [X]$.

Definition A.3.1

Let $F : \mathbb{R}^X \to \mathbb{R}^Y$. We say that

(i) F is unit-additive when $F(u + c\mathbf{1}_X) = F(u) + c\mathbf{1}_Y$;

(ii) F is unit-subadditive when $c \in \mathbb{R}_+$ implies $F(u + c\mathbf{1}_X) \leq F(u) + c\mathbf{1}_Y$;

(iii) F is order-preserving when $u \leq u'$ implies $F(u) \leq F(u')$;

(iv) F is nonexpansive for the infinity norm when $|F(u) - F(v)|_\infty \leq |u - v|_\infty$.

The notion of *abstract Shapley operators*, which is a generalization of Shapley operators, plays a crucial role in dynamic programming. Its formal definition is as follows:

Definition A.3.2

An abstract Shapley operator is a function $F : \mathbb{R}^X \to \mathbb{R}^X$ that is order-preserving and unit-additive.

One can easily see that the operator S^1 is an abstract Shapley operator, while for $\beta < 1$, S^β is the composition of S^1 with the homothetic contraction $u \mapsto \beta u$.

Nonexpansiveness of order-preserving operators

Theorem A.3.1 (Blackwell)

Assume $F : \mathbb{R}^X \to \mathbb{R}^X$ is unit-additive. Then the following two conditions are equivalent:

(i) F is order-preserving.

(ii) F is nonexpansive for the infinity norm.

Theorem A.3.2

Assume $F : \mathbb{R}^X \to \mathbb{R}^X$ is monotone. The following two conditions are equivalent:

(i) F is unit sub-additive.

(ii) F is nonexpansive for the infinity norm.

Hilbert's projective metric (1/3)

Given an $X \times X$ matrix A with positive coefficients, we cannot expect $u \mapsto Au$ to be a contraction mapping in general. For instance, if $A1_X = 1_X$ then every multiple of 1_X is a fixed point of A. However, we can show that A is a contraction mapping "up to a constant" for a well-chosen metric. We will make this statement precise using Hilbert's projective metric and Birkhoff's theorem.

Let a and b be two vectors of $\mathbb{R}^X_{++}$, the set of vectors with strictly positive entries. We define Hilbert's projective metric d_H by

$$d_H(a,b) = \log \max_{xx' \in [X \times X]} \left\{ \frac{a_x b_{x'}}{a_{x'} b_x} \right\}.$$

Hilbert's projective metric (2/3)

From setting $x = x'$, we see that $d_H(a, b) \geq 0$ for all a and b. d_H is not a distance since $d_H(a, \lambda a) = 0$ for any $\lambda > 0$; however, it satisfies the following modification of the axioms that define a distance:

- $d_H(a, b) = 0$ if and only if $b = \lambda a$ for some $\lambda > 0$,
- symmetry: $d_H(a, b) = d_H(b, a)$,
- triangle inequality: $d_H(a, b) \leq d_H(a, c) + d_H(c, b)$.

As a result, d_H becomes a distance when restricted to a subset of $\mathbb{R}^X$ that has no two colinear elements. Given an $X \times X$ matrix A with positive terms A_{xy}, we define $\theta(A) = \sup\{d_H(Au, Av) : u, v \in \mathbb{R}^X_{++}\}$. One can show that this quantity is finite and has the more convenient expression

$$\theta(A) = \max_{x,x',y,y'} \frac{A_{xy} A_{x'y'}}{A_{xy'} A_{x'y}}.$$

By taking $x = x'$ and $y = y'$ we see that $\theta(A) \geq 1$. Furthermore:

Theorem A.3.3 (Birkhoff)

We have

$$\sup\left\{\frac{d_H(Au, Av)}{d_H(u, v)} : u, v \in \mathbb{R}^X_{++}, u \neq \lambda v\right\} =: \kappa(A) = \frac{\sqrt{\theta(A)} - 1}{\sqrt{\theta(A)} + 1}.$$

Further, we can show that for any positive starting point u^0, the sequence defined by $u^k = A^k u^0 / (\mathbf{1}_X^\top A^k u^0)$ converges to a positive vector $u^\star$ such that $d_H(Au^\star, u^\star) = 0$, hence $u^\star$ is the unique positive eigenvector of A such that $\mathbf{1}_X^\top u^\star = 1$.

Theorem A.3.4

Let A be an $X \times X$ matrix with positive terms. Then:

(i) There exists a unique positive real eigenvalue λ (the Perron–Frobenius eigenvalue) that is strictly greater than the absolute value of any other eigenvalue of A. Hence λ is the spectral radius of A.

(ii) A has a unique (up to scalar multiplication) strictly positive eigenvector u (i.e. such that $u_x > 0$ for all $x \in [X]$) corresponding to the eigenvalue λ, called the right Perron–Frobenius eigenvector.

Theorem A.3.4 (continued)

(iii) $A^\top$ has a unique (up to scalar multiplication) strictly positive eigenvector v corresponding to the eigenvalue λ, called the left Perron–Frobenius eigenvector.

(iv) The Perron–Frobenius eigenvalue λ is simple, meaning its algebraic multiplicity is 1.

(v) The convergence $A^t/\lambda^t \to uv^\top/(v^\top u)$ holds as $t \to +\infty$.

Appendix A.3.5. Monotone comparative statics

Section content

- Isotonicity
- Lattices
- Submodular functions
- Topkis' theorem
- Stieltjes matrices

Given X a subset of $\mathbb{R}^n$, and Y a subset of $\mathbb{R}^m$, and $f : X \longmapsto Y$ a function from X to Y, one says that f is *isotone* if $x \leq y$ implies $f(x) \leq f(y)$. One says that f is *antitone* if $x \leq y$ implies $f(x) \geq f(y)$. One says that f is *inverse isotone* if $f(x) \leq f(y)$ implies $x \leq y$. When $n = m = 1$, f is inverse isotone if and only if f is strictly increasing, namely $x < y$ implies $f(x) < f(y)$. However, this relies on the fact that $\leq$ is a total order on $\mathbb{R}$, and does not extend in general to higher dimension. However, the following remark makes inverse isotone functions very useful:

Proposition A.3.1

Let $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^m$, and let $f : X \longmapsto Y$ be an inverse isotone function. Then f is injective.

For two vectors x and y in $\mathbb{R}^n$, we denote $x \wedge y$ the vector in $\mathbb{R}^n$ whose ith entry is $\min\{x_i, y_i\}$, and $x \vee y$ the vector in $\mathbb{R}^n$ whose ith entry is $\max\{x_i, y_i\}$.

Definition A.3.3

$X \subseteq \mathbb{R}^n$ is a sublattice (of $\mathbb{R}^n$) if for any $x, x' \in X$, $x \wedge x' \in X$ and $x \vee x' \in X$.

A lattice (of $\mathbb{R}^n$) is a subset of $\mathbb{R}^n$ such that for any x and x', the set of z such that $z \leq x$ and $z \leq x'$ hold simultaneously has a smallest largest element denoted $x \sqcap x'$; and the set of z such that $z \geq x$ and $z \geq x'$ hold simultaneously has a smallest element denoted $x \sqcup x'$. Clearly, a sublattice is a lattice; however, the converse fails to hold.

Veinott's lattice order

An important partial order on sublattices is the *lattice order*, or the *strong set ordering*.

Definition A.3.4 (lattice order)

Let X and X' be two subsets of $\mathbb{R}^n$. One says that X is smaller than X' in the lattice order, denoted $X \leq_L X'$, if for any $x \in X$ and $x' \in X'$, $x \wedge x' \in X$ and $x \vee x' \in X'$.

A partial order $\preceq$ should be reflexive, antisymmetric, and transitive. It is easy to see that $X \leq_L X$ holds if and only if X is a sublattice.

Proposition A.3.2

The relation $\leq_L$ is a partial order on sublattices.

Let $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^m$. Consider a function $f : X \times Y \to \mathbb{R}$. One says $f(x, y)$ has *increasing differences* in (x, y) if for $y \leq y'$, $f(x, y') - f(x, y)$ is nondecreasing in x.

Definition A.3.5 (submodular function)

If X is a sublattice of $\mathbb{R}^n$, one says that $f : X \to \mathbb{R}$ is *submodular* if for any x and x' in X, $f(x \wedge x') + f(x \vee x') \leq f(x) + f(x')$.

One says that f is *supermodular* if $-f$ is submodular.

One says that f is *modular* if f is both sub- and supermodular. If f is defined on a product of intervals, then f is modular if and only if $f(x) = \sum_{i=1}^{n} f_i(x_i)$.

If f is twice continuously differentiably, then it is submodular if and only if $\partial^2 f(x) \partial x_i \partial x_j \leq 0$ for all $i \neq j$, and supermodular if and only if the inequality is reversed.

Example A.5

The following are examples of sub- and supermodular functions:

(i) If $F : \mathbb{R}^2 \to \mathbb{R}$ is the c.d.f. of a bivariate random vector, then F is supermodular.

(ii) If $\varphi : \mathbb{R}^n \to \mathbb{R}$ is convex, then $f(x) = \varphi\left(\sum_{i=1}^n x_i\right)$ is supermodular.

(iii) If $a_{ij} \geq 0$, then $f(x) = \sum_{1 \leq i,j \leq n} a_{ij} \left|x_i - x_j\right|^2$ is submodular.

Topkis' theorem

Topkis' theorem is an important result in monotone comparative statics.

Theorem A.3.5

Let X be a sublattice of $\mathbb{R}^n$ and Y a subset of $\mathbb{R}^m$, and f a function $X \times Y \mapsto \mathbb{R} \cup \{-\infty\}$ such that:

(i) $f(., y)$ is supermodular for each $y \in Y$,

(ii) $f(x, y)$ has increasing differences in (x, y) on $X \times Y$,

(iii) $\arg\max_{x \in X} f(x, y)$ is nonempty for each $y \in Y$

Then $y \mapsto \arg\max_{x \in X} f(x, y)$ is isotone in the lattice order.

Consider f a submodular function. The function $(x, y) \mapsto x^\top y - f(x)$ is supermodular, and it is straightfoward to show that $f^*(y) = \sup_{x \in \mathbb{R}^n} \{x^\top y - f(x)\}$ is supermodular too. We have therefore:

Proposition A.3.3

Let X be a sublattice of $\mathbb{R}^n$, and $f : X \to \mathbb{R}$ be a submodular function. Then f^*, the Legendre-Fenchel transform of f, is supermodular.

When $n = 2$, the converse holds true in some sense: if f is convex and f^* is supermodular, then f is submodular. When $n > 2$, the converse no longer holds: supermodularity of f^* does not imply submodularity of f in general.

Example A.6

The following functions f are submodular and convex, and their Legendre-Fenchel transforms f^* are supermodular and convex:
(i) $f(x) = \log\left(1 + \sum_{i=1}^d \exp(x_i)\right)$, whose conjugate is $f^*(y) = \sum_{i=0}^d y_i \ln y_i$, where $y_0 := 1 - \sum_{i=1}^d y_i$.
(ii) $f(x) = \max_{i=1}^d \{x_i\}$, whose conjugate is $f^*(y) = 0$ if $y \geq 0$ and $\sum_{i=1}^d y_i = 1$ and , $f^*(y) = +\infty$ else.
(iii) Consider the constant elasticity of substitution (CES) function: $f(x) = \left(\sum_i x_i^r\right)^{1/r}$. Then f is submodular and convex for $r > 1$, and supermodular and concave for $r < 1$.

Proposition A.3.4

If $f:\mathbb{R}^d \to \mathbb{R}^d$ is submodular, then $y \mapsto \partial f^*(y)$ is isotone in the Veinott strong set order: if $x \in \partial f^*(y)$ and $x' \in \partial f^*(y')$ are such that $y \leq y'$, then $x \wedge x' \in \partial f^*(y)$ and $x \vee x' \in \partial f^*(y')$.

If $f:\mathbb{R}^d \to \mathbb{R}^d$ is submodular and strictly convex, then $y \mapsto \nabla f^*(y)$ is isotone.

Take a strictly convex and submodular quadratic functions, which is given by $f(x) = x^{\intercal} S x / 2$, where S is symmetric positive definite and is a Z-matrix.

Definition A.3.6

A Stieltjes matrix is a symmetric positive definite matrix with nonpositive off-diagonal entries.

Proposition A.3.5

Let S be a symmetric $d \times d$ matrix. Then S is a Stieltjes matrix if and only

$$x^{\top} S x^{+} \geq 0$$

holds for all $x \in \mathbb{R}^{d}$, with a strict inequality unless $x \leq 0$.

Appendix A.3.6. Z- and M- functions

Section content

- Z-functions
- Nonreversingness
- M-functions

Definition of Z-function

One says that a function $f : \mathbb{R}^X \to \mathbb{R}^Y$ is *isotone* as a synonymous for "order-preserving," i.e. if $x \leq y$ implies $f(x) \leq f(y)$. One says that $f : \mathbb{R}^X \to \mathbb{R}^Y$ is *inverse isotone* if $f(x) \leq f(y)$ implies $x \leq y$.

Definition A.3.7

One says that a function $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$ is a Z-function if any $\boldsymbol{q}_x$ is nonincreasing with respect to p_y for $y \neq x$.

Examples

The term "Z-function" has many synonyms, including *off-diagonally antitone* function, or function with the *gross substitutes* property.

(i) The function $\boldsymbol{q}(p) = Ap$ where $A_{xy} \leq 0$ for $x \neq y$. Such a square matrix whose off-diagonal terms are nonpositive is called a *Z-matrix*.

(ii) The function $\boldsymbol{q}$ defined by

$$\boldsymbol{q}_z(p) = \sum_{x \in [X]} \exp(p_z - p_x - c_{xz}) - \sum_{y \in [X]} \exp\left(p_y - p_z - c_{zy}\right), \quad \forall z \in [X].$$

(iii) If $\boldsymbol{q}$ is a Z-function, then $\tilde{\boldsymbol{q}}(p) := \boldsymbol{q}(p^+) - p^-$ is a Z-function.

(iv) If $f : \mathbb{R}^X \to \mathbb{R}$ is a differentiable function, then f is submodular if and only if $p \to \nabla(p)f$ is a Z-function.

A characterization of Z-functions

The following result offers a characterization of Z-functions in the differentiable case.

Proposition A.3.6

Let $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$ be a differentiable function. Then $\boldsymbol{q}$ is a Z-function if and only if its Jacobian $D\boldsymbol{q}\,(p)$ is a Z-matrix at each point $p \in \mathbb{R}^X$.

Definition A.3.8 (weak nonreversingness)

We say that $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$ is weakly nonreversing if

$$p \leq p' \text{ and } \boldsymbol{q}(p) \geq \boldsymbol{q}\left(p'\right) \quad \text{imply} \quad \boldsymbol{q}(p) = \boldsymbol{q}\left(p'\right).$$

Definition A.3.9 (strong nonreversingness)

We say that $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$ is strongly nonreversing if

$$p \leq p' \text{ and } \boldsymbol{q}(p) \geq \boldsymbol{q}\left(p'\right) \quad \text{imply} \quad p = p'.$$

Example of nonreversingness

(i) An isotone function is weakly nonreversing;

(ii) If $\boldsymbol{q}(p) = \nabla u(p)$, where $u : \mathbb{R}^X \to \mathbb{R}$ is smooth and convex, then $\boldsymbol{q}$ is weakly nonreversing; if in addition, u is strictly convex, then $\boldsymbol{q}$ is strongly nonreversing;

(iii) If $\boldsymbol{q}(p) = \arg\max_q \varphi(p, q)$ where $\varphi : \mathbb{R}^X \times \mathbb{R}^X \to \mathbb{R}^X$ satisfies increasing differences, and is such that the arg max is a single point, then $\boldsymbol{q}$ is weakly nonreversing;

(iv) If there is a vector $w \in \mathbb{R}_+^X$ such that $p \mapsto w^\top \boldsymbol{q}(p)$ is isotone, then $\boldsymbol{q}$ is weakly nonreversing;

(v) If $\boldsymbol{q}$ is inverse isotone, then $\boldsymbol{q}$ is strongly nonreversing;

(vi) If $\boldsymbol{q}$ is defined on the simplex $\Delta = \{p \in \mathbb{R}^X : \sum_{x \in [X]} p_x = 1\}$, then $\boldsymbol{q}$ is strongly nonreversing.

Definition A.3.10 (M0-function)

We say that q is an M0-function if it is a Z-function and weakly nonreversing.

Definition A.3.11 (M-function)

We say that q is an M-function if it is a Z-function and strongly nonreversing.

Theorem A.3.6

Let u be a convex and submodular function that is continuously differentiable. Then ∇u is an M0-function. Further, if u is strictly convex, then ∇u is an M-function.

Important characterizations of M0- and M-functions (1/2)

Theorem A.3.7

Let $\boldsymbol{q}$ be a Z-function. Then the following statements are equivalent:

(i) $\boldsymbol{q}$ is weakly nonreversing (i.e. $\boldsymbol{q}$ is an M0-function), and

(ii) $\boldsymbol{q}^{-1}$ is increasing in the strong set order, that is: if $\boldsymbol{q}(p) \leq \boldsymbol{q}(p')$, then $\boldsymbol{q}(p) = \boldsymbol{q}(p \wedge p')$ and $\boldsymbol{q}(p') = \boldsymbol{q}(p \vee p')$.

Proposition A.3.7

Let $f : \mathbb{R}^X \to \mathbb{R}$ be a convex submodular function that is C^1. Then $\boldsymbol{q}$ defined by $\boldsymbol{q}(p) = \nabla f(p)$ is an M0-function. Further, if f is strictly convex, then $\boldsymbol{q}$ is an M-function.

Important characterizations of M0- and M-functions (2/2)

Theorem A.3.8

Let $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$ be a Z-function. Then the following are equivalent:

(i) $\boldsymbol{q}$ is strongly nonreversing (i.e. $\boldsymbol{q}$ is an M-function), and

(ii) $\boldsymbol{q}^{-1}$ is an isotone function on its domain, that is: if $\boldsymbol{q}(p) \leq \boldsymbol{q}(p')$, then $p \leq p'$.

Let $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$ be a function for which we define the following possible assumptions:

Assumption A.2 (weak substitutes)

$\boldsymbol{q}$ is a Z-function, i.e. $\boldsymbol{q}_x(p)$ is nonincreasing in p_y for each $x \neq y$ in $[X]$.

Assumption A.3 (strong substitutes)

$\boldsymbol{q}_x(p)$ is decreasing in p_y for each $x \neq y$ in $[X]$.

Assumptions (2/2)

Assumption A.4

The aggregate supply $\sum_{x\in[X]} \boldsymbol{q}_x(p)$ is nondecreasing in p_x for each $x \in [X]$.

Assumption A.5

The aggregate supply $\sum_{x\in[X]} \boldsymbol{q}_x(p)$ is increasing in p_x for each $x \in [X]$.

Restating

Defining

$$\boldsymbol{q}_0(p) = 1 - \sum_{x \in [X]} \boldsymbol{q}_x(p),$$

we can restate the weak (respectively, strong) law of aggregate supply by saying that $\boldsymbol{q}_0(p)$ is non increasing (respectively, decreasing) in p_y for each $x \in [X]$.

Assumption A.6

For any $B \subseteq [X]$, for any p and $p' \in \mathbb{R}^X$ such that $p_x = p'_x$ for all $x \in B$ and $p_x > p'_x$ for $x \notin B$, there exists $x \in B \cup \{0\}$ such that $\boldsymbol{q}_x(p) < \boldsymbol{q}_x(p')$.

The connected strong substitutes assumption is a middle ground between weak substitutes, which assumes that $\boldsymbol{q}_x$ is weakly decreasing in p_y, and strong substitutes, which assumes that it is strictly decreasing.

Related implications

The following result provides important implications of weak substitutes and connected strong substitutes:

Theorem A.3.9

Consider a function $\boldsymbol{q} : \mathbb{R}^X \to \mathbb{R}^X$. Then:

(i) If $\boldsymbol{q}$ satisfies weak substitutes and the weak law of aggregate supply, then it is an M0-function.

(ii) If $\boldsymbol{q}$ satisfies weak substitutes, the weak law of aggregate supply, and connected strong substitutes, then it is an M-function.

(iii) If $\boldsymbol{q}$ satisfies weak substitutes and the strong law of aggregate supply, then it is an M-function.

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Appendix A.3: Numerical Analysis

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